Identification, Disturbance Rejection, and Learning-Based Control of a Simulated Gas Turbine

A comparative study of tracking, fuel-command reversal, and the limits of model-based improvement

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Second edition revised in response to a jury-style assessment. A retrospective, undergraduate-accessible master's-level research report with detailed methods, accepted and rejected experiments, explicit evidence roles, and bounded contribution claims.

Research direction	Volkan Aran
Implementation and writing	AI-assisted under human direction

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Evidence-only revision. No new plant simulations, identification fits, controller searches, or Monte Carlo runs were performed. Degree acceptance and independent mastery require a human university examination; neither is asserted by this report.

Abstract

This thesis-style report investigates a restricted engineering question: within an archived nonlinear gas-turbine simulation campaign, when do identified or learned models improve the tested control configurations, and which improvements can be attributed to model representation rather than feedback architecture, tuning, or scoring choices? The study is retrospective. It synthesizes existing experiments rather than claiming a newly balanced benchmark or a new control theory.

The plant is a single-shaft T-MATS model with fuel-flow actuation, filtered speed sensing, and optional shaft-power extraction. Numerical acceptance is established before identification or control scores are interpreted. The methods include Wiener and recursive ARX identification, sparse lag selection, filtered-inverse disturbance observers, alternative disturbance estimators, inverse Gaussian-process feedback, polynomial replacement, and PILCO-style offline PI gain optimization. Detailed derivations explain prediction, inversion, implicit feedback action, and an expected excess-variation objective for reducing command reversals.

The principal finding is that improved regression accuracy does not reliably predict improved closed-loop performance in this archive. At one target, derivative-normalized GP feedback has speed RMSE of approximately 1.752 rpm, while a matched filtered PID gives 1.749 rpm; a GP-specific tracking advantage is therefore not demonstrated there. A polynomial inverse remains competitive despite poorer nominal prediction. Adding a ringing penalty reduces the original learned policy's command reversal by about 86 percent in its 10 percent load selection case, but the resulting reversal remains slightly above base PI. The common-sensor audit reproduces prior traces exactly and exposes a reversal of controller ordering between actual-speed and sensed-speed ramp error.

In the final combined cycle, adding DOB98 with original PI gains reduces full-cycle RMSE from 5.408 to 4.546 rpm; subsequent PI retuning reduces it to 3.142 rpm. This is a configured-system improvement on the DOB tuning case, accompanied by greater command reversal and fuel saturation. An equally retuned PI-only comparator is absent, and the model identified near 9800 rpm spans only about 6.3 rpm in its training record. These facts prevent attribution of the complete improvement to the observer architecture or extrapolation to a robust engine-wide guarantee.

The master's-level contribution is an implementation-grounded comparative synthesis: explicit separation of prediction, selection, transfer and implementation checks; analysis of existing ablations; and a reproducible account of favorable, unfavorable and invalid outcomes. The expected excess-variation construction is presented as a technical adaptation whose originality remains unestablished. The revised manuscript narrows its claims accordingly, preserves the full study lineage, and identifies the additional evidence needed for stronger conclusions.

Revision scope and reading guide

Version 1 provided a complete chronological research narrative. The written jury-style review judged that the project had substantial master's-level potential but required a focused question, a literature-grounded contribution statement, clearer evidence roles, and stronger attribution discipline [19]. Version 2 addresses the documentary and analytical requirements. It does not relabel missing experiments as completed work.

Chapters 1–3 define the question, literature position, and retrospective protocol. Chapters 4–13 provide the physical explanation, mathematical methods and thematic evidence. Chapters 14–15 answer the question and state the defensible conclusions. Appendix A records each examination issue, its resolution or remaining limitation. The other appendices preserve detailed result tables, algorithms, evidence identifiers and the directory register. The original edition remains a separate historical artifact; numerical results are not replaced by newly generated simulations.

The primary reader is assumed to know elementary calculus, linear algebra, and introductory feedback control. Each advanced method is explained from its governing equations before its results are interpreted. Subsection details can be revisited after the main conclusions are understood. A chapter's historical use of the words "training" or "validation" must be read together with the evidence-role ledger in Chapter 3; a dataset may be held out from a regression fit while still used to select controller gains.

Authorship and examination status. Volkan Aran directed the project. Implementation, analysis and writing were developed with OpenAI Codex. The jury-style review and the response are AI-assisted documents, not independent human peer review. This report neither certifies candidate independence nor records an oral defence. Mathematical understanding, reproducibility, and authorship responsibility must be demonstrated to an actual examiner. The revision response therefore distinguishes a clarified manuscript from a fulfilled institutional requirement.

Notation and units

Symbol	Meaning
N, N_s, r	Actual shaft speed, sensed speed, and governed target; rpm.
u, W_f, I	Fuel command/flow and controller integral contribution; lbm/s.
$\dot{N}, \hat{a}$	Actual acceleration and causal estimated acceleration; rpm/s.
T_s, q	Sample period 0.015 s; one-sample delay $q = z^{-1}$.
T, P	Inlet absolute temperature and pressure; K and kPa for inverse models.
P_t, P_c, P_L	Turbine, signed compressor, and external extraction power; model hp.
K_p, K_i, K_d	Controller gains; units depend on speed or virtual-fuel error definition.
P_n, H, Q	Nominal plant, sensor approximation, and observer low-pass filter.
g, p_2	Learned inverse GP mean and full quadratic inverse.
μ, Σ	Gaussian mean vector and covariance matrix.
R, TV	Excess fuel variation and total variation; lbm/s, not consumed mass.
SM, PTO	Available compressor surge margin; external shaft power takeoff.
NR	No confirmed recovery within the stated record/band, where used in tables.

Symbols are reused locally in some historical derivations. Each chapter defines its state and normalization before use. In particular, the GP training matrix and controller gain must not be confused merely because both sometimes use the letter K.

Contents

Abstract i

Revision scope and reading guide ii

Notation and units iii

1 Research question, objectives, and bounded contribution 1

1.1 The engineering problem . . . 1
1.2 Primary question . . . 1
1.3 Subsidiary questions and observable answers . . . 1
1.4 What the study contributes . . . 2
1.5 Claims explicitly excluded . . . 2
1.6 Organization of the argument . . . 3

2 Literature position and assessment of originality 4

2.1 Review scope and method . . . 4
2.2 Simulation platform and engine-control context . . . 4
2.3 Identification and sparse model selection . . . 4
2.4 Disturbance observers and related estimators . . . 5
2.5 GP regression, feedforward and inverse coordinates . . . 5
2.6 PILCO and the restricted implementation used here . . . 5
2.7 Smooth actions and the excess-variation adaptation . . . 6
2.8 Contribution matrix . . . 6
2.9 The defensible gap . . . 6

3 Retrospective evidence protocol and comparison rules 7

3.1 What was and was not changed for Version 2 . . . 7
3.2 Evidence roles . . . 7
3.3 Evidence-role ledger . . . 7
3.4 What a matched comparison requires . . . 9
3.5 Primary response and phase weighting . . . 9
3.6 Fuel command, reversal and recovery . . . 10
3.7 Numerical and operating-domain acceptance . . . 10

3.8 Statistical interpretation 10
3.9 Why missing comparisons remain missing 11

4 Plant, sensing, metrics, and numerical acceptance 12
4.1 Physical plant and feedback foundations 12
4.2 Experimental design and the first numerical boundary 14
4.3 Identification primer: dynamics, regressors, and prediction 16

5 Identification quality and the limits of prediction 18
5.1 Wiener identification: a nonlinear output map after linear dynamics 18
5.2 Online acceleration identification with ten-term RLS 20
5.3 Realizing the estimator with ordinary Simulink blocks 24
5.4 Two-stage sparse RLS: conditioning and held-out prediction 28
5.5 Answer supported by this evidence 29

6 Disturbance estimation and the accepted load envelope 30
6.1 Understanding disturbance observers before the case studies 30
6.2 The first filtered-inverse DOB and governed-reference tests 32
6.3 External shaft-load rejection at 10000 rpm 37
6.4 The rejected 20 and 30 percent tests at 10000 rpm 39
6.5 Moving the shaft-load experiment to 9500 rpm 40
6.6 The 9000-rpm ramped-load envelope 42
6.7 LESO, UDE, and GPIO implementation and first comparison 50
6.8 Interpretation of the observer comparison 54

7 Gaussian-process prediction and its control interpretation 55
7.1 Gaussian-process regression explained from first principles 55
7.2 Forward exact GP: speed and fuel to acceleration 57
7.3 Inverse exact GP: desired motion to nominal fuel 57
7.4 Observer transfer and the direct GPR feedforward challenger 58

8 Inverse-coordinate feedback, feedforward, and gain selection 68
8.1 Protocol for the inverse-model controller comparisons 68
8.2 From inverse identification to feedback architecture 71
8.3 Dual inverse-model virtual-fuel control 73
8.4 Uncertainty scheduling: two competing hypotheses 75
8.5 Fixed-gain low-ringing design and four-way comparison 79

9 Environmental transfer, matched PID, and coverage 82

9.1 Four-input inverse GPR across ambient conditions 82

9.2 Three remedies and the emergence of R3 85

9.3 Moving speed and ambient conditions: coverage becomes decisive 89

9.4 Static feedforward clarified by an explicit ablation 93

10 Polynomial replacement and local tuning tradeoffs 96

10.1 A full quadratic inverse with unchanged R3 gains 96

10.2 Local Poly2 PID retuning at 9250 rpm 98

11 PILCO-style offline PI tuning and expected fuel reversal 104

11.1 PILCO-style policy learning: full derivation and first results 104

11.2 Learning smoother control with an expected ringing penalty 111

12 Frozen-controller acceleration and the combined cycle 121

12.1 Five-second no-load acceleration and the common-sensor audit 121

12.2 Combined acceleration and a five-second load pulse 124

13 Local DOB98 design, PI retuning, and attribution 128

13.1 A new inverse model at 9800 rpm and PI tuning after DOB engagement 128

13.2 Detailed interpretation of the final sensor and DOB results 131

14 Cross-study answers, attribution, and validity 135

14.1 RQ1: prediction quality is necessary for some uses, but not sufficient 135

14.2 RQ2: what the existing ablations identify 135

14.3 RQ3: objective choice and sensing change the result 137

14.4 RQ4: transfer is limited and multidimensional 138

14.5 Threats to validity and the remaining effect on claims 138

15 Conclusions, practical interpretation, and future work 140

15.1 Answer to the primary question 140

15.2 Conclusions organized by contribution 140

15.3 Implications for choosing among the existing configurations 141

15.4 A future protocol for the unresolved architecture question 141

15.5 A future protocol for the penalty and robustness questions 141

15.6 Master's revision status 142

A Response to the jury-style examination 143

A.1 Purpose and status conventions 143
A.2 Response to the six master's requirements 144
A.3 Defence questions and the reasoning an answer should contain 145
A.4 Preservation and reproducibility of this revision 145

B Complete early and fixed-controller result tables 147
B.1 Four fixed controllers 148

C Environmental transfer and inverse-slope audit 150

D Polynomial coefficients and full local gain search 152
D.1 All 9250 rpm tuning candidates 153

E Algorithm registry and reproducibility 154
E.1 Controller identities 154
E.2 Primary evidence register 154
E.3 September-paper V2 review evidence 155
E.4 Reproducing the historical September paper 155
E.5 Evidence exclusions and reader guidance 155

F Complete study-directory register 157

References 167

List of Figures

6.1 Overview. Archived study figure. 44
6.2 5% zoom. Archived study figure. 47
6.3 10% zoom. Archived study figure. 48
6.4 20% zoom. Archived study figure. 49
6.5 30% zoom. Archived study figure. 50
6.6 Overview. Archived study figure. 51
6.7 Zooms. Archived study figure. 52

7.1 Controller architecture. Archived study figure. 59
7.2 Six-controller response. Archived study figure. 63
7.3 Recovery zooms. Archived study figure. 64
7.4 Six-controller response. Archived study figure. 65
7.5 Recovery zooms. Archived study figure. 66
7.6 All loads. Archived study figure. 67

8.1 Four distinct evidential checks. Performance interpretation requires more than successful time integration. 71
8.2 Baseline speed PI and the early optional inverse correction. The load acts on the plant only. 72
8.3 Dual inverse queries create a fuel-domain error. The dashed direct setpoint branch is added in the later FF variant. 73
8.4 Uncertainty-to-gain calculation and integral compensation. The response-SD maximum is a design signal, not a certified loaded-plant error bound. 76
8.5 Implemented schedules. “Three SD” means three times the empirical reference SD; it is not a Gaussian tail-probability claim. 77
8.6 First load ramp at 9000 rpm / 10% load. The common fuel plateau does not reveal transient reversal; the enlarged application window does. Curves are regenerated from archived trajectories. 78
8.7 Realized response uncertainty and gain multiplier on the same case. Different gains change the trajectory and therefore the feedback-query SD; the two schedules do not necessarily see identical uncertainty histories. 78
8.8 Archived 9500 rpm / 5% fixed-gain tradeoff. The selected pair balances the declared ringing constraint and RMSE; it is not the minimum-RMSE point. 80

8.9 9500 rpm / 5% stepped load in the four-way experiment. The two inverse-controller traces coincide to numerical precision because the direct FF is constant and the integral offset compensates it. Archived legend names are retained. 81

9.1 Four-input exact-GP held-out fuel prediction and uncertainty from the archived environmental study. The aggregate contains 200 points across 33 represented environments. 83

9.2 First target load event with base PI and the four-input inverse PI. The enlarged window separates peak suppression from subsequent fuel oscillation and fine recovery. 84

9.3 Archived environmental RMSE comparison. Numerical scores must be interpreted together with the map-range and inverse-slope diagnostics, not as an unrestricted physical envelope. 84

9.4 R3 with static FF. Removing only the dashed direct FF branch gives the later feedback-only ablation. Replacing GP means and derivatives by Poly2 leaves the architecture intact. 86

9.5 Target load-application comparison of base PI, the previous inverse controller, and the three remedies. 88

9.6 Target unloading zoom. Removal transients are included in the ringing and recovery statistics; application-only plots would give an incomplete assessment. 89

9.7 Environmental transfer ratios from the archived remedy study. The complete numerical grid and fallback percentages are supplied in Appendix C. 90

9.8 Broad-cycle request and environmental route. The original 120 s route is retained in the later covered tests; only the speed excursion is narrowed. 91

9.9 Broad-cycle time-domain results. In the archived legend, "R3" includes static setpoint FF. The reported RMSE penalty must be read alongside 78.69% fallback and the domain flags. 92

9.10 Final covered-cycle zoom with base PI and the historical R3-with-FF configuration. Ramp and hold behavior must be evaluated separately. 93

9.11 Direct static FF and feedback/integral contributions in the covered-cycle ablation. Time-varying reference and ambient conditions prevent a single constant integral offset from reproducing the FF term throughout the cycle. 95

10.1 Matched model comparison. The polynomial uses the same training and held-out rows as the environmental GP; subsequent control trajectories are separate from the inverse-fit test. 96

10.2 Four-way covered-cycle comparison: base PI, R3 feedback only, R3 + static GPR FF, and R3 + static Poly2 FF. All use the same request and ambient route. 98

10.3 Covered-cycle zoom after replacing the inverse model by Poly2. Small differences are meaningful only within this matched experiment; the model-fit RMSE is a separate quantity. 99

10.4 All accepted Poly2 gain candidates at 9250 rpm, colored by the joint objective. The best RMSE-only point is not the joint selection. 101

10.5 9250 rpm / 10% load-pulse overview: base PI, original Poly2, and selected Poly2 gains. 101

10.6 First load application and removal at 9250 rpm. The retuned controller has a larger peak excursion than original Poly2 but a shorter fine tail and less excess fuel variation. 102

11.1 Policy-search objective and GP planning response. The shaded interval is model uncertainty, not a certified plant confidence band. 108

11.2 Full nonlinear 10% benchmark. Historical methods and new controller use matching time, request and disturbance profiles. 109

11.3 First pulse, emphasizing overshoot and command reversals. The new method is judged on these plant traces. 110

11.4 Frozen-policy 5% amplitude validation at the same operating point. 110

11.5 Measured 10% plant tradeoff and the declared selection score. These are candidate evaluations, not independent validation. 116

11.6 10% full test; only the controller gains differ between the two PILCO policies. . . . 118

11.7 First load pulse, showing the effect of the ringing penalty on physical speed and fuel command. 118

11.8 Frozen-policy 5% amplitude validation with the same timing and plant. 119

11.9 Ringing decomposed into six individual 10% load edges. Each bar is total fuel movement minus net movement in the same two-second window. 119

List of Tables

5.1 Wiener identification: a nonlinear output map after linear dynamics: Selected model and free-run fits. Archived values. . . . 18

5.2 Wiener identification: a nonlinear output map after linear dynamics: Accepted simulation checks. Archived values. . . . 19

5.3 Online acceleration identification with ten-term RLS: Results. Archived values. . . 21

5.4 Online acceleration identification with ten-term RLS: Scenario and validity. Archived values. . . . 21

5.5 Online acceleration identification with ten-term RLS: What works and what to improve. Archived values. . . . 23

5.6 Realizing the estimator with ordinary Simulink blocks: Experiment and results. Archived values. . . . 26

5.7 Realizing the estimator with ordinary Simulink blocks: Agreement with the previous implementation. Archived values. . . . 26

5.8 Two-stage sparse RLS: conditioning and held-out prediction: Study results. Archived values. . . . 28

6.1 The first filtered-inverse DOB and governed-reference tests: Results: accepted final configuration. Archived values. . . . 35

6.2 The first filtered-inverse DOB and governed-reference tests: Result files. Archived values. . . . 36

6.3 External shaft-load rejection at 10000 rpm: Comparison. Archived values. 38

6.4 External shaft-load rejection at 10000 rpm: Recovery after each edge. Archived values. 38

6.5 The rejected 20 and 30 percent tests at 10000 rpm: Validity and first-pulse results. Archived values. . . . 39

6.6 The rejected 20 and 30 percent tests at 10000 rpm: Saturation and interpretation. Archived values. . . . 40

6.7 Moving the shaft-load experiment to 9500 rpm: Results. Archived values. 41

6.8 Moving the shaft-load experiment to 9500 rpm: Recovery and limits. Archived values. 41

6.9 The 9000-rpm ramped-load envelope: Complete-run results. Archived values. . . . 43

6.10 The 9000-rpm ramped-load envelope: Time response at each ramp. Archived values. 44

6.11 The 9000-rpm ramped-load envelope: Ramp rates. Archived values. 45

6.12 LESO, UDE, and GPIO implementation and first comparison: Measured comparison. Archived values. . . . 51

6.13 LESO, UDE, and GPIO implementation and first comparison: Event recovery. Archived values. . . . 51

6.14 LESO, UDE, and GPIO implementation and first comparison: Actuation and model validity. Archived values. 53

7.1 Observer transfer and the direct GPR feedforward challenger: Matched 10% observer comparison. Archived values. 60

7.2 Observer transfer and the direct GPR feedforward challenger: All load scenarios. Archived values. 60

7.3 Observer transfer and the direct GPR feedforward challenger: First-invalid-sample diagnostics. Archived values. 66

8.2 Early-stage 9000 rpm / 10% shaft-load ramps. RMSE and peak are rpm; ringing is lbm/s. Validity counts cover all ten scenarios (D02). 76

8.3 Fixed-gain search at 9500 rpm / 5% load steps. RMSE in rpm, ringing in lbm/s, fine settling in seconds (D03). 79

8.4 9500 rpm / 5% steps at the final fixed fuel-domain gains 2/30. Units: rpm, rpm, lbm/s, s (D04). 80

9.1 Held-out inverse fuel prediction. RMSE/MAE in lbm/s; coverage is a fraction. The two GP rows use different datasets; only the 4-input GP and Poly2 share the same split (D01, D05, D13). 83

9.2 Nominal inverse slopes and ideal local equivalent gains for the 2/30 fuel-domain PI. Negative g_N reverses integral feedback sign (D08). 85

9.3 4000 m / ISA+5 / 9500 rpm / 10% steps. Units: rpm, rpm, lbm/s, s. R1–R3 are tuned at this target; the matched PID ablates inverse feedback normalization (D07, D09). 88

9.4 Frozen-gain environmental transfer. "Both better" means lower RMSE and ringing than the paired base PI. Rejected trajectories do not count as wins. Map-clean denominators use both trajectories (D07). 88

9.5 Broad 9025–9975 rpm cycle (D10). 91

10.1 Final covered 9450–9550 rpm cycle: same gains and frozen models. All four trajectories pass box/hull coverage, with no fallback or monitored map-speed overrun (D12, D19). 97

10.2 9250 rpm / 0 m / ISA 0 / 10% load steps. The retune changes only the gains (D15). 100

10.3 Frozen gains evaluated on 5% load steps at the same operating point. This is amplitude transfer, not an independent speed/environment test (D16). 102

10.4 Fuel reversal sensitivity at 9250 rpm. Columns change the duration of each of the six post-edge windows; entries are summed excess total variation in lbm/s. The 2 s column reproduces the published metric. These are reanalyses of the same saved trajectories, not new tuning results. 103

10.5 Worst persistent recovery in seconds at 9250 rpm. The last band is 92.5 rpm. Event intervals and the next-event persistence rule are held fixed while changing only the band. 103

11.1 10% load-amplitude comparison. 108

11.2 5% load-amplitude comparison. 108

11.3 All tested penalty weights. RMSE is in rpm; ringing is in lbm/s. The star marks the selected policy. 116

11.4 10% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band. 116

11.5 5% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band. . 117

13.1 The same ramp scored using different signals. Values are from the archived sensor-comparison CSV; no new simulation is involved. 132

13.2 All archived DOB98 PI candidates. RMSE is rpm; R_{seg} is lbm/s. Candidate 11 is selected by the stated objective. 134

B.1 All early scenarios: speed RMSE (rpm). R: ramped load; S: stepped load. A dash is a rejected full trajectory, not zero (D02). 147

B.2 All early scenarios: load-edge fuel ringing (lbm/s). R: ramped load; S: stepped load. A dash is a rejected full trajectory, not zero (D02). 147

B.3 Four fixed controllers: speed RMSE (rpm). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04). 148

B.4 Four fixed controllers: fuel ringing (lbm/s). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04). 148

B.5 Four fixed controllers: persistent recovery to 0.1 rpm (s). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04). 149

C.1 Complete 35-condition transfer audit, 9500 rpm / 10% load steps. The four middle ratios are speed RMSE relative to base PI. V: previous inverse PI; R3 includes static FF. A dash means rejected. FB: R3 fallback percentage (D07). 150

D.1 Full quadratic coefficients in standardized coordinates. All 15 terms are retained (D14). 152

D.2 All 34 tested Poly2 gain settings at 9250 rpm. All were numerically accepted; candidate 34 minimizes the stated joint score (D15). 153

Chapter 1

Research question, objectives, and bounded contribution

1.1 The engineering problem

A speed controller must supply enough fuel to accelerate a shaft and reject unknown external load without producing undesirable fuel-command motion. A PI controller already provides proportional correction and integral rejection of constant error. Adding a learned inverse, a disturbance estimator, or an optimized policy can help, but it can also alter feedback gains, amplify measurement dynamics, or improve a metric that does not represent the physical quantity of interest. The value of a more complicated model must therefore be tested inside the actual feedback loop.

This project originally developed through sequential requests: identification, observers, inverse regression, polynomial replacement, policy optimization, smoother control, ramp transfer, and a final local DOB. Sequential development is normal engineering practice, but its record is not automatically a controlled experiment comparing every method. A retrospective thesis must identify which comparisons isolate a mechanism and which compare complete designs with several differences.

1.2 Primary question

> Within the archived T-MATS simulation campaign, when do identified or learned models improve the tested controller configurations, and which improvements can the available evidence attribute to model representation, feedback architecture, gain selection, or measurement and scoring choices?

The phrase "within the archived campaign" is a scope condition. It prevents a few successful load cases from becoming a claim about all gas turbines, all operating conditions, or all learning-based controllers. "Tested configurations" means the implemented controller, its gains, initialization, filtering, limits and scenario. It is deliberately narrower than an entire method family.

1.3 Subsidiary questions and observable answers

RQ1. Prediction and inversion: Does high one-step or nominal regression accuracy establish a model suitable for free-running prediction and causal inversion? Evidence comes from the Wiener/ARX transfer tests, sparse estimator, inverse-zero audit and local DOB identification.

RQ2. Attribution: Which benefits survive existing controlled substitutions or ablations? Evidence comes from matched-gain feedforward tests, the local matched-PID diagnostic, unchanged-gain GP-to-polynomial replacement and DOB with original PI gains.

RQ3. Objective and sensing: How do tracking, command reversal and controller ordering change when the objective or the scored signal changes? Evidence comes from the ringing-penalty study, metric sensitivity, ramp tests and common-sensor audit.

RQ4. Transfer: How far do the selected configurations retain useful behavior outside their selection case? Evidence comes from independent predictor records, load-amplitude checks, environmental cases, moving-reference cycles and the final frozen-gain check.

These questions were formulated for the retrospective synthesis. They are not described as preregistered hypotheses governing the original campaign.

1.4 What the study contributes

C1: A traceable comparative record. The thesis connects accepted and rejected simulation studies to their models, controller definitions, performance windows and evidence roles. It distinguishes numerical validity, training coverage and control performance rather than treating them as one success criterion.

C2: An implementation-level attribution analysis. Existing ablations show where an apparent benefit can be reproduced by an integral offset, local equivalent PID action, gain selection or a simpler inverse. The mathematical explanations make those mechanisms inspectable. These are application-specific findings and pedagogical derivations, not claims to invent PI equivalence or linearization.

C3: A qualified tracking–reversal comparison. The archive demonstrates multiple tradeoffs rather than one universal winner. It documents the repair of a highly oscillatory learned policy, explains why that repair is not baseline superiority, and retains the final DOB's improved tracking alongside its increased actuator activity.

C4: A reproducible technical adaptation. The expected excess-variation objective is implemented using Gaussian moment propagation with command memory and an event anchor. Its calculation is documented, together with its limitations and verification. The present literature review does not establish a first-in-literature claim. C4 is therefore a reproducible implementation contribution, not an asserted new general theory.

The thesis does not claim that the number of controller variants, figures or pages constitutes originality. Its master's-level value lies in applying established knowledge carefully and deriving defensible conclusions from imperfect experimental evidence.

1.5 Claims explicitly excluded

The work does not establish universal superiority of GP, polynomial, PILCO-style, or observer control over PI. It does not establish an equal-budget optimum across controller families. There is no hardware validation, measured fuel-economy benefit, certified surge-protection envelope, or global robust-stability guarantee. Posterior GP uncertainty is not claimed to bound every physical disturbance. A successful deterministic run is not an independent statistical replicate of a population of engines.

The retained claims do not require missing experiments to be imagined. A comparison can be useful without being decisive. For example, the final DOB-plus-original-PI result demonstrates a change

after adding the configured observer to that baseline, while the missing retuned-PI-only branch leaves the observer's advantage over a comparably optimized conventional controller unresolved.

1.6 Organization of the argument

The literature chapter separates established building blocks from the present application. The evidence protocol fixes the vocabulary for fitting, selecting and evaluating them. The technical chapters then explain the sequence of models and controller behaviors. Finally, the cross-study synthesis answers RQ1–RQ4 using the most discriminating existing comparisons, rather than counting the number of favorable cases or averaging incompatible tests. Full historical tables remain in appendices to support audit without defining the thesis argument.

Chapter 2

Literature position and assessment of originality

2.1 Review scope and method

This is a targeted methodological comparison, not a systematic review with an exhaustive database search. The starting points are the methods actually implemented: T-MATS simulation, orthogonal identification, disturbance estimation, inverse GP control, gain scheduling and probabilistic policy search. Primary papers, institutional records and the official simulator documentation provide the comparison. The additional review was checked on 11 October 2026. No absence-of-prior-art conclusion follows merely because an exact phrase was not located.

Two questions organize the comparison. First, which algorithmic elements already exist? Second, what does the present turbine evidence add beyond applying those elements? A different operating speed or a renamed controller is not, by itself, a methodological contribution. Conversely, a carefully demonstrated negative result or a useful application-specific limitation can contribute to engineering understanding without inventing a new algorithm.

2.2 Simulation platform and engine-control context

T-MATS provides the thermodynamic component framework and numerical machinery used to construct the plant [2, 10]. The present project changes controller and experiment configurations around a particular inherited engine model; it does not develop a new gas-path simulator or validate new component maps. Model-based aircraft-engine control using online estimation also predates this work [18]. These sources establish context rather than direct performance comparators: their engines, outputs and objectives differ from the present shaft-speed tests.

The appropriate application claim is therefore specific. This archive documents how a set of identified and learned controllers behaves on the stated T-MATS configuration, including numerical and map limits. It does not claim to introduce model-based turbine control. A fair quantitative comparison with an external paper would require matching the plant, disturbance, actuator constraints and evaluation signal; the present literature chapter does not compare unrelated RMSE values.

2.3 Identification and sparse model selection

Orthogonal least-squares methods combine parameter estimation and structure selection for models linear in their parameters [15]. The sparse lag selection in this project is an application of that established family. Its relevant evidence is the recorded reduction in conditioning problems, smaller retained structure, and held-out performance relative to the ten-term model. A smaller covariance matrix is an implementation simplification, but it does not constitute a measured real-time speedup. The block realization and recurrence replay show that the estimator was implemented correctly; they do not create a new identification algorithm.

2.4 Disturbance observers and related estimators

Filtered inverse-model DOB control estimates an equivalent input disturbance and compensates it through the control channel. Its relationship to alternative disturbance-estimation methods is well established [16]. LESO and UDE likewise have prior formulations and tuning principles [12, 13]. The application choices here are the identified model, equilibrium correction, causal filter, compensation limit, startup procedure and gain selection.

The contribution is not "DOB plus PI" as a new architecture. It is the account of why a raw acceleration inverse was unsuitable, how the filtered implementation was constructed, and what the accepted cases demonstrate. The final 9800-rpm model is a different local nominal approximation, not a new class of observer. Its combined-cycle improvement must be assessed together with its narrow identification range and the missing equally retuned PI-only branch.

2.5 GP regression, feedforward and inverse coordinates

Gaussian-process regression supplies a posterior mean and variance conditional on specified statistical assumptions [1]. Prior work already applies GP feedforward to diesel-engine airpath control [4]; the related doctoral thesis establishes part of the project's intellectual lineage [5]. This turbine study concerns a different simulated plant and different test sequence. It neither duplicates those physical results nor claims GP feedforward for engine control as a new concept.

The more informative question is how an inverse is used. A map used only to supply a nominal command differs from a map queried using measured speed and acceleration inside feedback. The latter can change effective proportional, integral and derivative action. The present derivations expose that mechanism, and the matched-PID diagnostic tests whether a GP-specific benefit remains at a selected operating point. That diagnostic limits the novelty and performance claim even though the GP fit is excellent.

The uncertainty schedules are empirical design hypotheses. Gain scheduling itself has a substantial literature [7]. The presence of a GP standard deviation does not automatically determine whether gains should increase or decrease, especially when the posterior omits unknown-load mismatch and physical uncertainty. The thesis presents the measured tradeoffs without promoting the schedules into a general robustness rule.

2.6 PILCO and the restricted implementation used here

Original PILCO-style PI combines a learned probabilistic dynamics model, approximate long-horizon inference and analytical policy gradients, with iterative data collection and policy improvement [11]. Later work further develops GP-based data-efficient control [8]. In this project, historical transition data are fitted offline, feedforward coefficients are fixed, and only two PI gains are optimized with numerical gradients. A 40-point representative GP approximation and an affine unsaturated planning policy are used, while the nonlinear plant deployment includes fuel limits and conditional integration.

The accurate description is *PILCO-style GP-based offline PI tuning*. The study does not claim a reproduction of the full original learning loop, an improvement in sample efficiency over original PILCO, or analytic policy gradients. Its useful evidence concerns what this restricted probabilistic tuning procedure selected and how that policy transferred to the recorded plant scenarios.

2.7 Smooth actions and the excess-variation adaptation

CAPS regularizes temporal and spatial action variation to obtain smoother learned control [17]. It therefore precedes the broad idea of penalizing irregular learned commands. The present objective is different in detail: event-window total variation minus absolute net command change, evaluated in expectation under an augmented Gaussian state. CAPS and the present construction do not use identical policies or experiments, so the literature comparison establishes related purpose rather than numerical superiority.

For a Gaussian scalar Z with mean μ and variance v, the absolute first moment used by the project is

$$\mathbb{E}|Z| = \sqrt{\frac{2v}{\pi}} \exp\left(-\frac{\mu^2}{2v}\right) + \mu \operatorname{erf}\left(\frac{\mu}{\sqrt{2v}}\right), \qquad v > 0. \tag{2.1}$$

At zero variance it is $|\mu|$. This is a mathematical property of the Gaussian distribution, not a newly discovered moment formula. The implementation combines it with command-memory covariance and event-anchor resets. That combination is a candidate incremental adaptation; originality is not established by the present targeted search. Its value over ordinary total variation or a squared-increment cost was not tested under an equal selection protocol.

2.8 Contribution matrix

Element	Established basis	Contribution retained in this thesis
Wiener, ARX and sparse estimation	Existing identification and structure-selection methods	Recorded conditioning, prediction and realization comparisons for this plant.
Filtered inverse DOB	Existing residual estimation and low-pass compensation	Explicit inversion constraints, local implementations, partial observer/tuning ablation.
GP feedforward and inverse feedback	Existing regression and engine feedforward concepts	Analysis of feedback-coordinate effects and limits of prediction-based ranking.
R3 and matched PID	Local normalization and conventional PID action	Evidence that target improvement is not demonstrably GP-specific.
Polynomial replacement	Standard quadratic regression	Same-architecture challenger showing competitive control despite poorer fit.
PILCO-style PI search	Established probabilistic model-based policy evaluation	Offline two-gain adaptation with declared planning/deployment differences.
Expected excess variation	Known absolute moments and smooth-action motivation	Reproducible event-window adaptation; originality unestablished.
Complete comparative record	Established scientific reporting principles	Traceable evidence of metric-dependent and scenario-dependent outcomes.

2.9 The defensible gap

The gap addressed is *within this project*: why did more elaborate models fail to improve consistently over inherited PI? The answer separates prediction, feedback action, tuning, sensing and coverage. This application-specific synthesis supports a master's contribution. A broader novelty claim would require additional prior-art review and discriminating evidence.

Chapter 3

Retrospective evidence protocol and comparison rules

3.1 What was and was not changed for Version 2

The revision reorganizes and explains an existing record. No plant trajectory, model fit, controller gain, penalty-weight search or Monte Carlo sample was newly generated. Numerical values are copied from the archived reports and saved summaries. Derived percentages and algebraic explanations are interpretation of existing results. The original report, source evidence index and revised source manifest remain available to identify the provenance of those results.

The archive contains 69 result directories, not 69 independent replications. Some directories are pilots, failed runs, replays, aggregates or presentation outputs. The study register identifies them individually. Deterministically repeating a simulation checks repeatability of the implementation; it does not create an independent sample of engine-to-engine uncertainty.

3.2 Evidence roles

Fitting estimates model parameters from observations. **Selection** chooses a model order, gain set, architecture remedy or penalty weight using observed performance. **Held-out prediction** evaluates a predictor on a record excluded from the relevant fit. **Frozen-controller transfer** applies previously selected parameters to another condition without retuning. **Ablation** changes a specified component while retaining the other stated conditions. **Implementation verification** checks equations or code against a reference calculation. These labels describe different questions, not a ranking from bad to good.

A dataset can have more than one role at different stages. The third pulse may be excluded from a GP fit, but if a full test containing it is later used to choose a ringing penalty, that test is no longer held out from policy selection. A comparison can also be a transfer test for one controller and a selection case for another. The final combined cycle is exactly such a case.

3.3 Evidence-role ledger

The following ledger is the controlling interpretation of the historical terminology used in later chapters. “Independent” refers to the stated record or selection step, not to a new engine or a statistically independent set of every time sample.

Study	Fitting or selection role	Evaluation role	Permitted inference
Numerical baseline, steps, chirp and PRBS	Excitation feasibility; requested full records can fail	Accepted trajectories/prefixes assessed for residuals and finite signals	Defines usable data; failed full records are not successful controls.

Study	Fitting or selection role	Evaluation role	Permitted inference
Wiener models	Accepted chirp used for fitting; selection ramp chooses among candidates	Separate held-out ramp	Prediction within these records; not engine-wide validity.
Ten-term RLS	Online coefficient updates on the chirp	Frozen chirp and held-out ramp; block replay	Separates one-step adaptation, free-run transfer and implementation.
Sparse RLS	Offline lag-support selection, then coefficient estimation	Held-out ramp; conditioning and recurrence checks	Local comparison of retained structures, not a new online support learner.
Early DOB	Identified model, equilibrium constraint and filter design	Governed-reference and input-disturbance scenarios	Configured observer behavior under stated excitations.
PTO observer tests	Established configurations; scenario progression is adaptive	Operating-point/load checks, including invalid runs	Local envelope and paired differences; not a common-power cross-speed ranking.
Two-input forward/inverse GP	600 nominal training points	100 points from a separate experiment	Nominal prediction; temporal samples are not independent engine trials.
Virtual-fuel PI and FF retune	Architecture and gains evolve together	Fixed-load cases and same-gain FF ablations	Combined-design results; only the ablation isolates FF.
SD schedules and lower gains	Empirical scheduling direction and local gain selection	Accepted/rejected load cases	Tracking/reversal tradeoffs; no uncertainty-based safety guarantee.
Four-input GP	1400 training points across 35 ambient conditions	200 test points across 33 conditions	Nominal interpolation evidence; not loaded-plant uncertainty bounds.
R1–R3 remedies	80 candidates selected at the target case	Target table, frozen environmental grid, matched PID	Target is selection evidence; grid and matched PID answer distinct transfer/attribution questions.
Broad moving cycle	Frozen controller applied to expanded task	Coverage and performance audit	Stress/transfer evidence with out-of-domain exposure.
Covered moving cycle	Scenario narrowed after broad-cycle diagnostics	Matched controller and FF comparisons	Performance within the resulting tested coverage; not a blind generalization test.
GP-to-Poly2 substitution	Same nominal training data; gains initially retained	Same covered cycle	Representation comparison within one architecture.
9250-rpm Poly2 retune	28 initial candidates plus six refinements on 10% pulses	Frozen 5% pulses; saved-metric sensitivity	Local selection plus amplitude transfer; 9250 rpm is below old training minimum.
Original PILCO-style PI	4242 historical transitions; 40-point representative GP; two-gain optimization	Third-pulse GP holdout and nonlinear load comparison	Restricted offline model-based tuning; distinguish predictor holdout from policy evaluation.

Study	Fitting or selection role	Evaluation role	Permitted inference
Smooth PILCO-style PI	GP reused; penalty weight selected with 10% full-test results	5% case excluded from policy selection	Repair of ringing on selection case; limited amplitude transfer.
No-load ramp and sensor audit	Earlier controllers frozen	Five-second ramp; explicit equal-sensor replay	Task transfer and sensing audit, not new gain optimization.
Initial combined cycle	Earlier controllers frozen	Five-second ramp plus load at seconds 10–15	Common-scenario transfer comparison.
DOB98 local model	Fuel PRBS around 9800 rpm; least-squares fit	Separate local multisine	Local predictor fit over a few rpm.
DOB98 PI search	13 PI candidates with fixed DOB on combined cycle	Original-gain ablation; frozen 4 s ramp/5% pulse check	Main cycle is selection evidence; second case lacks a full comparator lineup.
Moment and algebra checks	No controller evidence added	Recurrence, inverse identity and GP Monte Carlo comparisons	Correctness of stated computations, not physical robustness.

3.4 What a matched comparison requires

The common plant, reference and load are necessary but insufficient for equal opportunity. A controller also depends on sensing, sample interval, initialization, gain-selection budget, clipping, anti-windup, and the objective used for selection. When those factors differ, the correct statement is "configuration A performed better than configuration B on this case." It is not "method A is intrinsically better."

The existing record contains several more controlled comparisons. Constant-reference FF tests match gains and integral offsets. The GP-to-polynomial substitution initially holds architecture and gains fixed. The common-sensor audit replays the same controllers with explicit common settings. The DOB original-gain result changes the observer while retaining the inherited PI gains. Each controls a particular source of ambiguity; none controls every source across the full project.

The matched filtered PID is a local diagnostic. Its near-equality with R3 at the target shows that a more conventional implementation can reproduce that target behavior. It does not prove all GP controllers are equivalent to PID at all operating points. Similarly, a favorable unchanged-gain polynomial result does not prove that every possible GP architecture is unnecessary.

3.5 Primary response and phase weighting

Unless a table explicitly states otherwise, tracking is scored using actual shaft speed N, while the controller uses sensed speed N_s. Define

$$E_N = \sqrt{\frac{1}{M}\sum_k (N_k - r_k)^2}, \qquad E_s = \sqrt{\frac{1}{M}\sum_k (N_{s,k} - r_k)^2}. \tag{3.1}$$

Neither value should be substituted for the other. In the deterministic simulator, actual speed is available for evaluation even though the controller receives a lagged signal. Hardware would require an independently justified measurement or estimate of the physical quantity.

For disjoint phases j with sample counts M_j, full-record RMSE obeys

$$E_{\text{full}}^2 = \sum_j \frac{M_j}{\sum_i M_i} E_j^2. \tag{3.2}$$

It is not the arithmetic mean of phase RMSEs. A design can worsen the ramp and improve the full score because the other phases are longer or their large deviations dominate the sum of squares. This is central to the final DOB retune. Phase endpoints in an actual replay must follow the original masks; a boundary sample should not silently be counted twice when reconstructing a disjoint decomposition.

3.6 Fuel command, reversal and recovery

For a prescribed event window $a{:}b$,

$$R_{a:b} = \sum_{k=a+1}^{b} |u_k - u_{k-1}| - |u_b - u_a|. \tag{3.3}$$

The triangle inequality gives $R_{a:b} \geq 0$. This is accumulated excess change in a flow signal, measured in lbm/s. It is not fuel mass, which would be approximately $T_s \sum_k u_k$. Nor is it a bound on slew, $\max_k |u_k - u_{k-1}|/T_s$. A sudden monotonic transition has zero excess variation but may have high slew. These quantities are reported as different performance dimensions.

Excess variation depends on event boundaries. Splitting a signal into several windows changes the subtracted endpoint terms. The final DOB objective uses segmented variation across ramp and load phases; the load-only table reports a subtotal. Both can be correct while numerically different. Noise and sampling also affect variation, so results are compared using the original common sample interval and stated windows.

Recovery needs an error band and persistence rule. Zero recovery under a wide band may mean the trajectory never left the band, not instantaneous correction. The saved sensitivity analysis shows that changing the band can reverse the original-versus-retuned Poly2 ordering. No universal settling-time superiority is inferred.

3.7 Numerical and operating-domain acceptance

Converged flow residuals establish satisfaction of the numerical equations at the prescribed tolerance. Compressor-map inclusion concerns the thermodynamic model's intended domain. GP box or convex-hull inclusion concerns nominal regressor coverage. Available surge margin is a simulated output. These checks do not imply one another, and none alone proves robust physical safety.

Failed full trajectories remain failed full trajectories. An accepted prefix can be used for an explicitly delimited identification task, as with the chirp, but cannot be reported as a successful completion of the requested full excitation. Invalid runs are retained in acceptance counts and shown as unscored where appropriate. They are not replaced by zero error or excluded without explanation.

3.8 Statistical interpretation

The nominal simulations are deterministic, and samples along a trajectory are temporally related. No p-value or confidence interval is constructed by pretending that every sample is an independent

experiment. Small differences such as the target R3/matched-PID gap are reported as observed numerical differences, not statistically significant population effects. Confidence in reproducibility comes from preserved inputs, trajectories, code and checks; confidence in generalization would require a separately designed range of conditions and uncertainty realizations.

3.9 Why missing comparisons remain missing

No equal-budget PI-only search, sensor-noise sweep, actuator-lag sweep, or independent-engine test was added. Version 2 addresses the jury's concern by limiting claims to the evidence that exists. If architectural superiority is later claimed, a new experiment must first equalize selection objectives, constraints and evaluation conditions. The proposed design in Chapter 15 is a future protocol, not a retroactive description of the present archive.

Chapter 4

Plant, sensing, metrics, and numerical acceptance

This chapter specifies the physical and numerical experiment. Numerical acceptance precedes controller ranking: a failed flow solution is not a valid response to score. Historical experiments are described retrospectively; no test was rerun for Version 2.

4.1 Physical plant and feedback foundations

4.1.1 From fuel to shaft speed

A gas turbine converts chemical energy into hot-gas energy and then shaft work. Air enters through the inlet, is compressed, receives heat in the burner, expands through the turbine, and exits through the exhaust system. The turbine drives the compressor on the same shaft. Their powers are individually large even when the shaft speed is constant. Steady speed means their signed net power balances the external extraction and losses; it does not mean either component has zero power.

The model is a single-shaft turbojet built from T-MATS components. Its gas-path sequence includes ambient/inlet, compressor, burner, turbine, duct, and nozzle. Component maps encode relationships such as flow, pressure ratio, efficiency, and corrected speed. These maps are part of the plant, not learned by the controller. T-MATS is the modeling framework; the particular component maps and model setup define the specific simulated engine [10].

Let angular speed be $\omega = 2\pi N/60$, where N is in rpm. Rotational mechanics gives

$$J\dot{\omega} = \tau_t + \tau_c - \tau_L, \qquad P = \tau\omega. \tag{4.1}$$

With compressor power entered using its signed convention, the project's shaft equation is

$$\dot{N} = \frac{60}{2\pi J}\, 5252.113 \frac{P_t + P_c - P_L}{N}, \qquad J = 30. \tag{4.2}$$

The numerical conversion belongs to the archived horsepower/rpm and inertia convention. It must not be combined blindly with SI torque. This equation explains the sign of a load transient: increasing P_L makes the net power negative, the shaft decelerates, and feedback raises fuel until turbine power restores balance. When the load is removed, the previously increased fuel initially creates excess power and overspeed.

4.1.2 Why the nonlinear solver matters

At each integration step, gas-path flows must agree across connected components. An inner Newton–Raphson calculation adjusts variables such as inlet flow, compressor map coordinate, and turbine map pressure ratio to reduce three flow residuals. The outer integration advances the shaft

state. Thus a smooth-looking speed trace can coexist with an internally failed flow solve. Reporting speed alone would miss this failure.

The baseline setup uses fixed-step ode4, $T_s = 0.015$ s, normalized flow tolerance 10^{-9}, and an iteration ceiling of 200. A residual below tolerance means the algebraic flow equations were solved to the prescribed numerical accuracy. It is not a measurement of physical model uncertainty. Conversely, failure to converge is not by itself proof that a real compressor surged. Map warnings, available surge margin, finite signals, and convergence each answer a different question.

The initial baseline archive verifies that the GPT copy matches the supplied standard model by SHA-256. Later experiments create named variants. Copying the plant allows a controller subsystem to change while retaining the gas path; it does not excuse checking sensor connections, initialization, solver settings, or actuator limits. Those details eventually became central to the ramp comparison.

4.1.3 Signals, units, and operating points

Three speed signals must remain separate: requested speed, governed reference r, and actual mechanical speed N. The sensor output N_s is the feedback measurement. A rate governor can make r a ramp even if the raw request is a step. The controller generally receives N_s, while performance tables primarily score $N - r$. In this report the feedback error is $e = r - N_s$ unless explicitly stated otherwise; signed plotted tracking error may instead be $N - r$.

Fuel flow $u = W_f$ is in lbm/s. Turbine and extraction powers are in hp in the model, with MW sometimes reported for interpretation. Temperatures from the station buses are absolute Rankine and convert as $T_K = (5/9)T_R$. Absolute pressure converts from psia to kPa by multiplication by 6.894757. Mixing absolute and gauge pressure, or Celsius and Kelvin, would corrupt an inverse model even if the regression code were correct. Altitude units are also recorded separately from the controller's temperature and pressure inputs.

Rated speed in the October tests is defined as 10000 rpm. Therefore 90%, 98%, and 100% denote 9000, 9800, and 10000 rpm. They are not model-fit percentages and are not compressor-map corrected-speed coordinates. A local model identified at 9800 rpm is therefore a model around a physical operating point, not a model with an asserted 98% accuracy.

4.1.4 PI action explained from the equation

An ordinary PI controller uses

$$u^* = K_p(r - N_s) + I, \qquad \dot{I} = K_i(r - N_s). \tag{4.3}$$

The proportional term responds immediately to the current error. The integral term remembers accumulated error and supplies the persistent fuel offset needed to balance an unknown constant load. If the shaft stays below target, the integral grows until the error can disappear. This is why adding a constant feedforward value at a fixed operating point need not improve a well-initialized PI controller: the integral can already provide that constant.

The inherited speed controller uses $K_p = 0.025$ and $K_i = 0.05$. Their units differ: K_p converts rpm into lbm/s, whereas K_i converts rpm-seconds into lbm/s. A virtual-fuel PI later uses different error units, so its numerical gains cannot be compared directly with these speed-PI gains. Initial integral state is also part of the controller. The standard starting value is not automatically the equilibrium fuel for every requested speed; preparation intervals allow the engine and controller to settle.

Fuel saturation clips a requested command into the allowable range, commonly $0.2 \leq u \leq 4$ lbm/s in the augmented studies. If an integrator keeps accumulating positive error while fuel is already at its maximum, it stores an unrealizable command. Conditional integration prevents updates that push farther into saturation and permits updates that unwind it. Some early studies intentionally retain the original PI without adding anti-windup; later implementations explicitly implement it. The report therefore does not assume one universal anti-windup law across the whole archive.

4.1.5 Sensor lag, differentiation, and causality

A first-order sensor satisfies approximately $\tau_s \dot{N}_s + N_s = N$. Its output cannot instantly follow changing true speed. With $\tau_s = 0.05$ s and a constant acceleration of 200 rpm/s, the continuous steady ramp lag is approximately $\tau_s \dot{N} = 10$ rpm. Sampling and signal timing can add to the observed lag. This simple calculation will help interpret the October ramp test.

Backward differencing gives a causal acceleration estimate $(N_{s,k} - N_{s,k-1})/T_s$. Causal means using only current and previous samples, never future measurements. Differencing amplifies rapid measurement changes, so filtered acceleration is used in later controllers. The ideal direct plant acceleration used for early identification is not automatically available to a deployed controller. Keeping those two information sets distinct prevents an unrealistically favorable controller comparison.

4.2 Experimental design and the first numerical boundary

4.2.1 What constitutes a valid comparison?

A comparison should hold plant, sensor, reference, load, sample interval, preparation, and scoring rules fixed while changing the intended controller feature. If gains also change, the result establishes the performance of the whole configured design; it does not isolate the contribution of a newly added model. An ablation changes one feature while holding the others fixed. A transfer test evaluates frozen parameters in another scenario. A tuning test supplies evidence used to select the parameters and cannot simultaneously serve as independent validation.

The archive contains baseline attempts, failed excitation studies, final accepted studies, intermediate pilot folders, analysis-only aggregates, and presentation artifacts. They are all part of the project history, but they are not equally independent evidence. The final directory register lists them explicitly. Repeated copies in the two Git repositories are distribution copies, not extra trials. Empty pilot directories supply no empirical support.

4.2.2 Metrics and their different meanings

For M samples and actual-speed error $e_k = N_k - r_k$,

$$\mathrm{RMSE} = \sqrt{M^{-1} \sum_{k=1}^{M} e_k^2}, \quad \mathrm{IAE} = T_s \sum_{k=1}^{M} |e_k|, \quad E_{\max} = \max_k |e_k|. \tag{4.4}$$

RMSE emphasizes large errors through squaring. IAE measures accumulated absolute deviation. Peak error addresses the worst excursion. A controller can improve one while worsening another. With unequal phase durations, full-record RMSE weights a long hold more than a short ramp. Phase-specific scores are therefore necessary in a combined cycle.

Model fit in the identification chapters is

$$\text{Fit} = 100\left(1 - \frac{\|y - \hat{y}\|_2}{\|y - \bar{y}\|_2}\right). \tag{4.5}$$

It is not R^2, which uses squared norms. The denominator changes across datasets, so a smaller RMSE can coexist with a lower fit percentage when the new reference varies less. A 99% one-step acceleration fit does not mean a controller tracks speed within 1%.

Fuel total variation is $\sum |u_{k+1} - u_k|$. Subtracting absolute net change gives excess variation, a command-reversal measure. Integrating u over time instead gives fuel mass. These are different physical quantities. Recovery time also needs a declared band and persistence rule; a trajectory that never leaves a wide band has zero recovery by that definition even if its fine regulation is visibly poor.

4.2.3 Baseline and step attempts

The accepted baseline holds 10000 rpm for 5 s, ramps toward 9000 rpm by 10 s, and runs to 10.5 s. Its 701 samples have maximum flow residual 9.9891×10^{-10} and at most 20 inner iterations. Final speed is 8998.3405 rpm, fuel 0.9970915 lbm/s, and thrust 4380.7032 lbf. The final speed is recorded at a finite time, not claimed to be a fully settled equilibrium.

A 10 rpm step from 10000 to 9990 rpm at 1.5 s completes the 3 s test with 201 samples, maximum residual 9.7587×10^{-10}, and at most 14 iterations. The earlier 100 rpm downward step reaches the 200-iteration ceiling and a residual around 0.37016, accompanied by singular-Jacobian or map warnings. Reducing the step demonstrates a numerically acceptable local experiment, not a universal permissible step-size limit. The failed run is retained as a diagnostic rather than scored as successful control.

4.2.4 Why the requested chirp could not be used in full

The requested chirp is

$$r(t) = 9500 - 500\cos\{2\pi(0.1t + 0.0075t^2)\}. \tag{4.6}$$

Its instantaneous frequency is $0.1 + 0.015t$ Hz, reaching 1 Hz at 60 s. A constant speed amplitude does not imply a constant acceleration demand: differentiating the phase shows that the required speed rate grows with frequency. The controller must generate increasingly rapid fuel changes even though the visible speed bounds remain 9000–10000 rpm.

After 30 s preparation, the first invalid sample occurs at chirp time 17.805 s, near 0.367075 Hz. Only the 1187 samples through 17.790 s are accepted. Subsequent negative fuel and solver failures are diagnostic data. Every later model trained on this record must be described as trained on the valid prefix, not on a successful 60 s, 0.1–1 Hz sweep. This restriction is preserved in the Wiener, RLS, and early GP narratives.

4.2.5 PRBS, rate limiting, and independent ramps

A pseudo-random binary sequence excites a system using deterministic switches between two levels. The project's 127-bit sequence follows $b_{k+7} = b_k \text{ xor } b_{k+3}$ with seed 0110101. The full-amplitude 9000/10000 rpm, 0.51 s/bit test fails at its first edge, leaving only 34 valid samples. Timing

adjustments in a repeat addressed breakpoint alignment, not the basic numerical difficulty of the large abrupt transition.

Later studies use governed binary requests or small-amplitude binary steps. Once a governor turns a step into a ramp, the applied reference is no longer a strict PRBS. This matters because the actual frequency content delivered to the plant is altered. The model-identification input is actual fuel, not simply the raw binary request.

Independent ramp-and-hold records supply accepted selection and validation data. Their rates include 50, 60, 95, 100, and 150 rpm/s. Holds expose equilibrium errors that a rapidly varying chirp can conceal. Separating the selection record from the final validation record allows model order or lag support to be chosen without consuming the final test. This design decision links the failed excitation phase to the more credible identification studies that follow.

4.3 Identification primer: dynamics, regressors, and prediction

4.3.1 A model is an approximation with an information pattern

A dynamic model predicts future output from stored state or past signals. A static regression predicts an output from a current input vector. Neither description alone specifies how the model will be used. In a one-step predictor, measured past outputs are available at every sample. In free-running prediction, the model must recursively use its own predicted outputs. An error can therefore accumulate rather than being corrected by the next measurement.

For an ARX model, with q a one-sample delay,

$$A(q)y_k = B(q)u_k + \epsilon_k, \quad A(q) = 1 + \sum_i a_i q^i, \quad B(q) = \sum_j b_j q^j. \tag{4.7}$$

Rearranging gives $\hat{y}_k = \phi_k^\top \theta$, where the regressor contains negative past outputs and delayed inputs. The signs are a convention inherited from moving the output terms to the right-hand side. A lag of ten samples here means 150 ms, not ten seconds. Normalizing signals improves numerical conditioning but requires restoring physical units before using coefficients in a controller.

4.3.2 Least squares and recursive least squares

Stacking observations gives $y = \Phi\theta + \epsilon$. Least squares minimizes $\|y - \Phi\theta\|^2$. If columns of Φ are almost redundant, many coefficient combinations predict nearly the same output. A high condition number warns that small perturbations can change individual coefficients substantially. Excellent prediction is therefore compatible with poorly identified parameters.

Recursive least squares updates a previous estimate using the new prediction error:

$$P_k^- = P_{k-1}/\lambda, \qquad L_k = \frac{P_k^- \phi_k}{1 + \phi_k^\top P_k^- \phi_k}, \tag{4.8}$$

$$\theta_k = \theta_{k-1} + L_k(y_k - \phi_k^\top \theta_{k-1}). \tag{4.9}$$

Here P is an information-related covariance matrix and λ discounts older data. The implementation uses a Joseph-form update,

$$P_k = (I - L_k\phi_k^\top)P_k^-(I - L_k\phi_k^\top)^\top + L_k L_k^\top, \tag{4.10}$$

followed by symmetrization. This is a numerical safeguard, not proof that P is a calibrated physical uncertainty interval. At $\lambda = 0.999$, the approximate memory scale $1/(1-\lambda)$ is 1000 samples or 15 s.

Publishing the prediction before updating with the current output prevents leakage. Otherwise the purported one-step prediction would already have used the answer. Similarly, a validity gate must stop parameter updates once the reference plant fails. In this project the failure latch freezes coefficients and history and returns undefined predictions for the failed remainder.

4.3.3 Wiener models and sparse lag selection

A Wiener model cascades linear dynamics with a static nonlinear output map. In this project a quadratic output nonlinearity can bend the linear dynamic response to represent a changing input-output relationship. It is a different modeling assumption from a static map of speed and acceleration to fuel. Its memory lives in the linear block, while its output nonlinearity has no independent dynamics.

Sparse lag selection asks which past samples add nonredundant information. The two-stage RLS study first screens a dictionary of possible delays, then estimates coefficients for the retained support. Because support is chosen using the full training record, this is offline structure selection followed by recursive coefficient replay; it is not a fully online algorithm that discovers structure causally. This distinction does not diminish the conditioning improvement, but it defines the claim accurately.

4.3.4 What identification can establish

The following chapters report progressively different successes: a useful Wiener free-run speed model, a highly accurate one-step acceleration predictor, a smaller and better-conditioned sparse predictor, and finally models suitable for filtered inversion. These are not interchangeable achievements. A model selected for one-step accuracy can have an unsuitable DC gain or a nonminimum-phase zero. Such defects become important only when the model is placed inside feedback, motivating the first disturbance-observer study.

Chapter 5

Identification quality and the limits of prediction

The first subsidiary question asks what can be inferred from a high prediction score. The following studies distinguish a fitted output map, online one-step prediction, frozen free-running prediction, and an implementable estimator. These are different achievements. Orthogonal term selection is an application of established identification ideas [15]; the contribution here is the recorded comparison and implementation.

5.1 Wiener identification: a nonlinear output map after linear dynamics

Evidence basis: `results/wiener_ramps_20260908_214558_458/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Fitted in MATLAB R2018b, System Identification Toolbox 9.9, using `nlhw`, `unitgain` on the input and `poly1d(2)` on the output.

5.1.1 Selected model and free-run fits

Table 5.1: Wiener identification: a nonlinear output map after linear dynamics: Selected model and free-run fits. Archived values.

Record	Fit, NRMSE (%)	RMSE (rpm)
Training: converged chirp prefix	94.0903	21.4845
Ramp/hold model-selection record	87.1724	39.5604
Held-out ramp/hold final validation	87.0954	41.9286

Selected orders **[nb nf nk] = [1 3 0]** at Ts=0.015 s. nb is the number of numerator coefficients excluding delay zeros; nf is denominator degree. Both are capped at 10 (the cap is per polynomial, not a total of 10 across both). Output nonlinearity degree is exactly 2. Maximum absolute linear-block pole is 0.9786791711; the selected linear block is stable. Maximum absolute final-validation error is 58.5106 rpm.

Fit = 100 * (1 - norm(y-yhat)/norm(y-mean(y))). These are free-run simulation scores with zero normalized initial states, not one-step prediction scores. Measured validation speed is not fed back to the predictor or used to estimate initial states. The coefficient-only implementation and MATLAB `compare` metric were checked against the saved predictions; the exported metrics were independently recomputed from the CSV files.

5.1.2 Training and validation separation

Training uses ONLY the first 1187 samples, 0-17.790 s, of the previous 9000-10000 rpm chirp. Every sample at and after the first solver failure at 17.805 s was excluded. Consequently the accepted chirp frequency coverage is approximately 0.1-0.36685 Hz, not the requested full 0.1-1 Hz. The discarded samples were not interpolated, repaired or used for fitting.

Ramp/hold records are separate simulations, each with 60 s retained after a 30 s preparation interval. The selection and final validation records use different levels, ramp rates and hold durations. All ramp profiles stay within 9000-10000 rpm. Across them, nonzero rate magnitudes are 50, 60, 95, 100 and 150 rpm/s. Actual fuel flow Wf (lbm/s), rather than the speed-demand signal, is the model input; mechanical shaft speed Nmech (rpm) is the output.

Only the chirp estimates model parameters and normalization. The selection ramp chooses model order and whether training-only refinement is retained. The final validation ramp is evaluated after that choice, without refitting. 44 candidate order/delay structures were tested, with up to 80 iterations per candidate and up to 200 for final refinement. The coarse search tests all nb/nf pairs 1-4 and equal nb/nf orders 5-10, with delays 0 and 1; local neighboring orders are then tested. "Best" means best selection fit among this search, not a proof of a global optimum or an exhaustive search of all orders. Candidate errors and scores are retained in candidate_scores.csv.

5.1.3 Accepted simulation checks

Table 5.2: Wiener identification: a nonlinear output map after linear dynamics: Accepted simulation checks. Archived values.

Record	Max normalized flow residual	Max iterations	Min fuel (lbm/s)	Min stability margin (%)
Selection	9.9944e-10	30	0.9337	16.6316
Validation	9.9968e-10	20	0.8548	15.9367

Both entire 90 s source runs pass the 1e-9 flow tolerance and 200-iteration limit. Recorded fuel remains positive and compressor stability margin remains positive. The original GPT model and PI controller were not changed to obtain these runs. These are noise-free synthetic closed-loop experiments. Fit on these trajectories does not establish validity for failed high-frequency chirps, abrupt full-range steps, untested operating conditions or measurement noise. The Wiener structure is an approximation of the gas-turbine dynamics.

5.1.4 Numerical model

Let u_n=(Wf-u0)/su; v=filter(B,F,u_n); Nmech=y0+sy*polyval(p,v), with zero filter initial states and coefficients ordered as MATLAB expects:

```
B = [1]
F = [1, -2.746098163319219, 2.550083469024761, -0.8028568299992156]
p = [-1.328480990017108e-07, 0.001843948442874731, -0.013415553542986841  # [quadratic,
linear, constant]
u0 = 0.9810869146816966 lbm/s
su = 1.108919409314424 lbm/s
y0 = 9000.000000000004 rpm
sy = 363.6986568026227 rpm
```

Leading zeros in B already implement nk. Do not add the delay again. Normalization is part of the saved model wrapper; it must be applied exactly. For other starting operating points, consistent filter initial states are needed.

5.1.5 Use in MATLAB

```
s = load('best_wiener_model.mat');
disp(s.bestModel)
% Full toolbox model, with the supplied normalized final validation data:
compare(s.zv, s.bestModel, Inf, compareOptions('InitialCondition','zero'))

% Coefficient-only implementation, physical fuel -> physical rpm:
d = readtable('validation_data.csv');
Nhat = predict_wiener_gpt(d.Wf, 'best_wiener_model.mat');
```

5.1.6 Files

- best_wiener_model.mat: selected idnlhw model, coefficients, scales, metrics, iddata estimation/selection/validation objects and predicted outputs.
- model_coefficients.json and fit_metrics.json: portable numerical model and scores.
- identification_data.mat: data tables, normalized iddata objects and scaling.
- training_chirp_valid.csv: only the accepted training segment.
- selection_data.csv and validation_data.csv: converged ramp/hold measurements.
- selection_profile.csv and validation_profile.csv: ramp breakpoints.
- *_predictions.csv: actual output, simulated output and residuals.
- candidate_scores.csv and search_checkpoint.mat: search results and candidate models.
- wiener_fit.png: training and held-out validation comparison and errors.
- experiments.mat: ramp data, metadata and original simulation-directory references.
- reproduction scripts and MATLAB logs: included for traceability. Simulation scripts require the original project layout and adjacent T-MATS installation.

5.2 Online acceleration identification with ten-term RLS

Evidence basis: `results/rls_acceleration_20260908_225810_094/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Created and simulated `GasTurbine_Dyn_Template_GPT_RLS.mdl` in MATLAB R2018b. The new model adds a discrete, online RLS observer to the GPT gas-turbine copy. It identifies actual fuel flow (lbm/s) to shaft acceleration (rpm/s), using the plant's direct `Ndot` signal. The original GPT file is unchanged.

The observer is very accurate for one-step prediction, but its frozen fuel-only model has moderate independent validation accuracy. These are different capabilities: the one-step predictor uses measured past acceleration; the frozen simulation does not.

5.2.1 Results

Fit is `100*(1-norm(reference-estimate)/norm(reference-mean(reference)))`, not R-squared. RMSE is in rpm/s. Chirp scores cover only the converged 0–17.790 s prefix, including initial learning transients.

Table 5.3: Online acceleration identification with ten-term RLS: Results. Archived values.

Evaluation	Fit	RMSE
Online RLS, prediction before current-sample update	99.8729%	0.7050
Persistence baseline: previous measured acceleration	97.1964%	15.5513
Online RLS, final five valid seconds	99.8779%	0.8860
Evolving coefficients, own predicted acceleration history	91.2627%	48.4640
Final frozen ARX, chirp free-run	88.3707%	64.5050
Final frozen ARX, independent 60 s ramp/hold free-run	62.6191%	30.5107

The evolving-coefficient free-run check still receives coefficient updates learned from reference data; it is not independent validation. The frozen ramp test uses no ramp acceleration for parameter estimation or prediction feedback. Free-run filters start from zero normalized state after the equilibrium preparation. No validation-based parameter tuning was performed. The smaller ramp RMSE despite its lower fit reflects the ramp's smaller acceleration variation.

5.2.2 Scenario and validity

The demanded speed is `9500-500*cos(2*pi*(0.1*t+0.0075*t^2))` rpm over 60 s: 9000–10000 rpm, with a linear frequency sweep from 0.1 to 1 Hz. A preceding 30 s preparation holds 10000 rpm to 5 s, ramps to 9000 rpm at 10 s and holds to 30 s. The simulation uses 0.015 s sampling and runs for 90 s total. The PI controller converts speed demand into actual fuel flow, which is the identification input.

The reference solver first fails the validity checks at chirp time **17.805 s** (absolute simulation time 47.805 s). Only 1187 samples through 17.790 s are usable, covering approximately 0.1–0.36685 Hz; 1185 RLS updates occur because near-zero initial regressors are skipped. Results do not establish accuracy over the complete requested 0.1–1 Hz sweep.

Updates require finite reference signals, flow residuals no greater than 1e-9, fewer than 200 inner solver iterations, nonnegative fuel and positive compressor stability margin. A failure latches the gate closed, freezes coefficients/covariance/history, and makes prediction output NaN. Restarting the simulation resets the latch. The full failed remainder is retained as diagnostic data, not training data.

The independent ramp profile has knots:

Table 5.4: Online acceleration identification with ten-term RLS: Scenario and validity. Archived values.

Time (s)	0	3	9	12	17	20	28	31	36	40	50	54	60
Demand (rpm)	9000	9000	9900	9900	9400	9400	9800	9800	9050	9050	10000	10000	10000

This gives ramps of +150, -100, +50, -150 and +95 rpm/s with intervening holds. Its entire 60 s evaluation passes the flow-residual and iteration checks; maximum absolute residual is 9.9971e-10.

5.2.3 Estimator implementation

Orders are `na=5, nb=5, nk=1`, exactly ten estimated coefficients, with no extra intercept. Define `y=Ndot/1000` and `u=(Wf-0.9810869146824126)/1`. The fuel offset is measured during preparation and frozen at the chirp start. Each sample forms

```
phi(k) = [-y(k-1), ..., -y(k-5), u(k-1), ..., u(k-5)]'
yhat(k|k-1) = phi(k)' * theta(k-1)
```

Thus the delay is one sample, 15 ms. Multiply normalized predictions by 1000 to recover rpm/s. The controller is unchanged; the estimator does not feed back into the plant. Its reference is direct shaft acceleration, avoiding numerical differentiation of speed in this experiment.

Initialization: `theta=0`, `P=10000*eye(10)`. Forgetting factor `lambda=0.999` gives approximately 1000 samples, or 15 s, of effective memory. For each accepted sample:

```
Pprior = P / lambda
K = Pprior*phi / (1 + phi'*Pprior*phi)
theta = theta + K*(y - phi'*theta)
J = eye(10) - K*phi'
P = J*Pprior*J' + K*K'
P = (P + P')/2
```

This Joseph-form covariance update improves numerical robustness. The Simulink S-function publishes predictions and coefficient states before its Update method, preventing current-output leakage into the reported one-step prediction. Covariance represents algorithm information under its assumed scaling; it is not a calibrated parameter confidence interval.

The green `RLS Acceleration Observer` subsystem contains the estimator. `Acceleration Reference vs RLS` displays reference and prediction. Workspace logs include `RLS_reference`, `RLS_yhat`, `RLS_error`, `RLS_theta`, `RLS_Pdiag`, `RLS_status`, `RLS_phi`, `RLS_u0` and `RLS_flowErrors`. Status is `[update_enabled, fault_latched, updates_before_sample]`.

5.2.4 Evolution and interpretation

The first 3 s prediction RMSE is 0.9218 rpm/s. The two-second rolling error decreases during early learning, then rises somewhat as chirp amplitude in acceleration and frequency increase. Its final-five-second fit remains 99.8779%. RLS lowers full-prefix RMSE about 22-fold versus persistence, so the improvement is substantial even against a strong smooth-signal baseline.

All sampled instantaneous ARX parameter sets have pole magnitudes below one; the final maximum magnitude is 0.978884. The adaptive own-output simulation also remains finite. Instantaneous pole stability alone is not a general stability guarantee for a time-varying model.

Several coefficients continue moving rather than settling. The normalized regressor matrix condition number is 1.0167e6 and its effective rank is 9 at a relative singular-value threshold of 1e-6. It is mathematically full rank, but the weakest direction is poorly identified. Total covariance trace rises

from 100000 to 107324.6 despite contraction in some directions. The high one-step fit therefore does not establish convergence of all ten physical dynamic parameters.

The ramp free-run plot shows underpredicted positive acceleration and nonzero acceleration during some holds. This reveals a low-frequency/equilibrium mismatch that is much less visible in one-step prediction. A single local linear model trained on a short oscillatory record does not fully reproduce the nonlinear fuel-to-acceleration dynamics across the operating range.

5.2.5 What works and what to improve

- **Works:** causal online updates, strong prediction accuracy, bounded simulated free-run behavior, explicit invalid-data protection, and a compact ten-coefficient estimator. The RLS arithmetic needs only a 10-by-10 covariance matrix and rank-one updates per sample; real-time execution timing was not benchmarked.

- **First priority:** obtain a fully converged excitation record spanning the required frequencies. Adjust demand amplitude/rate or controller/plant operating limits in a separately reviewed experiment. The current full-amplitude sweep cannot support a valid 60 s identification claim.

- **Improve excitation and conditioning:** collect longer valid chirps or multisine records at several equilibria, inspect singular values, and compare lower orders or resampling to avoid highly correlated adjacent 15 ms lags. Keep ten regressors if required, but consider stronger regularization or directional forgetting in poorly excited directions.

- **Improve free-run equilibrium behavior:** evaluate a speed-state formulation with acceleration derived from it, a physically constrained zero-DC fuel-to-acceleration model around equilibrium, or models scheduled by shaft speed. For a stable speed response, a settled fuel change should eventually give zero acceleration; the ramp holds expose this issue. Test any changes on a separate selection record, preserving the current ramp as a held-out benchmark.

- **Tune memory cautiously:** reducing lambda improves one-step fit slightly but greatly inflates covariance. The sensitivity results below do not justify choosing a smaller lambda solely for fit. Adaptation to a deliberately changing plant has not been tested.

- **For measured engine data:** direct Ndot is unavailable in many installations. Use a validated causal acceleration observer/filter and account for its delay. Closed-loop measurement noise can bias ordinary ARX least squares; investigate recursive output-error or instrumental-variable methods if noise tests show bias. The present ideal simulation does not establish noise robustness.

- **Maintain numerical safeguards:** covariance positivity and stability checks passed here. Stability projection or constrained parameter updates would be appropriate before deploying the estimator in control; the current observer only monitors the plant.

Table 5.5: Online acceleration identification with ten-term RLS: What works and what to improve. Archived values.

Forgetting factor	One-step fit	RMSE (rpm/s)	Final pole radius	Final covariance trace
1	99.8616%	0.7675	0.97829	38717.8
0.999 (implemented)	99.8729%	0.7050	0.97888	107324.6
0.995	99.8934%	0.5911	0.98095	4749947.8

5.2.6 Verification

- Offline replay reproduces every logged pre-update coefficient exactly.
- Final coefficients match exponentially weighted batch least squares to relative error 4.44e-8.
- Final coefficients match MATLAB System Identification Toolbox `recursiveLS` to relative error 4.81e-10.
- Covariance remains positive definite; final minimum eigenvalue is 1.1637e-4.
- No coefficient changes or updates occur after the reference-failure gate latches.
- Against the original valid chirp, maximum plant speed difference is 4.18e-11 rpm and fuel difference is 7.16e-13 lbm/s.
- Original GPT MDL SHA256 remains

`99D95A3F42A12DD42EE5C4353B8D3AE6DB2A54F7AC15A106FC72D0EEBBB31376`

.

5.2.7 Run and files

From the existing TMATSGPT folder, with its adjacent T-MATS libraries and standard example setup installed:

```
tmats_rls_setup;
open_system('GasTurbine_Dyn_Template_GPT_RLS');
sim('GasTurbine_Dyn_Template_GPT_RLS');
```

The model also runs from the Simulink Run button after setup. Changing the scenario requires updating `MWS.in` after loading the model; loading invokes its default setup callback. `run_tmats_rls_batch` runs the chirp plus ramp and exits MATLAB, for command-line use. `analyze_tmats_rls_batch` analyzes the latest run and also exits MATLAB; call `analyze_tmats_rls` interactively instead.

The batch workflow uses the existing `tmp/wiener_ramp_directory.txt` and previous chirp results for its validation profile and unchanged-plant assertion. The archive is a bundle for this existing T-MATS workspace, not a standalone T-MATS distribution. The model itself only requires the supplied setup/S-function/step files and adjacent T-MATS installation.

`rls_valid_evolution.csv` contains estimates, reference, error, all ten coefficients, covariance diagonals and rolling metrics. `full_chirp_gate_diagnostics.csv` preserves the complete chirp gate history. `frozen_ramp_validation.csv` contains the independent comparison. `final_arx_coefficients.json` contains A/B polynomials, physical scaling and offset. MAT files retain full simulation and analysis data. PNGs show estimation evolution, free-run comparisons and gate behavior.

5.3 Realizing the estimator with ordinary Simulink blocks

Evidence basis: `results/rls_blocks_acceleration_20260909_063846_871/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

The new model is `GasTurbine_Dyn_Template_GPT_RLS_Blocks.mdl`. Its complete acceleration observer is implemented using basic Simulink blocks; there are **zero S-functions and zero MATLAB Function blocks inside the observer**. The T-MATS reference plant retains its existing component S-functions. MATLAB R2018b successfully compiled and simulated the chirp and independent ramp/hold experiments.

5.3.1 Implementation

Double-click the green `RLS Acceleration Observer` subsystem. The top level contains input sampling, normalization, ten explicit scalar history delays, coefficient/covariance states, update switches, output scaling and logging. The nested `RLS matrix arithmetic` subsystem exposes every calculation. `Validity and update gate` exposes the acceptance checks and latched fault logic. The inventory contains 157 blocks including subsystem and mask internals.

Ordinary Unit Delay blocks hold theta, P, the histories, fuel offset and update count. Matrix Product blocks, transpose Math Function blocks, Sum and Gain blocks implement the RLS equations. Switch blocks accept the new theta/P only when enabled. No custom executable estimator code is hidden behind the subsystem masks. The `.m` setup file initializes parameters; the `.m` builder constructs the diagram. Neither executes the estimator during simulation.

The configuration is unchanged: actual fuel flow in lbm/s to direct shaft acceleration Ndot in rpm/s, ARX(5,5,1), ten regressors, Ts=0.015 s, lambda=0.999, initial theta=0 and P=10000*I. Normalize acceleration by 1000 and subtract the preparation fuel equilibrium, 0.9810869146824126 lbm/s, from input fuel. The output remains in rpm/s.

```
phi(k) = [-y(k-1), ..., -y(k-5), u(k-1), ..., u(k-5)]'
yhat(k|k-1) = phi(k)' * theta(k-1)
Pprior = P / lambda
K = Pprior*phi / (1 + phi'*Pprior*phi)
thetaNext = theta + K*(y - phi'*theta)
J = I - K*phi'
Pnext = J*Pprior*J' + K*K'
Pnext = (Pnext + Pnext')/2
```

The Unit Delays ensure predictions use the coefficients from the preceding update. Joseph-form covariance arithmetic preserves the same algorithm as the previous implementation. The matrices are visible signal values, not a call to the earlier `tmats_rls_step` helper. That helper is used only for independent offline verification.

5.3.2 Experiment and results

After 30 s of preparation, the command is `9500-500*cos(2*pi*(0.1*t+0.0075*t^2))` rpm for 60 s. It oscillates between 9000 and 10000 rpm, sweeping 0.1–1 Hz. Actual fuel flow produced by the existing PI controller is the RLS input.

The unchanged reference plant first becomes invalid at **17.805 s of chirp time**. The gate freezes the observer and makes its prediction NaN from that point. There are 1187 valid samples through 17.790 s and 1185 coefficient updates. Consequently, the chirp metrics establish performance only over approximately 0.1–0.36685 Hz, not the full requested frequency range.

Fit is `100*(1-norm(y-yhat)/norm(y-mean(y)))`, not R-squared.

Table 5.6: Realizing the estimator with ordinary Simulink blocks: Experiment and results. Archived values.

Evaluation	Fit	RMSE (rpm/s)
Online, pre-update one-step chirp prediction	99.8729%	0.7050
Persistence baseline, previous measured acceleration	97.1964%	15.5513
Last five valid seconds, online prediction	99.8779%	0.8860
Own-output simulation with evolving learned coefficients	91.2627%	48.4640
Final frozen model, chirp free-run	88.3707%	64.5050
Final frozen model, independent 60 s ramp/hold free-run	62.6192%	30.5107

The frozen ramp simulation uses no ramp output samples for adaptation or output feedback. Its demand includes ramps of +150, -100, +50, -150 and +95 rpm/s and holds within 9000–10000 rpm. All ramp flow residuals pass the 1e-9 criterion; their maximum is 9.9971e-10. The profile is included as `validation_profile.csv`.

The evolving-coefficient own-output simulation still uses parameters learned online from reference acceleration, so it is not an independent validation result. Frozen free-run filters start with zero normalized state following equilibrium preparation. Lower RMSE on the ramp despite lower percentage fit reflects its smaller acceleration variation.

5.3.3 Agreement with the previous implementation

Every sample was compared against the previous S-function simulation, including invalid-output locations and the entire frozen remainder.

Table 5.7: Realizing the estimator with ordinary Simulink blocks: Agreement with the previous implementation. Archived values.

Logged quantity	Maximum absolute difference
Acceleration prediction	3.2557e-8 rpm/s
Coefficients	3.8216e-9
Covariance diagonal	2.5763e-5
Regressor	2.2204e-16
Update/fault/count status	0
Fuel offset	0

These small differences are consistent with floating-point operation ordering in matrix arithmetic; covariance entries are on the order of thousands to tens of thousands. The independent numerical checks also passed: offline replay coefficient error 3.13e-9, weighted-batch relative coefficient error 4.42e-8, and MATLAB System Identification Toolbox `recursiveLS` relative coefficient error 9.81e-10. Final covariance is positive definite, with minimum eigenvalue 1.1637e-4.

Reference speed differs from the original valid chirp by at most 4.18e-11 rpm and fuel by 7.16e-13 lbm/s. The original GPT and previous RLS model files were preserved. `observer_block_inventory.csv` and `basic_blocks_audit.json` record the implementation audit.

5.3.4 Evolution, successes and improvements

The one-step estimate tracks the reference closely, reducing RMSE about 22-fold relative to persistence. First-three-second RMSE is 0.9218 rpm/s; final-five-second RMSE is 0.8860 rpm/s. `rls_evolution.png` shows the output/error, ten coefficient trajectories, covariance diagonals, rolling error and pole magnitudes. `rls_free_run_and_gate.png` distinguishes one-step performance from independent fuel-only simulation.

All sampled parameter sets have pole magnitudes below one, with a final maximum of 0.978884. No adaptive free-run divergence occurs in the valid segment. This is useful evidence for this run, not a general guarantee of time-varying stability.

The block implementation makes the states, update order, gating and numerical calculations inspectable without reading S-function code. It reproduces the earlier estimator rather than changing identification performance. Runtime or deployment speed was not benchmarked, and the complete T-MATS model's code-generation suitability was not tested.

The principal identification limitations are unchanged:

1. **Insufficient valid excitation:** the reference failure prevents learning from the complete frequency sweep. Obtain a longer converged excitation record before drawing conclusions across 0.1–1 Hz. 2. **Weak coefficient determination:** regressor condition number is 1.0167e6, with effective rank 9 at relative threshold 1e-6. Some coefficients keep moving and covariance trace increases from 100000 to 107324.6. A high prediction fit does not establish convergence of all ten parameters. Compare lower orders, less correlated sampling, or regularization on a separate selection set. 3. **Moderate free-run generalization:** the 62.6192% ramp fit and nonzero estimated acceleration on some holds expose low-frequency/equilibrium mismatch. Consider a physically constrained speed-state model, a zero-DC acceleration model around equilibrium, or speed-scheduled local models. 4. **Memory tuning:** the repeated sensitivity calculation gives one-step fits of 99.8616%, 99.8729% and 99.8934% at lambda=1, 0.999 and 0.995. The smallest lambda inflates covariance trace to about 4.75e6. Its slight prediction gain does not justify selecting it without further validation. 5. **Measured-data robustness:** this experiment uses ideal direct Ndot. Real speed differentiation needs causal filtering/observation and delay treatment; closed-loop measurement noise may require methods beyond ordinary ARX least squares.

5.3.5 Run and deliverables

From the existing TMATSGPT folder:

```
tmats_rls_setup;
open_system('GasTurbine_Dyn_Template_GPT_RLS_Blocks');
sim('GasTurbine_Dyn_Template_GPT_RLS_Blocks');
```

To inspect just the arithmetic:

```
open_system('GasTurbine_Dyn_Template_GPT_RLS_Blocks/RLS Acceleration Observer/RLS matrix
arithmetic');
```

The Simulink Run button also works after setup. The model's load callback installs the default scenario, so set custom MWS inputs after loading. The completed model does not depend on `tmats_rls_arx_sfun.m` to simulate.

`TMATS_RLS_BLOCKS_USAGE.md` explains the workflow. Command-line batch scripts rebuild, simulate and analyze; they exit MATLAB when finished. They use the existing earlier-result files for comparisons. The archive is intended for the current adjacent T-MATS installation, not as a standalone engine distribution.

The bundle includes the actual MDL model, setup/builder/analysis source, both raw MAT simulations, coefficients, full evolution and validation data, numerical metrics, block inventory, block diagrams and result plots.

5.4 Two-stage sparse RLS: conditioning and held-out prediction

Evidence basis: `results/two_stage_rls_20260912_125027_844/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Stage 1 used reorthogonalized forward selection over 20 candidates (output/fuel delays 1–10), with relative residual-energy threshold 1e-06. Stage 2 used RLS with lambda 0.9990 and P0 10000. The smallest stable candidate within 0.010 normalized MSE of the best selection score was chosen. No final-validation outputs entered support selection.

Selected **8 terms**: output delays **[1 2 4 6]**, fuel delays **[1 2 5 10]** (samples; multiply by 15 ms). Candidate path contains 9 admissible models.

Table 5.8: Two-stage sparse RLS: conditioning and held-out prediction: Study results. Archived values.

Evaluation	Selected fit	Previous ten-term fit
Chirp one-step	99.8720%	99.8729%
Chirp frozen free-run	87.3880%	88.3707%
Ramp final validation free-run	65.2666%	62.6191%

Selected training one-step RMSE: 0.710104 rpm/s; training free-run RMSE: 69.956083 rpm/s. Final validation RMSE: 28.349800 rpm/s versus previous 30.510694 rpm/s. Selection-ramp fit: 66.8846%.

Selected design-matrix condition number 15571.7 versus previous 1.01674e+06. Final maximum pole magnitude 0.97774209. Stage-2 covariance has 64 entries versus 100; this is theoretical storage/work reduction, not a measured whole-system speedup. The 20-candidate selection stage and longer lag buffer add overhead during learning.

Validation fit changes by +2.6475 percentage points, with 7.08% lower validation RMSE. Covariance storage decreases by 36% and conditioning improves by a factor of 65.29. Training free-run fit changes by -0.9827 percentage points. The result is a modest generalization improvement with a smaller, better-conditioned model, not a uniformly more accurate model.

The selected sample delays correspond to output delays `[15 30 60 90]` ms and fuel delays `[15 30 75 150]` ms. Stage 1 stopped before ten terms because remaining candidates failed the declared redundancy test. Candidate-path CSV retains rejected unstable candidates; the plot shows stable candidates only.

5.4.1 Reproduction and verification

Run `results = analyze_two_stage_rls;` in this project, or pass this folder's `offline_inputs.mat` as an argument. The function does not exit MATLAB or launch Simulink. Raw data are reused

offline. Offline saved chirp and separate selection/validation ramps; selection Ndot reconstructed by verified shaft power law.

Selection acceleration uses Ndot = 60/(2*pi*30) * 5252.113 * (compressor_power/N + turbine_power/N), using signed power in hp, N in rpm and zero external load. This was checked against both chirp and validation direct Ndot logs; maximum difference 5.59e-10 rpm/s.

Weighted augmented QR and RLS coefficients agree to relative error 1.58e-11; MATLAB recursiveLS agreement 1.57e-11. Covariance minimum eigenvalue 0.000147. The repeated baseline prior predictions match previous logs within 2.55e-08 rpm/s.

5.4.2 Interpretation and limits

Only the valid chirp prefix 0–17.790 s (1187 samples) was used. Reference failure begins at 17.805 s; no failed samples enter the dictionary. Because the support is chosen using the full prefix, its replay one-step fit is an offline training measure, not a fully causal online structure-selection result. Weights still predict before each current-sample update.

The final coefficients were frozen for the 60 s validation ramp, using zero normalized filter states after equilibrium preparation. Greedy forward selection is not an exhaustive search. The predetermined redundancy threshold can exclude useful correlated dynamic terms; this run does not tune it against final validation. Sparse linear structure cannot by itself guarantee correct equilibrium or nonlinear behavior.

A(z^-1), including zero coefficients at skipped lags:

`[1 -1.8943373173 0.84354606889 0 0.0730433076284 0 -0.0213582767346 0 0 0 0]`

.

B normalized:

`[0 0.361940693762 -0.497137697864 0 0 0.147924362846 0 0 0 0 -0.0127326672965]`

.

Fuel offset 0.9810869146824126 lbm/s; acceleration scale 1000 rpm/s. Coefficient ordering follows selected dictionary indices `[11 1 2 4 12 15 20 6]`.

The polynomial convention is `A(z^-1)*y = B(z^-1)*u + e`, where `y=Ndot/1000` and `u=Wf-Wf0` in the stated units. Missing lags have exactly zero coefficients. The final lag set is frozen for this offline experiment; online support switching was not simulated.

Files include the candidate path, full input data, weight/covariance evolution, comparison tables, coefficients, MAT results and plot. For the mathematical proposal and literature see `TWO_STAGE_RLS_PROPOSAL.md` in the project.

5.5 Answer supported by this evidence

Strong one-step prediction does not establish equally strong free-running transfer. Sparse selection improves the recorded conditioning and held-out acceleration result, while block replay verifies the realization. Neither result establishes robustness under untested noise or plant changes. The selected lag support was determined offline; the later coefficient recursion does not make support selection an online discovery procedure.

Chapter 6

Disturbance estimation and the accepted load envelope

This chapter addresses whether an identified model can be inverted usefully. It separates the residual-estimation idea from the particular filter, model correction, and accepted plant envelope. DOB, LESO, UDE, and GPIO are established method families, not inventions of this study [16, 12, 13]. Equal nominal pole settings do not imply equal closed-loop bandwidth.

6.1 Understanding disturbance observers before the case studies

6.1.1 Disturbance rejection as a residual calculation

Suppose a nominal linear plant P_n predicts the measured response to fuel. If the real response differs, the discrepancy may arise from an external load, model error, sensor approximation, or initialization. A disturbance observer expresses that discrepancy in input units. It does not separately identify every physical cause.

For the idealized relationship $y = P_n(u + d)$, formal inversion gives $P_n^{-1}y - u = d$. Direct inversion is usually unsuitable: differentiating or advancing measurements amplifies noise or requires future samples, while an unstable inverse cannot be implemented reliably. The practical construction is

$$\hat{d} = \frac{Q}{P_n}y - Qu, \qquad u = u_{\mathrm{PI}} - \gamma\hat{d}. \tag{6.1}$$

Q is a low-pass filter chosen so that Q/P_n is causal and stable. The subtraction sign follows from the chosen input-disturbance convention. A negative equivalent fuel disturbance requires a positive compensating fuel correction. An external resisting shaft load often appears this way, but its hp value is not numerically equal to the inferred disturbance in lbm/s.

Under exact modeling, no saturation, and a constant γ, the relationship is $u = u_{\mathrm{PI}} - \gamma Qd$. At low frequency, $Q \simeq 1$, so a fraction γ of the disturbance is cancelled. At high frequency, Q attenuates the estimate. This simplified derivation explains the design intent. It does not establish robust stability when $P \neq P_n$, the operating point changes, or the fuel limit is reached.

6.1.2 Zeros, poles, and an equilibrium constraint

A stable forward model has poles inside the unit circle. Its inverse exchanges poles and zeros, so a forward zero outside that circle becomes an unstable inverse pole. The early eight-term acceleration model has just such a zero, near 1.00026356. This is not fixed by claiming that its prediction fit is high.

A separate physical observation is that a constant fuel perturbation around a stable settled speed should eventually produce zero acceleration. The raw identified numerator has a small nonzero sum $B(1)$, giving a spurious acceleration DC gain. The project changes the last numerator coefficient so

that $B_{\mathrm{eq}}(1) = 0$, allowing

$$B_{\mathrm{eq}}(q) = (1-q)C(q), \qquad P_N(q) = \frac{T_s C(q)}{A(q)}. \tag{6.2}$$

The factor $(1-q)$ cancels the corresponding discrete integration factor when converting acceleration dynamics to speed dynamics. This is an explicit physical modeling constraint. It improves the inversion structure, but the altered equilibrium gain still requires empirical evaluation. A small coefficient correction can have a large structural consequence near a zero at unity.

6.1.3 Why an equation residual is not always a load estimate

The direct ARX residual $(Aa - Bu)/b_1$ is causal and easy to calculate. If the true acceleration is driven by $u + d$, however, that residual is $B(q)d/b_1$, not d. For the raw model, $B(1)/b_1$ is approximately -1.47×10^{-5}. A constant disturbance is nearly suppressed by this mapping. It is therefore possible for a residual to be an excellent detector of equation mismatch and a poor estimate of a sustained physical input disturbance. The later filtered speed inverse addresses a different estimation problem.

6.1.4 Three alternative observer structures

The literature-inspired comparison normalizes sensed speed into a signal x so that the approximate first-order relation is $\dot{x} = v + d$. Its gain normalization uses the identified leading coefficient, not a newly identified exact continuous plant gain. An observer estimates d from x and the known command v.

A linear extended-state observer (LESO) uses an estimated speed state and disturbance state:

$$\dot{\hat{x}} = v + \hat{d} + 2\omega_o(x - \hat{x}), \tag{6.3}$$

$$\dot{\hat{d}} = \omega_o^2(x - \hat{x}). \tag{6.4}$$

The estimation error polynomial is $(s + \omega_o)^2$ under the nominal model. Increasing ω_o makes the nominal observer faster, but it also gives more weight to rapid measurement changes and unmodeled behavior.

A first-order uncertainty and disturbance estimator (UDE) can be realized without directly differentiating measured speed:

$$\dot{z} = -\omega_o z - \omega_o^2 x - \omega_o v, \qquad \hat{d} = z + \omega_o x. \tag{6.5}$$

This realization requires one dynamic state. It is a filtered residual estimator, not a direct measurement of the unknown load.

A generalized proportional-integral observer (GPIO) adds an estimated disturbance-rate state:

$$\dot{\hat{x}} = v + \hat{d} + 3\omega_o(x - \hat{x}), \tag{6.6}$$

$$\dot{\hat{d}} = \hat{h} + 3\omega_o^2(x - \hat{x}), \tag{6.7}$$

$$\dot{\hat{h}} = \omega_o^3(x - \hat{x}). \tag{6.8}$$

The nominal error polynomial is $(s+\omega_o)^3$. The additional state represents locally varying disturbance rather than guaranteeing better performance in every metric. All three are discretized through exact zero-order-hold matrix exponentials in the archived implementation. Their nominal pole frequency is shared at 0.5 Hz, but equal pole frequency does not mean equal frequency response or equal noise sensitivity.

6.1.5 Reading the PTO envelope correctly

PTO means externally extracted shaft power. A load expressed as a percentage is incomplete unless its reference power is stated. At 9000 rpm, the 10% observer-comparison load is about 0.879124 MW; at 9500 rpm it is about 1.428820 MW. The higher-speed test therefore applies about 62.5% more absolute power even though both are called 10% tests. Comparing their raw errors does not isolate speed dependence.

Load ramps also differ from load steps. The 9000-rpm envelope uses finite rise times of 0.3, 1.5, and 3 s within its pulse sequence. A load edge may move the compressor into a difficult region before the speed controller can respond. The fact that the 9000-rpm cases remain accepted through 30% load does not imply that 30% is acceptable at 9500 rpm. Similarly, an invalid 20% case and a valid 30% case in a broader method matrix do not prove a monotone stability boundary: nonlinear path dependence and numerical convergence require case-specific inspection.

The following archived-study chapters retain the detailed equations, numerical checks, and rejected cases. Their reproduction commands describe what produced the historical record; they were not executed to prepare this report.

6.2 The first filtered-inverse DOB and governed-reference tests

Evidence basis: `results/dob_refined_20260912_175532_449/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

6.2.1 Scope and source

This study adapts Volkan Aran, *Flexible and Robust Control of Heavy Duty Diesel Engine Airpath Using Data Driven Disturbance Observers and GPR Models*, PhD thesis, Sabanci University, 2019, Chapter 4, particularly equations (4.11)-(4.16) and Chapter 6.1. Public source: `https://research.sabanciuniv.edu/39260/2/10279446_VolkanAran.pdf`

The thesis distinguishes nominal linear dynamics from lumped model error, nonlinearity, and disturbances. It estimates a control-equivalent residual and filters it before subtractive compensation. Here the outer controller remains the sample plant's PI (Kp=0.025, Ki=0.05), rather than the thesis's sliding-mode controller. This is an adaptation and nonlinear simulation study, not a reproduction of the diesel-engine experiment or a stability proof.

The eight-term frozen ARX from `analyze_two_stage_rls.m` is the starting model. The standard GPT model is not edited. `GasTurbine_Dyn_Template_GPT_DOB.mdl` contains ordinary Simulink blocks in its DOB subsystem; the original T-MATS plant still uses its component S-functions. Online coefficient adaptation is disabled so that observer/controller interactions can be evaluated first.

6.2.2 Derivation and the direct-residual comparator

Let q denote a one-sample delay, Ts=0.015 s, u=Wf-u0 in lbm/s, a=dN/dt in rpm/s, and u0=0.9810869146824126 lbm/s. The identified physical model is

```
A(q) a_k = B(q) u_k + epsilon_k,
```

```
A(q) = 1 - 1.8943373173 q + 0.8435460689 q^2
         + 0.0730433076 q^4 - 0.0213582767 q^6,
B(q) = 361.9406938 q - 497.1376979 q^2
         + 147.9243628 q^5 - 12.7326673 q^10.
```

The thesis equation (4.13) can be adapted to delayed ARX regressors without introducing a noncausal inverse. Isolating b1 u_(k-1) gives the equivalent input residual

```
dres_k = [A(q) a_k - B(q) u_k] / b1,
dhat_res = Qres(q) dres,
ucmd = sat(uPI - gamma dhat_res).
```

This is a delayed residual-based comparator, not an exact copy of the thesis's current-state approximation (4.15). Acceleration is obtained causally from sensed speed by backward difference in this branch. With autoregressive output terms, the residual is an equation disturbance. It is not generally the physical fuel disturbance: for an additive fuel disturbance d, it is B(q)d/b1. In particular, B(1)/b1 is approximately -1.47e-5, so near-perfect one-step prediction does not make this residual a useful constant-fuel-disturbance estimate.

6.2.3 Equilibrium-consistent nominal model and filtered inverse

The raw B polynomial has a zero at 1.00026356, outside the unit circle. Its direct inverse would be unstable. In addition, B(1)=-0.00530855346 produces the small spurious nonzero acceleration DC gain identified in the preceding report.

The following is an explicit project-specific modeling constraint, not a formula claimed from the thesis. Keep A and all other B coefficients fixed and set

```
Beq_10 = B_10 - B(1),   hence Beq(1)=0.
```

This changes the last physical numerator coefficient by +0.00530855346, about 0.0417% of its magnitude. Writing Beq(q)=(1-q) C(q), a backward-integration approximation gives the nominal speed model

```
PN(q) = Ts C(q) / A(q).
```

Its DC gain is about 335.425 rpm/(lbm/s). This equilibrium constraint removes an inversion defect but does not establish that the resulting DC gain accurately matches the nonlinear engine throughout the speed range. Its largest remaining zero magnitude is approximately 0.952616.

The observer uses the same sensed speed available to the PI. Approximate the existing 0.05 s sensor with H(q)=(1-p)q/(1-pq), p=exp(-Ts/0.05), and define Pn=PN H. Use a second-order low-pass with two delays:

```
Q(q) = (1-r)^2 q^2 / (1-rq)^2,  r=exp(-2 pi fc Ts),
dhat = [Q/Pn](Nsensed-9000) - Q(ucmd-u0),
dtrim = last dhat sample before engagement,
dc = dhat - dtrim,
h(t) = clip((t-30)/2, 0, 1),
ucmd = sat_[0.2,4](uPI - h(t) sat_[-0.15,0.15](gamma dc)).
```

Final settings are gamma=0.25 and fc=0.5 Hz. The pre-engagement trim is latched using a Switch and Unit Delay. It removes the initial operating-point bias; it does not remove model mismatch that changes with operating point.

The two advances in Pn^-1 cancel analytically against the two Q delays. The resulting Q/Pn is causal and stable. Its implementation uses discrete transfer function blocks with coefficient lengths padded so they represent polynomials in q correctly. fc names the pole-frequency parameter, not the cascade's exact -3 dB frequency. The sensor discretization and acceleration integration are approximations; nonlinear simulation remains the relevant comparison.

The observer sees saturated commanded fuel before an unknown test disturbance is added. It does not see the injected disturbance or direct plant acceleration. Compensation ramps in over two seconds after 30 s preparation. Both controllers use identical fuel limits and demand conditioning. Saturation activity must be reported; anti-windup is not added to the unchanged PI in this experiment.

6.2.4 Experiment and acceptance

The previous unconditioned 9000/10000 rpm PRBS at 0.51 s/bit failed at its first edge. Therefore this comparison distinguishes raw binary demand from applied speed demand. A 150 rpm/s governor makes the full-range case a rate-limited PRBS, not a strict binary plant input. The 60 s evaluation uses a truncated maximal length sequence, 10.005 s/bit, after 30 s preparation. A separate 9495/9505 rpm PRBS at 0.51 s/bit tests strict binary demand with the governor effectively bypassed. The 127-bit recurrence is b(k+7)=xor(b(k),b(k+3)).

Matched positive/negative fuel offsets and a sinusoidal fuel disturbance are injected after the command limit. Nominal and disturbed runs use paired command sequences. Tracking RMSE, IAE, peak error and fuel variation are evaluated only after preparation; numerical validity is checked over the entire simulation. Every accepted sample requires positive fuel, finite speed/fuel, flow residual at most 1e-9, fewer than 200 solver iterations, and positive compressor margin. Results from failed trajectories cannot substantiate control improvement.

6.2.5 Reproduction

```
build_tmats_dob_model;
outputDirectory = run_tmats_dob_study('refined');
[MWS,DOB] = tmats_dob_setup;
open_system('GasTurbine_Dyn_Template_GPT_DOB');
```

`run_tmats_dob_batch.m` is the isolated batch entry point. Each experiment saves native MAT data, CSV signals and numerical metrics under `results/dob_*`. The saved model's default setup reproduces the accepted disturbed full-range comparison, with the rate-limited demand. `analyze_tmats_dob(outputDirectory)` regenerates plots and checks; `verify_tmats_dob_saved` rebuilds the model and checks default replay against the latest case 2. Call `tmats_dob_setup(struct('gain',0.25,'fc',0.5))` to recompute filter coefficients when changing settings. Set `DOB.gain=0` after loading for the baseline, or set `DOB.disturbance(:,2)=0` for the nominal case. A changed fc requires recomputing the coefficients, not merely editing the scalar field after setup.

6.2.6 Results: accepted final configuration

The final results are in `results/dob_refined_20260912_175532_449`. All first seven cases passed the numerical acceptance checks for the entire 90 s simulation, including preparation. Maximum inner iterations were 20; all residuals stayed below 1e-9. Fuel remained positive and all operating comparisons stayed clear of the shared fuel limits. The minimum compressor margin was 14.9054% in the full-range pair. No solver tolerances were relaxed.

Table 6.1: The first filtered-inverse DOB and governed-reference tests: Results: accepted final configuration. Archived values.

Experiment / measure	Existing PI	PI + DOB	Improvement
Rate-limited 9000-10000 rpm, disturbed: tracking RMSE (rpm)	1.42705	1.20710	15.41%
Same test: IAE (rpm s)	54.0001	43.4278	19.58%
Same test: peak tracking error (rpm)	8.12493	7.21189	11.24%
Rate-limited nominal: tracking RMSE (rpm)	1.32431	1.08438	18.12%
Strict 9495/9505 rpm PRBS, disturbed: RMSE (rpm)	2.29021	2.26952	0.90%
Paired disturbance-induced speed RMS (rpm)	0.63657	0.60068	5.64%
Paired disturbance-induced peak (rpm)	4.20270	4.17338	0.70%

The paired rejection measure uses

```
deltaN_PI  = N_PI,disturbed  - N_PI,nominal,
deltaN_DOB = N_DOB,disturbed - N_DOB,nominal,
```

on identical 60 s demand histories. This removes nominal tracking behavior from the comparison, though nonlinear interactions still remain. The 15.41% tracking improvement must not be described as a 15.41% disturbance-rejection improvement: the latter is only 5.64% by this paired RMS measure.

The full-range disturbance sequence is +0.08 lbm/s over 7.5-15 s, -0.08 lbm/s over 20-27.5 s, and a 0.06 lbm/s, 0.2 Hz sine over 35-50 s. The clock here starts after preparation. Injection is downstream of the known command. These tests cover matched additive actuator/fuel disturbances, not independent shaft-load, thermal, ambient, or sensor disturbances.

The original ARX equation-residual comparator gives 1.42594 rpm RMSE versus 1.42705 rpm for PI: only about 0.077% improvement. This confirms that simply feeding back the small ARX prediction innovation does not yield a strong input disturbance observer for this model.

What the result establishes and what it does not

1. **Useful bounded augmentation.** The existing PI gains were preserved. All accepted final comparisons improved tracking, although the strict small PRBS improvement is modest. Known-command total variation changed from 12.2658 to 12.2716 lbm/s in the disturbed full-range test, about +0.047%; this is a sampled variation measure, not an actuator wear or fuel-economy measurement. 2. **Model mismatch dominates the broad-range estimate.** The raw lumped

estimate approaches roughly 1 lbm/s during high-speed operation, far larger than the injected 0.08 lbm/s offset. The correction reaches its 0.15 lbm/s limit for about 50.44% of the full-range disturbed evaluation. It must not be interpreted as a calibrated estimate of physical fuel error. Around the fixed 9500 rpm operating point, the trim-removed estimate follows the injected offset much more closely, and compensation does not reach its limit. 3. **A governor is essential in the full-range example.** Raw PRBS tracking RMSE is 383.611 rpm for PI and 383.789 rpm for PI+DOB, because the applied demand is intentionally limited to 150 rpm/s. The small applied-demand error does not mean that instantaneous 1000 rpm steps are tracked. In new unconditioned 0.51 s/bit tests, both controllers first fail flow convergence at 0.525 s, one sample after the first edge. Those three-second records are diagnostic only. Their complete-trajectory error scores in the machine-readable table must not be used to rank controllers. 4. **Bounded correction matters.** The initial gamma=0.5 design improved the disturbed full-range case more, but its nominal pair failed at 40.755 s and the strict small case worsened before trim compensation was added. Those settings were rejected. The final gamma=0.25 setting passed the intended tests. Refinement used these evaluation cases; they are comparative design experiments, not an untouched statistical validation set. 5. **No general robustness or deployment claim.** The tests are deterministic, noise-free sample-engine simulations. They do not establish closed-loop robustness over an operating envelope, sensor-noise tolerance, load rejection, fixed-point suitability, or worst-case embedded execution time. The model computes both observer branches for comparison; a deployed implementation should retain only the chosen branch after validation.

6.2.7 Numerical and implementation verification

The analytical Q/Pn and block coefficients agree to 1.22e-14 over 1001 points from DC to Nyquist. Maximum inverse pole magnitude is 0.953969. This checks the filter realization, not stability of the uncertain nonlinear closed loop. The DOB contains 32 ordinary blocks, zero S-functions and zero MATLAB Function blocks. The new top-level speed scope displays applied demand and sensed speed. Default-model replay after layout changes agrees with accepted case 2 to 4.92e-11 rpm and 5.11e-15 lbm/s.

The reference GPT model SHA-256 remains

`99D95A3F42A12DD42EE5C4353B8D3AE6DB2A54F7AC15A106FC72D0EEBBB31376`

.

Result files

Table 6.2: The first filtered-inverse DOB and governed-reference tests: Result files. Archived values.

File in final result directory	Contents
`case_01` / `case_02` MAT and CSV	Disturbed full-range, PI / PI+DOB
`case_03` / `case_04`	Nominal full-range, PI / PI+DOB
`case_05` / `case_06`	Strict small PRBS, PI / PI+DOB
`case_07`	Delayed thesis-style ARX residual comparator
`case_08` / `case_09`	Failed strict full-range diagnostic trajectories
`metrics.csv`, `metrics.json`	Tracking, limits, fuel and validity measures
`verification.json`, `default_replay.json`	Mathematical/block checks and replay
`comparison_1.png` to `comparison_3.png`	Speed, error, fuel and observer histories
`disturbance_rejection.png`	Paired disturbance-induced speed comparison
`dob_blocks.png`	Actual basic-block observer implementation

6.2.8 Recommended next development

First fit equilibrium-consistent local fuel-to-speed models, with measured operating-point gains, rather than relying on a small post-fit coefficient adjustment in an acceleration ARX model. Preserve a new untouched PRBS/load record for assessment. Then test causal sensing with noise, shaft-load and ambient disturbances, filter/compensation bandwidth sensitivity, and observer disable/fallback behavior. Add PI anti-windup before broadening the operating range into fuel saturation. Only after these checks should fixed-point or single-precision code generation and target timing be assessed. Online RLS updates should remain disabled until pole/zero acceptance, bumpless transfer, and closed-loop robustness checks are in place.

6.3 External shaft-load rejection at 10000 rpm

Evidence basis: `results/dob_pto_20260912_182940_001/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

The original PI is compared against the existing frozen-model PI+DOB, without retuning. The speed demand stays at 10000 rpm throughout. Fuel-injection disturbances are disabled.

6.3.1 Physical definition and scenario

A 60 s unloaded preparation is followed by 90 s evaluation. The reference is mean gross turbine shaft power over preparation seconds 55-60: **27896.507327 hp** (20802.421931 kW). Mean compressor load is 27896.507327 hp; unloaded net shaft power is 0.000000000 hp. Thus 10% is taken from gross turbine power, which balances compressor load at steady speed, not from near-zero net power.

Each extraction pulse is fixed at **2789.650733 hp = 2080.242193 kW** for both controllers. It is not recomputed as a percentage of the changing turbine output under load. Sudden on/off edges occur at 10.005/15.000, 30.000/40.005, and 60.000/75.000 s after preparation. These represent three applications of the same 10% additional load, with 4.995, 10.005 and 15.000 s holds.

The installed Shaft block power port receives positive extraction in hp. Its equation is

```
dN/dt = 60/(2*pi*J) * (T_turbine + T_compressor - 5252.11*P_PTO/N),
```

with the original J=30. The turbine and compressor torque conversions use 5252.113 in the component code. The observer sees sensed speed and known fuel command only; the load profile and plant acceleration are not observer inputs.

6.3.2 Comparison

PI gains remain Kp=0.025, Ki=0.05. DOB gain=0.25, Q pole-frequency parameter=0.5 Hz, compensation bound=0.15 lbm/s. Trim is latched at 30 s and compensation ramps in over two seconds, well before the evaluation.

Table 6.3: External shaft-load rejection at 10000 rpm: Comparison. Archived values.

Measure	PI	PI + DOB
RMSE (rpm)	4.821036	4.627883
IAE (rpm s)	103.1914	86.3594
Maximum speed droop (rpm)	38.23095	37.97716
Maximum overspeed (rpm)	40.12642	40.12642
Fuel min (lbm/s)	2.746102	2.746102
Fuel max (lbm/s)	4	4
Fuel mass in 90 s (lbm)	295.7642	295.7748
Fuel-limit fraction	0.005499083	0.006498917
DOB correction-limit fraction	0	0.3187802
Maximum flow residual	9.983553e-10	9.998794e-10
Maximum iterations	14	14
Minimum compressor margin (%)	7.635762	7.642796

Both full trajectories pass numerical/physical checks: PI=1, PI+DOB=1. Tracking RMSE change is 4.006% improvement; maximum droop change is 0.664% improvement.

The main benefit is faster recovery rather than suppression of the initial mechanical shock. The load-removal overspeed is essentially unchanged. The fuel command reaches its 4 lbm/s ceiling for 0.550% of PI evaluation samples and 0.650% of DOB samples. DOB correction reaches its bound for 31.878% of samples; the inherited PI has no added anti-windup. These limits should be considered before increasing observer gain or extraction power.

6.3.3 Recovery after each edge

Recovery means the error remains within +/-1 rpm until the next load edge, with at least 0.5 s of confirmed dwell. NaN means recovery was not established before that edge; 0 means the response never left the band.

Table 6.4: External shaft-load rejection at 10000 rpm: Recovery after each edge. Archived values.

Edge time (s)	Event	PI peak abs. error (rpm)	DOB peak abs. error (rpm)	PI recovery (s)	DOB recovery (s)
10.005	Apply load	38.23095	37.97716	1.695	1.365
15.000	Remove load	40.12567	40.12605	1.680	1.140
30.000	Apply load	38.23095	37.97716	1.695	1.365
40.005	Remove load	40.12642	40.12642	1.680	1.140
60.000	Apply load	38.23095	37.97716	1.695	1.365
75.000	Remove load	40.12642	40.12642	1.680	1.140

6.3.4 Verification and reproduction

The power and speed profiles match exactly between controllers. The independent shaft torque-balance reconstruction matches logged acceleration to 1.556e-12 rpm/s for PI and 1.566e-12 rpm/s for PI+DOB. Acceptance requires finite signals, positive fuel and compressor margin, residual <=1e-9, and fewer than 200 iterations throughout preparation and evaluation.

Run `run_tmats_pto_batch` from TMATSGPT to build the new `_DOB_PTO.mdl` copy, measure steady turbine power, and simulate both controllers. Run `analyze_tmats_pto` to regenerate this report and the figures. Open the model after `tmats_pto_setup`; its default uses the measured reference saved in `tmp/dob/pto_reference.json`.

Saved evidence: `trim.mat/csv`, `reference.json`, `case_01.mat/csv`, `case_02.mat/csv`, `metrics.json/csv`, `events.csv`, `pto_comparison.png`, and `pto_transient_detail.png`. Existing GPT and DOB model files are preserved.

This is one deterministic, noise-free sample-engine operating point. The estimate is fuel-equivalent lumped model error, not a direct calibrated power measurement. The experiment tests mechanical load rejection but does not qualify the observer for hardware or establish broad closed-loop robustness.

6.4 The rejected 20 and 30 percent tests at 10000 rpm

Evidence basis: `results/PTO_20_30_comparison/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Both requested levels were simulated with the existing PI and PI+DOB settings. All four complete trajectories failed the numerical validity checks. Consequently, full-run RMSE, recovery time and peak errors cannot be used as valid controller comparisons.

6.4.1 Unchanged experiment

Steady unloaded turbine power is 27896.5073265 hp. Loads are fixed at 20% and 30% of that gross shaft power, not a percentage of instantaneous or net power. Both controllers receive identical physical power extraction at the Shaft power port. Preparation lasts 60 s; evaluation lasts 90 s. Pulses occur at 10.005-15.000, 30.000-40.005, and 60.000-75.000 s after preparation. Demand is constant at 10000 rpm; fuel-injection disturbances are disabled.

PI gains remain 0.025 and 0.05. DOB gain remains 0.25, pole-frequency parameter 0.5 Hz and correction limit +/-0.15 lbm/s. Fuel limits remain 0.2-4 lbm/s. No anti-windup, increased fuel authority, or observer retuning was introduced.

6.4.2 Validity and first-pulse results

The first loaded interval is 10.005 <= t < 15 s. Its minimum speed is reported only when all samples leading through this interval pass validity checks.

Table 6.5: The rejected 20 and 30 percent tests at 10000 rpm: Validity and first-pulse results. Archived values.

Load	Extraction	Controller	Full trial	First invalid time*	Minimum speed during first load
20%	4.160484 MW	PI	FAIL	18.210 s	9660.157 rpm
20%	4.160484 MW	PI+DOB	FAIL	18.210 s	9660.158 rpm
30%	6.240727 MW	PI	FAIL	10.770 s	Not valid
30%	6.240727 MW	PI+DOB	FAIL	10.770 s	Not valid

*Time after preparation. The first load is removed at 15 s. Later apparent solver recovery does not rehabilitate states propagated through invalid samples.

6.4.3 Saturation and interpretation

Table 6.6: The rejected 20 and 30 percent tests at 10000 rpm: Saturation and interpretation. Archived values.

Load	Controller	First loaded interval at fuel ceiling	Peak pre-limit PI demand during that interval
20%	PI	97.90%	81.661 lbm/s
20%	PI+DOB	97.90%	81.650 lbm/s
30%	PI	N/A	N/A
30%	PI+DOB	N/A	N/A

The fuel actuator reaches its ceiling while speed falls under the applied load. At 20%, the large difference between pre-limit PI demand and delivered command, followed by overspeed after unloading before convergence fails, is consistent with integrator windup in the unchanged PI. At 30%, convergence fails only 0.765 s after application; unloading and recovery cannot be assessed from that invalid trajectory. The bounded DOB correction cannot create additional actuator authority. The recorded full trajectories also leave the numerically usable map/convergence envelope. Thus the present controller configuration does not demonstrate acceptable 10000 rpm regulation at either requested load.

The previous 10% experiment passed and reduced RMSE from 4.821 to 4.628 rpm, but that improvement cannot be extrapolated to these higher loads. The new figure shows only the common valid prefix of each pair; the full diagnostic traces remain in the per-load directories.

Before repeating a performance comparison, establish feasible shaft extraction at the available fuel/thermal limits and add saturation-aware PI anti-windup. Those are proposed follow-up changes, not changes made in this repeat test. Increasing fuel limits without checking the plant envelope is not justified by these results.

6.4.4 Verification and reproduction

Both runs contain all 10001 samples through 150 s. Demand/load profiles match exactly across controllers. The independent shaft torque balance continues to match the logged acceleration; this verifies disturbance routing but does not validate failed flow solves. Validity requires finite signals, positive fuel and compressor margin, flow residual <=1e-9, and fewer than 200 iterations.

Run `run_tmats_pto_20_30_batch` from TMATSGPT, or use `out=run_tmats_pto_study(0.2); analyze_tmats_pto(out);` and repeat with `0.3`. The existing PTO model is reused. Its no-argument setup still defaults to 10%; the new batch explicitly passes 0.2 and 0.3.

Per-load reports label full-trial error metrics as diagnostic. Their recovery tables are also diagnostic after the first invalid sample. `events.csv` contains `trajectory_valid_through_event` to identify which event windows remain valid.

- Results: `results/dob_pto_20pct_20260912_183919_021`
- Results: `results/dob_pto_30pct_20260912_184142_273`

6.5 Moving the shaft-load experiment to 9500 rpm

Evidence basis: `results/PTO_9500rpm_comparison/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

The preceding 20% and 30% scenarios were repeated at 9500 rpm without retuning PI or DOB. Each load is recalculated using steady turbine power at 9500 rpm.

6.5.1 Reference and scenario

Steady turbine shaft power is **19160.793887 hp** (14.288202 MW). The corresponding unloaded fuel flow is 1.849471 lbm/s. Power is averaged over seconds 55-60 after convergence to the operating point. Gross turbine power balances compressor load; the percentages are not taken from near-zero net shaft power.

Preparation starts at the standard 10000 rpm initial state, holds through 5 s, ramps the demand to 9500 rpm at 10 s, then holds through 60 s. The speed demand stays at 9500 rpm during the 90 s evaluation. Identical mechanical load pulses are applied at 10.005-15.000, 30.000-40.005, and 60.000-75.000 s after preparation. The injected fuel disturbance is zero.

The original PI gains (0.025, 0.05), DOB gain 0.25, Q pole-frequency parameter 0.5 Hz, correction limit +/-0.15 lbm/s, and fuel limits 0.2-4 lbm/s are unchanged. The eight-term model and its equilibrium correction are unchanged; trim removal handles the initial observer offset.

6.5.2 Results

Table 6.7: Moving the shaft-load experiment to 9500 rpm: Results. Archived values.

Load	Extraction	Controller	Status	First invalid time*	RMSE	Maximum droop
20%	2.857640 MW	PI	PASS	None	6.430 rpm	53.397 rpm
20%	2.857640 MW	PI+DOB	PASS	None	6.203 rpm	53.013 rpm
30%	4.286460 MW	PI	FAIL	10.155 s	Not valid	Not valid
30%	4.286460 MW	PI+DOB	FAIL	10.155 s	Not valid	Not valid

*Time after preparation; the first extraction starts at 10.005 s. Complete-run performance is suppressed if any earlier state was propagated through an invalid solve.

6.5.3 Recovery and limits

Table 6.8: Moving the shaft-load experiment to 9500 rpm: Recovery and limits. Archived values.

Load	Controller	First application recovery	First removal recovery	Peak fuel in valid prefix
20%	PI	1.815 s	1.875 s	3.3335 lbm/s
20%	PI+DOB	1.665 s	1.320 s	3.3602 lbm/s
30%	PI	Not valid	Not valid	3.5686 lbm/s
30%	PI+DOB	Not valid	Not valid	3.5926 lbm/s

Recovery requires remaining within +/-1 rpm until the next edge, with at least 0.5 s confirmed dwell. Invalid full trajectories do not receive recovery-time claims. Valid-prefix fuel maxima use only samples before the first invalid point.

At **20%**, the DOB reduces RMSE by **3.527%**. Application recovery changes from 1.815 to 1.665 s; removal recovery changes from 1.875 to 1.320 s. The benefit is mainly faster recovery rather than a large reduction of the initial speed excursion.

At **30%**, the complete test does not pass. PI first fails at 10.155 s; PI+DOB first fails at 10.155 s. Fuel at the first failed solve is 3.7489 lbm/s (PI) and 3.7769 lbm/s (PI+DOB), below the 4 lbm/s ceiling. Minimum reported compressor margin in the valid prefix is 0.879% and 0.625%, respectively. Thus initial failure is a flow/map convergence limitation before fuel saturation; it does not establish physical surge. Later apparent recovery and complete-run error metrics remain diagnostic. The current controller/model combination has not established regulation performance for this condition.

6.5.4 Comparison with the 10000 rpm tests

At 10000 rpm, the previous 20% and 30% loads were 4.160484 and 6.240727 MW, respectively, and both levels failed. The new tests change both operating point and absolute load as requested. Therefore an improvement at the same percentage must not be described as improvement at the same absolute extracted power.

6.5.5 Verification, files, and reproduction

Each run contains 10001 samples at 15 ms through 150 s. The same source profile reaches the shaft power port in both controllers. The torque-balance reconstruction is checked against logged acceleration. Validity requires finite signals, positive fuel and compressor margin, flow residual <=1e-9, and fewer than 200 solver iterations over the entire run. The observer receives sensed speed and known fuel only.

Batch reproduction uses `run_tmats_pto_9500_batch`. For one load level, call `run_tmats_pto_study(0.2,9500)` and then `analyze_tmats_pto(out)`, assigning the returned result to `out`; use 0.3 for the larger load. For an already loaded model, `tmats_pto_setup([],0.2,9500)` loads the measured speed-specific power reference. The no-argument setup retains the original 10000 rpm, 10% case.

No model rebuild or controller retuning is necessary. The original model files are unchanged. The new `pto_reference_9500rpm.json` cache does not replace the 10000 rpm power reference. These are deterministic noise-free sample-engine tests, not a deployment qualification.

- Source result directory: `results/dob_pto_9500rpm_20pct_20260912_195059_980`
- Source result directory: `results/dob_pto_9500rpm_30pct_20260912_195120_747`

6.6 The 9000-rpm ramped-load envelope

Evidence basis: `results/PTO_9000rpm_ramp_comparison/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Steady gross turbine shaft power: **11789.245211 hp** (8.791239 MW). Steady fuel: 0.981087 lbm/s. Percentages use this frozen reference, not near-zero net shaft power.

6.6.1 Scenario and formulation

Each amplitude uses three separate ramp/hold/ramp pulses. Application starts at 10.005, 30 and 60 s; removal starts at 15, 40.005 and 75 s. Corresponding rise and fall durations are 0.30, 1.50 and 3.00 s. The 90 s evaluation follows 60 s preparation (10000 rpm initially, demand ramp to 9000 rpm during seconds 5–10). Sampling is 15 ms. Thus ramps contain 20, 100 and 200 sample intervals.

For amplitude α and measured turbine power P0, each pulse is P_L(t)=αP0[sat((t-t_on)/T_r)-sat((t-t_off)/T_r)], where sat(x)=min(1,max(0,x)). The three nonoverlapping pulses are summed and applied as actual shaft power extraction. The speed demand stays at 9000 rpm; injected fuel disturbance is zero.

PI gains remain Kp=0.025 and Ki=0.05. The frozen eight-term identified model, equilibrium correction, DOB gain 0.25, Q pole-frequency parameter 0.5 Hz, compensation bound ±0.15 lbm/s and fuel bounds 0.2–4 lbm/s are unchanged. No controller is retuned between amplitudes or ramp rates. The observer uses sensed speed and known fuel, without access to injected shaft load.

6.6.2 Complete-run results

Table 6.9: The 9000-rpm ramped-load envelope: Complete-run results. Archived values.

Load	MW	Controller	Valid	RMSE rpm	Peak absolute error rpm	First failure s
5%	0.440	PI	True	0.594	5.325	—
5%	0.440	PI+DOB	True	0.517	5.162	—
10%	0.879	PI	True	1.193	10.677	—
10%	0.879	PI+DOB	True	1.038	10.351	—
20%	1.758	PI	True	2.540	22.553	—
20%	1.758	PI+DOB	True	2.237	22.329	—
30%	2.637	PI	True	3.973	35.359	—
30%	2.637	PI+DOB	True	3.664	35.258	—

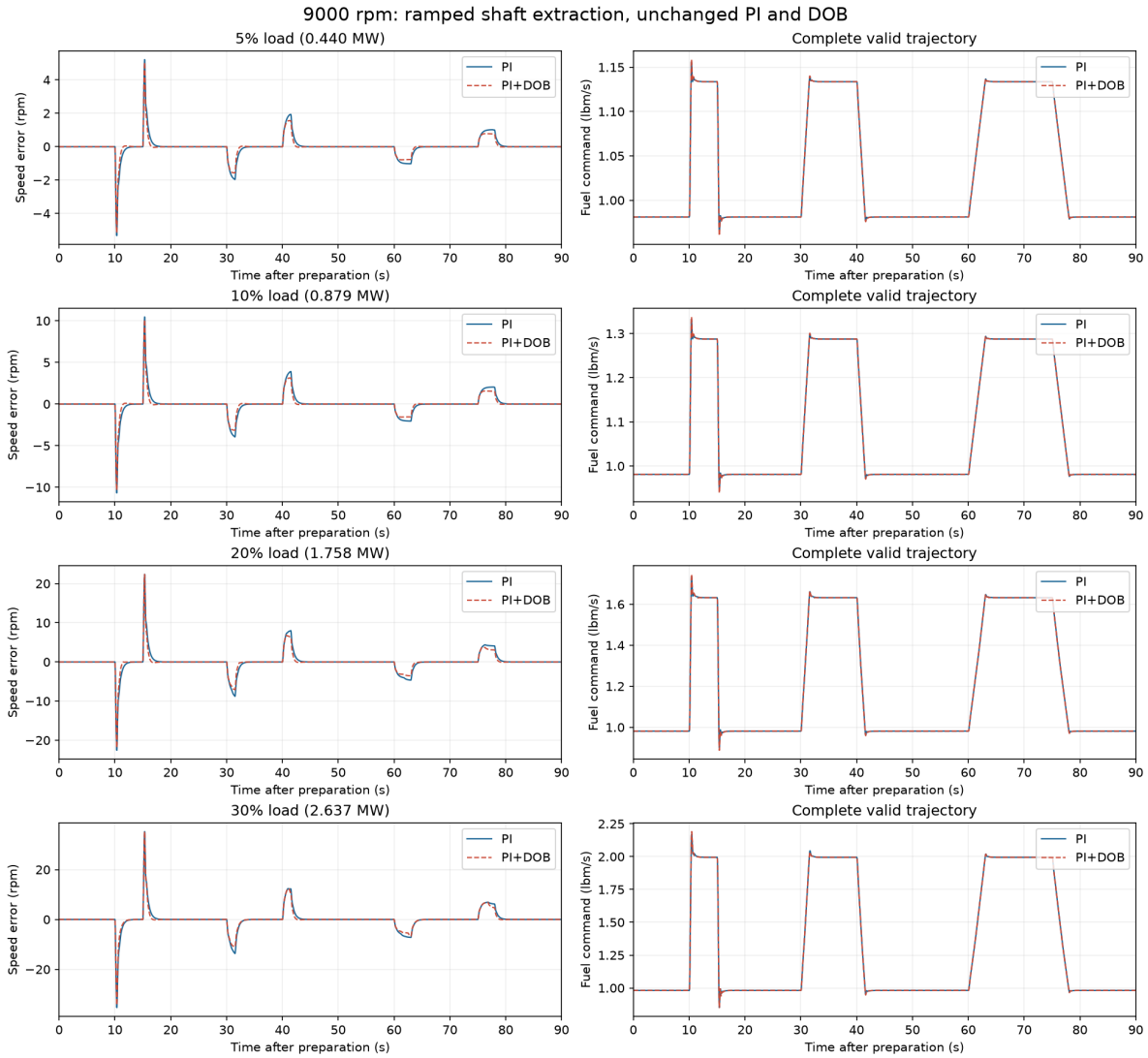


Figure 6.1: Overview. Archived study figure.

6.6.3 Time response at each ramp

Recovery is the first sample after the last excursion outside ±1 rpm, confirmed by at least 0.5 s within the band through the next ramp start. After-ramp recovery is max(0, recovery from start - ramp duration). A zero means recovery occurred by ramp completion. Metrics are withheld for an event if any earlier plant solve was invalid. A dash can also mean no confirmed settling.

Table 6.10: The 9000-rpm ramped-load envelope: Time response at each ramp. Archived values.

Load	Ramp s	Edge	Controller	Peak error rpm	Recovery from start s	After ramp s
5%	0.30	application	PI	5.325	1.035	0.735
5%	0.30	removal	PI	5.215	1.050	0.750
5%	1.50	application	PI	1.965	1.785	0.285
5%	1.50	removal	PI	1.932	1.785	0.285
5%	3.00	application	PI	1.020	3.045	0.045
5%	3.00	removal	PI	1.020	2.985	0.000
5%	0.30	application	PI+DOB	5.162	0.825	0.525
5%	0.30	removal	PI+DOB	5.055	0.825	0.525
5%	1.50	application	PI+DOB	1.565	1.665	0.165
5%	1.50	removal	PI+DOB	1.567	1.650	0.150
5%	3.00	application	PI+DOB	0.784	0.000	0.000

Load	Ramp s	Edge	Controller	Peak error rpm	Recovery from start s	After ramp s
5%	3.00	removal	PI+DOB	0.782	0.000	0.000
10%	0.30	application	PI	10.677	1.350	1.050
10%	0.30	removal	PI	10.457	1.380	1.080
10%	1.50	application	PI	3.950	2.130	0.630
10%	1.50	removal	PI	3.901	2.145	0.645
10%	3.00	application	PI	2.054	3.315	0.315
10%	3.00	removal	PI	2.042	3.315	0.315
10%	0.30	application	PI+DOB	10.351	1.005	0.705
10%	0.30	removal	PI+DOB	10.140	1.035	0.735
10%	1.50	application	PI+DOB	3.145	1.845	0.345
10%	1.50	removal	PI+DOB	3.134	1.860	0.360
10%	3.00	application	PI+DOB	1.565	3.165	0.165
10%	3.00	removal	PI+DOB	1.566	3.150	0.150
20%	0.30	application	PI	22.553	1.695	1.395
20%	0.30	removal	PI	22.380	1.740	1.440
20%	1.50	application	PI	8.786	2.490	0.990
20%	1.50	removal	PI	7.988	2.490	0.990
20%	3.00	application	PI	4.649	3.705	0.705
20%	3.00	removal	PI	4.363	3.660	0.660
20%	0.30	application	PI+DOB	21.914	1.170	0.870
20%	0.30	removal	PI+DOB	22.329	1.215	0.915
20%	1.50	application	PI+DOB	7.096	2.055	0.555
20%	1.50	removal	PI+DOB	6.823	2.040	0.540
20%	3.00	application	PI+DOB	3.536	3.360	0.360
20%	3.00	removal	PI+DOB	3.771	3.345	0.345
30%	0.30	application	PI	35.359	1.875	1.575
30%	0.30	removal	PI	35.257	1.950	1.650
30%	1.50	application	PI	13.677	2.685	1.185
30%	1.50	removal	PI	12.372	2.700	1.200
30%	3.00	application	PI	7.239	3.900	0.900
30%	3.00	removal	PI	6.909	3.870	0.870
30%	0.30	application	PI+DOB	34.333	1.680	1.380
30%	0.30	removal	PI+DOB	35.258	1.410	1.110
30%	1.50	application	PI+DOB	11.032	2.580	1.080
30%	1.50	removal	PI+DOB	12.372	2.190	0.690
30%	3.00	application	PI+DOB	6.730	3.870	0.870
30%	3.00	removal	PI+DOB	6.901	3.480	0.480

6.6.4 Ramp rates

Table 6.11: The 9000-rpm ramped-load envelope: Ramp rates. Archived values.

Load	0.30 s ramp (MW/s)	1.50 s ramp (MW/s)	3.00 s ramp (MW/s)
5%	1.465	0.293	0.147
10%	2.930	0.586	0.293
20%	5.861	1.172	0.586
30%	8.791	1.758	0.879

These magnitudes apply to both rising and falling ramps. The source is sampled and held at 15 ms.

6.6.5 Interpretation

- **5%:** observer RMSE changes by -13.03% relative to PI. Peak absolute error changes from 5.325 to 5.162 rpm. Minimum compressor margin is 29.404% / 29.347% (PI / DOB). DOB compensation clips for 0.00% of evaluation samples.

- **10%:** observer RMSE changes by -13.02% relative to PI. Peak absolute error changes from 10.677 to 10.351 rpm. Minimum compressor margin is 24.429% / 24.322% (PI / DOB). DOB compensation clips for 0.00% of evaluation samples.
- **20%:** observer RMSE changes by -11.92% relative to PI. Peak absolute error changes from 22.553 to 22.329 rpm. Minimum compressor margin is 14.445% / 14.238% (PI / DOB). DOB compensation clips for 26.98% of evaluation samples.
- **30%:** observer RMSE changes by -7.77% relative to PI. Peak absolute error changes from 35.359 to 35.258 rpm. Minimum compressor margin is 5.526% / 5.231% (PI / DOB). DOB compensation clips for 31.98% of evaluation samples.

At 30% load, increasing the application ramp from 0.30 to 3.00 s reduces PI peak error from 35.359 to 7.239 rpm and DOB peak error from 34.333 to 6.730 rpm. However, on the 3.00 s application ramp the observer improves recovery after completion by only 0.030 s (0.900 to 0.870 s). On the 0.30 s removal ramp, the peak is effectively unchanged (35.257 versus 35.258 rpm), while after-ramp recovery improves from 1.650 to 1.110 s. This supports a recovery benefit, not uniform peak suppression. At 5% with a 3.00 s ramp, DOB error stays within ±1 rpm throughout both events.

All eight trajectories pass the numerical checks, with no fuel saturation. At 30%, DOB peak fuel is 2.191 lbm/s and minimum reported compressor margin is 5.231%. The lower margin and frequent compensation clipping motivate constraint-aware tuning before extending the load envelope. Compared with the earlier 9500 rpm step tests, both the operating point, absolute load and disturbance shape have changed: 30% is now 2.637 MW rather than 4.286 MW. The valid result cannot be attributed solely to observer performance or solely to ramp shaping.

The DOB is bounded equivalent-input compensation around the existing PI loop. It estimates the combined effect of external load and nominal-model mismatch; its estimate is not a calibrated shaft-power measurement. Slower ramps generally give the controller more time to compensate, but the event table is the evidence for this plant: peak rejection, settling and global RMSE need not improve together. The three ramp rates share a trajectory with different preceding histories and hold lengths, so these are representative scenarios rather than independent rate-identification experiments.

The fixed model was identified from fuel-to-acceleration data and corrected for zero equilibrium acceleration before observer synthesis. Its inverse and low-pass filtering introduce a bandwidth tradeoff: stronger/faster compensation can improve rejection while amplifying sensor noise and model error. Compensation clipping limits authority. PI integral action provides steady regulation; no anti-windup was added for these comparisons.

Next steps are independent runs for each rate with identical settling windows, gain scheduling across operating points, sensor-noise and delay tests, anti-windup and explicit fuel, temperature, and compressor-margin constraints, then fixed-step embedded timing and hardware-in-the-loop assessment. If flow convergence fails, first establish a numerically valid operating envelope; an invalid trajectory is not proof of physical instability or surge.

6.6.6 Verification and reproduction

The two controllers receive identical sampled power profiles, verified against the ramp formula. Each run has 10001 samples through 150 s. The mechanical torque balance is reconstructed from logged turbine, compressor and external powers and checked against shaft acceleration. Validity requires finite signals, positive fuel and compressor margin, flow residual $\leq$1e-9 and fewer than 200

solver iterations. All plots show only the prefix before the first invalid solve. Full raw trajectories are retained for diagnosis.

Run `run_tmats_pto_9000_ramp_batch` in MATLAB, then `summarize_pto_9000_ramps.py` in Python for the report. A single level is `out=run_tmats_pto_study(0.10,9000,[0.30 1.50 3.00]);`. Omitting the third argument preserves the earlier step-load study. The existing PTO Simulink model is unchanged.

6.6.7 5% response zooms

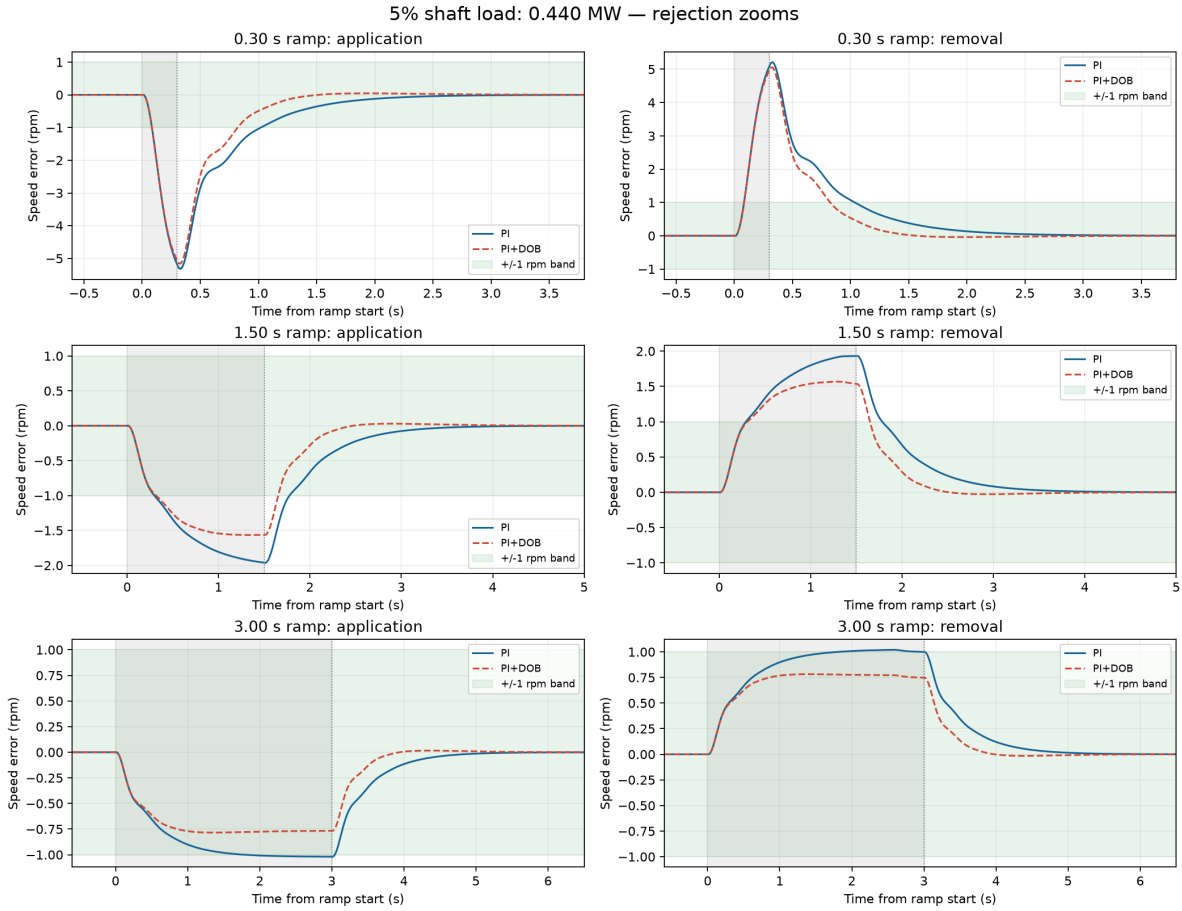


Figure 6.2: 5% zoom. Archived study figure.

6.6.8 10% response zooms

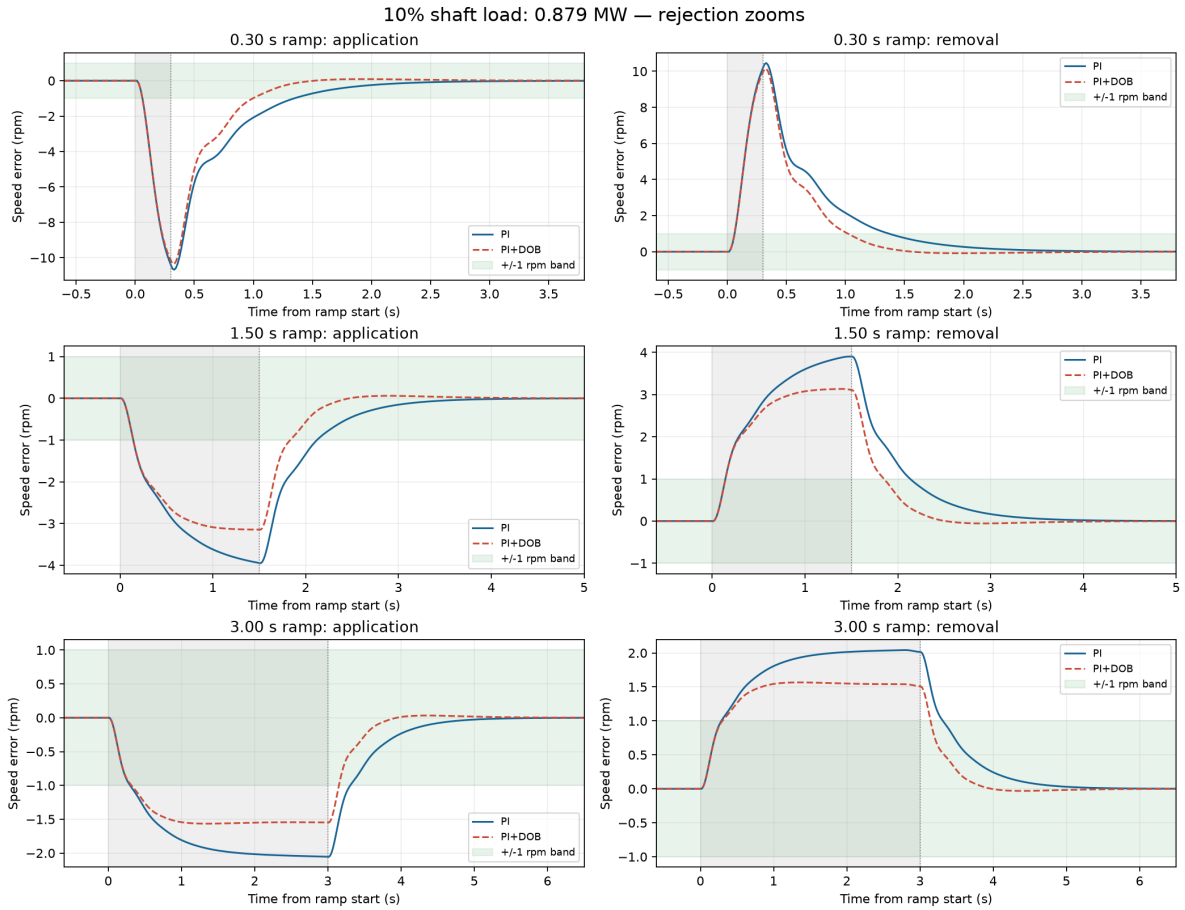


Figure 6.3: 10% zoom. Archived study figure.

6.6.9 20% response zooms

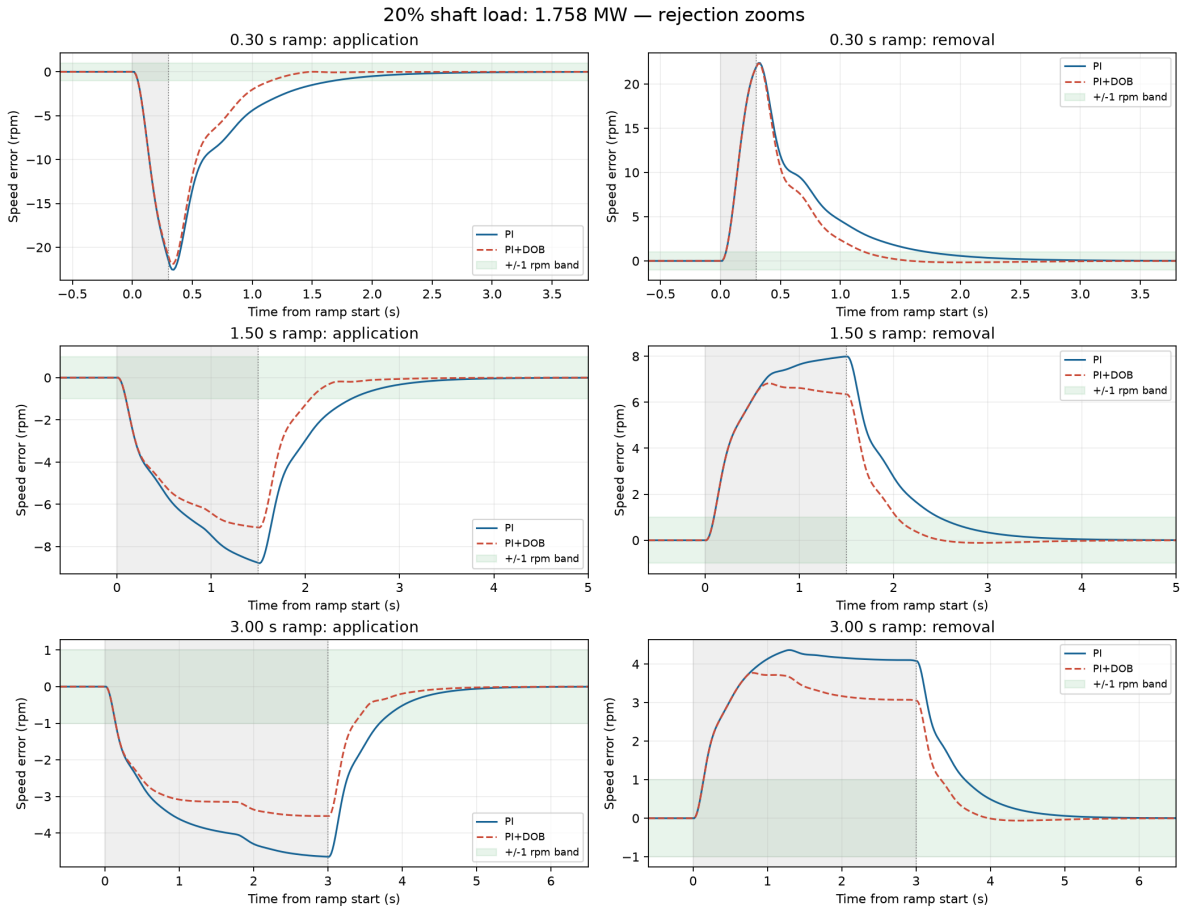


Figure 6.4: 20% zoom. Archived study figure.

6.6.10 30% response zooms

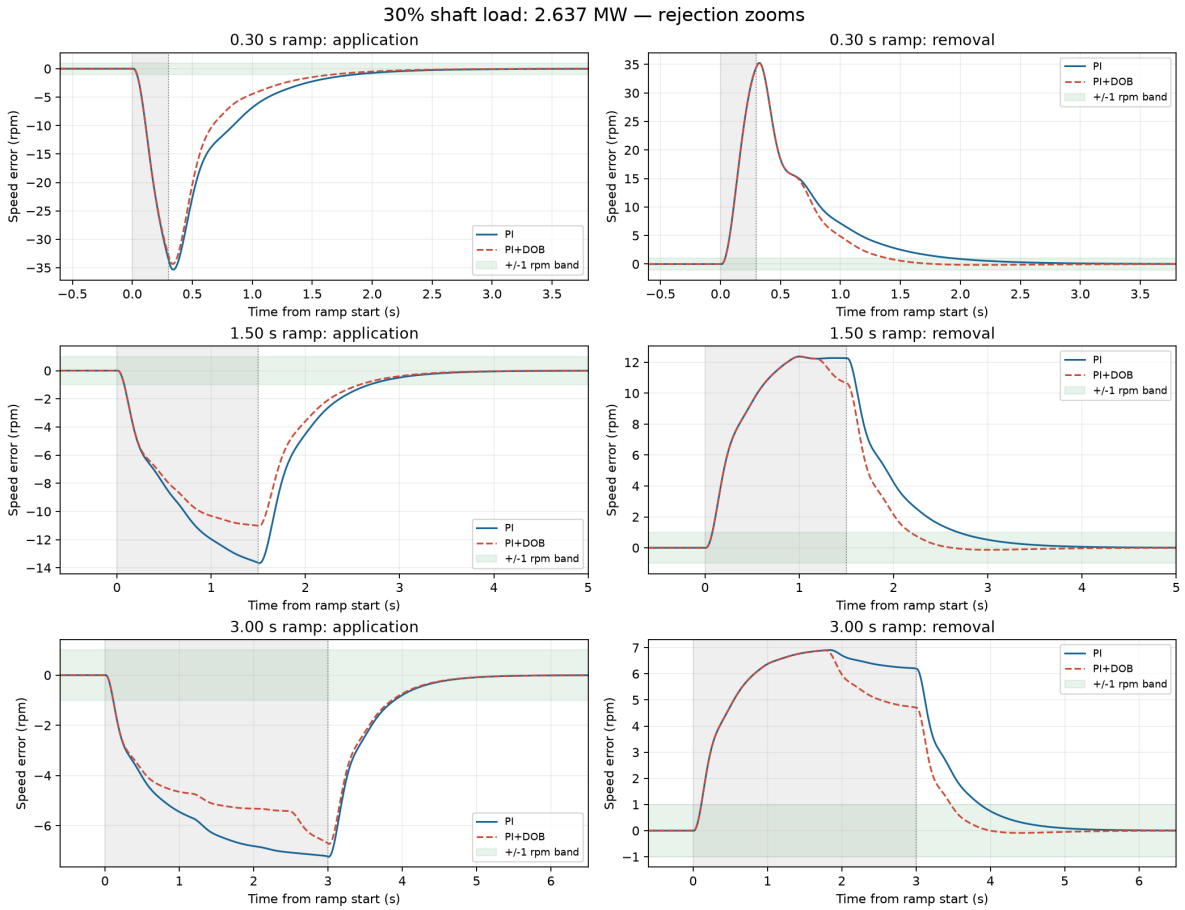


Figure 6.5: 30% zoom. Archived study figure.

6.7 LESO, UDE, and GPIO implementation and first comparison

Evidence basis: `results/literature_observers_9000rpm_10pct_20260912_211600_355/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Load is **0.879124 MW** (10% of 11789.245211 hp gross turbine power). The previous 0.30, 1.50 and 3.00 s ramp/hold/ramp scenario is repeated with five controllers. PI gains, 0.25 compensation gain, ±0.15 lbm/s correction bound, fuel limits and preparation are unchanged. The observer pole-frequency parameter is 0.5 Hz for every method; actual estimator bandwidths differ. No test-based tuning is performed.

The three candidates are LESO (two states), UDE (one state), and ramp GPIO (three states). See the accompanying formulation for equations, primary literature, implementation details and comparison limitations.

6.7.1 Measured comparison

Table 6.12: LESO, UDE, and GPIO implementation and first comparison: Measured comparison. Archived values.

Method	Valid	RMSE rpm	Reduction vs PI	Peak error rpm	IAE rpm·s	Peak fuel lbm/s
PI	True	1.193	0.000%	10.677	36.774	1.333
DOB	True	1.038	13.021%	10.351	28.341	1.338
LESO	True	1.039	12.903%	10.349	28.694	1.338
UDE	True	0.969	18.746%	9.377	27.587	1.336
GPIO	True	0.951	20.301%	9.866	27.577	1.344

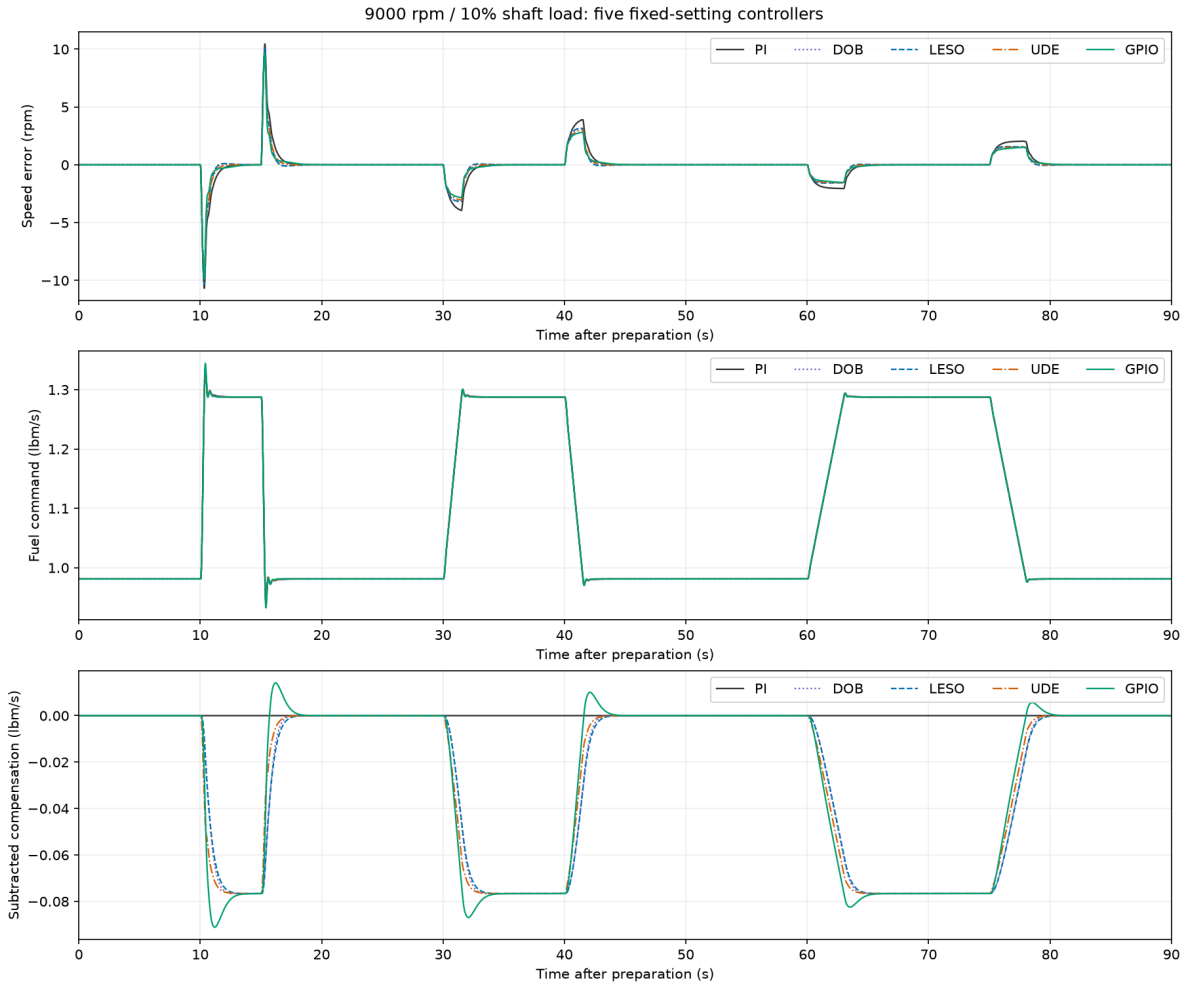


Figure 6.6: Overview. Archived study figure.

6.7.2 Event recovery

Recovery requires remaining within ±1 rpm to the next ramp, with at least 0.5 s confirmed dwell. After-ramp recovery subtracts the ramp duration, bounded below by zero. Failed propagated trajectories receive no performance claim.

Table 6.13: LESO, UDE, and GPIO implementation and first comparison: Event recovery. Archived values.

Ramp s	Edge	Method	Peak rpm	Recovery from start s	After ramp s
0.30	apply	PI	10.677	1.350	1.050
0.30	apply	DOB	10.351	1.005	0.705
0.30	apply	LESO	10.349	1.020	0.720

Ramp s	Edge	Method	Peak rpm	Recovery from start s	After ramp s
0.30	apply	UDE	9.377	1.080	0.780
0.30	apply	GPIO	9.866	0.915	0.615
0.30	remove	PI	10.457	1.380	1.080
0.30	remove	DOB	10.140	1.035	0.735
0.30	remove	LESO	10.137	1.050	0.750
0.30	remove	UDE	9.205	1.095	0.795
0.30	remove	GPIO	9.670	0.930	0.630
1.50	apply	PI	3.950	2.130	0.630
1.50	apply	DOB	3.145	1.845	0.345
1.50	apply	LESO	3.171	1.845	0.345
1.50	apply	UDE	3.050	1.920	0.420
1.50	apply	GPIO	2.850	1.785	0.285
1.50	remove	PI	3.901	2.145	0.645
1.50	remove	DOB	3.134	1.860	0.360
1.50	remove	LESO	3.158	1.875	0.375
1.50	remove	UDE	3.022	1.920	0.420
1.50	remove	GPIO	2.817	1.800	0.300
3.00	apply	PI	2.054	3.315	0.315
3.00	apply	DOB	1.565	3.165	0.165
3.00	apply	LESO	1.577	3.165	0.165
3.00	apply	UDE	1.545	3.165	0.165
3.00	apply	GPIO	1.528	3.150	0.150
3.00	remove	PI	2.042	3.315	0.315
3.00	remove	DOB	1.566	3.150	0.150
3.00	remove	LESO	1.578	3.150	0.150
3.00	remove	UDE	1.536	3.150	0.150
3.00	remove	GPIO	1.513	3.150	0.150

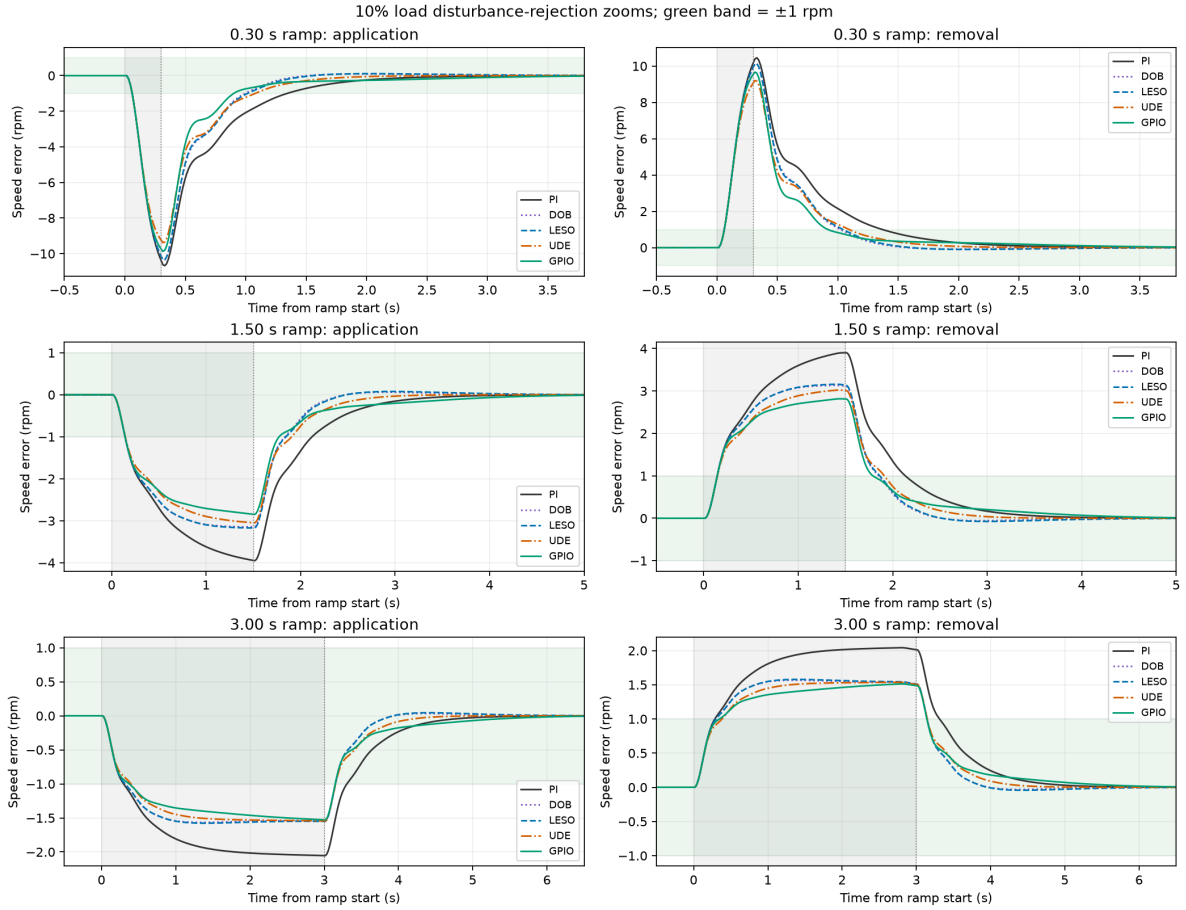


Figure 6.7: Zooms. Archived study figure.

6.7.3 Actuation and model validity

Table 6.14: LESO, UDE, and GPIO implementation and first comparison: Actuation and model validity. Archived values.

Method	Fuel limit %	Compensation clip %	Fuel total variation lbm/s	Minimum compressor margin %
PI	0.000	0.000	2.116	24.429
DOB	0.000	0.000	2.136	24.322
LESO	0.000	0.000	2.143	24.302
UDE	0.000	0.000	2.144	24.389
GPIO	0.000	0.000	2.174	24.161

6.7.4 Assessment

GPIO has the lowest RMSE in this fixed-setting test: 0.9509 rpm. This is a measured result for the chosen nominal gain, observer poles and limits, not a universal ranking or optimally tuned benchmark. The existing DOB uses the fuller identified model; new observers use only its first fuel-to-acceleration coefficient as an approximate input gain.

Practical recommendation: start with UDE if implementation simplicity and peak rejection are the priorities; retain GPIO as the performance candidate when faster recovery justifies two extra observer states. LESO is a simpler-model substitute for the existing DOB here, but does not improve its overall RMSE. This recommendation is conditional on passing subsequent noise and robustness tests.

For the 0.30 s application ramp, post-ramp recovery is 1.050 s (PI), 0.705 s (DOB), 0.720 s (LESO), 0.780 s (UDE), and 0.615 s (GPIO). Thus UDE has the lowest peak, but is slower to enter the ± 1 rpm band than DOB and GPIO. For the 1.50 s ramp, GPIO application peak is 2.850 rpm versus 3.950 rpm for PI, with post-ramp recovery 0.285 versus 0.630 s. At 3.00 s, after-ramp recovery differs by only one sample or less among the augmented methods.

Relative to the existing DOB, UDE lowers RMSE by about 6.58% and GPIO by 8.37%. Neither fuel nor compensation saturation occurs. GPIO has the greatest fuel total variation (2.174 versus 2.116 lbm/s for PI) and the highest peak fuel (1.344 versus 1.333 lbm/s), illustrating a small actuation tradeoff. These total-variation values sum absolute sample-to-sample command changes; they are not fuel consumption.

UDE has the smallest state count and no raw derivative, but its direct speed term makes measurement-noise testing important. LESO is a simple general-purpose candidate. GPIO represents disturbance slope explicitly, at the cost of an additional state and stronger high-frequency correction. A shaft-power ramp is not necessarily a ramp in equivalent-fuel disturbance because the plant and sensor contribute to the residual.

Next compare separately tuned settings using an independent tuning case, then hold out load profiles for validation. Include sensor noise, delay variation, gain mismatch, fuel constraints and compressor margin. Do not choose a deployed method using speed RMSE alone: peak excursions, recovery, actuator variation and numerical/physical limits matter.

6.7.5 Verification and reproduction

All five actual Simulink trajectories contain 10001 samples through 150 s. Checks confirm identical shaft profiles, 9000 rpm evaluation demand, zero fuel disturbance and mechanical torque balance. The three new estimators are independently replayed from logged sensed speed and applied fuel with their exact discrete state equations. PI and existing DOB responses reproduce the preceding 10% study. Detailed numerical errors are in verification.json.

Run `out=run_tmats_literature_comparison;` in MATLAB, then `summarize_literature_observers.py` in Python. The new model is `GasTurbine_Dyn_Template_GPT_DOB_PTO_ALT.mdl`; original models are preserved. Load the model first, then select an alternative with `tmats_literature_setup(2)` (LESO), `(3)` (UDE) or `(4)` (GPIO) before simulation; its PreLoadFcn initializes LESO by default. Observer pole stability does not establish nonlinear closed-loop robustness. No noisy or hardware experiment is claimed.

6.8 Interpretation of the observer comparison

The accepted runs provide local evidence that these configured observers can reduce the recorded disturbance error. The rejected high-load cases restrict the scope of that evidence. Since load percentages are defined relative to operating-point turbine power, identical percentages at 9000 and 9500 rpm are not identical physical extraction powers. The tables compare controllers within a scenario; they do not estimate a universal observer ranking.

Chapter 7

Gaussian-process prediction and its control interpretation

The predictor is introduced before its use in feedback. A GP posterior describes uncertainty conditional on the kernel, likelihood, and training distribution [1]. It is not automatically a bound on unknown external load or a certificate for feedback stability. The two-input nominal tests below therefore answer a regression question before the later chapters answer a control question.

7.1 Gaussian-process regression explained from first principles

7.1.1 From a fitted curve to a distribution over functions

Regression seeks a relationship between inputs and outputs. In the forward study the input is speed and fuel, and the output is acceleration. In the inverse study the input is speed and acceleration, and the output is fuel. A Gaussian process specifies correlated possible function values rather than choosing only a fixed polynomial form. Its mean gives a point prediction; its covariance describes uncertainty under its assumptions.

For training inputs $X = [x_1, \ldots, x_n]$ and outputs y, the model is $y_i = f(x_i) + \epsilon_i$, with Gaussian noise variance σ_n^2. A squared-exponential kernel is

$$k(x, x') = \sigma_f^2 \exp\left[-\frac{1}{2}\sum_j \frac{(x_j - x'_j)^2}{\ell_j^2}\right]. \tag{7.1}$$

The length scales ℓ_j determine how rapidly similarity decays in each input direction. Different scales per input are called automatic relevance determination (ARD). They account for different smoothness and scaling, but they do not by themselves establish a causal importance ranking of physical variables.

Let $K_{ij} = k(x_i, x_j)$, $k_* = k(X, x_*)$, and $K_n = K + \sigma_n^2 I$. Conditioning the joint Gaussian gives

$$\mu_* = m(x_*) + k_*^\top K_n^{-1}(y - m(X)), \tag{7.2}$$

$$\sigma_{f,*}^2 = k(x_*, x_*) - k_*^\top K_n^{-1} k_*. \tag{7.3}$$

Response variance adds σ_n^2. The distinction matters when a controller schedules gains from prediction uncertainty. The project records which standard deviation is used. In computation, Cholesky factorization solves these systems; explicitly forming an inverse is unnecessary and less attractive numerically. The standard Gaussian conditioning framework is described by Rasmussen and Williams [1].

7.1.2 What fitting the hyperparameters does

The negative log marginal likelihood balances data mismatch against model complexity through a quadratic term and $\log|K_n|$. A very flexible model can follow training data closely, but uncertainty

and held-out errors are needed to assess generalization. The word exact in the early GP studies refers to the full Gaussian-process inference calculation on the chosen training set. It does not mean an exact physical model of the turbine or an exact representation of all available samples.

Representative input selection makes the training set cover its observed geometry. Selecting points using only inputs avoids choosing points because their test-output error looks favorable. Nevertheless, closed-loop trajectories are temporally correlated and occupy particular paths. A held-out point from a similar deterministic run is a narrower validation than an independent engine, noisy sensor, or changed ambient condition.

7.1.3 Forward and inverse regression are different fits

For $\dot{N} = f(N, u)$, an inverse relation $u = g(N, \dot{N})$ may be useful locally. Training g independently is not algebraically inverting the fitted function f. The mapping can also become ambiguous if unobserved thermodynamic states, external load, or environment affect acceleration. In the later four-input inverse, temperature and pressure account for some environmental dependence, but unmeasured load and internal transient states are still not explicit inputs.

At constant speed, a nominal zero-load inverse predicts nominal equilibrium fuel. An unknown external load requires extra fuel. If the controller never receives the load, evaluating the nominal inverse at the unchanged target cannot anticipate that extra demand. Feedback or a disturbance observer must infer the mismatch after it affects measured behavior.

7.1.4 Prediction accuracy versus useful derivatives

Feedback depends on how a model changes when its inputs change. A local Taylor expansion gives

$$g(N + \delta N, a + \delta a) \simeq g(N, a) + g_N \delta N + g_a \delta a. \tag{7.4}$$

Small average output error does not guarantee correctly signed or well-scaled g_N and g_a. A nearly flat or negative speed derivative can turn a nominally restoring virtual-fuel signal into a weak or misleading feedback signal. This motivates the later derivative audit, separated acceleration feedback, and R3 normalization.

For example, at constant reference the dual-query error $g(r, 0) - g(N_s, \hat{a})$ is approximately $g_N(r - N_s) - g_a \hat{a}$. Proportional action on this quantity already contains a derivative-like term. Integral action on it can also alter the effective proportional behavior. Thus a controller called virtual-fuel PI can behave locally like a coupled PID. The label PI alone does not reveal the complete feedback structure.

7.1.5 Uncertainty and coverage have limits

A Gaussian posterior interval is conditional on the selected kernel, hyperparameters, noise model, training set, and included inputs. It need not include unknown load bias. A point inside each individual input's minimum and maximum can still lie outside the joint training cloud. A convex-hull check is stronger than separate range checks, but even a point inside the hull may be far from dense data. None of these geometric tests is a closed-loop stability certificate.

The forward and inverse GP studies below establish strong local prediction performance. The subsequent control chapters deliberately ask a harder question: whether acting on those predictions, their derivatives, or their uncertainty improves the actual nonlinear closed-loop response.

7.2 Forward exact GP: speed and fuel to acceleration

Evidence basis: `results/exact_gpr_20260913_145742_458/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Inputs: speed N [rpm], fuel Wf [lbm/s]. Output: dN/dt [rpm/s], predictive standard deviation [rpm/s], 95% response interval, and latent standard deviation.

Exact maximum-likelihood fit and exact prediction with ARD squared-exponential kernel. 600 training points; exactly 100 independent-experiment test points. Training uses valid chirp and selection-ramp records; validation uses a separate ramp. Training samples selected by input-only farthest-point coverage. No test outputs used for fitting or selection.

100 target speeds are linspace(9000,10000,100). Each selects the nearest unique valid recorded validation sample, without synthesizing labels. Actual speed range 9000.000000 to 9998.885249 rpm; maximum target mismatch 1.6856 rpm. Training speed 9000.0000 to 9999.8978 rpm; fuel 0,183863 to 4,194426 lbm/s.

RMSE 0.645908 rpm/s; MAE 0.510063 rpm/s; R2 0.99996430; nominal 95% response-interval coverage 90.0%.

Checks: power-law acceleration versus direct plant output max error 2.41e-10 rpm/s; exported Cholesky posterior versus MATLAB mean/SD errors 2.97e-09 / 2.73e-10 rpm/s.

Labels use signed compressor and turbine power and J=30: the shaft power-balance equation derived in the foundations. No finite-difference derivative is used. These records assume zero external shaft load and fixed ambient conditions. A two-input model cannot identify arbitrary changes in load, ambient conditions or hidden dynamic states. Exact refers to GP inference, not an exact physical model.

Uncertainty: latent variance = k(x,x)-||L\k(X,x)||^2; response variance adds fitted noise variance. Hyperparameters are fixed at their likelihood estimate; intervals are model-conditional, not guaranteed physical error bounds or a calibrated safety bound. No test-based interval rescaling. Test points come from one trajectory and are not statistically independent replicates.

References: Volkan Aran (2019), Flexible and Robust Control of Heavy Duty Diesel Engine Airpath Using Data Driven Disturbance Observers and GPR Models, sec. 5.1, eqs. 5.5-5.14, and sec. 6.2.1. Uses per-input length scales of eq. 5.8. Thesis reference [92]: Rasmussen and Williams, Gaussian Processes for Machine Learning (2006), ch. 2, `https://gaussianprocess.org/gpml/chapters/RW2.pdf` . MATLAB: `https://www.mathworks.com/help/stats/fitrgp.html` and `https://www.mathworks.com/help/stats/regressiongp.predict.html` . This adapts the GP methodology to forward turbine acceleration; it does not reproduce the thesis diesel inverse controller.

7.3 Inverse exact GP: desired motion to nominal fuel

Evidence basis: `results/inverse_gpr_20260913_150824_698/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

Inputs: N [rpm], dN/dt [rpm/s]. Output: Wf [lbm/s], response standard deviation, 95% response interval, and latent function standard deviation. This is a separately trained inverse regression, not an algebraic inverse of the forward GP.

Exact ARD squared-exponential GP with maximum-likelihood hyperparameters, zero mean on standardized responses, exact fitting and exact prediction. Uses the same 600 training observations and 100 held-out validation observations as the forward study. Training samples retain the original speed/fuel space-filling selection for direct comparison. Training records: valid chirp plus selection ramp; test record: separate validation ramp. No test-based tuning or interval rescaling.

Test target speeds span 9000-10000 rpm. Actual recorded speeds: 9000.000000 to 9998.885249 rpm; maximum target mismatch 1.685586 rpm. Test acceleration range: -151.003100 to 222.589628 rpm/s.

Fuel RMSE: 0.001048216 lbm/s (0.000475463 kg/s). MAE: 0.000832021 lbm/s. R2: 0.999997241. Nominal 95% interval coverage: 97.0% (97/100).

Exported Cholesky posterior vs MATLAB maximum mean/SD errors: 1e-11 / 4.59e-13 lbm/s. Scalar and vector API checks passed.

Training input ranges: N 9000.000000 to 9999.897804 rpm; acceleration -1153.470131 to 1076.907740 rpm/s. Training fuel range 0.183863 to 4.194426 lbm/s. The rectangular ranges do not imply every input combination is supported.

Uncertainty uses k(x,x)-||L\k(X,x)||^2 for latent variance and adds fitted noise variance for response variance. All outputs are rescaled to lbm/s. Hyperparameter and input-measurement uncertainty are not integrated. Test acceleration is the plant acceleration, not a noisy numerical derivative or a forward-GP prediction.

Application: Wf_hat = g(N, desired_acceleration) is a candidate feedforward estimate under the original fixed ambient and zero external shaft-load conditions. Closed-loop control and varying load are not validated here. Predictions are not clipped to actuator limits. The test samples are from one trajectory; exact GP inference does not guarantee an exact or globally unique physical inverse.

Method follows Volkan Aran (2019), Flexible and Robust Control of Heavy Duty Diesel Engine Airpath Using Data Driven Disturbance Observers and GPR Models, section 5.1 equations 5.5-5.14 and inverse feedforward modeling section 6.2.1; thesis reference [92], Rasmussen and Williams, Gaussian Processes for Machine Learning, chapter 2. This adapts the methodology to T-MATS shaft dynamics.

Source forward study: `results/exact_gpr_20260913_145742_458`.

7.4 Observer transfer and the direct GPR feedforward challenger

Evidence basis: `results/Observer_9000_9500_GPR_FF_Comparison/REPORT.md`. The following detailed account is adapted from the archived study; references to simulation and verification describe historical work.

GPR_FF is the inverse-GP feedforward requested as GPT FF. It augments the original PI; it is not a disturbance observer. All 60 runs are new Simulink simulations. The previous five controllers at 10% load reproduce the archived speed histories within 1e-6 rpm.

7.4.1 Controller

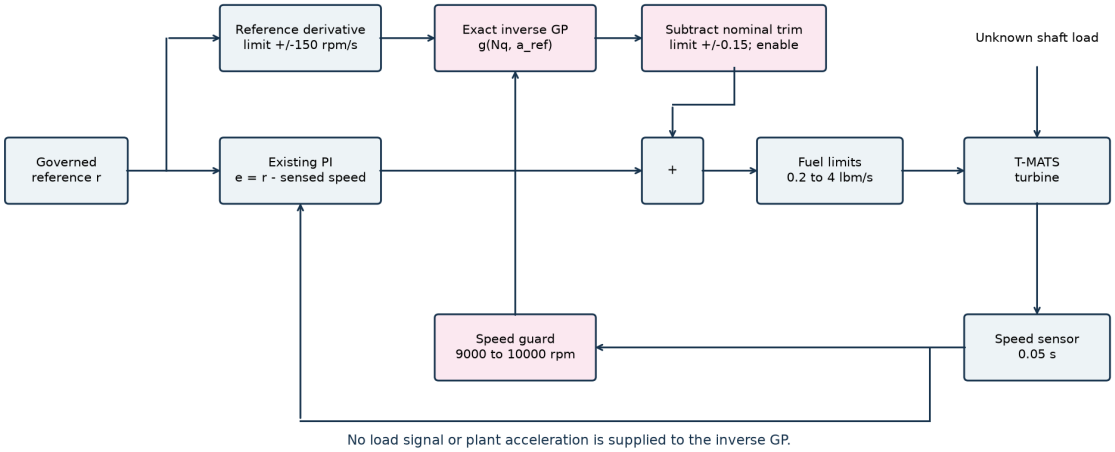


Figure 7.1: Controller architecture. Archived study figure.

```
a_ref[k] = clip((r[k]-r[k-1])/0.015, -150, 150)
Nq[k] = clip(N_sensed[k], 9000, 10000)
delta_u_FF[k] = rho(t) * clip(g(Nq[k],a_ref[k])-g(N_nominal,0), -0.15, 0.15)
u[k] = clip(u_PI[k] + delta_u_FF[k], 0.2, 4)
```

The original continuous PI retains Kp=0.025, Ki=0.05 and integrator initialization 3. The sensor time constant remains 0.05 s. rho ramps from 0 to 1 over absolute seconds 30-32. No observer compensation is active in GPR_FF. No shaft-load value, future disturbance schedule, actual plant acceleration, or inverse-model refit is supplied to the controller.

The observer branches retain their inherited compensation gain 0.25 and pole-frequency parameter 0.5 Hz. The physical inverse-GP correction uses unit gain before the same +/-0.15 lbm/s limit; it is not bandwidth-matched to the observers. No noise, delay, or gain-tuning comparison is claimed.

This is state-scheduled nominal feedforward: sensed speed schedules the inverse model. It is not purely reference-only feedforward; that would remain constant throughout these constant-reference tests. There is no extra speed-error-to-acceleration gain. The trim subtraction prevents double-counting equilibrium fuel already supplied by the inherited PI integrator. The speed guard prevents extrapolation below the 9000 rpm training boundary, but can make the 9000 rpm response asymmetric.

The inverse model is the frozen 600-point exact ARD squared-exponential GP. Online mean evaluation is O(600); its 100-point held-out inverse fuel RMSE was 0.001048216 lbm/s and 95% response interval coverage was 97%. Fuel uncertainty is evaluated offline on up to 100 valid samples per control run; it is not used as a control gain or as a bound on loaded-plant error.

7.4.2 Matched 10% observer comparison

Table 7.1: Observer transfer and the direct GPR feedforward challenger: Matched 10% observer comparison. Archived values.

Speed	Load	Shape	Method	Valid	RMSE rpm	Peak rpm	IAE rpm s	Failure s
9000	10%	ramps	PI	True	1.193	10.677	36.774	N/A
9000	10%	ramps	DOB	True	1.038	10.351	28.341	N/A
9000	10%	ramps	LESO	True	1.039	10.349	28.694	N/A
9000	10%	ramps	UDE	True	0.969	9.377	27.587	N/A
9000	10%	ramps	GPIO	True	0.951	9.866	27.577	N/A
9000	10%	ramps	GPR_FF	True	1.208	11.058	36.774	N/A
9500	10%	ramps	PI	True	2.125	18.642	65.863	N/A
9500	10%	ramps	DOB	True	1.850	18.147	50.799	N/A
9500	10%	ramps	LESO	True	1.853	18.155	51.429	N/A
9500	10%	ramps	UDE	True	1.730	16.574	49.411	N/A
9500	10%	ramps	GPIO	True	1.698	17.393	49.392	N/A
9500	10%	ramps	GPR_FF	True	2.192	19.938	65.864	N/A

7.4.3 Findings

At 9000 rpm, GPR_FF RMSE is 1.207770 rpm versus PI 1.193143 rpm (+1.23% change; positive is worse). Peak error is 11.057788 rpm. Speed guard active: 34.04% of evaluation samples. GPR_FF +/-1 rpm recovery at 9000 rpm is faster than PI in 3/6 events, equal in 3/6, and slower in 0/6. Thus the larger peaks and RMSE coexist with faster threshold recovery; these metrics must not be collapsed into a single blanket ranking. At 9500 rpm, GPR_FF RMSE is 2.191826 rpm versus PI 2.125136 rpm (+3.14% change; positive is worse). Peak error is 19.937677 rpm. Speed guard active: 0.00% of evaluation samples. GPR_FF +/-1 rpm recovery at 9500 rpm is faster than PI in 6/6 events, equal in 0/6, and slower in 0/6. Thus the larger peaks and RMSE coexist with faster threshold recovery; these metrics must not be collapsed into a single blanket ranking.

A nominal inverse learned at zero external shaft load cannot anticipate an unknown load. Accurate offline inverse regression does not imply improved rejection. In these tests a_ref is zero during evaluation, so the FF branch only schedules nominal fuel with sensed speed; the PI integrator supplies the additional load-dependent fuel. A learned increase in equilibrium fuel with speed can reduce effective restoring feedback when added to the unchanged PI. Results are reported without retuning to favor the GP.

Regression of unclipped FF correction against sensed speed on the matched runs gives slopes 0.0015974 and 0.0020183 lbm/s per rpm at 9000 and 9500 rpm, respectively. These are descriptive trajectory fits, not independent linearizations; the 9000 rpm fit excludes the active speed guard.

7.4.4 All load scenarios

Table 7.2: Observer transfer and the direct GPR feedforward challenger: All load scenarios. Archived values.

Speed	Load	Shape	Method	Valid	RMSE rpm	Peak rpm	IAE rpm s	Failure s
9000	5%	ramps	PI	True	0.594	5.325	18.309	N/A
9000	5%	ramps	DOB	True	0.517	5.162	14.109	N/A
9000	5%	ramps	LESO	True	0.517	5.161	14.285	N/A
9000	5%	ramps	UDE	True	0.483	4.677	13.735	N/A
9000	5%	ramps	GPIO	True	0.473	4.920	13.730	N/A
9000	5%	ramps	GPR_FF	True	0.601	5.515	18.309	N/A
9000	10%	ramps	PI	True	1.193	10.677	36.774	N/A

Speed	Load	Shape	Method	Valid	RMSE rpm	Peak rpm	IAE rpm s	Failure s
9000	10%	ramps	DOB	True	1.038	10.351	28.341	N/A
9000	10%	ramps	LESO	True	1.039	10.349	28.694	N/A
9000	10%	ramps	UDE	True	0.969	9.377	27.587	N/A
9000	10%	ramps	GPIO	True	0.951	9.866	27.577	N/A
9000	10%	ramps	GPR_FF	True	1.208	11.058	36.774	N/A
9000	20%	ramps	PI	True	2.540	22.553	78.110	N/A
9000	20%	ramps	DOB	True	2.237	22.329	60.838	N/A
9000	20%	ramps	LESO	True	2.240	22.335	61.314	N/A
9000	20%	ramps	UDE	False	N/A	N/A	N/A	61.530
9000	20%	ramps	GPIO	True	2.072	21.457	60.111	N/A
9000	20%	ramps	GPR_FF	True	2.571	23.707	78.110	N/A
9000	30%	ramps	PI	True	3.973	35.359	121.624	N/A
9000	30%	ramps	DOB	True	3.664	35.258	104.370	N/A
9000	30%	ramps	LESO	True	3.668	35.258	105.042	N/A
9000	30%	ramps	UDE	True	3.530	35.237	103.631	N/A
9000	30%	ramps	GPIO	True	3.501	35.258	103.625	N/A
9000	30%	ramps	GPR_FF	True	4.022	37.310	121.624	N/A
9500	5%	ramps	PI	True	1.017	8.905	31.538	N/A
9500	5%	ramps	DOB	True	0.885	8.667	24.317	N/A
9500	5%	ramps	LESO	True	0.886	8.670	24.619	N/A
9500	5%	ramps	UDE	True	0.827	7.905	23.660	N/A
9500	5%	ramps	GPIO	True	0.811	8.281	23.651	N/A
9500	5%	ramps	GPR_FF	True	1.049	9.513	31.538	N/A
9500	10%	ramps	PI	True	2.125	18.642	65.863	N/A
9500	10%	ramps	DOB	True	1.850	18.147	50.799	N/A
9500	10%	ramps	LESO	True	1.853	18.155	51.429	N/A
9500	10%	ramps	UDE	True	1.730	16.574	49.411	N/A
9500	10%	ramps	GPIO	True	1.698	17.393	49.392	N/A
9500	10%	ramps	GPR_FF	True	2.192	19.938	65.864	N/A
9500	20%	ramps	PI	True	4.453	39.270	137.688	N/A
9500	20%	ramps	DOB	True	4.148	39.270	120.437	N/A
9500	20%	ramps	LESO	True	4.152	39.270	121.213	N/A
9500	20%	ramps	UDE	True	4.024	39.270	119.694	N/A
9500	20%	ramps	GPIO	True	3.985	39.270	119.688	N/A
9500	20%	ramps	GPR_FF	True	4.593	41.941	137.689	N/A
9500	30%	ramps	PI	False	N/A	N/A	N/A	10.380
9500	30%	ramps	DOB	False	N/A	N/A	N/A	10.380
9500	30%	ramps	LESO	False	N/A	N/A	N/A	10.380
9500	30%	ramps	UDE	False	N/A	N/A	N/A	10.380
9500	30%	ramps	GPIO	False	N/A	N/A	N/A	10.365
9500	30%	ramps	GPR_FF	False	N/A	N/A	N/A	10.380
9500	20%	steps	PI	True	6.430	54.100	137.688	N/A
9500	20%	steps	DOB	True	6.203	54.100	120.508	N/A
9500	20%	steps	LESO	True	6.207	54.100	121.530	N/A
9500	20%	steps	UDE	True	5.989	54.100	119.694	N/A
9500	20%	steps	GPIO	True	6.016	54.100	119.688	N/A
9500	20%	steps	GPR_FF	True	6.712	56.329	137.689	N/A
9500	30%	steps	PI	False	N/A	N/A	N/A	10.155
9500	30%	steps	DOB	False	N/A	N/A	N/A	10.155
9500	30%	steps	LESO	False	N/A	N/A	N/A	10.155
9500	30%	steps	UDE	False	N/A	N/A	N/A	10.140
9500	30%	steps	GPIO	False	N/A	N/A	N/A	10.155
9500	30%	steps	GPR_FF	False	N/A	N/A	N/A	10.155

At both speeds, 5%, 10%, 20%, and 30% gross-turbine-power extraction uses three ramp/hold/ramp pulses (0.30, 1.50, 3.00 s). The earlier 9500 rpm 20% and 30% step-load cases are repeated too. Application times are 10.005, 30, 60 s; removal times 15, 40.005, 75 s after 60 s preparation. The 9500 rpm ramped 5/20/30% cases extend the previous matrix. Loads are speed-specific fractions, not equal absolute powers.

7.4.5 Validity and recovery

Validity requires finite logged signals, positive fuel, positive compressor margin, flow residual <=1e-9, and fewer than 200 iterations. Full-run scores are withheld after any invalid solve; plots show only the valid prefix. A later numerically converged sample does not rehabilitate the trajectory. Shaft balance and command reconstruction are verified independently. Identical load profiles are checked across methods.

Recovery is the first sample after the last excursion outside +/-1 rpm with at least 0.5 s dwell before the next load edge. After-ramp recovery subtracts ramp duration and is bounded below by zero. Event scores are withheld if any preceding sample is invalid. All 360 event records are in events.csv.

7.4.6 Reproduce

`run_tmats_gpr_ff_benchmarks` builds GasTurbine_Dyn_Template_GPR_FF.mdl and runs the matrix. Then run `summarize_tmats_gpr_ff.py`. The previous models and trained inverse GP remain separate. The new Level-2 MATLAB S-function is a desktop simulation implementation, not an embedded code-generation claim.

References: Aran (2019), thesis sections 5.1 and 6.2.1; Rasmussen and Williams, Gaussian Processes for Machine Learning, chapter 2. Previous report derivations for the inherited PI/DOB/LESO/UDE/GPIO remain applicable.

7.4.7 9000 rpm plots

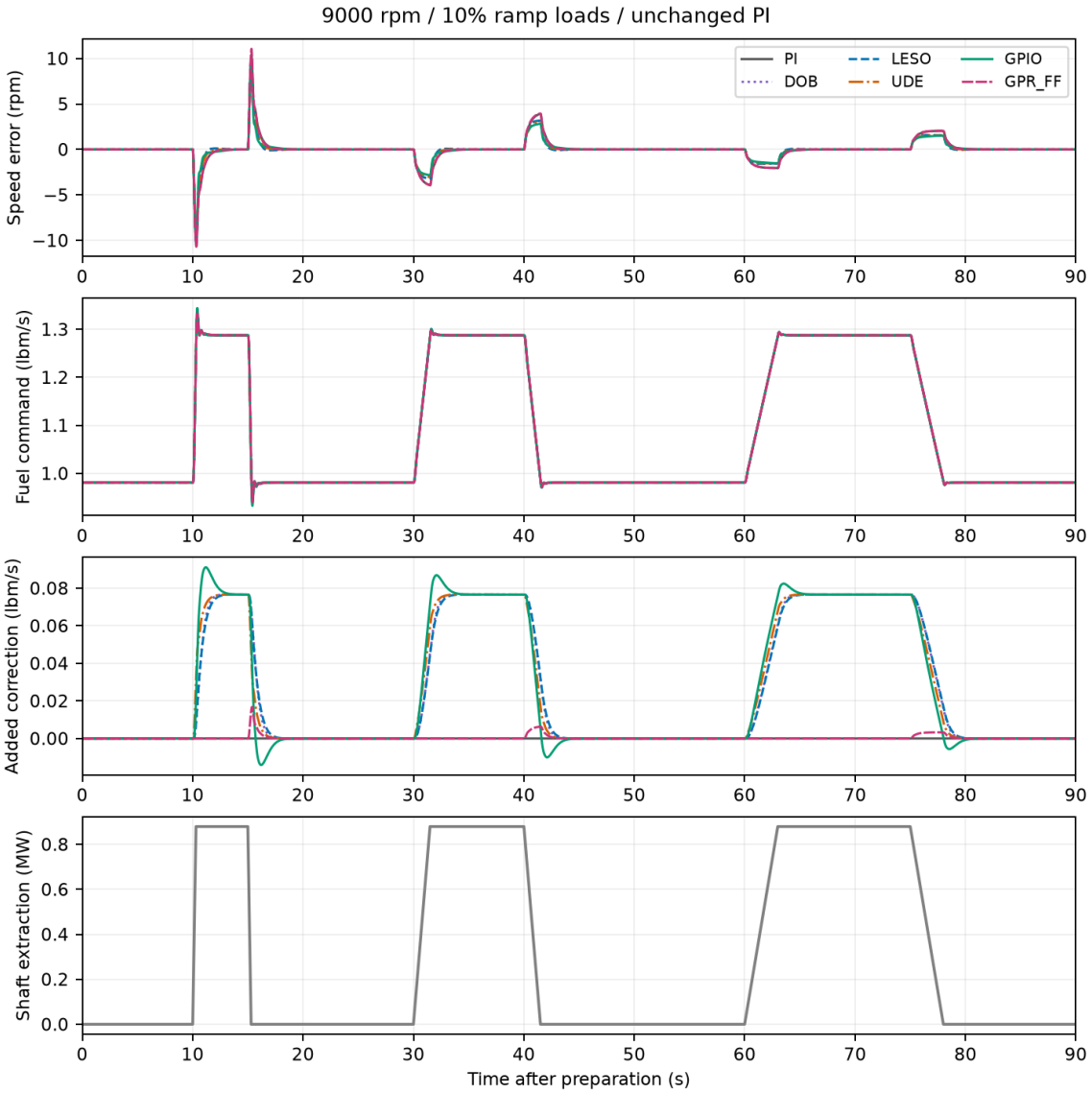


Figure 7.2: Six-controller response. Archived study figure.

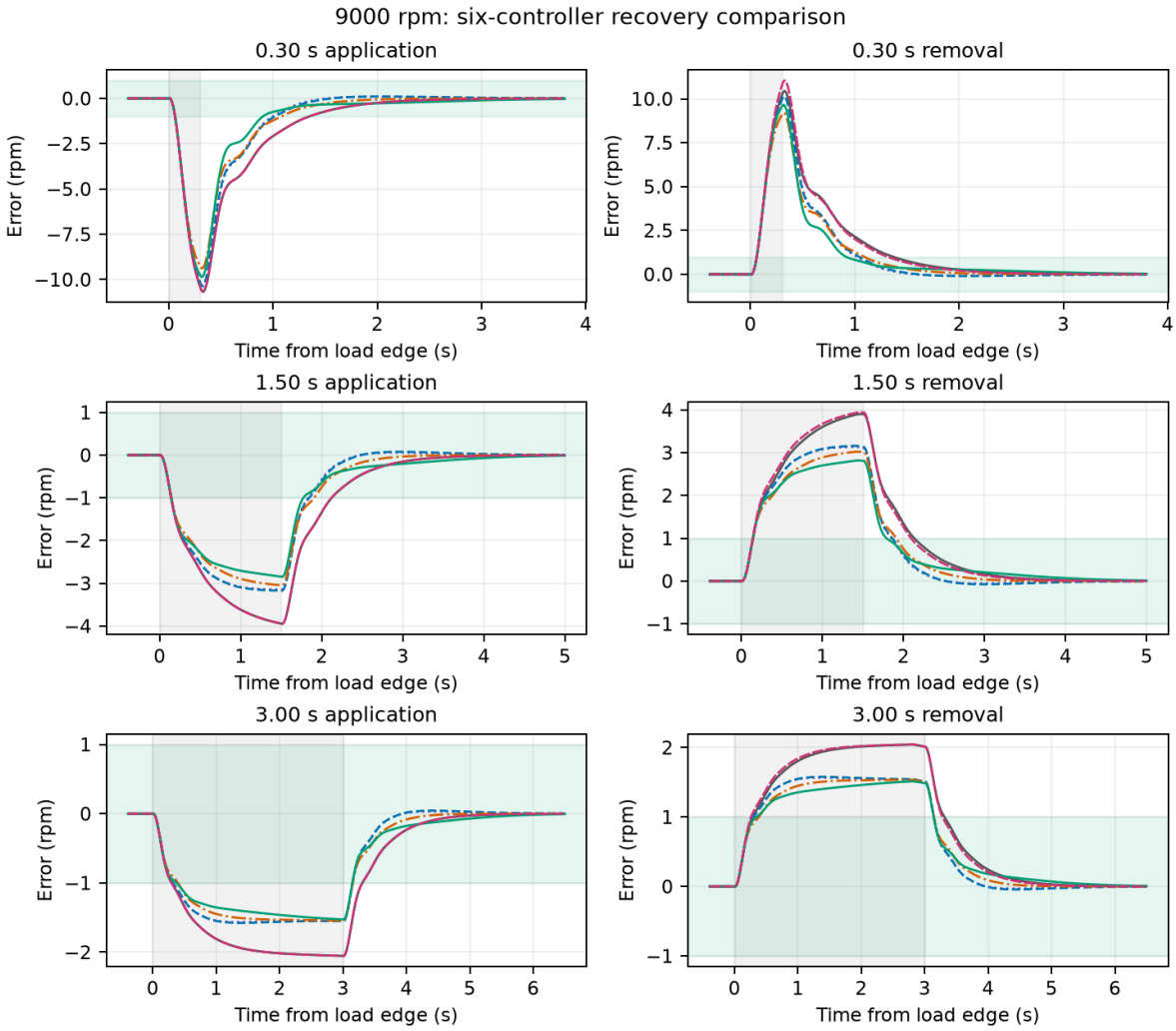


Figure 7.3: Recovery zooms. Archived study figure.

7.4.8 9500 rpm plots

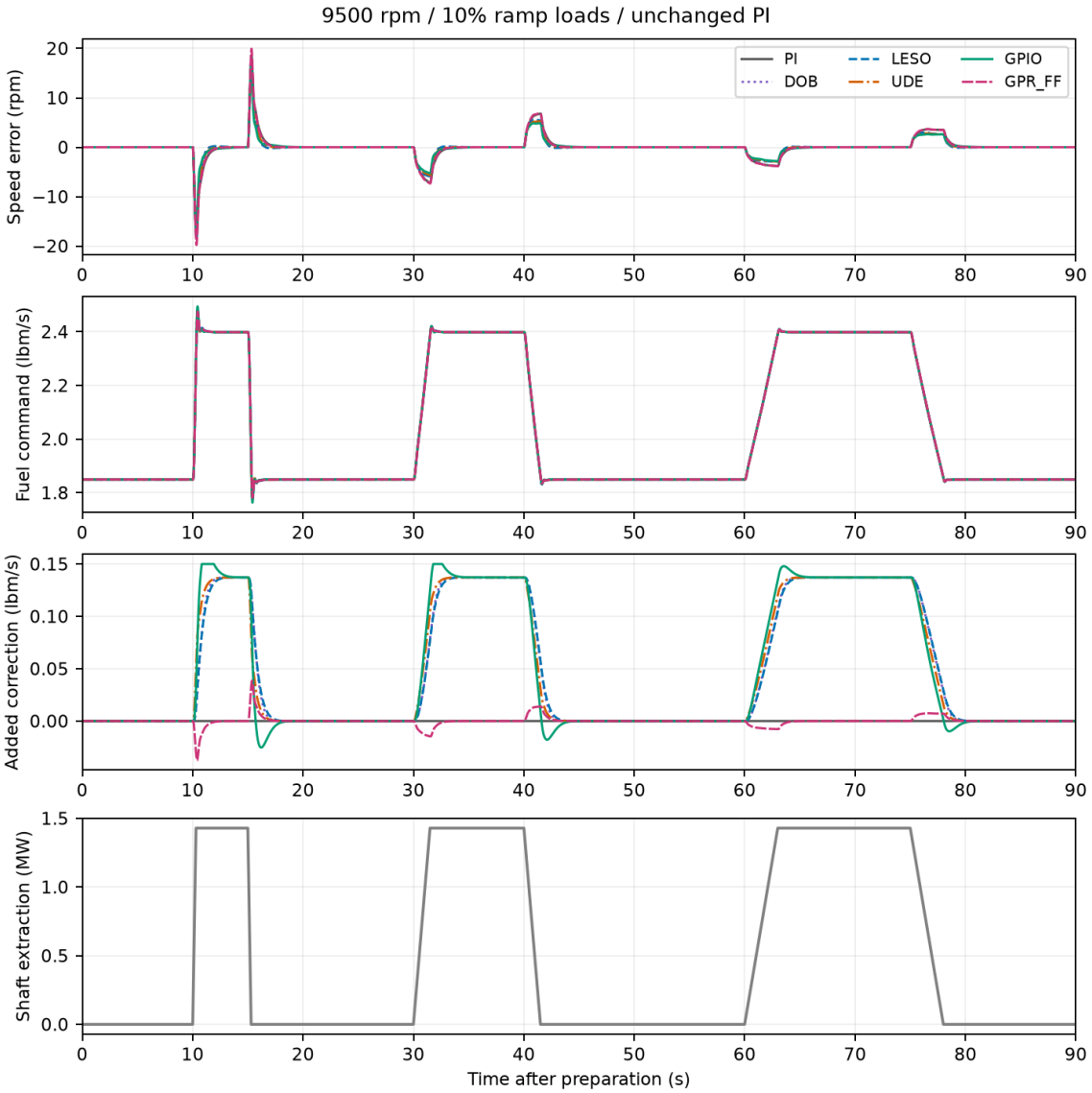


Figure 7.4: Six-controller response. Archived study figure.

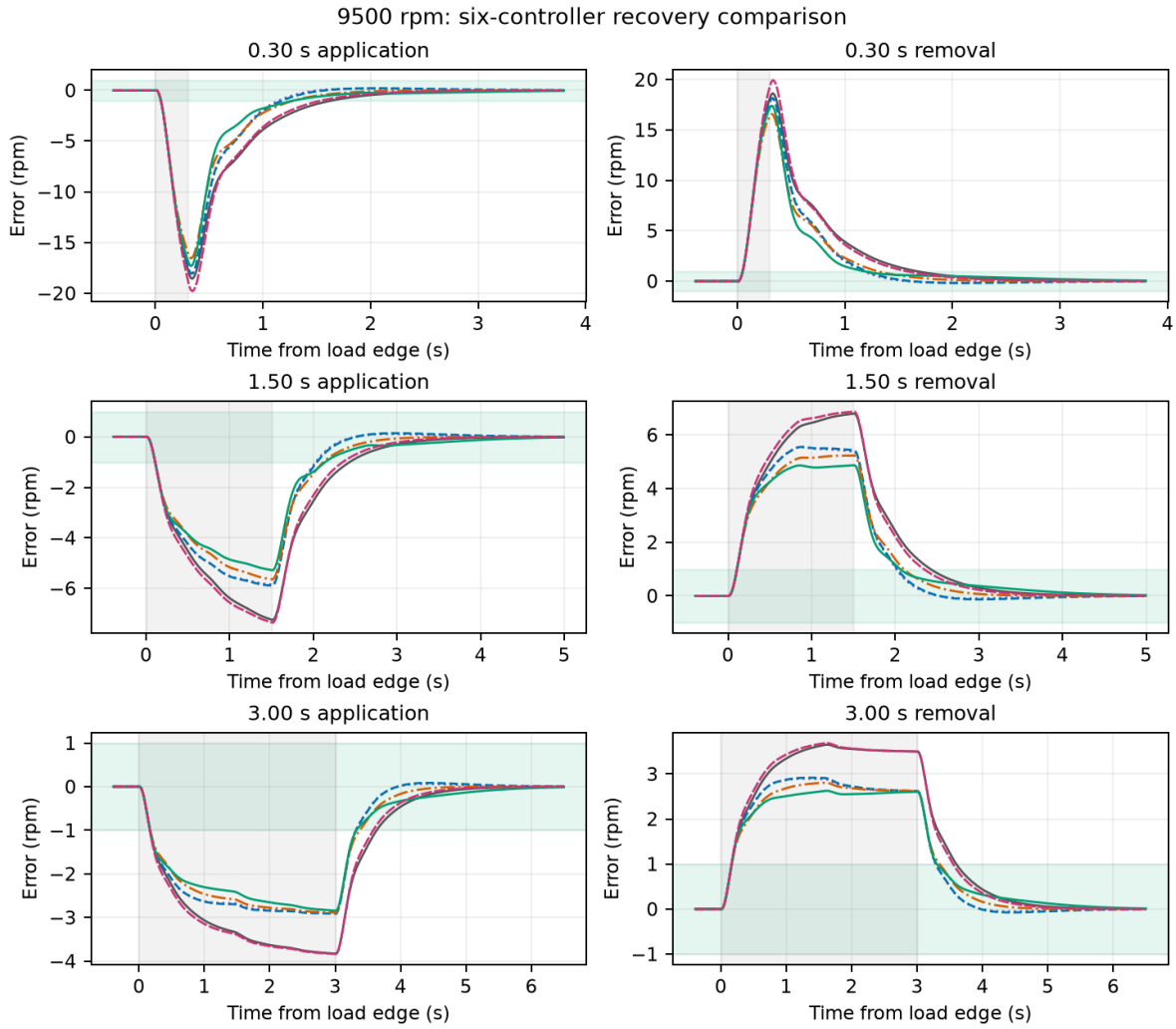


Figure 7.5: Recovery zooms. Archived study figure.

7.4.9 First-invalid-sample diagnostics

Table 7.3: Observer transfer and the direct GPR feedforward challenger: First-invalid-sample diagnostics. Archived values.

Scenario	Method	Time s	Flow residual	Iterations	Margin %
9000rpm_20pct_ramps	UDE	61.530	1.45e-05	200	25.679
9500rpm_30pct_ramps	PI	10.380	0.00119	200	0.000
9500rpm_30pct_ramps	DOB	10.380	0.00304	200	0.000
9500rpm_30pct_ramps	LESO	10.380	0.00332	200	0.000
9500rpm_30pct_ramps	UDE	10.380	0.000294	200	0.000
9500rpm_30pct_ramps	GPIO	10.365	8.86e-05	200	0.000
9500rpm_30pct_ramps	GPR_FF	10.380	0.000656	200	0.000
9500rpm_30pct_steps	PI	10.155	0.00647	200	0.000
9500rpm_30pct_steps	DOB	10.155	0.00804	200	0.000

Scenario	Method	Time s	Flow residual	Iterations	Margin %
9500rpm_30pct_steps	LESO	10.155	0.00801	200	0.000
9500rpm_30pct_steps	UDE	10.140	0.00154	200	0.000
9500rpm_30pct_steps	GPIO	10.155	0.0107	200	0.000
9500rpm_30pct_steps	GPR_FF	10.155	0.00119	200	0.000

7.4.10 Wider load tests

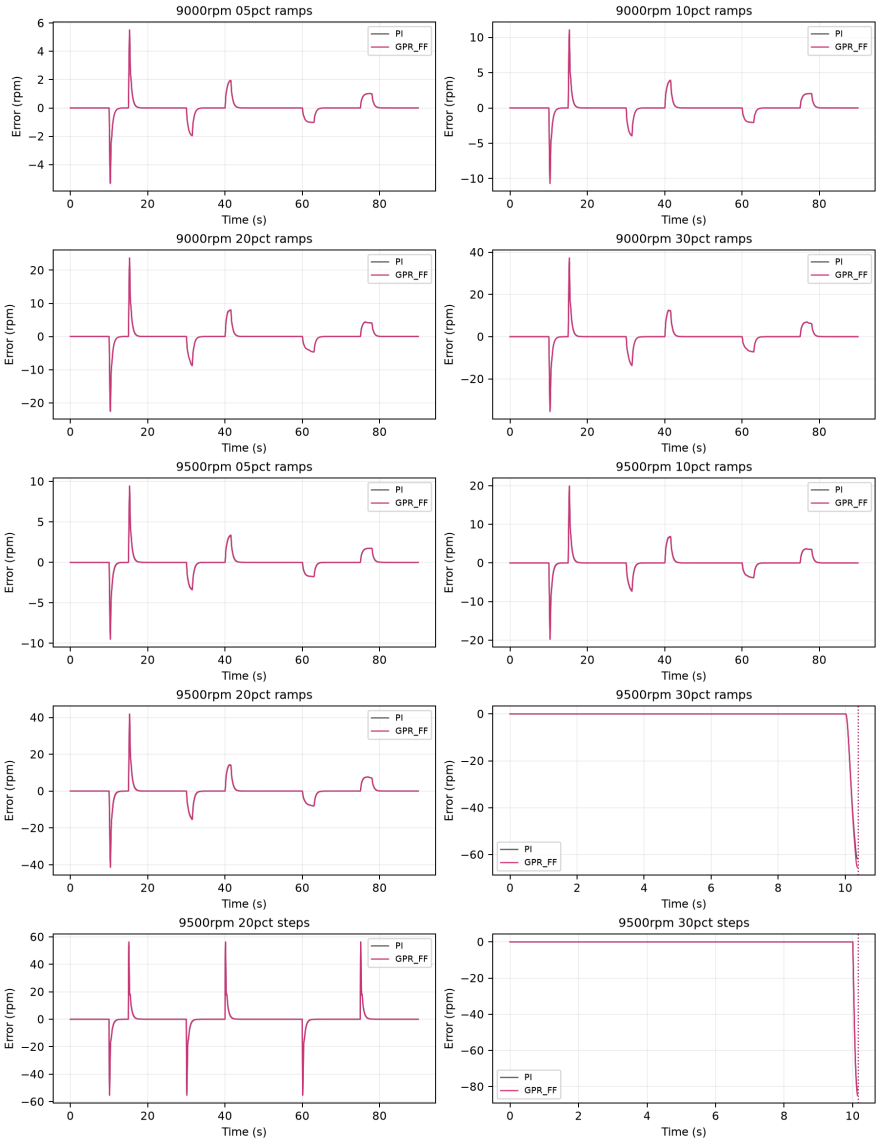


Figure 7.6: All loads. Archived study figure.

Chapter 8

Inverse-coordinate feedback, feedforward, and gain selection

The central attribution question is whether improvement belongs to the inverse model, the feedback coordinates, or gain selection. This chapter retains the exact algorithm distinctions and the fixed-reference feedforward equivalence. Its conclusions concern the September controller family; the later PILCO-style chapter uses a separate dynamics GP.

8.1 Protocol for the inverse-model controller comparisons

8.1.1 Simulation plant and physical interpretation

T-MATS supplies modular thermodynamic components and iterative solution machinery for steady and dynamic propulsion-system models [2, 3]. The local study uses its inherited nonlinear gas-turbine configuration. The controller actuates fuel flow and observes a filtered shaft-speed signal. External shaft power is applied to the plant, not supplied to the inverse model or controller as a measured feedforward signal.

A physical bookkeeping equation in consistent SI units is

$$J\dot{\omega} = \tau_t - \tau_c - \tau_L, \qquad \omega = \frac{2\pi N}{60}, \qquad \tau_L = \frac{P_L}{\omega}. \tag{8.1}$$

Equation (8.1) expresses the shaft-power mechanism; it is not a replacement for the archived nonlinear T-MATS component model. The simulation interface expresses load power in hp and fuel in lbm/s. Multiplication by 0.45359237 converts fuel flow to kg/s. A load fraction is a fraction of the condition-specific unloaded turbine shaft-power reference, not a fraction of fuel flow or a prescribed speed decrease.

8.1.2 Notation and controller timing

Symbol	Meaning	Unit
N, N_s, r	Actual speed, sensed speed, governed request	rpm
$\hat{a}, a_r$	Filtered sensed acceleration, explicit reference acceleration	rpm/s
T, P	Sample-delayed inlet values used by the online inverse	K, kPa
$u, u^\star, I$	Limited fuel, unsaturated fuel, integral contribution	lbm/s
g, p_2	GP and full quadratic inverse means	lbm/s
e_v, e_0	Dynamic virtual-fuel error, static inverse error	lbm/s
e_N^g, a^g	Derivative-normalized speed error and acceleration	rpm, rpm/s
g_N, g_a	Partial derivatives of the inverse mean	fuel/rpm, fuel/(rpm/s)
T_s, τ_s, τ_a	Sample interval, speed-sensor and acceleration-filter constants	s

R_L, R_C	Load-edge ringing and cycle-segment fuel reversal	lbm/s

The controller interval is $T_s = 0.015$ s. The inherited speed sensor has time constant $\tau_s = 0.05$ s, and the inverse controllers use $\tau_a = 0.03$ s for acceleration estimation. The reported tests have no newly introduced sensor noise or actuator lag. Actual plant acceleration is recorded for identification and validation; the online feedback acceleration is reconstructed from sensed speed.

For four-input online controllers, the inlet measurements pass through one-sample Unit Delay blocks after conversion to K and kPa. Writing current plant inlet values as $T_{\text{in}}, P_{\text{in}}$, the query notation used throughout the controller equations is

$$T[k] = T_{\text{in}}[k-1], \qquad P[k] = P_{\text{in}}[k-1], \tag{8.2}$$

with initialized values 288.15 K and 99.298 kPa. Offline identification rows use their contemporaneous recorded inlet values. Thus replay of online queries must shift logged inlet signals by one sample; the archived cycle coverage audit does so. At changing ambient conditions this timing distinction is part of the implemented algorithm, even though it is small at the 15 ms sample interval.

At each sample, define $p = \exp(-T_s/\tau_a)$ and update

$$\hat{a}[k] = p\hat{a}[k-1] + (1-p)\frac{N_s[k] - N_s[k-1]}{T_s}, \quad D_a(z) = \frac{(1-p)(1-z^{-1})}{T_s(1-pz^{-1})}. \tag{8.3}$$

Here $p \simeq 0.60653$. The filter is causal and preserves a finite high-frequency gain, but adds phase effects to the sensor dynamics. Treating $\hat{a}$ as an ideal, delay-free derivative is appropriate only for local interpretation, not for reproducing the simulations.

8.1.3 Fixed-speed shaft-load tests

Fixed-speed runs last 150 s, with 60 s preparation and a 90 s evaluation interval. The inherited startup begins at 10000 rpm and approaches the requested operating speed. The early inverse controller takes over at 30 s; later environment/R3/Poly2 configurations use 45 s. The baseline speed PI supplies startup control, while the new integral state tracks the corresponding command decomposition.

With evaluation time $t_e = t - 60$, load applications begin at approximately 10.005, 30, and 60 s and removals at 15, 40.005, and 75 s. The early ramp protocol uses ramp durations 0.3, 1.5, and 3 s for the three pulses; the step protocol uses zero ramp duration. Loads of 5, 10, 20, and 30% appear in the initial studies. Later environmental comparisons use 10% steps. The final 9250 rpm tuning uses 10% steps, followed by a 5% amplitude check.

For a ramped pulse with rise time $\rho > 0$, the normalized load shape is

$$b(t_e) = \text{sat}_{[0,1]}\left(\frac{t_e - t_{\text{on}}}{\rho}\right) - \text{sat}_{[0,1]}\left(\frac{t_e - t_{\text{off}}}{\rho}\right), \qquad P_L(t_e) = fP_{\text{ref}}\sum_j b_j(t_e). \tag{8.4}$$

The zero-ramp case is the corresponding rectangular pulse. The logged timing includes the sample-grid conventions of the implementation; event windows are aligned to those conventions.

8.1.4 Metrics and their limits

For the M evaluation samples, actual-speed tracking is scored as

$$\mathrm{RMSE}_N = \sqrt{\frac{1}{M}\sum_{k=1}^{M}(N[k]-r[k])^2}, \quad \mathrm{IAE}_N = T_s\sum_{k=1}^{M}|N[k]-r[k]|, \quad E_{\max} = \max_k |N[k]-r[k]|. \tag{8.5}$$

RMSE weights larger errors more strongly, whereas IAE also captures small persistent tails. Both are measured against actual shaft speed, not sensed speed; this distinction matters during ramps.

For a fuel sequence in a window W, define excess total variation

$$R(W) = \sum_{k\in W,\ k>k_0} |u[k]-u[k-1]| - |u[k_1]-u[k_0]| \geq 0. \tag{8.6}$$

The load-test ringing metric R_L sums this quantity over six two-second windows after each step or after each load ramp finishes. A monotonic change contributes zero; reversals contribute positive variation. This is a fuel-rate reversal measure with units lbm/s, not fuel mass consumed, frequency-weighted vibration, or a spectral oscillation amplitude. Window length and sample interval are part of its definition. The fixed-load "peak fuel slew" is likewise evaluated only within those same six two-second windows:

$$S_L = \max_j \max_{k,k-1\in W_j} \frac{|u[k]-u[k-1]|}{T_s}. \tag{8.7}$$

It is not the maximum across the complete evaluation trace. Cycle peak slew, in contrast, uses the entire scored cycle. These different domains must be retained when comparing tables.

The later cyclic test uses $R_C = \sum_j R(S_j)$ over all 16 ramp/hold segments. It also reports hold-only excess variation. Moving ambient conditions can require legitimate fuel reversals, so R_C is best called a reversal proxy rather than a pure ringing measure. R_L and R_C are not directly comparable across chapters.

For an event beginning at t_j, recovery to a band ϵ is the first sample after which $|N-r| \leq \epsilon$ persists until the next event boundary. The recorded value is the worst across the six load edges, or across the eight cycle holds. If a trajectory never leaves the band, recovery is zero. If it fails to enter persistently before the next boundary, the result is NR. The 1% window is $\epsilon = 0.01r$; a supplementary 0.1 rpm window distinguishes fine settling when all responses remain inside the broad percentage band.

8.1.5 Acceptance is not coverage

A complete scored load run requires finite valid plant signals, positive fuel and surge margin, converged flow residuals, fewer than 200 solver iterations, complete duration, and pre-test speed settling within 0.1 rpm. A failed trajectory has its full-run performance scores withheld. A brief accepted prefix is not a valid full-run comparison.

Numerical acceptance, compressor-map range, and learned-model coverage are separate questions. The logged plant-map audit uses actual speed and current inlet temperature, $N_{c,\log} = N/(10000\sqrt{T_{\mathrm{in}}/288.15})$, with a working range $[0.5, 1.05]$. The online fallback guard instead uses sensed speed and sample-delayed temperature, $N_{c,\mathrm{guard}} = N_s/(10000\sqrt{T/288.15})$. They are distinct signals, so zero logged map overrun and zero fallback need not coincide. A simulation can converge while leaving that range. Likewise, a GP query can lie in its training rectangle while leaving the joint training hull. The paper therefore reports these layers separately rather than treating convergence as evidence of physical or statistical validity.

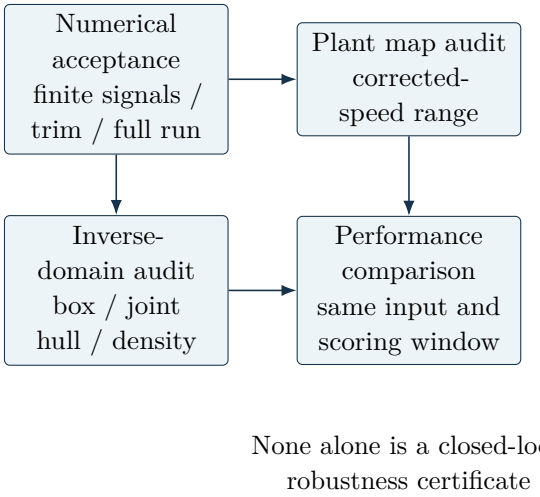


Figure 8.1: Four distinct evidential checks. Performance interpretation requires more than successful time integration.

8.2 From inverse identification to feedback architecture

8.2.1 Two-input inverse GP

The initial inverse is fitted directly to observations $x_i = [N_i, a_i]^\mathsf{T}$ and fuel w_i. It reuses 600 training and 100 held-out observations from the earlier identification dataset in the 9000–10000 rpm band. The forward GP does not generate the inverse targets. A separately regressed inverse need not be the algebraic inverse of a separately fitted forward model.

Let $z_j = (x_j - \mu_{x,j})/s_{x,j}$ and $y_i = (w_i - \mu_w)/s_w$. With an ARD squared-exponential kernel,

$$k(z, z') = \sigma_f^2 \exp\left[-\frac{1}{2}\sum_{j=1}^{d}\frac{(z_j - z'_j)^2}{\ell_j^2}\right], \qquad d = 2 \text{ or } 4. \tag{8.8}$$

The exact posterior is exported through $LL^\mathsf{T} = K + \sigma_n^2 I_n$ and $\alpha = L^{-\mathsf{T}}L^{-1}\boldsymbol{y}$:

$$g(x) = \mu_w + s_w \boldsymbol{k}_x^\mathsf{T}\alpha, \tag{8.9}$$

$$\sigma_g^2(x) = s_w^2\left[k(z,z) - \|L^{-1}\boldsymbol{k}_x\|^2\right], \tag{8.10}$$

$$\sigma_{\text{resp}}^2(x) = \sigma_g^2(x) + s_w^2\sigma_n^2. \tag{8.11}$$

These are the standard exact-regression quantities [1]; the study's standardization, exported parameters, and query semantics specify their implementation. Small negative variance from roundoff is clipped to zero. Matrix factorization is offline. A mean query sums kernel terms with frozen weights; it does not solve the complete training system anew. Computing exact uncertainty using the triangular factor is more expensive than evaluating the mean.

The initial held-out inverse RMSE is 0.00104822 lbm/s and its nominal 95% response interval covers 97 of 100 held-out outputs (D01). This assesses nominal regression on the saved split, not loaded-plant inverse accuracy or future closed-loop reliability. Acceleration-input uncertainty, unmodeled shaft load, and hyperparameter uncertainty are not propagated by this response interval.

8.2.2 Base speed PI

The inherited comparator acts directly on $e_N = r - N_s$:

$$u^{\star}_{\mathrm{PI}} = K_{pN} e_N + I_N, \qquad \dot{I}_N = K_{iN} e_N, \quad K_{pN} = 0.025,\ K_{iN} = 0.05. \tag{8.12}$$

The gains carry speed-to-fuel units. They cannot be compared numerically with gains multiplying a fuel-valued inverse error. The existing baseline is a fixed comparator, not the best possible conventional PI/PID. This point later becomes decisive when a matched-gain PID is introduced.

8.2.3 The first GPR feedforward addition

The early base-PI-plus-GPR branch uses a correction around an equilibrium fuel estimate:

$$a_r[k] = \mathrm{sat}_{[-150,150]} \frac{r[k] - r[k-1]}{T_s}, \tag{8.13}$$

$$N_q[k] = \mathrm{sat}_{[9000,10000]}\, N_s[k], \tag{8.14}$$

$$\Delta u_{\mathrm{FF}}[k] = \eta[k]\, \mathrm{sat}_{[-0.15,0.15]}\{g(N_q[k], a_r[k]) - g(N_0, 0)\}, \tag{8.15}$$

$$u[k] = \mathrm{sat}_{[0.2,4]}\{u_{\mathrm{PI}}[k] + \Delta u_{\mathrm{FF}}[k]\}. \tag{8.16}$$

The enable factor η ramps from zero to one over 30–32 s. This branch deliberately subtracts equilibrium fuel because the PI integral already supplies it. It is state-scheduled nominal feedforward: its speed input is sensed speed, not exclusively the request. During constant-speed load tests $a_r = 0$, so it has no direct knowledge of the disturbance. The speed clipping and correction limit are specific to this early branch and must not be attributed to the later R3 static FF implementation.

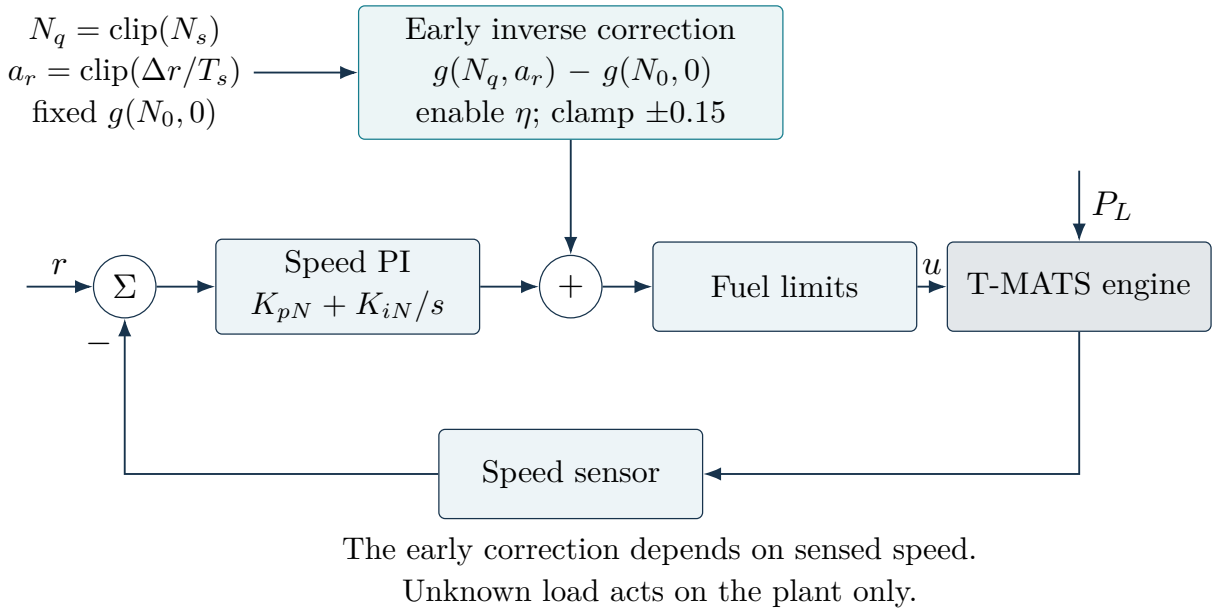


Figure 8.2: Baseline speed PI and the early optional inverse correction. The load acts on the plant only.

The broader historical benchmark also compared disturbance observers. Those observer models motivated the control workspace but are not the subject of this GPR-to-Poly2 lineage. The four-controller fixed-gain comparison in Chapter 8.5 retains the relevant base PI and base PI + FF results under a common protocol; it avoids conflating this state-scheduled correction with the later direct setpoint-feedforward law.

8.3 Dual inverse-model virtual-fuel control

8.3.1 Algorithm definition

The central initial architecture evaluates the same inverse twice:

$$v_r = g(r, a_r), \quad v_y = g(N_s, \hat{a}), \quad e_v = v_r - v_y, \qquad u^\star = K_p e_v + I. \tag{8.17}$$

The original virtual-fuel gains are $K_p = 4$ and $K_i = 40\ \mathrm{s}^{-1}$. Because e_v has fuel-flow units, K_p is dimensionless. For constant-speed disturbance tests, $a_r = 0$ and v_r is constant. The same-model evaluation gives exact cancellation when the two input pairs coincide, but does not cancel all state-dependent inverse error.

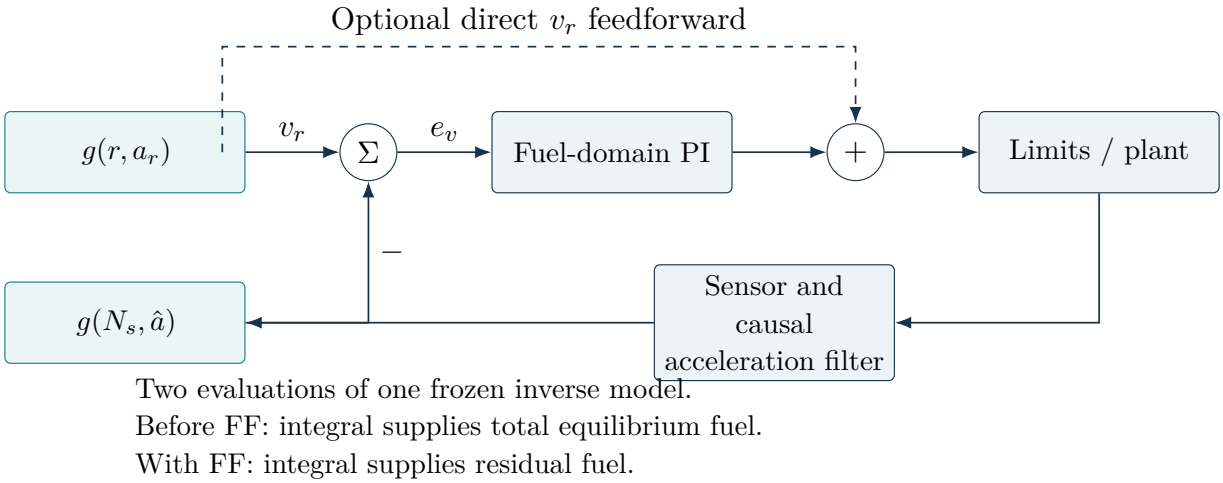


Figure 8.3: Dual inverse queries create a fuel-domain error. The dashed direct setpoint branch is added in the later FF variant.

The term "virtual fuel" denotes a nominal motion coordinate. It is not an online estimate of actual injected fuel under an unknown external load. Nor is the map from speed and acceleration to a scalar fuel value a globally invertible coordinate transformation: different motion pairs may produce the same value.

8.3.2 Conditional integration and startup

The active output is $u = \mathrm{sat}_{[u_{\min}, u_{\max}]}\, u^\star$, with limits 0.2 and 4 lbm/s. Define

$$\gamma(u^\star, e_i) = \mathbf{1}[(u^\star < u_{\max} \ \vee \ e_i < 0) \wedge (u^\star > u_{\min} \ \vee \ e_i > 0)]\,. \tag{8.18}$$

Here e_i is the error actually integrated and $u^\star$ contains every enabled FF and feedback contribution. All implemented integral gains are positive. The general update is $I[k+1] = I[k] + \gamma(u^\star[k], e_i[k]) T_s K_i e_i[k]$. Use $e_i = e_v$ for the original virtual PI, R1, and both SD schedules; use $e_i = e_0$ for R2; and use the selected normalized or fallback speed error for R3 and Poly2. For the original controller this reduces to $I[k+1] = I[k] + \gamma(u^\star[k], e_v[k]) T_s K_i e_v[k]$. The later shorthand γ always denotes this architecture-specific gate. This conditional integration suppresses increments that drive the command farther into saturation, while permitting unwinding. It is not back-calculation. Integrator windup is a classical implementation concern [6]; the exact gate here is established by the controller source.

Before handover, the integral tracks $u_{\text{startup}} - K_p e_v$. When direct FF is present it instead tracks $u_{\text{startup}} - v_r - K_p e_v$. Output evaluation precedes state update, so this sampled tracking is approximately bumpless, rather than an algebraic guarantee of identical commands during a changing signal. The archived evaluation starts well after handover.

8.3.3 Why the integral must carry unknown load

Consider an illustrative local shaft model

$$\dot{n} = -\lambda n + b\delta u - d_a, \qquad n = N - N_0, \quad b > 0. \tag{8.19}$$

If the nominal inverse is exact for the unloaded model,

$$g(N_0 + n, \dot{n}) \simeq u_0 + \frac{\lambda}{b} n + \frac{1}{b}\dot{n} = u_0 + \delta u - \frac{d_a}{b}. \tag{8.20}$$

At recovered speed, $n = \dot{n} = 0$ and $v_y = v_r = u_0$, but actual fuel remains $u_0 + d_a/b$. In the original controller the integral supplies that total equilibrium fuel. With direct FF it supplies the residual d_a/b. Thus virtual-fuel feedback returning to its nominal level while actual fuel remains elevated is expected behavior, not evidence that the load has disappeared.

8.3.4 Local equivalent PID and hidden gain coupling

Let $c_N = \partial g/\partial N$ and $c_a = \partial g/\partial a$ at the operating point. Locally,

$$e_v \simeq c_N(r - N_s) + c_a(a_r - \hat{a}). \tag{8.21}$$

For ideal matching derivatives, the resulting continuous-time controller is

$$C_v(s) = \left(K_p + \frac{K_i}{s}\right)(c_N + c_a s), \tag{8.22}$$

$$K_{d,\text{eq}} = K_p c_a, \quad K_{p,\text{eq}} = K_p c_N + K_i c_a, \quad K_{i,\text{eq}} = K_i c_N. \tag{8.23}$$

The virtual PI therefore creates coupled speed-domain proportional, integral, and derivative-like action. Raising K_i also changes an effective proportional term through integrated acceleration; changing K_p changes derivative-like feedback. This coupling is a plausible mechanism for the observed tracking/ringing tradeoff. The actual sampled sensor/filter loop must still be simulated; the ideal correspondence is not an exact implementation identity.

For the local plant in Eq. (8.19), with constant request and no direct perturbation from FF, the ideal characteristic polynomial becomes

$$(1 + bK_p c_a)s^2 + (\lambda + bK_p c_N + bK_i c_a)s + bK_i c_N. \tag{8.24}$$

This expression exposes why an unfavorable speed slope is consequential: for positive b, K_i, c_a, K_p, a negative c_N gives a negative constant term and an unstable root in this simplified model. Excellent function-value prediction does not guarantee an appropriate derivative sign for feedback.

If $c_N = \lambda/b$ and $c_a = 1/b$, the ideal load response satisfies

$$\frac{n(s)}{d_a(s)} = -\frac{s}{(s + \lambda)((1 + K_p)s + K_i)}. \tag{8.25}$$

The plant pole remains in the disturbance channel even when a nominal inverse suggests cancellation in a reference channel. The ratio c_a/c_N provides a local time-scale diagnostic; it is not a universally identified plant time constant. This distinction helps explain later slow recovery without claiming that an inverse map alone eliminates plant dynamics.

8.3.5 Initial evidence and the FF attribution test

At 9000 rpm with 10% ramped load, the original inverse PI gives RMSE 0.744078 rpm, peak error 4.46486 rpm, and ringing 0.716899 lbm/s (D02). The later FF-retuned version gives 0.504458 rpm and 1.32761 lbm/s respectively: lower tracking error, but more reversal. This is already inconsistent with the broad claim that inverse enhancement necessarily improves both objectives.

The direct-FF law is $u^\star = v_r + K_p e_v + I_{\text{res}}$. If v_r is constant and $I_{\text{res}} = I_{\text{old}} - v_r$, its command is identical to the previous no-direct-FF law at equal gains, including the conditional-integration decision. The archived same-gain ablation over four 9000 rpm cases gives maximum speed difference 1.37×10^{-7} rpm and fuel difference 7.93×10^{-9} lbm/s (D21). The observed improvement after selecting gains 3/60 is therefore attributed to retuning in these constant-reference tests. That result does not deny possible FF benefits when the request or ambient conditions move; it establishes what this particular test can identify.

Eight gain pairs were tested on the four 9000 rpm ramp amplitudes and the selection was frozen before the 9500 rpm cases. The 3/60 choice improved RMSE by about 32% relative to the original virtual PI on the mutually valid ramp cases, but reduced the count of valid scenarios from nine to seven out of ten. A target-only tracking gain was not a general robustness improvement.

8.4 Uncertainty scheduling: two competing hypotheses

8.4.1 What uncertainty is being scheduled

The decreasing schedule implements the hypothesis that inverse-model unreliability should reduce feedback aggressiveness. Each sample evaluates the response standard deviations at both inverse queries and forms

$$q = \frac{\max(\sigma_r, \sigma_y)}{\sigma_{\text{ref}}}, \qquad \sigma_{\text{ref}} = \text{median}_{x_i \in \mathcal{D}_{\text{train}}} \sigma_{\text{resp}}(x_i) = 0.00117869140424 \text{ lbm/s}. \tag{8.26}$$

The reference is computed from the frozen training locations, not fitted to the disturbance results. The maximum enforces the intended rule that either inverse output can trigger a gain change. It is a heuristic aggregation of marginal uncertainties, not the posterior uncertainty of the difference. For correlated GP predictions,

$$\text{Var}(v_r - v_y) = \text{Var}(v_r) + \text{Var}(v_y) - 2\,\text{Cov}(v_r, v_y). \tag{8.27}$$

Ignoring that covariance changes the meaning of the scheduler input. The implementation intentionally uses the maximum, but it should not be described as a calibrated uncertainty bound on the control error.

8.4.2 Decreasing-gain algorithm

The requested three-standard-deviation attenuation is implemented as

$$\alpha_\downarrow(q) = 10^{-(q/3)^2}, \quad K_p = 3\alpha_\downarrow(q), \quad K_i = 60\alpha_\downarrow(q). \tag{8.28}$$

At $q = 3$, the multiplier is 0.1; at $q = 6$, it is 10^{-4}. It approaches zero smoothly. Because response SD includes the fitted observation-noise contribution, q need not approach zero on familiar inputs.

At $q = 1$, the multiplier is already about 0.7743. The implemented rule therefore derates nominal feedback as well as unusually uncertain feedback.

The accumulated integral is retained when gains become small. Only subsequent correction increments shrink. Otherwise reducing gains would also remove the already learned fuel needed to support a steady load. A proportional gain change is compensated by

$$I^+ = I^- + (K_p^- - K_p^+)e_v, \tag{8.29}$$

before forming the new command. This cancels a discontinuity attributable solely to changing K_p at a fixed current error; it does not cancel the desired response to a changing error. Conditional integration is then applied to the complete FF-plus-feedback command.

8.4.3 Increasing-gain challenger

The alternative hypothesis is that higher uncertainty might justify stronger correction. The requested symmetric bands are interpreted through a nonnegative SD magnitude:

$$x = \mathrm{sat}_{[0,1]} \frac{q-1}{2}, \qquad \alpha_\uparrow(q) = 1 + 3x^2 - 2x^3. \tag{8.30}$$

The factor is one for $q \le 1$, 1.5 at $q = 2$, and two for $q \ge 3$, with zero endpoint slope. The base gains remain 3/60 and the same integral transfer, FF, filter, and limits are used. Thus the comparison changes scheduling direction without a new base-gain optimization.

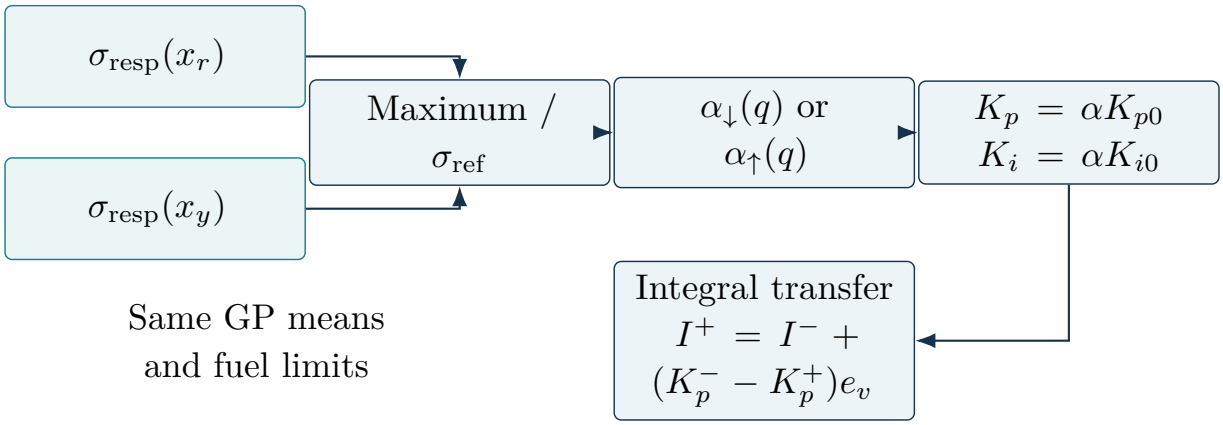


Figure 8.4: Uncertainty-to-gain calculation and integral compensation. The response-SD maximum is a design signal, not a certified loaded-plant error bound.

8.4.4 Measured effects on ringing and tracking

Method	RMSE	Peak	Ringing	Valid/10
Initial inverse PI	0.7441	4.465	0.7169	9/10
FF + retuned PI	0.5045	3.605	1.328	7/10
SD down	0.6797	4.896	0.5075	8/10
SD up	0.5021	3.588	1.356	7/10

Table 8.2: Early-stage 9000 rpm / 10% shaft-load ramps. RMSE and peak are rpm; ringing is lbm/s. Validity counts cover all ten scenarios (D02).

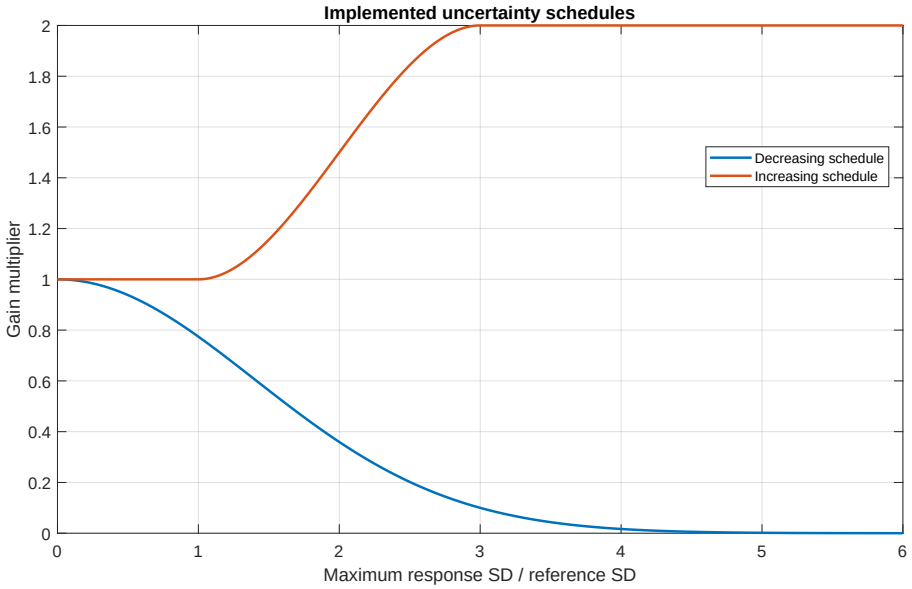


Figure 8.5: Implemented schedules. "Three SD" means three times the empirical reference SD; it is not a Gaussian tail-probability claim.

Table 8.2 illustrates a clear tradeoff. At 9000 rpm / 10% ramps, decreasing scheduling reduces ringing from 1.32761 to 0.507542 lbm/s compared with the fixed 3/60 controller, approximately 61.8%. RMSE increases from 0.504458 to 0.679744 rpm, approximately 34.7%. Relative to the original 4/40 inverse PI, the decreasing schedule improves both quantities in this example, but its peak speed error is larger. "Less ringing" is therefore true for a specified comparator and test, not an unconditional controller property.

The increasing schedule produces RMSE 0.502092 rpm and ringing 1.35649 lbm/s on the same case. It barely improves tracking relative to fixed 3/60 and slightly increases ringing. The realized uncertainty remains close enough to the reference scale that this experiment mainly tests a modest gain perturbation, not prolonged operation near the factor-two limit. Figure 8.7 distinguishes the mathematical schedule from the gain trajectory actually visited by the test.

Across all ten scenarios, the original inverse PI is accepted in 9/10, fixed FF retuning in 7/10, SD-down in 8/10, and SD-up in 7/10. In the 9500 rpm / 20% step case, SD-down restores validity relative to fixed 3/60. Compared with the original inverse PI, ringing falls from 61.8435 to 13.3352 lbm/s and minimum surge margin rises from 0.673% to 4.958%, while RMSE worsens from 3.09509 to 3.75782 rpm. Both 9500 rpm / 30% scheduled scenarios remain invalid. Complete early tables are retained in Appendix B.

8.4.5 Why neither uncertainty rule is a stability argument

Gain scheduling is a controller-design problem involving the dynamics and the scheduling variable, not just a smooth scalar function [7]. Here the GP SD describes confidence conditional on the nominal regression model. An unknown load can cause a systematic inverse mismatch while the query remains close to training inputs; derivative errors or sensor delay can also matter without a large response SD. Conversely, a query with larger SD does not by itself reveal whether more or less feedback gain is stabilizing.

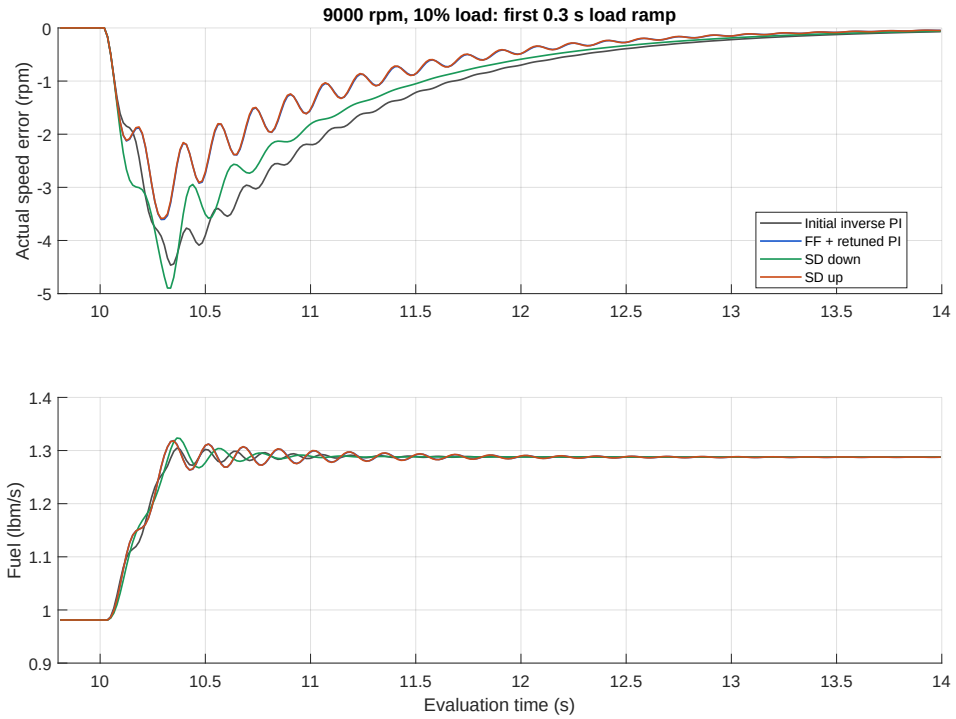


Figure 8.6: First load ramp at 9000 rpm / 10% load. The common fuel plateau does not reveal transient reversal; the enlarged application window does. Curves are regenerated from archived trajectories.

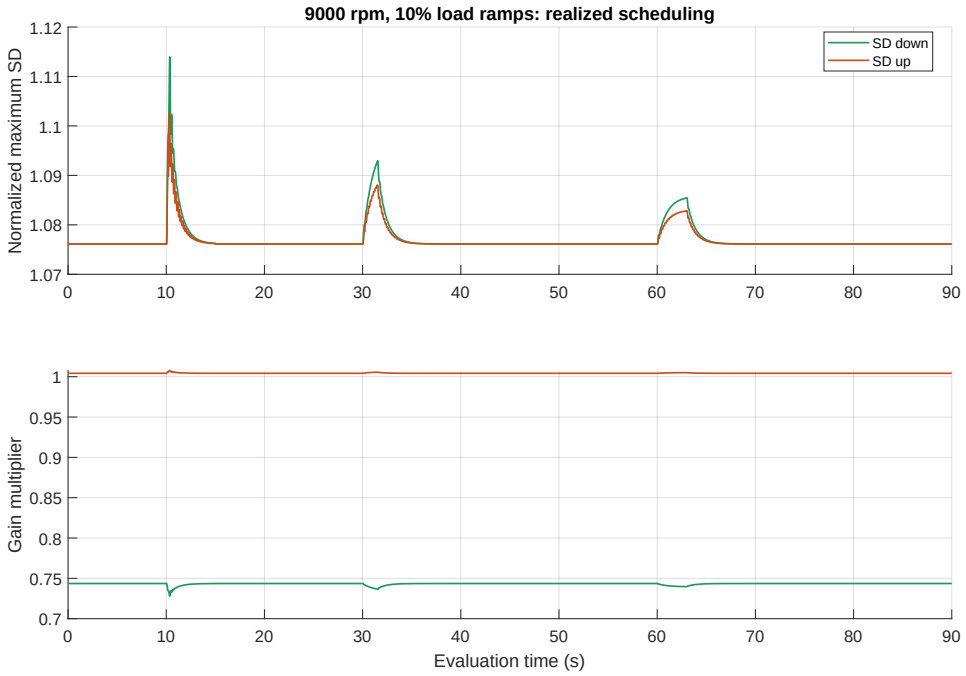


Figure 8.7: Realized response uncertainty and gain multiplier on the same case. Different gains change the trajectory and therefore the feedback-query SD; the two schedules do not necessarily see identical uncertainty histories.

The decreasing schedule is therefore best interpreted as a tested gain-attenuation heuristic. Its success in reducing reversal is compatible with ordinary gain reduction; it does not prove that SD is the best scheduling signal. The increasing schedule is a useful counterexample to the assumption that more confidence or less confidence uniquely determines a gain direction. A more rigorous future design would separate local plant gain, inverse-slope uncertainty, measurement noise, and out-of-domain detection, then assess scheduled-loop stability and performance explicitly. Those steps were not implemented in this campaign.

8.5 Fixed-gain low-ringing design and four-way comparison

8.5.1 Reframing the objective

After observing ringing and mixed scheduling results, the design returns to fixed parameters. At 9500 rpm / 5% load steps, twelve pairs are tested with direct inverse FF. The initial 3/60 setting has RMSE 0.711646 rpm but ringing 25.7817 lbm/s. The selection rule first requires ringing no greater than 10% of that reference value, then chooses the lowest RMSE among eligible accepted candidates. It selects $K_p = 2$, $K_i = 30$.

This is a constrained performance choice, not a universal gain optimum. The selected RMSE is 0.930388 rpm, about 30.7% higher than 3/60 on the tuning case, but ringing is 2.48776 lbm/s, a 90.35% reduction. Still lower gains can nearly eliminate reversal while degrading tracking more severely. For example, 0.5/5 yields ringing 0.00351 lbm/s but RMSE 4.7812 rpm. A small ringing value alone is not a satisfactory fuel-control objective.

K_p	K_i	RMSE	Ringing	Fine settling
3	60	0.7116	25.78	3.525
3	40	0.7208	6.649	2.94
2.5	40	0.7358	5.076	2.925
2	40	0.7651	4.844	2.91
2	30	0.9304	2.488	3.165
1.5	30	0.9669	2.495	3.15
1.5	20	1.338	0.9591	3.525
1	20	1.392	1.109	3.495
1	10	2.518	0.02655	4.17
0.5	10	2.644	0.1944	4.11
0.5	5	4.781	0.003511	4.905
0.25	5	4.957	0.01043	4.83

Table 8.3: Fixed-gain search at 9500 rpm / 5% load steps. RMSE in rpm, ringing in lbm/s, fine settling in seconds (D03).

8.5.2 Matched four-controller experiment

The subsequent experiment reruns base PI, base PI + the early GPR correction, inverse PI, and inverse PI + direct GPR FF over eleven cases. The inverse cases share gains 2/30, and the baseline cases retain 0.025/0.05. The eleventh case is the new 9500 rpm / 5% step tuning condition. The other cases cover the inherited 9000/9500 rpm ramp and large-step scenarios.

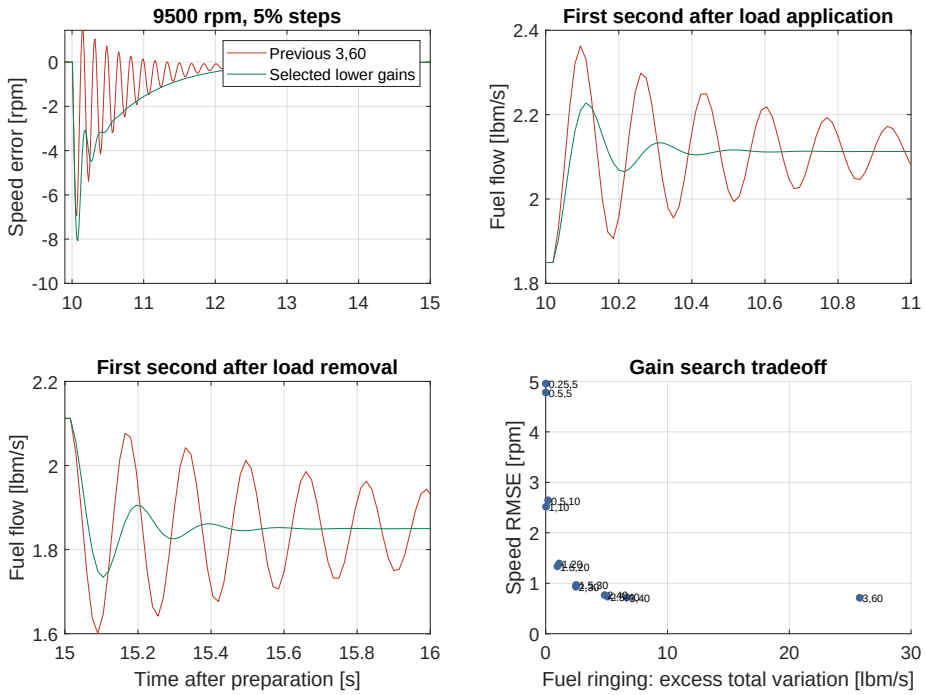


Figure 8.8: Archived 9500 rpm / 5% fixed-gain tradeoff. The selected pair balances the declared ringing constraint and RMSE; it is not the minimum-RMSE point.

Method	RMSE	Peak	Ringing	Fine settling
Base PI	1.484	12.87	1.195	2.265
Base PI + FF	1.548	13.35	1.089	2.1
Inverse PI	0.9304	8.076	2.488	3.165
Inverse PI + FF	0.9304	8.076	2.488	3.165

Table 8.4: 9500 rpm / 5% steps at the final fixed fuel-domain gains 2/30. Units: rpm, rpm, lbm/s, s (D04).

The equality between inverse PI and inverse PI + FF at the same gains is an essential control experiment. It isolates gain changes from direct constant feedforward. By contrast, the base-PI correction depends on sensed speed, so it is not an identical algebraic offset and can slightly change the disturbance response even when the request is constant.

All eleven-case RMSE, ringing, and fine-settling tables appear in Appendix B.1. Invalid full trajectories remain marked as unavailable. The four-way comparison is more informative than a two-way comparison between the original PI and a simultaneously changed model, gain set, and FF law: it identifies which changes are distinguishable under the test excitation.

8.5.3 What this stage establishes

The fixed-gain study shows that ringing can be controlled by a deliberate objective and conservative gains without depending on posterior uncertainty. It also exposes a limitation of the original fuel-coordinate design: improvements in reversal can come at the cost of a slow tail, and the same numerical gains need not imply the same local speed-domain gains when inverse slopes change. This motivates expanding the inverse to ambient variables while keeping the selected fuel-domain gains fixed, followed by a closer examination of the resulting dynamics.

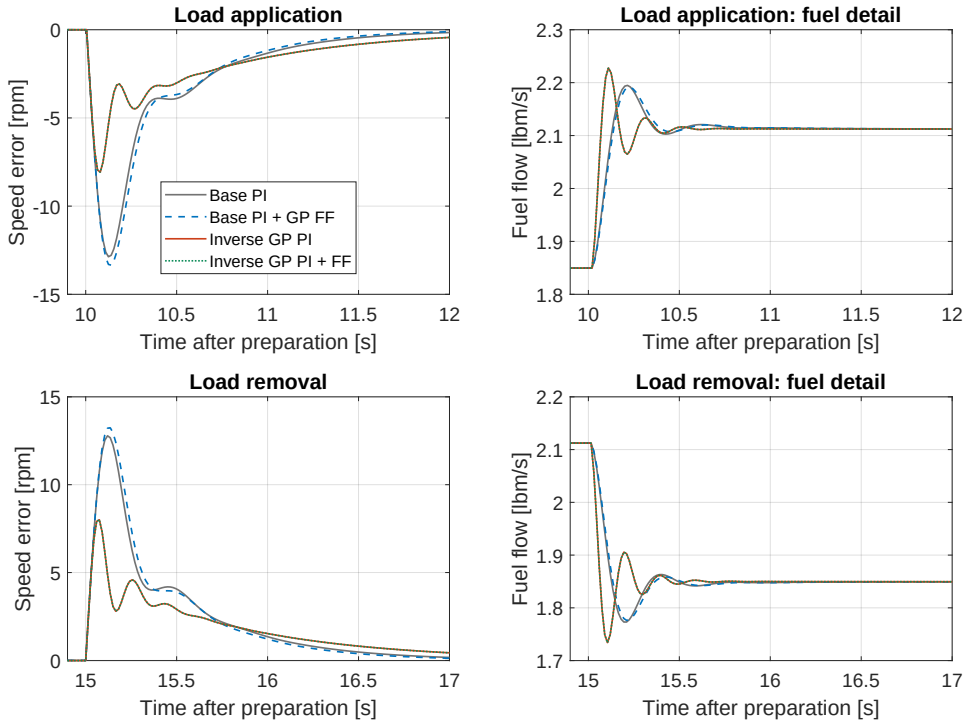


Figure 8.9: 9500 rpm / 5% stepped load in the four-way experiment. The two inverse-controller traces coincide to numerical precision because the direct FF is constant and the integral offset compensates it. Archived legend names are retained.

Chapter 9

Environmental transfer, matched PID, and coverage

The most informative comparator is not always the inherited PI. A matched filtered PID asks whether the GP representation contributes beyond equivalent local feedback action. Coverage diagnostics ask a separate question: whether a query lies within a stated training envelope. Neither substitutes for the other.

9.1 Four-input inverse GPR across ambient conditions

9.1.1 Why inlet temperature and pressure enter the model

Fuel required for a given shaft motion varies with the thermodynamic inlet condition. A two-input inverse trained at one ambient condition cannot represent that dependence explicitly. The expanded input is

$$x = [N,\ a,\ T,\ P]^{\mathsf{T}}, \qquad g : \mathbb{R}^4 \to \mathbb{R}. \tag{9.1}$$

The training grid combines altitudes $\{0, 2500, 5000, 7500, 10000\}$ m with ISA departures $\{0, -10, +10, +20, -20, +30, -30\}$ degrees Celsius, giving 35 conditions. Mach remains zero. Actual simulated inlet T and P are used, rather than substituting altitude or ISA labels directly into the GP. The T-MATS interfaces convert altitude to feet and temperature departure to degrees Fahrenheit; the regression inputs are converted to K and kPa.

Pressure at the inverse input is not simply standard sea-level ambient pressure. For example, the saved sea-level ISA-0 inlet pressure is about 99.299 kPa, reflecting the actual simulated inlet signal. Using 101.325 kPa by assumption would query a different point and would not reproduce the archived controller.

9.1.2 Data generation, split, and exact fit

Each environment contributes 40 space-filling training samples selected from valid unloaded trajectories between 45 and 90 s, for 1400 total. Selection spreads points in standardized speed/acceleration coordinates within each environment. The model uses exact fitting and exact prediction with an ARD squared-exponential kernel, no separate basis function, and explicit input/output standardization. The posterior is frozen before evaluating test labels.

The 200 held-out samples come from later trajectory portions at or after 105 s, with no overlap in environment/source-row identities. They are *200 total test observations*, not 200 training observations or 200 per environment. The archived test set covers 33 of the 35 environments; the 10000 m / ISA 0 and 10000 m / ISA+10 conditions have no selected held-out rows. Thus the reported test aggregate must not be described as uniform validation of every ambient grid point. Time-disjoint samples from the same simulator are also weaker evidence of generalization than independent hardware or altered-plant experiments.

Inverse model	Train/test	Test RMSE	MAE	R^2	95% coverage
2-input exact GP	600/100	0.00104822	0.000832021	0.999997	0.97
4-input exact GP	1400/200	0.00161248	0.000906799	0.999992	0.935
4-input Poly2	1400/200	0.0622207	0.0475266	0.987788	N/A

Table 9.1: Held-out inverse fuel prediction. RMSE/MAE in lbm/s; coverage is a fraction. The two GP rows use different datasets; only the 4-input GP and Poly2 share the same split (D01, D05, D13).

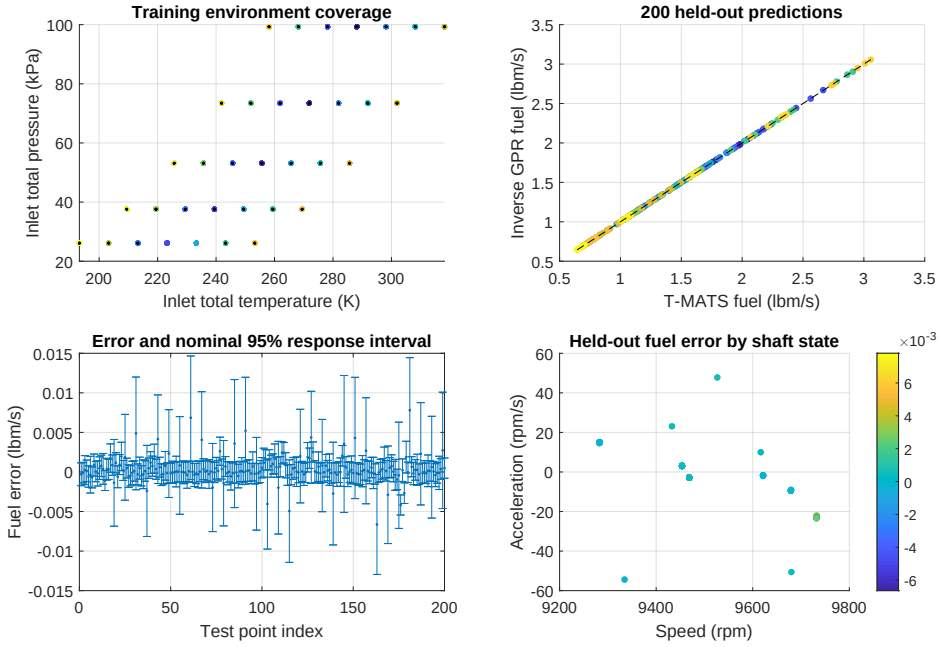


Figure 9.1: Four-input exact-GP held-out fuel prediction and uncertainty from the archived environmental study. The aggregate contains 200 points across 33 represented environments.

The four-input GP achieves test RMSE 0.00161248 lbm/s, maximum error 0.00780312 lbm/s, and 93.5% empirical coverage of the nominal 95% response interval (D05). The exported mean and SD agree with the fitted MATLAB predictor to approximately 10^{-11} and 10^{-12} lbm/s. These checks establish numerical consistency and nominal prediction performance, not accurate derivatives everywhere or robustness under load.

9.1.3 Transfer target and unexpected response

The requested new target is 4000 m / ISA+5 / 9500 rpm with 10% shaft-power steps. The four-input inverse PI retains gains 2/30 and direct static FF, with no uncertainty scheduling. Inlet conditions approach the target during preparation; the controller takes over at 45 s, and scoring begins at 60 s.

At this target, base PI gives RMSE 2.53147 rpm and ringing 2.03043 lbm/s. The previous inverse PI + FF gives RMSE 1.81870 rpm but ringing 3.81687 lbm/s. RMSE improves by approximately 28.2%, while ringing increases by approximately 88.0%. The inverse controller also fails to reach the persistent 0.1 rpm window in every event interval, despite remaining within the broad 1% window. The expectation of simultaneous improvement is therefore only partly met.

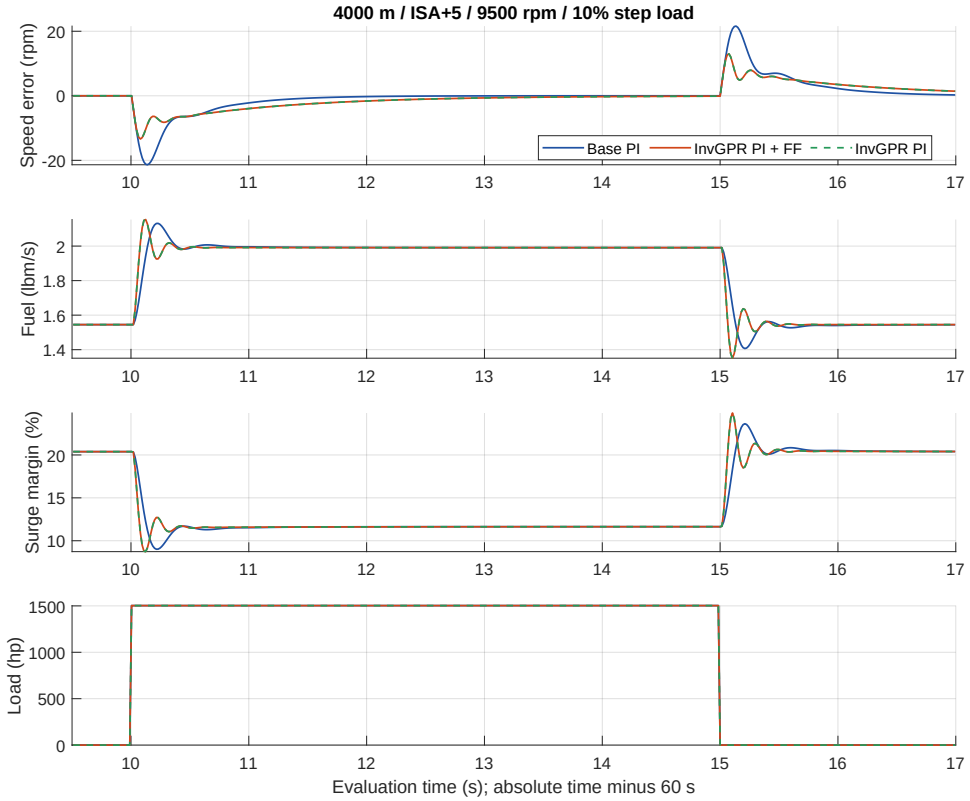


Figure 9.2: First target load event with base PI and the four-input inverse PI. The enlarged window separates peak suppression from subsequent fuel oscillation and fine recovery.

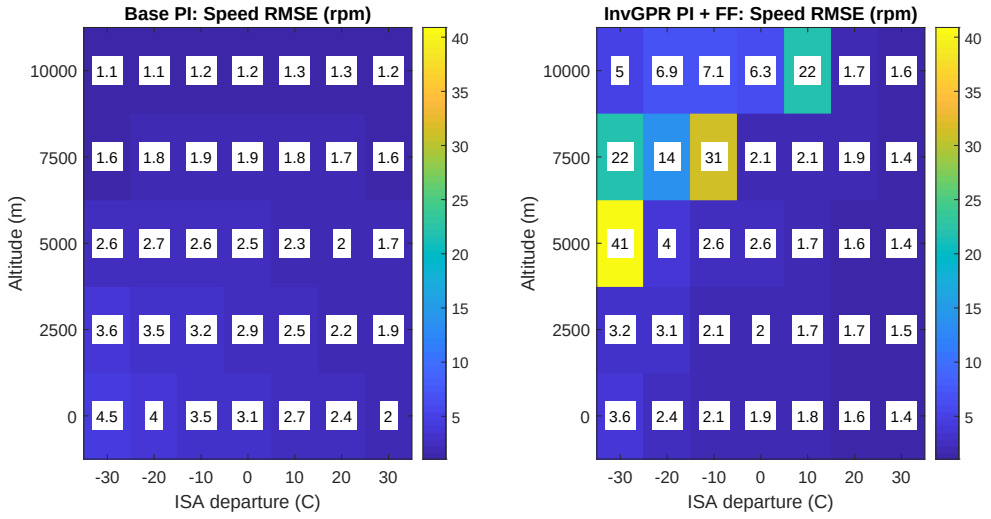


Figure 9.3: Archived environmental RMSE comparison. Numerical scores must be interpreted together with the map-range and inverse-slope diagnostics, not as an unrestricted physical envelope.

9.1.4 Diagnosis: model derivatives and integral coupling

The target inverse derivatives are approximately $g_N = 0.001458$ and $g_a = 0.001579$ in their physical units, giving $g_a/g_N \simeq 1.083$ s. With the 2/30 fuel-domain gains, Eq. (8.23) produces a substantial acceleration-dependent proportional contribution. In several colder/high-altitude locations the static inverse speed slope becomes negative, although the aggregate fuel-prediction score remains excellent.

Condition	g_N	g_a	$K_{p,eq}$	$K_{i,eq}$	$K_{d,eq}$
0 m / ISA 0	0.0020313	0.0015339	0.05008	0.060939	0.0030678
4000 m / ISA+5	0.0014576	0.0015786	0.050274	0.043727	0.0031573
10000 m / ISA-30	-2.2839e-05	0.0013411	0.040187	-0.00068518	0.0026822

Table 9.2: Nominal inverse slopes and ideal local equivalent gains for the 2/30 fuel-domain PI. Negative g_N reverses integral feedback sign (D08).

These observations support three diagnoses. First, fitting output values does not impose physically useful monotonicity in speed. Second, integrating the full dynamic inverse error couples acceleration into the integral path and can sustain a slow tail. Third, the inherited sensor/filter dynamics and coupled effective gains can produce fuel reversals even when peak speed error improves. The diagnoses are supported by derivative and trajectory audits; their relative causal contributions are not separated by a complete factorial experiment.

9.2 Three remedies and the emergence of R3

9.2.1 Shared decomposition

For the same frozen four-input inverse, define

$$v_r = g(r, 0, T, P), \qquad v_0 = g(N_s, 0, T, P), \qquad v_a = g(N_s, \hat{a}, T, P), \tag{9.2}$$

$$e_0 = v_r - v_0, \qquad \Delta v_a = v_a - v_0, \qquad e_v = e_0 - \Delta v_a. \tag{9.3}$$

All three remedies preserve the plant, static FF, acceleration filter, limits, and conditional anti-windup. They differ in the error integrated and the acceleration feedback gain. The comparison is useful because it changes interpretable controller mechanisms rather than simply fitting a more complex inverse.

9.2.2 R1: retune the original virtual-fuel PI

R1 retains e_v in both proportional and integral paths:

$$u^\star = v_r + K_p e_v + I, \qquad I^+ = I + \gamma T_s K_i e_v. \tag{9.4}$$

The selected gains are 1.5/22. This lowers aggressiveness without resolving inverse-slope sign or acceleration/integral coupling. It is the simplest remedy and serves as an important control against attributing every benefit to a new architecture.

9.2.3 R2: separate static and acceleration feedback

R2 integrates only e_0 and attenuates acceleration fuel independently:

$$u^\star = v_r + K_p(e_0 - \beta \Delta v_a) + I, \qquad I^+ = I + \gamma T_s K_i e_0. \tag{9.5}$$

The selected values are $K_p = 16$, $K_i = 40$, and $\beta = 0.05$. Its ideal local controller has terms $K_p c_N$, $K_p \beta c_a$, and $K_i c_N$ without the $K_i c_a$ contribution in Eq. (8.23). It therefore decouples one source of implicit proportional action. However, the static integral path still inherits the sign and magnitude of c_N; changing that architecture alone does not make a negative static slope benign.

9.2.4 R3: normalize by inverse derivatives

R3 uses local setpoint derivatives

$$g_N = \left.\frac{\partial g}{\partial N}\right|_{(r,0,T,P)}, \qquad g_a = \left.\frac{\partial g}{\partial a}\right|_{(r,0,T,P)} \tag{9.6}$$

to form

$$e_N^g = \frac{e_0}{g_N}, \qquad a^g = \frac{\Delta v_a}{g_a}, \qquad u^\star = v_r + K_p e_N^g - K_d a^g + I, \quad I^+ = I + \gamma T_s K_i e_N^g. \tag{9.7}$$

The selected values are $K_p = 0.025$, $K_i = 0.1$, and $K_d = 0.002$. These gains act on approximate speed and acceleration coordinates, so they have conventional speed-to-fuel units rather than the fuel-domain units of R1/R2. Normalization makes local gains less sensitive to inverse slope magnitude, and the separate derivative coefficient prevents acceleration action from being determined implicitly by the virtual proportional gain.

The GP derivatives are evaluated analytically. For standardized inputs, a mean derivative has the form

$$\frac{\partial g}{\partial x_j} = \frac{s_w}{s_{x,j}\ell_j^2} \sum_i \alpha_i k(z_i, z)(z_{ij} - z_j). \tag{9.8}$$

Input-scale factors are essential: derivatives with respect to standardized speed cannot be used as though they were derivatives per rpm.

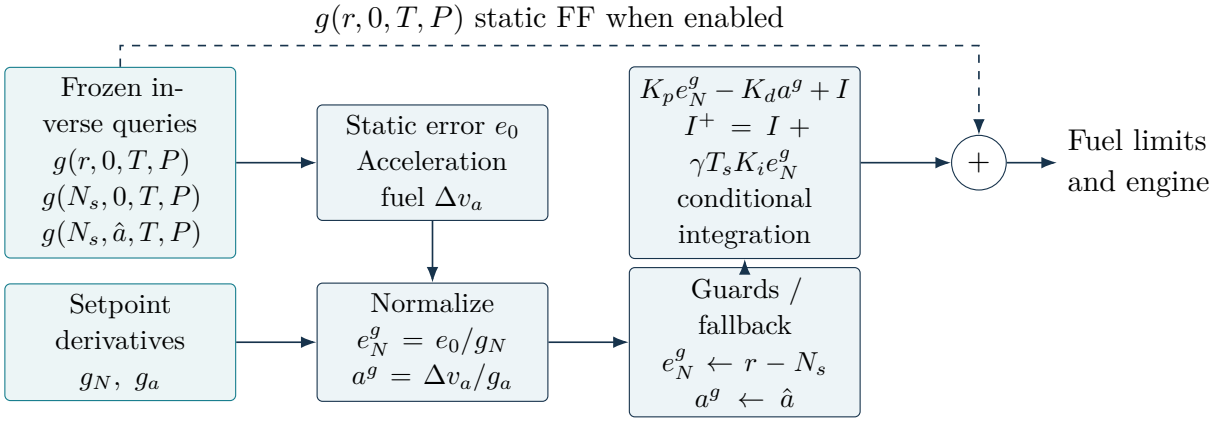


Figure 9.4: R3 with static FF. Removing only the dashed direct FF branch gives the later feedback-only ablation. Replacing GP means and derivatives by Poly2 leaves the architecture intact.

9.2.5 Guards, fallback, and local interpretation

Fallback substitutes $e_N^g = r - N_s$ and $a^g = \hat{a}$ if either derivative is at or below 10^{-5} in its own physical units, if the controller coordinate $N_{c,\text{guard}}$ leaves $[0.5, 1.05]$, or if

$$\frac{e_N^g}{r - N_s} \notin [0.25, 4] \quad \text{when} \quad |r - N_s| > 0.01 \text{ rpm}. \tag{9.9}$$

The derivative thresholds are implementation guards, not dimensionless universal constants. Fallback retains static inverse FF; it replaces the feedback coordinates only. A map-range trigger does not validate extrapolation of the plant map, and a fallback event is not evidence that the inverse model has been corrected.

The decision is recomputed at every sample, with no hysteresis, dwell time, or blending. The current integral and static FF are retained; there is no integral reset to compensate for a change of feedback coordinates. This differs from the proportional-gain compensation used in the SD schedules. At fixed signals and integral, switching from finite normalized coordinates to raw coordinates would change the unsaturated command by

$$\Delta u^\star = K_p[(r - N_s) - e_N^g] - K_d[\hat{a} - a^g]. \tag{9.10}$$

This identity explains a possible transition discontinuity; it is not a calculation performed at a singular denominator, nor evidence that switching caused the measured ringing. Transition smoothness, chattering, and switched-loop robustness require separate validation. The training-box flag is logged only, and joint-hull coverage is audited offline; neither is an online fallback trigger or domain-containment mechanism.

Near a well-behaved operating point, $e_0 \simeq g_N(r - N_s)$ and $\Delta v_a \simeq g_a \hat{a}$, so R3 approaches ordinary filtered speed PID with static FF. For the ideal first-order local plant of Eq. (8.19), its feedback characteristic polynomial is

$$(1 + bK_d)s^2 + (\lambda + bK_p)s + bK_i. \tag{9.11}$$

Positive coefficients are encouraging in this simplified model, but sensor lag, sampled differentiation, nonlinear FF variation, guards, saturation, and model error remain outside that expression. In particular, treating a larger derivative coefficient as universally more damping is not justified for every lagged realization. The nonlinear simulation and matched-controller tests remain the relevant performance evidence.

9.2.6 Selection and target results

Eighty candidate configurations across R1–R3 were evaluated at the 4000 m / ISA+5 target. Within each family, selection minimizes

$$J = \max\left(\frac{\text{RMSE}}{\text{RMSE}_{\text{base}}}, \frac{R_L}{R_{L,\text{base}}}\right) \tag{9.12}$$

over accepted candidates. The selections are frozen before the 35-condition transfer grid. The target is thus a tuning result, not a held-out demonstration.

R1 improves ringing compared with the previous inverse PI, but leaves poor fine recovery. R2 gives the lowest target ringing, 0.686124 lbm/s, with a moderate RMSE improvement over base PI. R3 offers a stronger joint balance: RMSE 1.75198 rpm and ringing 1.45899 lbm/s, both below base PI. All target methods remain within the 1% speed window, so that metric is zero and cannot rank their fine behavior.

Method	RMSE	Peak	Ringing	Fine settling
Base PI	2.531	21.59	2.03	2.46
Previous inverse PI	1.819	13.4	3.817	NR
R1	2.439	15.34	1.798	NR
R2	2.323	16.77	0.6861	1.53
R3 + static FF	1.752	14.39	1.459	1.38
Matched PID + FF	1.749	14.26	1.429	1.38

Table 9.3: 4000 m / ISA+5 / 9500 rpm / 10% steps. Units: rpm, rpm, lbm/s, s. R1–R3 are tuned at this target; the matched PID ablates inverse feedback normalization (D07, D09).

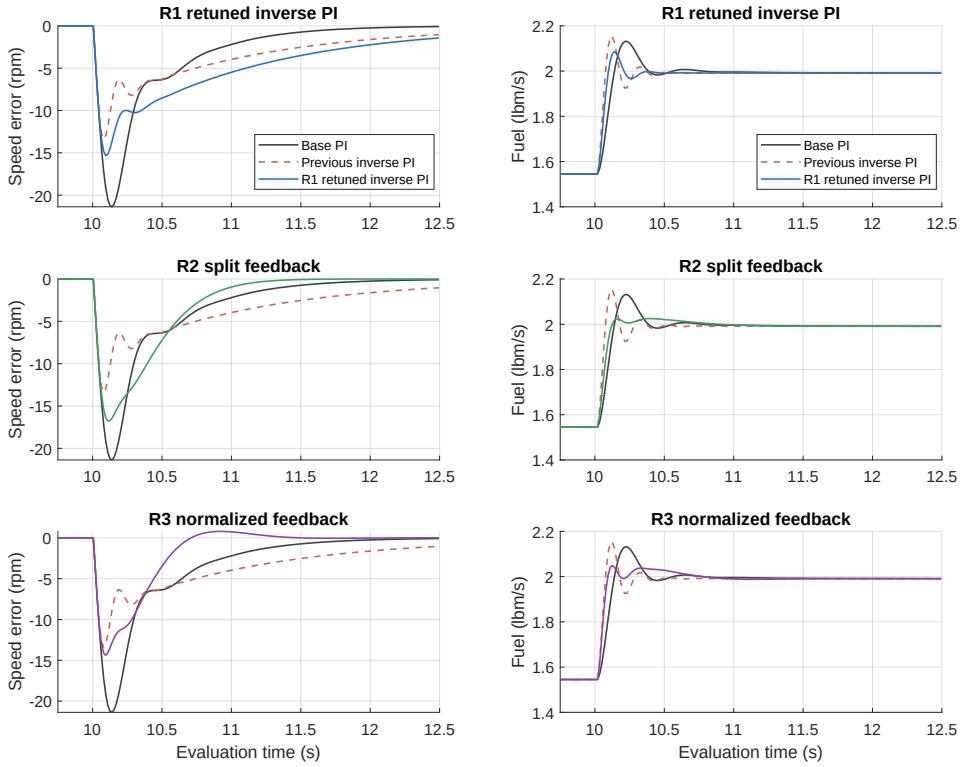


Figure 9.5: Target load-application comparison of base PI, the previous inverse controller, and the three remedies.

9.2.7 Frozen environmental transfer and the matched-PID test

Method	Accepted	Both better	Both / map clean	Fallback cases
Previous inverse	35/35	0/35	0/25	0
R1	35/35	7/35	7/25	0
R2	27/35	9/35	9/25	0
R3 + FF	35/35	35/35	25/25	10

Table 9.4: Frozen-gain environmental transfer. "Both better" means lower RMSE and ringing than the paired base PI. Rejected trajectories do not count as wins. Map-clean denominators use both trajectories (D07).

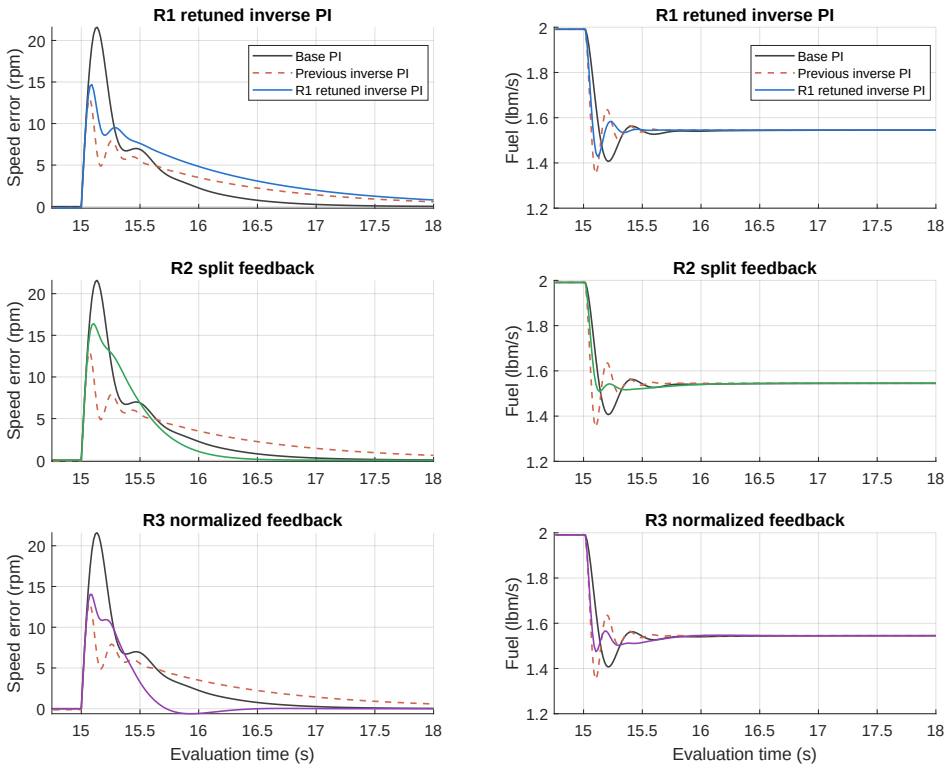


Figure 9.6: Target unloading zoom. Removal transients are included in the ringing and recovery statistics; application-only plots would give an incomplete assessment.

R3 improves both RMSE and ringing in all 35 environmental load cases, including all 25 paired conditions without the monitored map-speed overrun. It invokes fallback in ten conditions. R1 achieves both improvements in seven cases and R2 in nine; eight R2 trajectories are rejected. This is a meaningful transfer result for R3, but it combines normalized inverse feedback and guarded conventional feedback, and includes plant-map caveats in part of the grid.

The forced-fallback diagnostic uses the same 0.025/0.1/0.002 gains with raw speed error and filtered acceleration, retaining static FF. It gives RMSE 1.74912 rpm and ringing 1.42916 lbm/s at the target, slightly better than R3's 1.75198 and 1.45899. The benefit over the inherited base PI therefore cannot be assigned uniquely to GPR. The baseline has different integral/derivative action, and the matched filtered PID largely reproduces the gain. This result strongly supports the architectural diagnosis and limits a GP-superiority claim.

9.3 Moving speed and ambient conditions: coverage becomes decisive

9.3.1 The requested 120-second cycle

The next experiment retains R3 and compares it with base PI under repeated acceleration/deceleration ramps and moving ambient conditions. Four 30 s cycles follow 60 s preparation. Each cycle accelerates for 5 s, holds for 10 s, decelerates for 5 s, and holds for 10 s. The initial broad speed limits are 9025 and 9975 rpm, giving a ramp magnitude of 190 rpm/s. The request governor is raised to 200 rpm/s for this broad test because its inherited 150 rpm/s limit would

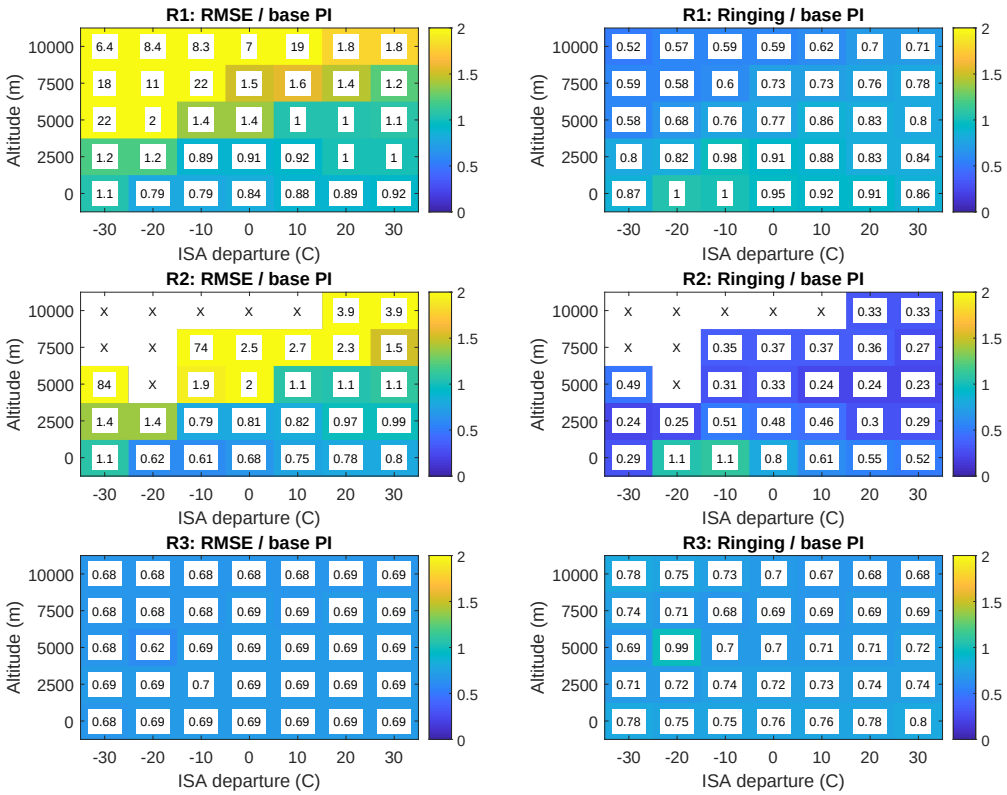


Figure 9.7: Environmental transfer ratios from the archived remedy study. The complete numerical grid and fallback percentages are supplied in Appendix C.

otherwise stretch the prescribed five-second ramps. A preliminary governor-limited diagnostic is excluded from the final comparison.

Let $\phi = t_e \bmod 30$, N_L be the lower speed, and $\Delta N = N_H - N_L$. The governed request is constructed from

$$r(t_e) = \begin{cases} N_L + \Delta N \phi/5, & 0 \le \phi < 5, \\ N_H, & 5 \le \phi < 15, \\ N_H - \Delta N(\phi - 15)/5, & 15 \le \phi < 20, \\ N_L, & 20 \le \phi < 30. \end{cases} \tag{9.13}$$

Altitude and ISA departure are linearly interpolated through the knots $(t_e, h, \Delta T_{\mathrm{ISA}}) = (0, 0, -20)$, $(60, 10000, +20)$, and $(120, 0, 0)$, with time in seconds and altitude in metres. Actual inlet temperature and pressure follow the simulated atmosphere and inlet. There are no added shaft-load pulses or additive fuel disturbances in this cycle. It tests reference and ambient tracking, not the same disturbance-rejection channel as the preceding load campaign.

9.3.2 Broad-cycle result and its evidential limitations

Base PI gives full RMSE 3.66889 rpm; R3 gives 6.13779 rpm, a 67.3% increase. The cycle-segment reversal metric decreases from 6.75506 to 4.24234 lbm/s, a 37.2% reduction. Both runs pass numerical acceptance and remain in the 1% hold-speed window, but R3 uses fallback for 78.69% of evaluation samples. Its combined inverse query set is outside the training input range

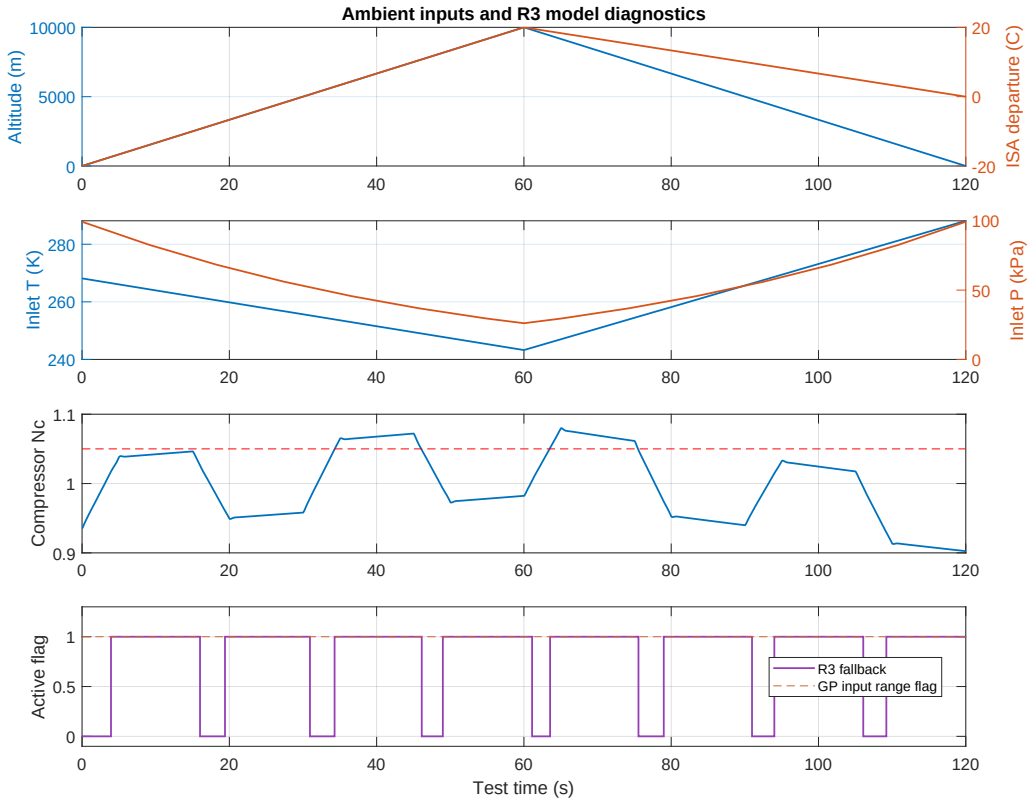


Figure 9.8: Broad-cycle request and environmental route. The original 120 s route is retained in the later covered tests; only the speed excursion is narrowed.

Metric	Base PI	R3 + static FF
Full RMSE (rpm)	3.66889	6.13779
Ramp RMSE (rpm)	5.81171	9.82203
Hold RMSE (rpm)	1.80799	2.85889
Segment reversal (lbm/s)	6.75506	4.24234
Hold reversal (lbm/s)	4.4652	2.5795
Peak error (rpm)	18.266	23.8176

Table 9.5: Broad 9025–9975 rpm cycle (D10).

throughout evaluation, and about 19.8% of each trajectory exceeds the monitored compressor-map corrected-speed range.

These facts materially change the interpretation. The broad cycle is not a test of continuously in-domain GP-normalized control. It is largely a test of the guarded hybrid architecture under extrapolation, with further plant-map concerns. It cannot establish that a well-covered GP controller intrinsically tracks worse than PI, although it does establish that this implemented broad-cycle response has worse RMSE.

9.3.3 Constraining the cycle to GP coverage

The first narrowed attempt uses 9400–9600 rpm. Its input minima/maxima pass, but brief joint-hull excursions remain. It is therefore superseded by 9450–9550 rpm, with the same five-second ramps

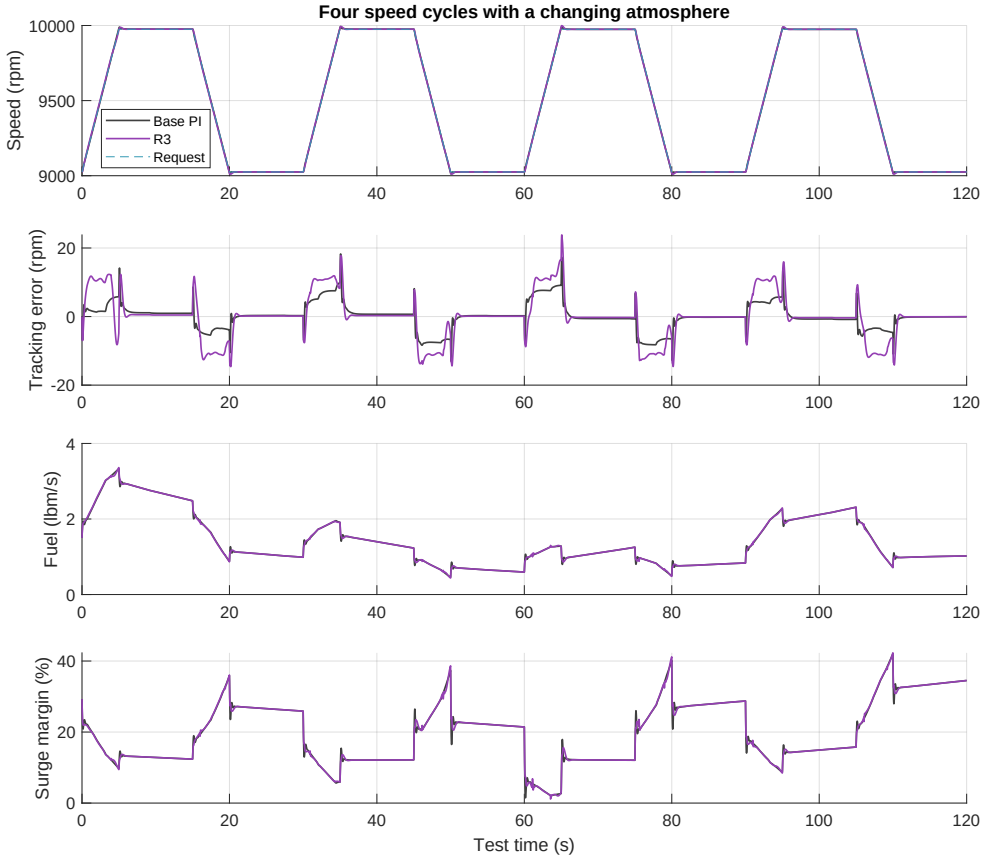


Figure 9.9: Broad-cycle time-domain results. In the archived legend, "R3" includes static setpoint FF. The reported RMSE penalty must be read alongside 78.69% fallback and the domain flags.

and ten-second holds. The final ramp magnitude is 20 rpm/s, well below the restored 150 rpm/s governor limit. The ambient route is unchanged.

The coverage audit evaluates all three R3 query branches at every one of the 8001 evaluation samples, giving 24003 queries per method. Base-PI queries are reconstructed offline solely to assess the same domain; they are not used in its controller. The audit separately checks:

1. strict componentwise training bounds, $x_j \in [\min_i x_{ij}, \max_i x_{ij}]$;
2. membership in the joint convex hull of standardized four-input training points, using verified half-spaces and a 10^{-7} numerical tolerance;
3. nearest-neighbor distance in GP kernel-scaled coordinates as a local-density diagnostic.

All 24003 queries per method pass the final strict box and hull checks. No fallback or monitored map-speed overrun occurs. However, the maximum nearest-neighbor distance is approximately 0.4446, compared with the training leave-one-out 95th-percentile distance 0.1851; about 59.46% of queries exceed that training reference (D11). Hull membership therefore does not mean uniformly dense local support or negligible model uncertainty.

9.3.4 Why the covered result still does not beat base PI in full RMSE

For the final covered case, base PI / R3-with-FF RMSE is 0.591513 / 0.702711 rpm. R3 improves hold RMSE from 0.496482 to 0.337547 rpm and decreases segment fuel reversal from 0.723611 to 0.387541 lbm/s, but ramp RMSE increases from 0.745638 to 1.11847 rpm. The overall penalty is thus localized to moving-reference portions rather than an unexplained uniform loss of regulation quality.

The sensor introduces a useful scale for interpreting this result. Under a sustained ramp of slope v, a first-order sensor approximately lags actual speed by $\tau_s v$. At 20 rpm/s this is about 1 rpm; at 190 rpm/s it is about 9.5 rpm. A controller that tightly aligns sensed speed with the request can consequently exhibit actual-speed error during a ramp. This is a local diagnostic, not proof that sensor lag alone accounts for every difference. The static FF also uses zero acceleration and therefore does not supply an explicit trajectory-acceleration fuel term.

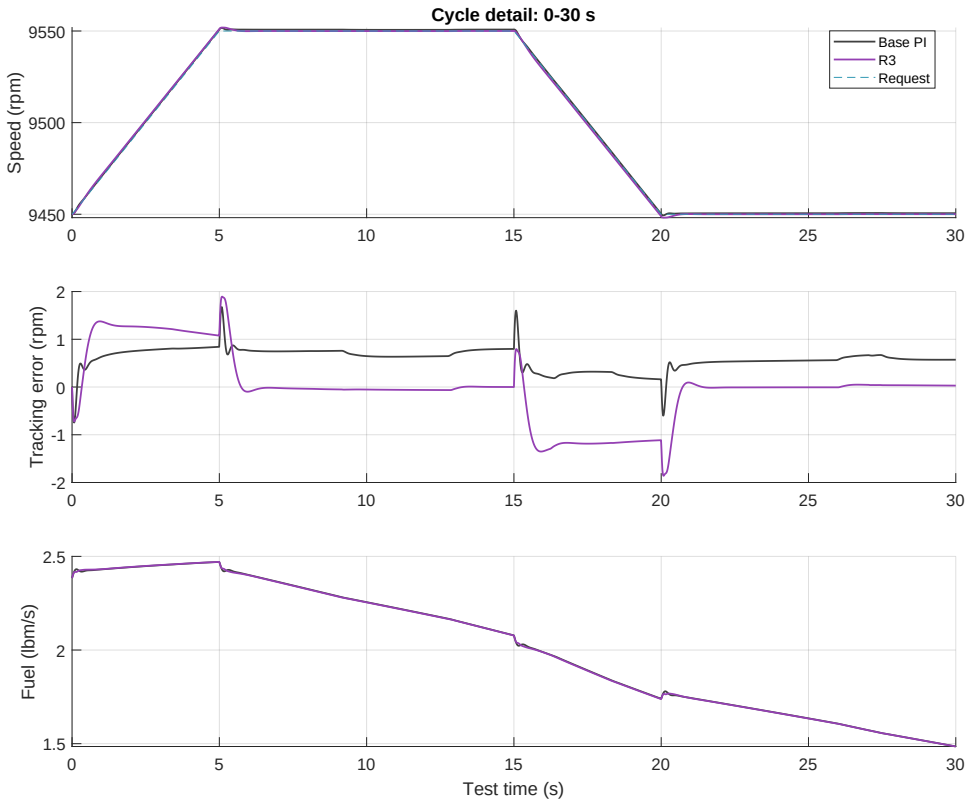


Figure 9.10: Final covered-cycle zoom with base PI and the historical R3-with-FF configuration. Ramp and hold behavior must be evaluated separately.

All eight holds remain within their 1% windows. That broad criterion is useful as a tolerance check, but too loose to distinguish sub-rpm hold regulation and ramp timing. Reporting it alone would conceal the main result.

9.4 Static feedforward clarified by an explicit ablation

9.4.1 Resolving the historical R3 name

The previous remedy and cycle implementations called the complete controller "R3," but their command already included $g(r, 0, T, P)$ directly. Adding that term a second time would double-count nominal fuel. The subsequent comparison therefore explicitly creates two versions:

$$\text{R3-FB:} \quad u^\star = K_p e_N^g - K_d a^g + I, \tag{9.14}$$

$$\text{R3-GP:} \quad u^\star = g(r, 0, T, P) + K_p e_N^g - K_d a^g + I. \tag{9.15}$$

Both still evaluate the inverse setpoint to form e_0. "Feedback only" means the direct FF term is absent from the command sum; it does not mean the inverse setpoint query disappears. Startup integral tracking is adjusted accordingly.

The requested static FF uses the moving speed setpoint with acceleration input fixed at zero. Measured acceleration remains in feedback only. A preliminary desired-acceleration FF attempt was superseded and is excluded from the final comparison and this paper's result tables. This makes the implemented interpretation of "dynamic setpoint input to static FF" precise: the setpoint value changes with time, but the inverse acceleration coordinate for FF is zero.

9.4.2 Same model, same gains, different command decomposition

The covered speed/ambient cycle and the 0.025/0.1/0.002 gains remain fixed. R3-GP exactly reproduces the earlier covered "R3" trajectory, confirming that the new naming has not created a new algorithm inadvertently. R3-FB removes only the direct static FF term. The later Poly2 comparison reruns and preserves these same three GP/base cases, permitting one consolidated covered-cycle table in the next chapter.

R3-FB gives full RMSE 0.639320 rpm, hold RMSE 0.394958 rpm, and reversal 0.378265 lbm/s. R3-GP gives 0.702711, 0.337547, and 0.387541 respectively. Adding static FF therefore improves hold RMSE by approximately 14.5% but worsens full RMSE by 9.9% and increases reversal by 2.5%. It also improves peak error and IAE. There is no contradiction: the feedforward changes the allocation and timing of fuel during moving-reference segments, and squared error emphasizes the larger ramp deviations.

9.4.3 How this differs from the constant-reference equivalence

If v_r varies with time, define $J = I + v_r$. Then $\dot{J} = K_i e + \dot{v}_r$ for the FF controller, whereas a no-FF controller with ordinary integration has only $\dot{J} = K_i e$. A time-varying feedforward term is therefore not generally removable by one initial integral offset. Its derivative contains reference and ambient contributions:

$$\dot{v}_r = g_N(r, 0, T, P)\dot{r} + g_T(r, 0, T, P)\dot{T} + g_P(r, 0, T, P)\dot{P}. \tag{9.16}$$

This is a mathematical explanation of why FF can change the cycle even though it had no independent effect in the earlier fixed-reference test. It is not a recommendation to inject this derivative directly; the implementation remains the static inverse evaluated at the changing request.

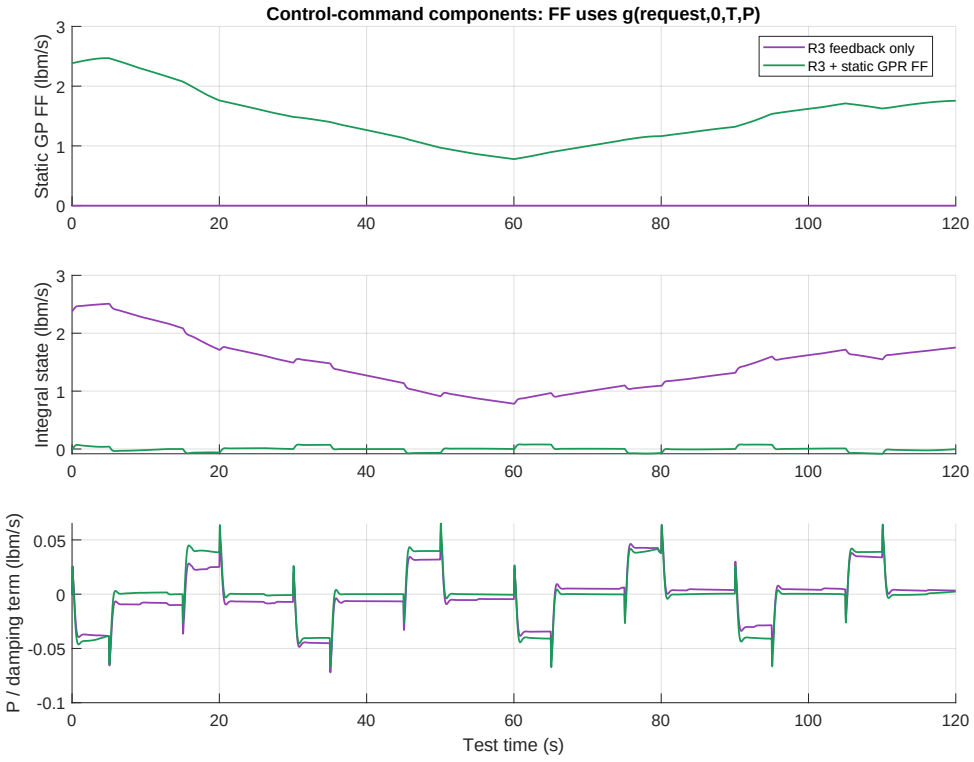


Figure 9.11: Direct static FF and feedback/integral contributions in the covered-cycle ablation. Time-varying reference and ambient conditions prevent a single constant integral offset from reproducing the FF term throughout the cycle.

Chapter 10

Polynomial replacement and local tuning tradeoffs

A simpler inverse is an explicit challenger to the assumption that lower regression error yields better control. First the representation changes with gains retained; only afterwards are gains retuned locally. The distinction makes the initial replacement a more informative model comparison than the final tuned-versus-inherited comparison.

10.1 A full quadratic inverse with unchanged R3 gains

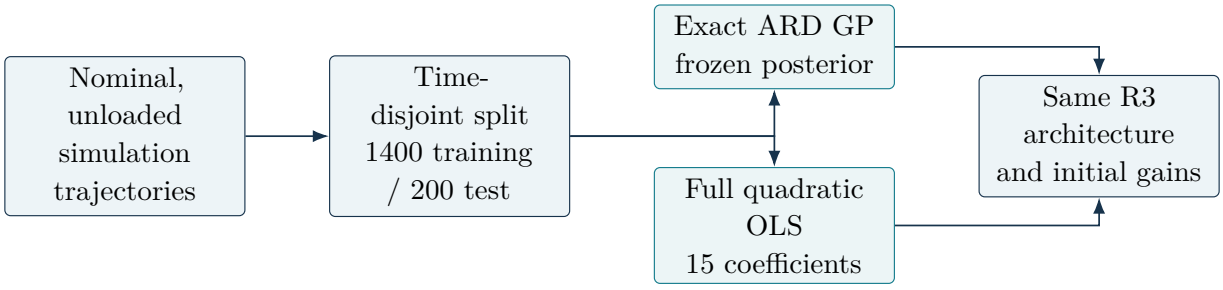
10.1.1 Controlled replacement of the regression model

The polynomial challenger asks whether the same architecture needs the representational complexity of an exact GP. It retains the four physical inputs, the 1400/200 training/test split, the sample/filter settings, guards, anti-windup, FF structure, and initial R3 gains. Only the inverse mean and its derivatives are replaced.

For $z_j = (x_j - \mu_j)/s_j$, the full second-order model is

$$p_2(x) = \beta_0 + \sum_{j=1}^{4} \beta_j z_j + \sum_{j=1}^{4} \beta_{jj} z_j^2 + \sum_{j<\ell} \beta_{j\ell} z_j z_\ell. \tag{10.1}$$

There are 15 terms: one intercept, four linear, four squared, and six pairwise interactions. The coefficients minimize $\|\Phi\beta - \boldsymbol{w}\|_2^2$ by ordinary least squares. This is linear regression in a nonlinear feature basis, with no regularization, coefficient constraints, or polynomial-specific gain tuning at this stage. Polynomial regression is a standard model class [9]; Eq. (10.1) and the saved coefficients define the exact version tested here.



Test labels assess prediction only. Controller tuning uses separate loaded trajectories. Model complexity changes first; controller gains change in the subsequent 9250 rpm study.

Figure 10.1: Matched model comparison. The polynomial uses the same training and held-out rows as the environmental GP; subsequent control trajectories are separate from the inverse-fit test.

The input means are approximately $(9515.1465, -1.43521, 255.68644, 57.91711)$ and scales $(138.04299, 35.10059, 30.44901, 26.08974)$ in rpm, rpm/s, K, and kPa. The training design has rank

15 and condition number 13.58. Full coefficients appear in Appendix D. Unlike the GP export, this fitted polynomial produces fuel directly through the saved physical-output coefficients.

Analytic physical-input derivatives follow from

$$\frac{\partial p_2}{\partial x_j} = \frac{1}{s_j}\left(\beta_j + 2\beta_{jj}z_j + \sum_{\ell \neq j}\beta_{j\ell}z_\ell\right), \tag{10.2}$$

where the cross coefficient is understood symmetrically. The R3 normalization and static FF use these derivatives and values exactly as they used the GP quantities.

10.1.2 Prediction quality versus control quality

The polynomial test RMSE is 0.0622207 lbm/s, versus 0.00161248 for the four-input GP, about 38.6 times larger. Its training RMSE is 0.0598142 lbm/s and test R^2 is 0.987788. A high R^2 therefore coexists with a materially larger absolute fuel prediction error. No probabilistic predictive interval is fitted or used for Poly2, and no GP uncertainty scheduling remains in this controller.

Metric	Base PI	R3 feedback only	R3 + GPR FF	R3 + Poly2 FF
Full RMSE (rpm)	0.591513	0.63932	0.702711	0.676972
Ramp RMSE (rpm)	0.745638	0.955251	1.11847	1.07267
Hold RMSE (rpm)	0.496482	0.394958	0.337547	0.333108
Segment reversal (lbm/s)	0.723611	0.378265	0.387541	0.385818
Hold reversal (lbm/s)	0.403316	0.229514	0.236351	0.234662
Peak error (rpm)	2.17604	2.19926	2.0073	2.02413

Table 10.1: Final covered 9450–9550 rpm cycle: same gains and frozen models. All four trajectories pass box/hull coverage, with no fallback or monitored map-speed overrun (D12, D19).

On the same covered cycle, R3-P2 gives full RMSE 0.676972 rpm and reversal 0.385818 lbm/s. Relative to R3-GP, these are improvements of approximately 3.66% and 0.44%. Hold RMSE and IAE also improve slightly, whereas peak error and maximum fuel slew are slightly higher. Base PI still has the lowest full RMSE, and R3-FB still has the lowest overall reversal among the four. Thus the polynomial is competitive in this matched test without becoming the universal winner.

10.1.3 Why a poorer inverse fit can still control well

Normalization uses relative changes and local slopes. For a smooth static inverse, with $e = r - N_s$,

$$\frac{g(r,0) - g(r-e,0)}{g_N(r,0)} = e - \frac{1}{2}\frac{g_{NN}(r,0)}{g_N(r,0)}e^2 + O(e^3). \tag{10.3}$$

The leading term is the ordinary speed error regardless of the absolute slope magnitude, provided the denominator is well behaved. Therefore a model with a moderate absolute fuel bias can still yield useful normalized feedback on a narrow trajectory. Curvature and derivative ratios, rather than only fuel prediction RMSE, become relevant local error quantities.

An exact invariance makes this point stronger. If $\tilde{g} = cg + d$ with constant $c > 0$ and d, the normalized static and acceleration differences are unchanged wherever the same guard decisions apply. Additive offsets cancel and the multiplicative factor cancels against the derivative. Direct FF is not invariant, but a constant FF offset can be absorbed by the integral at a fixed condition. The

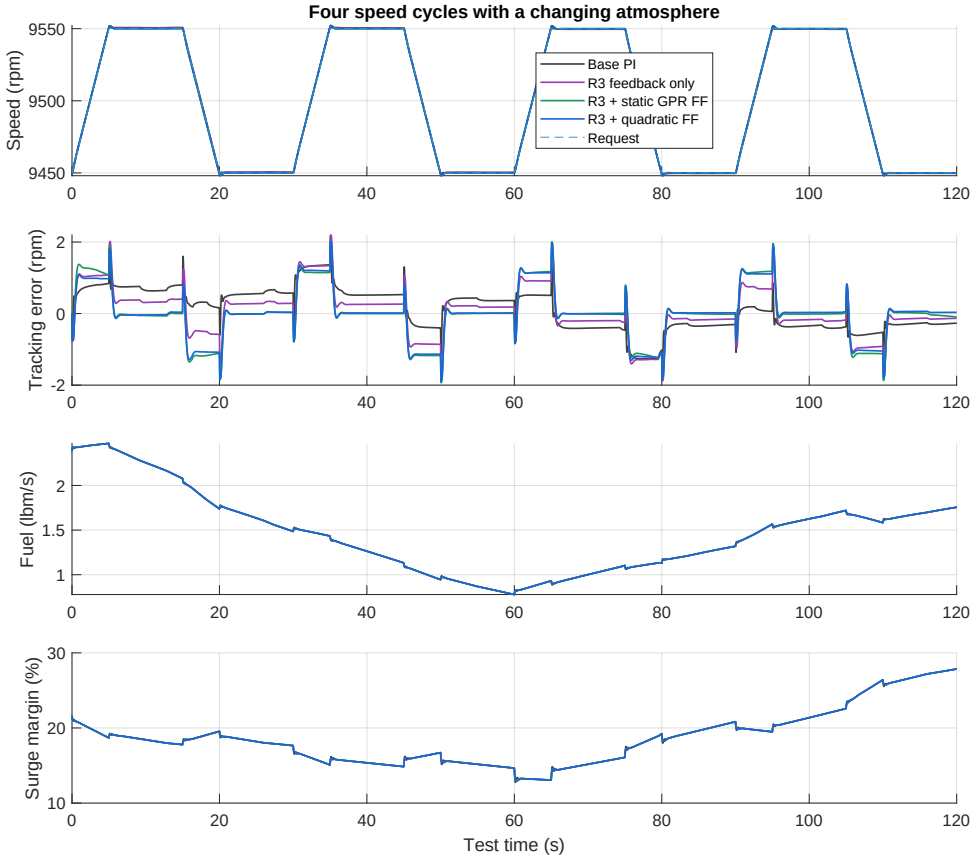


Figure 10.2: Four-way covered-cycle comparison: base PI, R3 feedback only, R3 + static GPR FF, and R3 + static Poly2 FF. All use the same request and ambient route.

real polynomial is not assumed to be an affine transformation of the GP; this argument explains a mechanism by which absolute fit error can be partially irrelevant to normalized feedback.

Other explanations remain plausible. The smoother, low-order surface may have favorable local derivatives on the constrained path, and all controllers operate close to the same nominal trajectory. The available experiments do not separate these possibilities or establish superiority in sparse/extrapolating regions. A regression model should therefore be assessed with derivative checks and controller tests in addition to held-out output prediction.

10.1.4 Computational implication

For n training points and four inputs, a GP mean requires a sum of n kernel contributions; exact variance with a saved dense Cholesky factor adds a triangular solve. Poly2 requires only its fixed 15-term expression and elementary derivatives. This is a structural reduction in arithmetic and model storage. The campaign does not report embedded execution-time, worst-case latency, compiler behavior, or hardware resource measurements, so it does not establish a real-time deployment speedup quantitatively. The implementations remain desktop Simulink S-functions.

10.2 Local Poly2 PID retuning at 9250 rpm

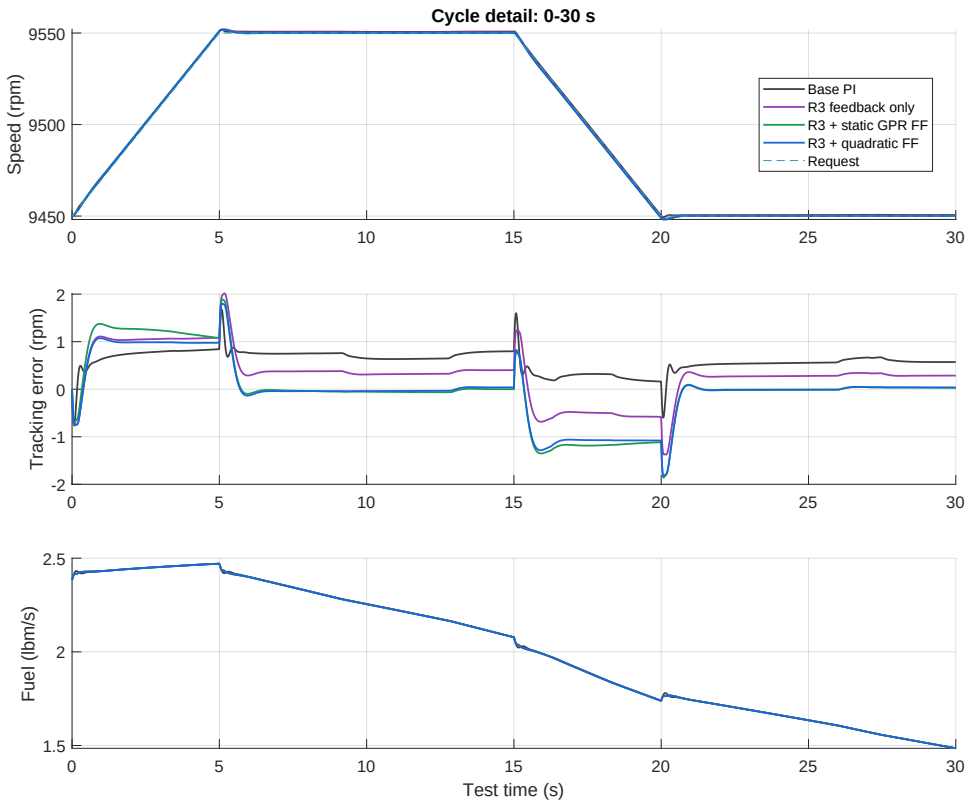


Figure 10.3: Covered-cycle zoom after replacing the inverse model by Poly2. Small differences are meaningful only within this matched experiment; the model-fit RMSE is a separate quantity.

10.2.1 Fixed model and declared objective

The latest task changes the tuning point to 9250 rpm, 0 m, ISA 0. The inverse coefficients, R3 structure, static FF, acceleration filter, limits, and guards remain fixed. The original Poly2 gains are $(0.025, 0.1, 0.002)$. Following the existing fixed-speed protocol, the tuning excitation is a 10% shaft-load pulse sequence; the condition-specific unloaded reference is 15376.79195 hp, giving 1537.67919 hp pulses.

The coarse search includes the original setting and a $3 \times 3 \times 3$ grid:

$$K_p \in \{0.015, 0.025, 0.035\}, \quad K_i \in \{0.08, 0.14, 0.20\}, \quad K_d \in \{0.001, 0.002, 0.003\}. \tag{10.4}$$

After these 28 settings, six coordinate refinements change one gain at a time by $\pm 25\%$ around the best grid point. The objective is

$$J_{9250} = \max\left(\frac{\text{RMSE}}{\text{RMSE}_{\text{original Poly2}}}, \frac{R_L}{R_{L,\text{original Poly2}}}\right). \tag{10.5}$$

The comparator is now the original Poly2 at the same condition, rather than base PI as in the remedy selection. This difference in normalization is intentional and must be stated. The base PI is rerun for context but excluded from candidate selection.

10.2.2 Selected gains and measured tradeoff

All 34 Poly2 candidates and the base-PI comparator pass numerical acceptance. Candidate 34 minimizes the tested joint score at $J = 0.928166$:

$$\boxed{K_p = 0.025, \qquad K_i = 0.14, \qquad K_d = 0.00125.} \tag{10.6}$$

Proportional action is unchanged; integral gain increases by 40%, and the derivative coefficient decreases by 37.5%. The best tested setting is not a global optimum and was not selected under an explicit peak-error or robustness constraint.

Metric	Base PI	Original Poly2	Retuned Poly2
RMSE (rpm)	2.4209	1.6658	1.54614
Peak error (rpm)	21.4062	14.1133	15.8246
IAE (rpm s)	51.1162	28.5348	21.6607
Ringing (lbm/s)	2.07023	1.6968	1.29904
Fine recovery (s)	2.475	1.365	0.93
1% recovery (s)	0	0	0
Edge-window peak slew (lbm/s^2)	4.8175	8.71872	7.27362
Minimum SM (%)	18.2802	19.6313	18.9394

Table 10.2: 9250 rpm / 0 m / ISA 0 / 10% load steps. The retune changes only the gains (D15).

Compared with original Poly2, RMSE decreases from 1.66580 to 1.54614 rpm (7.2%), ringing from 1.69680 to 1.29904 lbm/s (23.4%), and worst fine recovery from 1.365 to 0.930 s (31.9%). Relative to base PI, the retuned controller reduces RMSE by approximately 36.1% and ringing by 37.3%. These are meaningful local improvements under identical load power and evaluation timing.

The cost is a peak-error increase from 14.1133 to 15.8246 rpm (12.1%) relative to the original Poly2 setting. The result should therefore not be described as uniformly better. Reduced acceleration feedback and stronger integral action are consistent with allowing a larger initial excursion while shortening the tail, but the exact transient reflects the complete sampled nonlinear loop. The measured tradeoff, rather than a generic damping intuition, justifies the conclusion.

The fuel-demand tradeoff also depends on the comparator. The selected setting reduces two-second-window peak fuel slew from 8.71872 to 7.27362 lbm/s^2 relative to original Poly2, but its slew remains 51.0% above base PI's 4.81750 lbm/s^2, despite the 37.3% reduction in fuel reversal. Fewer reversals therefore do not imply lower instantaneous actuator-rate demand. Fuel magnitude limits are inactive in these three evaluation traces; this does not test a physical valve rate constraint. Peak speed errors are 0.2314%, 0.1526%, and 0.1711% of 9250 rpm for base PI, original Poly2, and retuned Poly2, respectively. These effect sizes describe simulation behavior, not compliance with a supplied product requirement.

10.2.3 Frozen-gain amplitude transfer

After selecting the gains, the load amplitude is halved to 5% while timing, environment, and request remain fixed. The three comparator cases are simulated again and the result is not fed back into gain selection. Retuned Poly2 improves RMSE from 0.801760 to 0.743853 rpm (7.2%) and ringing from 0.900486 to 0.654400 lbm/s (27.3%) relative to the original gains. Peak error increases from 6.96899 to 7.80644 rpm, again about 12%. The same qualitative tradeoff therefore persists at half

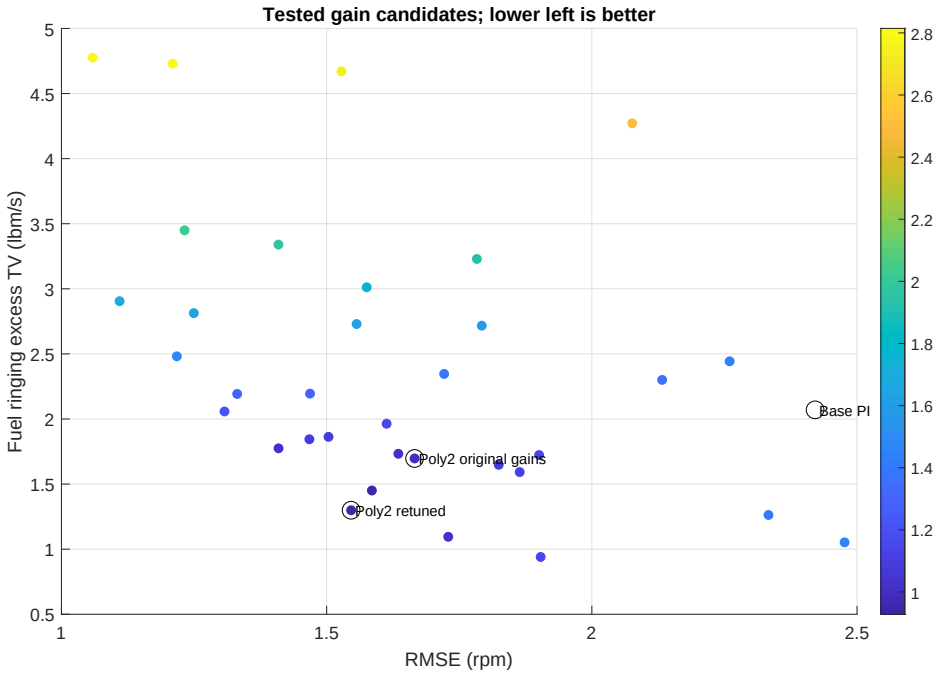


Figure 10.4: All accepted Poly2 gain candidates at 9250 rpm, colored by the joint objective. The best RMSE-only point is not the joint selection.

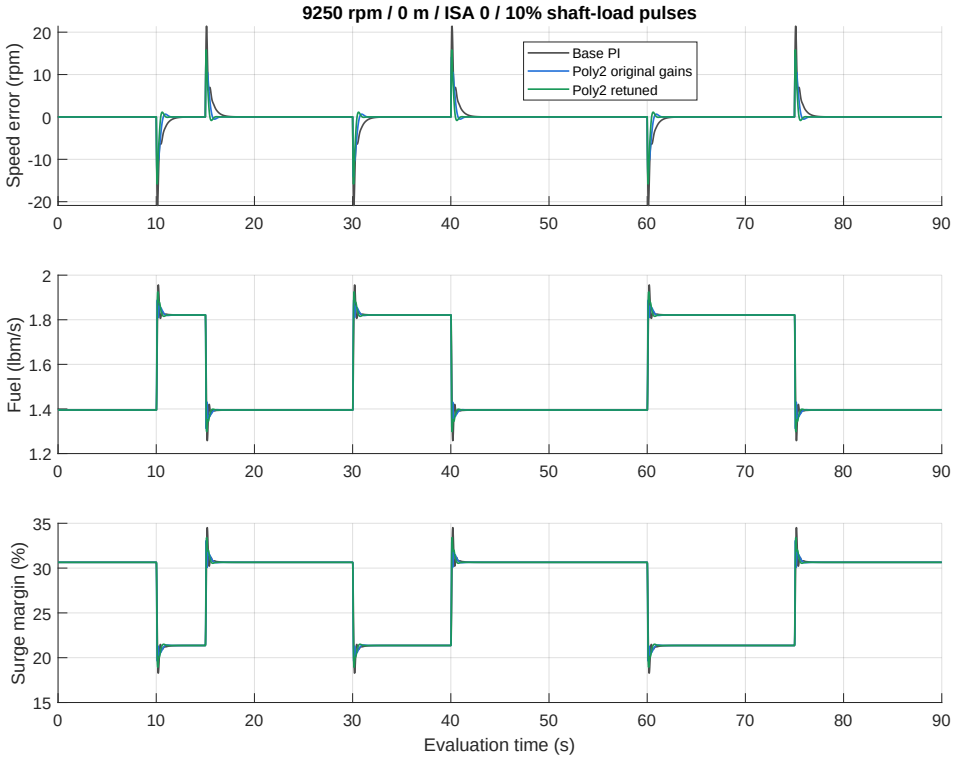


Figure 10.5: 9250 rpm / 10% load-pulse overview: base PI, original Poly2, and selected Poly2 gains.

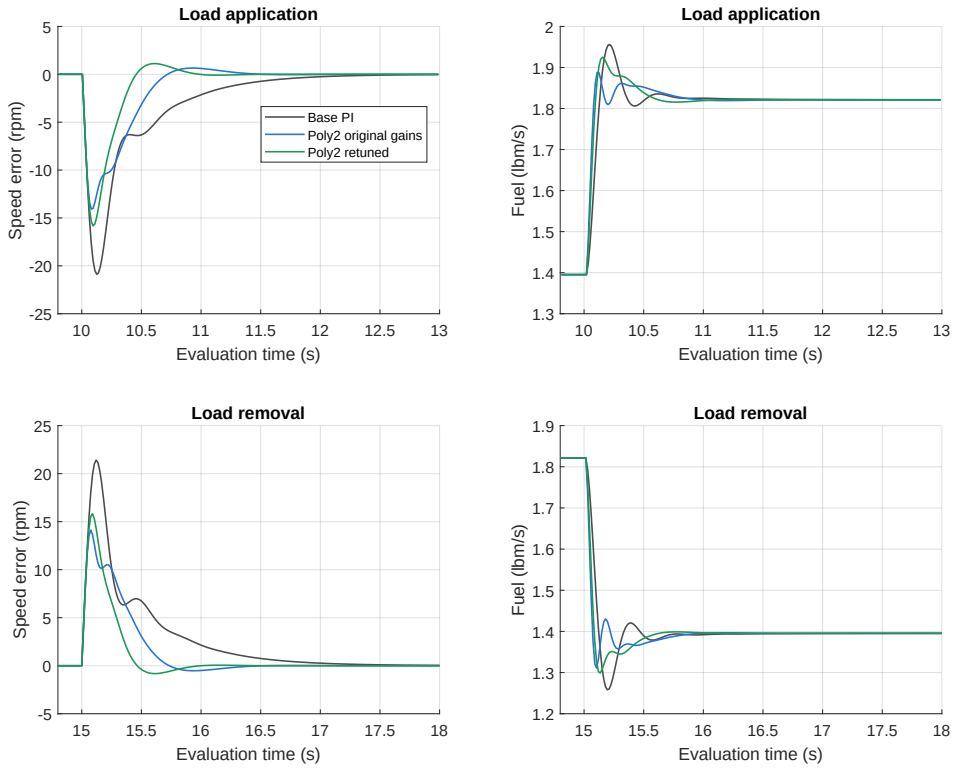


Figure 10.6: First load application and removal at 9250 rpm. The retuned controller has a larger peak excursion than original Poly2 but a shorter fine tail and less excess fuel variation.

Metric	Base PI	Original Poly2	Retuned Poly2
RMSE (rpm)	1.16915	0.80176	0.743853
Peak error (rpm)	10.3922	6.96899	7.80644
IAE (rpm s)	24.6277	13.6364	10.2683
Ringing (lbm/s)	1.05827	0.900486	0.6544
Fine recovery (s)	2.115	1.26	0.87
1% recovery (s)	0	0	0

Table 10.3: Frozen gains evaluated on 5% load steps at the same operating point. This is amplitude transfer, not an independent speed/environment test (D16).

disturbance amplitude. This is useful amplitude-transfer evidence, but it is not an independent altitude, temperature, speed, model-error, or noise test.

10.2.4 Sensitivity to ringing windows and recovery bands

The referee revision reanalyzes the saved base, original-Poly2, and retuned-Poly2 trajectories at both load amplitudes. Models, gains, event timing, and the original objective are unchanged. Only the reporting window or recovery band changes; the results do not enter candidate selection. Retuned Poly2 has less summed excess variation than original Poly2 and base PI for each tested window of 0.5, 1, 2, and 3 s at both amplitudes. Thus the local reversal ranking is not peculiar to the original two-second choice over this tested range. This is a window-sensitivity check on deterministic traces, not evidence about different sampling rates, noisy measurements, or physical

Load	Method	0.5 s	1 s	2 s	3 s
10%	Base PI	1.8745	2.0314	2.0702	2.0743
10%	Original Poly2	1.3387	1.6813	1.6968	1.6979
10%	Retuned Poly2	1.0556	1.2946	1.299	1.2991
5%	Base PI	0.95405	1.0402	1.0583	1.06
5%	Original Poly2	0.74255	0.8939	0.90049	0.90089
5%	Retuned Poly2	0.55481	0.65287	0.6544	0.65441

Table 10.4: Fuel reversal sensitivity at 9250 rpm. Columns change the duration of each of the six post-edge windows; entries are summed excess total variation in lbm/s. The 2 s column reproduces the published metric. These are reanalyses of the same saved trajectories, not new tuning results.

actuator motion. The recovery ordering is less universal. At 10% load and a 0.1 rpm band, retuned

Load	Method	0.1 rpm	0.5 rpm	1 rpm	1% speed
10%	Base PI	2.475	1.71	1.365	0
10%	Original Poly2	1.365	1.08	0.63	0
10%	Retuned Poly2	0.93	0.81	0.675	0
5%	Base PI	2.115	1.35	1.02	0
5%	Original Poly2	1.26	0.63	0.555	0
5%	Retuned Poly2	0.87	0.405	0.375	0

Table 10.5: Worst persistent recovery in seconds at 9250 rpm. The last band is 92.5 rpm. Event intervals and the next-event persistence rule are held fixed while changing only the band.

Poly2 recovers in 0.930 s versus 1.365 s for original Poly2. At 0.5 rpm it remains faster (0.810 versus 1.080 s), but at 1 rpm it is slightly slower (0.675 versus 0.630 s). At 5% load it is faster at all three fine bands tested. Every method remains inside the 1% speed band in both tests, hence the zero entries. The larger initial peak and shorter final tail can therefore reverse a settling-time ranking when the acceptance band changes. Recovery claims must name that band explicitly.

10.2.5 Coverage and implementation qualification

The original four-input training-speed minimum is approximately 9264.23275 rpm and maximum 9750.59551 rpm. The requested 9250 rpm setpoint is 14.23 rpm below that lower limit. Static setpoint queries are consequently outside the old speed interval throughout evaluation, with load deceleration potentially moving measured speed farther below. The task intentionally honors the specified speed rather than silently moving the test into the previous coverage envelope.

The accepted tuning trajectories have zero monitored map-speed overrun, and the selected controller uses no fallback, but neither observation converts the extrapolating inverse query into in-domain validation. This is a local nonlinear simulation retune of a frozen model. The new gains have not yet been rerun on the full covered ambient cycle or the 35-condition load grid; combining their local improvement with earlier broad transfer results would overstate the evidence.

Independent trace replay checks all 35 accepted tuning/comparator trajectories. Recomputed RMSE and ringing agree with the stored summaries; maximum proportional-term and command reconstruction errors are approximately 2.54×10^{-14} and 3.31×10^{-14} lbm/s, and the integral recurrence error is about 1.12×10^{-15} lbm/s (D20). These checks validate arithmetic and logging consistency, not model adequacy. The configured model and earlier defaults remain archived.

Chapter 11

PILCO-style offline PI tuning and expected fuel reversal

This chapter studies a restricted adaptation of probabilistic policy evaluation, not a new PILCO algorithm. A frozen quadratic supplies feedforward, an offline GP predicts dynamics, and two PI gains are optimized numerically. The first experiment and the later ringing penalty are retained together so that the repair of an aggressive policy is not mistaken for a demonstrated advantage over conventional PI. The relationship to original PILCO and CAPS is evaluated in Chapter 2.

11.1 PILCO-style policy learning: full derivation and first results

Archived October study. All simulations, searches, and checks described below predate this report.

Chapter overview

A Gaussian-process (GP) transition model and analytic Gaussian moment propagation are used to optimize a proportional-integral controller around an existing quadratic feedforward map. The benchmark is the previous 9250 rpm, sea-level, ISA-zero test with shaft-load pulses. Three historical controllers and the initial PI policy provide comparators. The selected gains are frozen before new nonlinear T-MATS simulations at 10% and 5% load amplitudes. This implementation is a PILCO-style, offline, single-model policy-improvement experiment: it uses numerical policy gradients, a fitted linear mean, a shared-kernel GP subset, and an expected quadratic cost. It is not an exact reproduction of the original PILCO training loop. All reported plant metrics come from saved nonlinear simulation traces, not GP rollout predictions.

11.1.1 Experiment, provenance, and main result

The chosen test is the most recent fixed-operating-point study in this workspace. The reference is 9250 rpm, altitude is 0 m, ISA deviation is 0 K, and Mach number is zero. Each simulation includes 60 s preparation and 90 s evaluation. The sample period is $h = 0.015$ s. Turbine trim power is 15376.79195 hp; the 10% disturbance is 1537.67919 hp. Relative to the evaluation start, load pulses occupy $[10.005, 15)$, $[30, 40.005)$, and $[60, 75)$ s. Five-percent validation halves only the pulse magnitude.

The existing base PI, original Poly2, and retuned Poly2 traces are read from the previous study without alteration. The latter two include acceleration feedback (PID-like), so they are explicitly labelled as such. The new policy uses ordinary sensed-speed error and no derivative feedback.

The new nonlinear simulation has speed RMSE 1.1973 rpm, peak error 14.389 rpm, and fuel ringing 14.8252 lbm/s. Compared with retuned Poly2, tracking RMSE falls by 22.6%, but ringing grows to 11.4 times its previous value. This is a tracking–actuation tradeoff, not an overall improvement.

Scope of the feedforward claim. The quadratic coefficients are frozen from the prior inverse fit. PILCO learns the two PI gains, not the fifteen polynomial coefficients. At one fixed speed and ambient condition, feedforward is a constant and its offset is absorbed by bumpless integral initialization. Consequently this benchmark cannot identify a quadratic-map advantage over other constant feedforward values. The static map also slightly extrapolates: 9250 rpm is below its previous training-speed minimum of approximately 9264.23 rpm.

11.1.2 Step 1: write the deployed control law

Let N_s denote sensed shaft speed, r the governed reference, T inlet temperature in kelvin, and P inlet pressure in kPa. The old inverse model uses inputs $v = [N, \dot{N}, T, P]^\top$, standardized as $z_j = (v_j - \bar{v}_j)/s_j$. Its full quadratic is

$$p_2(v) = c_0 + \sum_{j=1}^{4} c_j z_j + \sum_{j=1}^{4} c_{jj} z_j^2 + \sum_{i<j} c_{ij} z_i z_j. \tag{11.1}$$

There are $1 + 4 + 4 + 6 = 15$ coefficients. Setting desired acceleration to zero defines static feedforward. The actual controller is

$$u_{\mathrm{ff,k}} = p_2(r_k, 0, T_k, P_k), \qquad e_k = r_k - N_{s,k}, \tag{11.2}$$

$$u_k^* = u_{\mathrm{ff,k}} + K_p e_k + I_k, \qquad u_k = \mathrm{clip}(u_k^*, 0.2, 4), \tag{11.3}$$

$$I_{k+1} = I_k + hK_i e_k \qquad \text{when integration is permitted.} \tag{11.4}$$

Fuel is measured in lbm/s, K_p in (lbm/s)/rpm and K_i in (lbm/s)/(rpm s). Conditional integration permits an update if ($u^* < 4$ or $e < 0$) and ($u^* > 0.2$ or $e > 0$). Before 45 s the old controller supplies the command and the new integral tracks $I = u_{\mathrm{old}} - u_{\mathrm{ff}} - K_p e$. This gives a bumpless handover. Existing plant, sensor, request governor, load profile, solver and fuel limits are inherited by the new model copy.

11.1.3 Step 2: construct the dynamics-learning state

Use a dimensionless augmented state and a normalized load:

$$x = \begin{bmatrix} (N - 9250)/20 \\ (N - N_s)/5 \\ (I - I_0)/0.3 \end{bmatrix}, \quad d = \frac{P_{\mathrm{load}}}{1537.6791946}, \quad I_0 = u_0 - u_{\mathrm{ff}}. \tag{11.5}$$

The trim fuel $u_0 = 1.39529998$ lbm/s. The second state represents sensor lag; discarding it would make fast feedback dynamics less nearly Markov. At fixed reference,

$$\tilde{u} = (u - u_0)/0.3 = b^\top x, \qquad b^\top = \left[-\frac{20K_p}{0.3}, \frac{5K_p}{0.3}, 1 \right]. \tag{11.6}$$

The GP input is $z = [x_1, x_2, \tilde{u}, d]^\top = Bx + a(d)$ and its output is the two physical-state increments $y = [\Delta x_1, \Delta x_2]^\top$. The integrator transition is known exactly, so learning it would waste data. With

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -20hK_i/0.3 & 5hK_i/0.3 & 1 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad x^+ = Dx + Ey, \tag{11.7}$$

the feedback integral remains correlated with the turbine state during planning.

11.1.4 Step 3: fit the probabilistic transition model

Training uses transitions from the first two pulses of each historical controller: absolute windows $[69.9, 78]$ and $[89.9, 103]$ s. There are 4242 transitions. The third-pulse window $[119.9, 138]$ s is held out for one-step dynamics checks. These trajectories share a deterministic plant and operating point; this is a temporal split, not independent hardware validation.

Fit a linear mean by least squares on training data only, $m(z) = a + Az$, and standardize each residual output with scale s_j. A deterministic farthest-point selection retains 40 representative GP inputs. For both standardized residual outputs, use the shared squared-exponential kernel

$$k(z, z') = \sigma_f^2 \exp\left[-\tfrac{1}{2}(z - z')^\top \Lambda^{-1}(z - z')\right], \qquad \Lambda = \mathrm{diag}(\ell_1^2, \ldots, \ell_4^2). \tag{11.8}$$

Let $K_n = K + \sigma_n^2 \mathcal{I} + 10^{-9}\mathcal{I}$ and let R contain standardized residuals. Cholesky solves minimize the summed negative log marginal likelihood, up to constants:

$$\mathcal{L} = \tfrac{1}{2}\,\mathrm{tr}(R^\top K_n^{-1} R) + \frac{2}{2}\log|K_n|, \qquad \beta = K_n^{-1} R. \tag{11.9}$$

The hyperparameter bounds in log units are $[-1, 4]$ for length scales, $[-3, 3]$ for signal standard deviation and $[-7, 0]$ for noise standard deviation. Conditional on a deterministic input, residual outputs are independent and share covariance hyperparameters. Uncertain inputs induce output cross-covariance. The linear mean is treated as fixed: uncertainty in its estimated coefficients is not integrated out.

11.1.5 Step 4: propagate a Gaussian through the policy

Suppose $x \sim \mathcal{N}(\mu, \Sigma)$. For the unsaturated policy, the input moments are exact:

$$m_z = B\mu + a(d), \qquad S_z = B\Sigma B^\top, \qquad C_{xz} = \Sigma B^\top. \tag{11.10}$$

The load is a known deterministic scenario input during planning, not a measured feedforward input to the deployed policy. A full Gaussian may mathematically cross the fuel limits. Planning here uses the unsaturated affine policy; the nonlinear plant evaluation applies clipping and anti-windup. This is a local approximation, justified only to the extent that the evaluated commands remain inside the limits.

11.1.6 Step 5: derive exact uncertain-input GP moments

Write $m = m_z$, $S = S_z$, and let z_i denote training inputs. Completing the square in the Gaussian integral of one kernel gives

$$q_i = \int k(z, z_i)\mathcal{N}(z \mid m, S)\, dz \tag{11.11}$$

$$= \sigma_f^2 \frac{|\Lambda|^{1/2}}{|S + \Lambda|^{1/2}} \exp\left[-\tfrac{1}{2}(z_i - m)^\top (S + \Lambda)^{-1}(z_i - m)\right]. \tag{11.12}$$

For two kernels, define $c_{ij} = (z_i + z_j)/2$ and $\delta_{ij} = z_i - z_j$. Their product separates into a distance factor and a Gaussian centered at c_{ij}. Therefore

$$Q_{ij} = \mathbb{E}[k(z, z_i)k(z, z_j)] \tag{11.13}$$

$$= \sigma_f^4 \exp[-\tfrac{1}{4}\delta_{ij}^\top \Lambda^{-1}\delta_{ij}] \frac{|\Lambda/2|^{1/2}}{|S + \Lambda/2|^{1/2}} \cdot \exp[-\tfrac{1}{2}(c_{ij} - m)^\top (S + \Lambda/2)^{-1}(c_{ij} - m)]. \tag{11.14}$$

Let g be the standardized residual GP output. The law of total expectation and total covariance now give

$$\bar{g} = \beta^\top q, \tag{11.15}$$

$$V_g = \beta^\top Q\beta - \bar{g}\bar{g}^\top + \mathcal{I}\left[\sigma_f^2 - \mathrm{tr}(K_n^{-1}Q) + \sigma_n^2\right]. \tag{11.16}$$

The last term includes observation/model-residual noise in each predicted transition. Propagating this as temporally independent noise does not represent a single fixed GP function sample across the entire horizon. This standard moment-propagation approximation can accumulate model uncertainty differently from trajectory sampling.

The Gaussian kernel identity also yields

$$G = (S + \Lambda)^{-1}\sum_i (z_i - m)q_i\beta_i^\top H, \qquad H = \mathrm{diag}(s_1, s_2). \tag{11.17}$$

Then $\mathrm{Cov}(z, Hg) = SG$. Restoring the linear mean and residual scales,

$$\bar{y} = a + Am + H\bar{g}, \tag{11.18}$$

$$V_y = ASA^\top + HV_gH + ASG + G^\top SA^\top, \tag{11.19}$$

$$C_{xy} = C_{xz}(A^\top + G). \tag{11.20}$$

The cross terms matter: dropping C_{xy} loses the dependence between current state and predicted state increment. All inverses above involve a positive-definite kernel length-scale matrix added to S, so a singular S from deterministic scenario inputs is allowed.

11.1.7 Step 6: close the moment recursion

Substitution into $x^+ = Dx + Ey$ gives

$$\mu^+ = D\mu + E\bar{y}, \tag{11.21}$$

$$\Sigma^+ = D\Sigma D^\top + EV_yE^\top + DC_{xy}E^\top + EC_{xy}^\top D^\top. \tag{11.22}$$

Each transition is then approximated by a Gaussian with these moments. This preserves first and second moments of the one-step mapping under the Gaussian input assumption; it does not preserve multimodality of the true multistep distribution. Initial planning moments are $\mu_0 = 0$ and $\Sigma_0 = \mathrm{diag}(10^{-6}, 10^{-6}, 10^{-8})$.

11.1.8 Step 7: derive and optimize the expected cost

For a 5 s planning horizon at the same 15 ms sample period, a unit load pulse is applied for $1 \le t < 2.5$ s. The objective is

$$J(\theta) = h\sum_k \mathbb{E}[x_{1,k}^2 + 0.01\tilde{u}_k^2] \tag{11.23}$$

$$= h\sum_k \left[\mu_{1,k}^2 + \Sigma_{11,k} + 0.01\{(b^\top\mu_k)^2 + b^\top\Sigma_k b\}\right]. \tag{11.24}$$

Use $\theta = [\log K_p, \log K_i]^\top$, so gains remain positive. Bounds are $0.005 \le K_p \le 0.07$ and $0.01 \le K_i \le 0.25$. The initial policy is $(0.025, 0.05)$. L-BFGS-B repeatedly evaluates the complete Gaussian rollout and estimates parameter gradients numerically. The conceptual chain rule is

$$\frac{dJ}{d\theta} = \sum_k \left(\frac{\partial J}{\partial \mu_k}\frac{d\mu_k}{d\theta} + \frac{\partial J}{\partial \Sigma_k} : \frac{d\Sigma_k}{d\theta} + \frac{\partial J}{\partial \theta}\right). \tag{11.25}$$

For implementation, this derivative is approximated by finite differences of the full objective rather than hand-coded analytic sensitivities. The learned result is

$$\boxed{K_p = 0.06460568, \qquad K_i = 0.17102610, \qquad K_d = 0.} \tag{11.26}$$

Expected cost changes from 0.464463 to 0.141314. This cost is dimensionless and is not the reported full-test speed RMSE or fuel-ringing metric. Gains are frozen immediately after optimization; no 5% validation result is used in selection.

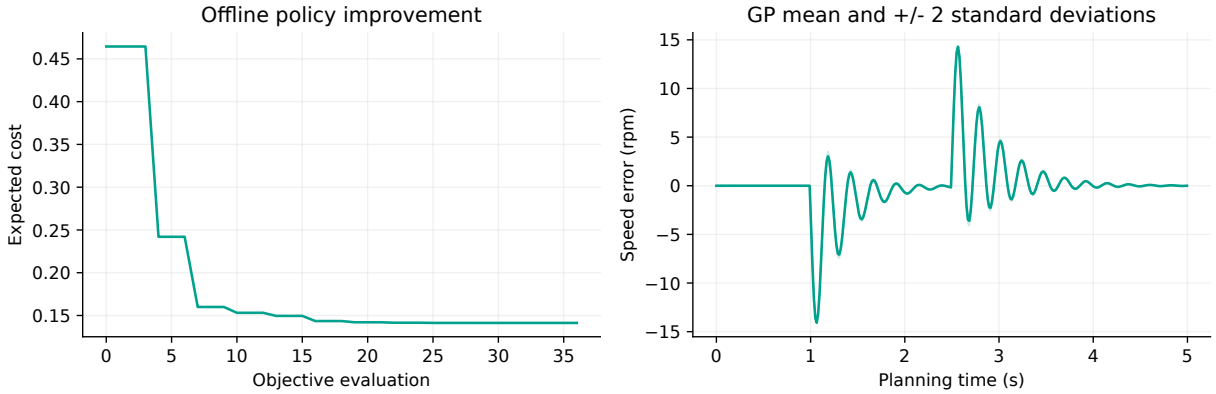


Figure 11.1: Policy-search objective and GP planning response. The shaded interval is model uncertainty, not a certified plant confidence band.

11.1.9 Step 8: nonlinear plant comparison and independent checks

All three new simulations use the copied T-MATS plant: selected policy at 10%, selected policy at 5%, and initial quadratic-feedforward PI at 10%. Historical traces supply the three earlier comparators. The initial-policy ablation helps distinguish gain learning from merely adding a constant feedforward value. File hashes and replay checks are saved with the study.

Controller	RMSE rpm	Peak rpm	Ringing lbm/s	Settling s
Base PI	2.4209	21.406	2.0702	2.475
Original Poly2 + PID	1.6658	14.113	1.6968	1.365
Retuned Poly2 + PID	1.5461	15.825	1.2990	0.930
PILCO FF+PI	1.1973	14.389	14.8252	1.920

Table 11.1: 10% load-amplitude comparison.

Controller	RMSE rpm	Peak rpm	Ringing lbm/s	Settling s
Base PI	1.1692	10.392	1.0583	2.115
Original Poly2 + PID	0.8018	6.969	0.9005	1.260
Retuned Poly2 + PID	0.7439	7.806	0.6544	0.870
PILCO FF+PI	0.5930	7.008	7.9471	1.695

Table 11.2: 5% load-amplitude comparison.

RMSE uses every sample in $[60, 150]$ s. Peak error is maximum absolute speed error in that interval. IAE uses the same rectangular sample sum as the prior report. Fuel ringing sums, over six two-second event windows, total variation minus absolute net fuel change:

$$R = \sum_{j=1}^{6} \left[\sum_{k \in W_j} |u_{k+1} - u_k| - |u_{\mathrm{last}(j)} - u_{\mathrm{first}(j)}| \right]. \tag{11.27}$$

This is a reversal metric in lbm/s, not fuel consumed. Settling requires persistent entry into ± 0.1 rpm until the next event. Numerical acceptance checks full duration, finite signals, positive fuel, positive surge margin, flow residual at most 10^{-9}, fewer than 200 solver iterations, and pre-test speed within 0.1 rpm.

The analytic GP covariance agrees with a 50,000-input Monte Carlo integration to 0.31% relative Frobenius error; input/output cross-covariance error is 1.57%. Held-out one-step speed-increment RMSE ranges from 0.000432 to 0.000709 rpm. Independent command and integral replay errors are below 10^{-8}. All evaluated new traces pass the existing numerical acceptance checks. The selected 10% run has minimum surge margin 14.800% and fuel saturation fraction 0.000%.

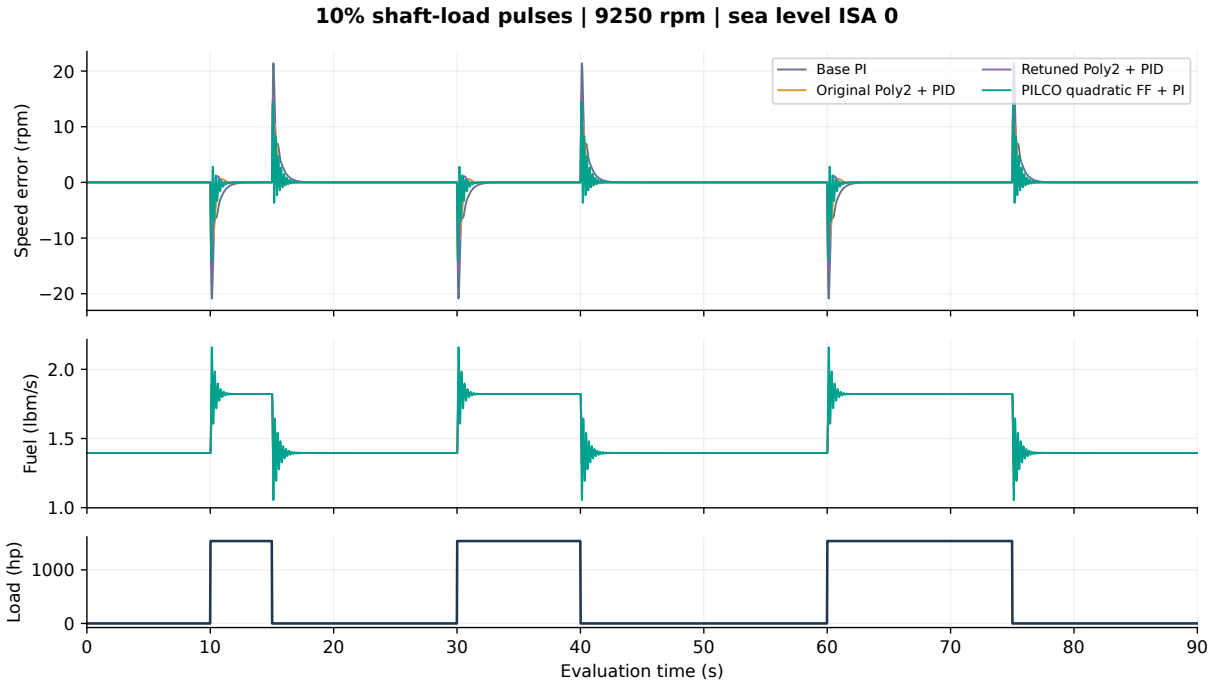


Figure 11.2: Full nonlinear 10% benchmark. Historical methods and new controller use matching time, request and disturbance profiles.

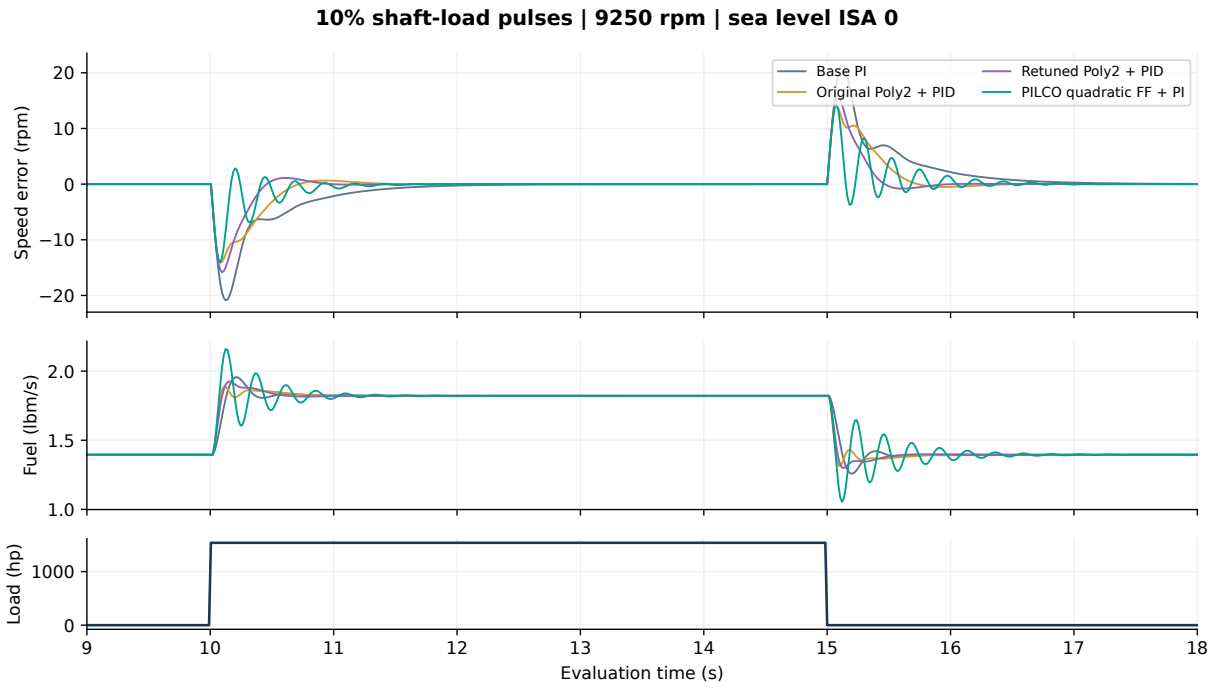


Figure 11.3: First pulse, emphasizing overshoot and command reversals. The new method is judged on these plant traces.

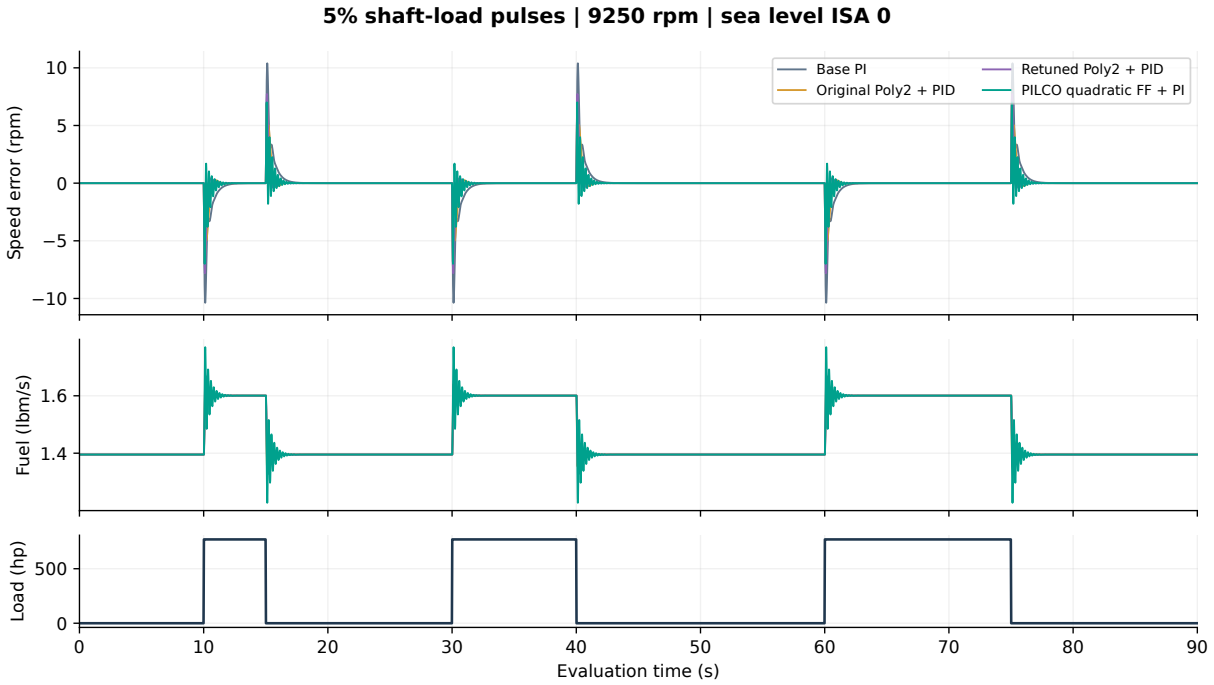


Figure 11.4: Frozen-policy 5% amplitude validation at the same operating point.

11.1.10 Interpretation and limitations

Relative to Base PI, RMSE changes by -50.5% and ringing by +616.1%. Relative to Retuned Poly2 + PID, RMSE changes by -22.6% and ringing by +1041.2%. The initial feedforward PI has RMSE

2.4406 rpm and ringing 2.0296 lbm/s. These quantities describe different tradeoffs; minimizing expected tracking cost does not directly minimize fuel reversals.

The original PILCO method uses GP dynamics, uncertainty-aware long-horizon prediction and policy search with analytic gradients [11]. This study retains GP moment matching but changes the policy family, cost, mean function, training source and gradient implementation. It performs one offline improvement round using existing experiments; it does not claim learning from scratch or a repeated collect–refit–improve loop. Forty representative GP points are a subset approximation, and a low held-out one-step error does not establish calibrated long-horizon uncertainty. The moment check validates the algebra against integration, not the physical accuracy of the GP.

The local state omits thermodynamic internal states; the linear mean and GP can absorb only their effects visible in the training trajectories. The deployment sensor dynamics are inherited from the plant, while learning identifies their observed effect. No guarantee of global stability, robustness at other speeds or ambient conditions, or hardware safety follows from these simulations. A multi-speed test with a frozen validation protocol would be required to assess the quadratic feedforward surface itself. The video is an animation of recorded signals; rotor speed is visually scaled and is not a blade-resolved turbine simulation.

11.1.11 Reproduction and delivered files

The source scripts are `pilco_tmats.py` (GP and policy learning), `tmats_pilco_pi_sfun.m` (deployed controller), `tmats_pilco_setup.m`, `run_tmats_pilco.m` (plant runs), and `build_pilco_deliverables.py` (verification, figures, paper and video). The ready-to-open model is `GasTurbine_PILCO_Quadratic_PI.mdl`. It requires the existing adjacent T-MATS library and the frozen fit files in this workspace.

Run Python learning first, then execute `run_tmats_pilco` in MATLAB R2018b or a compatible release. Finally run the artifact builder. The results directory contains policy JSON, GP parameters, search history, prediction moments, MAT/CSV plant traces, comparison metrics, replay verification and a hash manifest. The paper source and figures are included alongside the PDF. Re-running overwrites only this study's outputs; preserve them first if a second independent run is desired.

11.2 Learning smoother control with an expected ringing penalty

Archived October study. All simulations, searches, and checks described below predate this report.

Chapter overview

The preceding PILCO-style controller reduced speed-error RMSE but produced much larger fuel-command reversals. This study modifies policy learning to penalize expected excess total variation of fuel. A previous-command state and an event-start anchor retain the correlations needed to evaluate that penalty under Gaussian moment propagation. Seven coarse penalty weights and three intermediate refinements are optimized with the unchanged GP dynamics model and checked on the nonlinear gas-turbine plant. A declared joint tracking/ringing score selects a policy using only 10% load tests, after which a 5% test evaluates the frozen gains. The method remains an offline PILCO-style adaptation with numerical policy gradients and a frozen quadratic feedforward map. The report separates probabilistic predictions, plant measurements, and limitations.

11.2.1 Outcome and why the objective changed

The selected penalty is $\lambda = 0.2$, with $K_p = 0.02479762$ and $K_i = 0.05385920$. In the nonlinear 10% test, ringing falls by 85.9% relative to the previous PILCO policy, from 14.8252 to 2.0838 lbm/s. Speed RMSE changes from 1.1973 to 2.3877 rpm. The smoother command therefore has a measurable tracking cost.

The previous learned gains were $K_p = 0.06460568$, $K_i = 0.17102610$, $K_d = 0$. Their training objective rewarded small speed error and modest fuel magnitude, but neither term distinguished a smooth fuel trajectory from one repeatedly reversing around its equilibrium. A fast, aggressive PI policy could therefore score well despite oscillation. This was visible in the previous full nonlinear 10% test: RMSE was 1.19728 rpm and ringing was 14.82523 lbm/s, compared with 1.54614 rpm and 1.29904 lbm/s for retuned Poly2 with derivative feedback.

The present change acts in *policy learning*. No post-processing is applied to plant traces or video curves, and no extra low-pass filter is inserted into the fuel command. The deployed controller remains quadratic feedforward plus ordinary speed-error PI; changing its optimized gains changes the actual nonlinear plant response.

11.2.2 The experiment and what is held fixed

The reference is 9250 rpm at sea level, ISA zero and Mach zero. There are 60 s of preparation and 90 s of evaluation at $h = 0.015$ s. The three 10% shaft-load pulses occupy evaluation intervals $[10.005, 15)$, $[30, 40.005)$ and $[60, 75)$ s. Their magnitude is 1537.67919 hp, obtained from the same 15376.79195 hp trim as the preceding studies. Validation uses 768.83960 hp with identical timing. The numerical solver, speed sensor, fuel limits, feedforward coefficients and startup procedure are unchanged.

The dynamics GP is loaded from the previous study without refitting. Its training comprises 4242 transitions from the first two pulse windows of base PI, original Poly2 and retuned Poly2. A 40-point representative subset models residuals around a fitted linear mean. The original third-pulse holdout remains outside GP fitting. However, the 10% full-test traces now participate in penalty-weight selection; they are tuning results, not independent validation. Five-percent data are never used to choose the policy in this study.

11.2.3 Step 1: the deployable controller

For standardized inverse-model inputs $z_j = (v_j - \bar{v}_j)/s_j$ with $v = [N, \dot{N}, T, P]^\top$, retain the full quadratic

$$p_2(v) = c_0 + \sum_{j=1}^{4} c_j z_j + \sum_{j=1}^{4} c_{jj} z_j^2 + \sum_{i<j} c_{ij} z_i z_j. \tag{11.28}$$

At each sample, the actual control law is

$$u_{\mathrm{ff},k} = p_2(r_k, 0, T_k, P_k), \qquad e_k = r_k - N_{s,k}, \tag{11.29}$$

$$u_k^* = u_{\mathrm{ff},k} + K_p e_k + I_k, \qquad u_k = \mathrm{clip}(u_k^*, 0.2, 4), \tag{11.30}$$

$$I_{k+1} = I_k + hK_i e_k \quad \text{if conditional integration permits it.} \tag{11.31}$$

The integral updates if ($u^* < 4$ or $e < 0$) and ($u^* > 0.2$ or $e > 0$), allowing correction out of saturation. Before 45 s, the old command drives the plant and the new integral tracks $I = u_{\mathrm{old}} - u_{\mathrm{ff}} - K_p e$ for bumpless transfer. Gains have units (lbm/s)/rpm and (lbm/s)/(rpm s).

Only K_p and K_i are learned. The polynomial's fifteen coefficients are fixed. At this single setpoint and ambient condition, feedforward is constant and an integral offset can compensate its value. The experiment therefore assesses PI tuning, not the benefit of the quadratic surface across speeds. The setpoint remains slightly below the old polynomial training-speed minimum of 9264.23 rpm.

11.2.4 Step 2: distinguish magnitude, slew, total variation, and ringing

Fuel magnitude describes how much fuel is commanded. Slew is the local rate $(u_k - u_{k-1})/h$. Total variation counts accumulated command motion, while excess variation removes the unavoidable net motion between endpoints:

$$\mathrm{TV}_{a:b}(u) = \sum_{k=a+1}^{b} |u_k - u_{k-1}|, \tag{11.32}$$

$$R_{a:b}(u) = \mathrm{TV}_{a:b}(u) - |u_b - u_a| \geq 0. \tag{11.33}$$

The inequality follows from the triangle inequality. For a monotone trajectory $[0, 0.5, 1]$, TV and net change both equal 1, so ringing is zero. For $[0, 1, 0, 1]$, TV is 3 and net change is 1, so ringing is 2. A direct excess-TV penalty discourages reversals without penalizing a required monotone fuel transition as strongly as plain TV would.

The metric has units lbm/s because it sums changes in a fuel-flow command. It is neither integrated fuel consumption nor the integral of squared fuel slew. It is sensitive to sample rate and noise. This study keeps the sample rate fixed for every comparison. Penalizing ringing does not automatically bound peak slew: the report separately measures that quantity.

11.2.5 Step 3: augment the state to preserve temporal correlation

Use normalized physical and controller states

$$x = \begin{bmatrix} (N - 9250)/20 \\ (N - N_s)/5 \\ (I - I_0)/0.3 \end{bmatrix}, \quad I_0 = u_0 - u_{\mathrm{ff}}, \quad \tilde{u} = \frac{u - u_0}{0.3} = b^\top x, \tag{11.34}$$

where $u_0 = 1.39529998$ lbm/s and $b^\top = [-20K_p/0.3,\ 5K_p/0.3,\ 1]$ for the unsaturated fixed-reference policy. To evaluate fuel increments and endpoint changes, introduce

$$\xi = \begin{bmatrix} x \\ p \\ a \end{bmatrix}, \qquad p = \tilde{u}_{k-1}, \qquad a = \tilde{u}_{\text{event start}}, \qquad \bar{b} = \begin{bmatrix} b \\ 0 \\ 0 \end{bmatrix}. \tag{11.35}$$

The five-dimensional covariance includes the cross-covariance between current state, previous command and event-start anchor. Ignoring these correlations would generally overestimate incremental-command uncertainty. In particular,

$$\mathrm{Var}(\tilde{u}_k - p_k) = \mathrm{Var}(\tilde{u}_k) + \mathrm{Var}(p_k) - 2\,\mathrm{Cov}(\tilde{u}_k, p_k). \tag{11.36}$$

At the beginning of each planning event, set the anchor to the current policy output. This is a deterministic linear transformation $\xi \leftarrow T\xi$, where T is identity except its fifth row is $\bar{b}^\top$. Thus both moments transform: $m \leftarrow Tm$ and $S \leftarrow TST^\top$. Resetting only the anchor mean would be incorrect.

11.2.6 Step 4: propagate the unchanged GP and augmented moments

The normalized GP input is

$$z = [x_1, x_2, \tilde{u}, d]^\top = B\xi + c(d), \qquad d = P_{\text{load}}/1537.6791946. \tag{11.37}$$

With $\xi \sim \mathcal{N}(m, S)$, its input moments and state/input cross-covariance are $m_z = Bm + c$, $S_z = BSB^\top$, and $C_{\xi z} = SB^\top$. The GP returns two increments $y = [\Delta x_1, \Delta x_2]^\top$. Integration and command memory are known transitions:

$$\xi^+ = D\xi + Ey, \qquad D = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -20hK_i/0.3 & 5hK_i/0.3 & 1 & 0 & 0 \\ -20K_p/0.3 & 5K_p/0.3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{11.38}$$

The fourth state becomes the current command, and the fifth retains the event anchor. The GP residual uses a shared squared-exponential kernel,

$$k(z, z') = \sigma_f^2 \exp[-\tfrac{1}{2}(z - z')^\top \Lambda^{-1}(z - z')], \qquad K_n = K + \sigma_n^2 \mathcal{I} + 10^{-9}\mathcal{I}. \tag{11.39}$$

For Gaussian input, let $q_i = \mathbb{E}[k(z, z_i)]$, $Q_{ij} = \mathbb{E}[k(z, z_i)k(z, z_j)]$ and $\beta = K_n^{-1}R$, where R contains standardized training residuals. Then

$$q_i = \sigma_f^2 \frac{|\Lambda|^{1/2}}{|S_z + \Lambda|^{1/2}} \exp[-\tfrac{1}{2}(z_i - m_z)^\top (S_z + \Lambda)^{-1}(z_i - m_z)], \tag{11.40}$$

$$\mu_g = \beta^\top q, \tag{11.41}$$

$$V_g = \beta^\top Q\beta - \mu_g\mu_g^\top + \mathcal{I}[\sigma_f^2 - \operatorname{tr}(K_n^{-1}Q) + \sigma_n^2]. \tag{11.42}$$

The two-kernel integral Q uses the midpoint $(z_i + z_j)/2$ and covariance $\Lambda/2$; the full expression and previous validation are available in the preceding policy-learning chapter. Restoring residual output scales and the fitted linear mean yields $\mu_y, V_y, C_{\xi y}$. The recursion is

$$m^+ = Dm + E\mu_y, \tag{11.43}$$

$$S^+ = DSD^\top + EV_yE^\top + DC_{\xi y}E^\top + EC_{\xi y}^\top D^\top. \tag{11.44}$$

Each propagated state is approximated as Gaussian. GP noise is treated as independent at successive transitions. This is uncertainty propagation under a model approximation, not a probabilistic safety guarantee or a temporally coherent GP function draw.

11.2.7 Step 5: derive expected absolute command motion

For a scalar Gaussian $Z \sim \mathcal{N}(\mu, v)$, split the integral at zero:

$$\mathbb{E}|Z| = \int_{-\infty}^{0} (-z)\mathcal{N}(z \mid \mu, v)\, dz + \int_{0}^{\infty} z\mathcal{N}(z \mid \mu, v)\, dz \tag{11.45}$$

$$= \sqrt{\frac{2v}{\pi}} \exp\left(-\frac{\mu^2}{2v}\right) + \mu \operatorname{erf}\left(\frac{\mu}{\sqrt{2v}}\right) \equiv F(\mu, v). \tag{11.46}$$

For zero variance, take the continuous limit $F(\mu, 0) = |\mu|$. For zero mean and positive variance, $F(0, v) = \sqrt{2v/\pi}$, which is nonzero. This exposes why penalizing only $|\mathbb{E}[\Delta u]|$ would miss uncertain command motion.

Define $w_\Delta = \bar{b} - e_4$ and $w_A = \bar{b} - e_5$. After a transition,

$$\mathbb{E}|\tilde{u}_k - \tilde{u}_{k-1}| = F(w_\Delta^\top m_k, w_\Delta^\top S_k w_\Delta), \tag{11.47}$$

$$\mathbb{E}|\tilde{u}_b - \tilde{u}_a| = F(w_A^\top m_b, w_A^\top S_b w_A). \tag{11.48}$$

Linearity of expectation permits summing absolute-increment expectations even when increments are correlated. For each event window,

$$\mathcal{R}_{a:b} = \sum_{k=a+1}^{b} F(w_\Delta^\top m_k, w_\Delta^\top S_k w_\Delta) - F(w_A^\top m_b, w_A^\top S_b w_A). \tag{11.49}$$

The corresponding physical expected ringing is $0.3\mathcal{R}$ lbm/s. No extra factor of h belongs in this TV sum: $h\sum|\Delta u/h| = \sum|\Delta u|$. An independent Monte Carlo test checks the folded-normal formula, and a particle rollout cross-checks the complete augmented calculation on the same GP.

11.2.8 Step 6: form the objective and search penalty weights

The former tracking and magnitude costs are retained:

$$J_{\text{track}} = h\sum_k (m_{1,k}^2 + S_{11,k}), \tag{11.50}$$

$$J_{\text{fuel}} = 0.01h\sum_k [(\bar{b}^\top m_k)^2 + \bar{b}^\top S_k \bar{b}], \tag{11.51}$$

$$J_\lambda = J_{\text{track}} + J_{\text{fuel}} + \lambda \sum_{\text{events}} \mathcal{R}_{a:b}. \tag{11.52}$$

The 5 s planning trajectory applies a unit load for $1 \le t < 2.5$ s. On the 15 ms grid, transitions begin at 1.005 and 2.505 s. The load-on event closes immediately before unloading, and the unloading event closes at the end of the horizon. Its event windows differ from the fixed two-second windows used in the plant metric. This difference is declared; the plant metric is never replaced by a model prediction.

The coarse weights are $\lambda \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$. Candidate zero retains the previous tracking-only gains. Positive weights use bounded L-BFGS-B on $[\log K_p, \log K_i]$, warm-started from the preceding weight. After inspecting the coarse 10% tradeoff, one refinement adds weights 0.15, 0.20 and 0.25 in the 0.1–0.3 bracket, warm-started from the nearest available weight. This refinement uses tuning data only and is recorded before the added simulations; it does not use 5% validation. Bounds remain $0.005 \le K_p \le 0.07$ and $0.01 \le K_i \le 0.25$. Policy gradients are finite differences of the full probabilistic rollout. This finite set and local optimization do not establish a global optimum.

For each weight, the resulting gains are evaluated in the complete nonlinear 10% test. The selection score, declared before these evaluations, is

$$S_j = \max\left(\frac{\text{RMSE}_j}{2.4209039065}, \frac{R_j}{2.0702280417}\right). \tag{11.53}$$

The denominators are the historical base PI metrics. The smallest score among numerically accepted candidates wins; the score gives equal priority to the worse of these two relative metrics. It does not minimize settling time, peak error, surge margin, or fuel slew. Gains are frozen before 5% validation.

λ	K_p	K_i	RMSE	Ringing	Score
0	0.06461	0.17103	1.1973	14.8252	7.161
0.001	0.06206	0.16260	1.2054	13.0728	6.315
0.003	0.05751	0.15035	1.2343	10.6050	5.123
0.01	0.04640	0.13319	1.3609	6.8917	3.329
0.03	0.03820	0.10339	1.5716	4.6845	2.263
0.1	0.02949	0.07057	1.9952	2.8858	1.394
0.15	0.02643	0.06097	2.2146	2.3771	1.148
0.2*	0.02480	0.05386	2.3877	2.0838	1.007
0.25	0.02355	0.04890	2.5381	1.8762	1.048
0.3	0.02216	0.04512	2.6968	1.6841	1.114

Table 11.3: All tested penalty weights. RMSE is in rpm; ringing is in lbm/s. The star marks the selected policy.

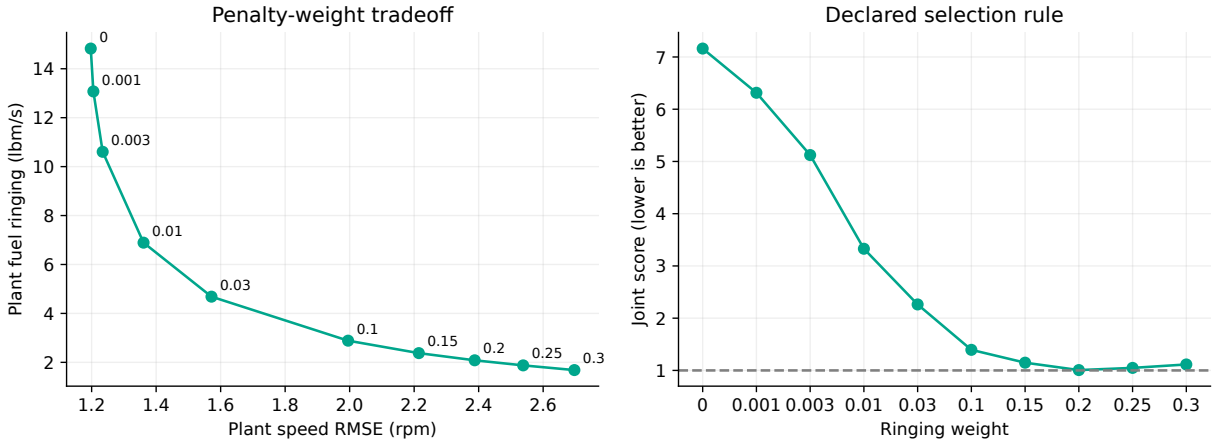


Figure 11.5: Measured 10% plant tradeoff and the declared selection score. These are candidate evaluations, not independent validation.

11.2.9 Step 7: nonlinear results and interpretation

The deployed gains are frozen at $K_p = 0.02479762$, $K_i = 0.05385920$ and $K_d = 0$.

Controller	RMSE (rpm)	Peak (rpm)	Ringing	Settling (s)
Base PI	2.4209	21.406	2.0702	2.475
Original Poly2	1.6658	14.113	1.6968	1.365
Retuned Poly2	1.5461	15.825	1.2990	0.930
Previous PILCO	1.1973	14.389	14.8252	1.920
Smooth PILCO	2.3877	21.593	2.0838	2.220

Table 11.4: 10% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band.

Controller	RMSE (rpm)	Peak (rpm)	Ringing	Settling (s)
Base PI	1.1692	10.392	1.0583	2.115
Original Poly2	0.8018	6.969	0.9005	1.260
Retuned Poly2	0.7439	7.806	0.6544	0.870
Previous PILCO	0.5930	7.008	7.9471	1.695
Smooth PILCO	1.1533	10.486	1.0615	1.905

Table 11.5: 5% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band.

Relative to Base PI, smooth PILCO changes RMSE by -1.4%, ringing by +0.7%, and fine settling from 2.475 to 2.220 s.

Relative to Retuned Poly2, smooth PILCO changes RMSE by +54.4%, ringing by +60.4%, and fine settling from 0.930 to 2.220 s.

Relative to Previous PILCO, smooth PILCO changes RMSE by +99.4%, ringing by -85.9%, and fine settling from 1.920 to 2.220 s.

Its sampled peak fuel slew is 4.763 lbm/s^2, versus 10.514 for previous PILCO. Minimum surge margin changes from 14.800% to 18.201%; the selected 10% run has 0.000% fuel saturation. These diagnostics complement the objective rather than entering its selection score.

With the gains frozen, the 5% test gives RMSE 1.1533 rpm and ringing 1.0615 lbm/s, a 86.6% ringing reduction from previous PILCO. This supports amplitude transfer of the smoothing effect at this operating point.

What the video compares. Four synchronized windows show base PI, retuned Poly2 + PID, previous PILCO, and smooth PILCO. Original Poly2 remains in the report tables but is omitted from the four-window video to make room for the new method. Each window has its own speed-error and fuel plots, matching axes, current speed, current fuel, a shaft-load indicator and full-test metrics. Orange shading marks load-on intervals. The video includes the full 10% test, slowed load application/removal, frozen-gain 5% validation and a final scorecard. All plotted values are recorded plant signals without visual smoothing.

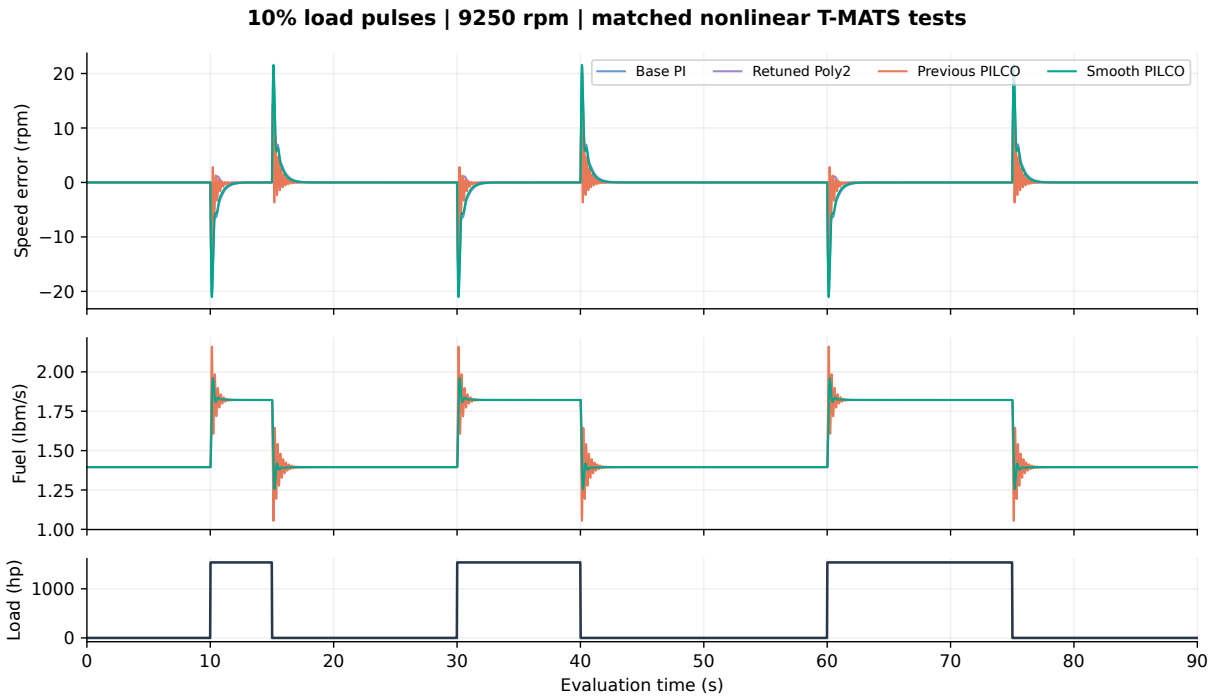


Figure 11.6: 10% full test; only the controller gains differ between the two PILCO policies.

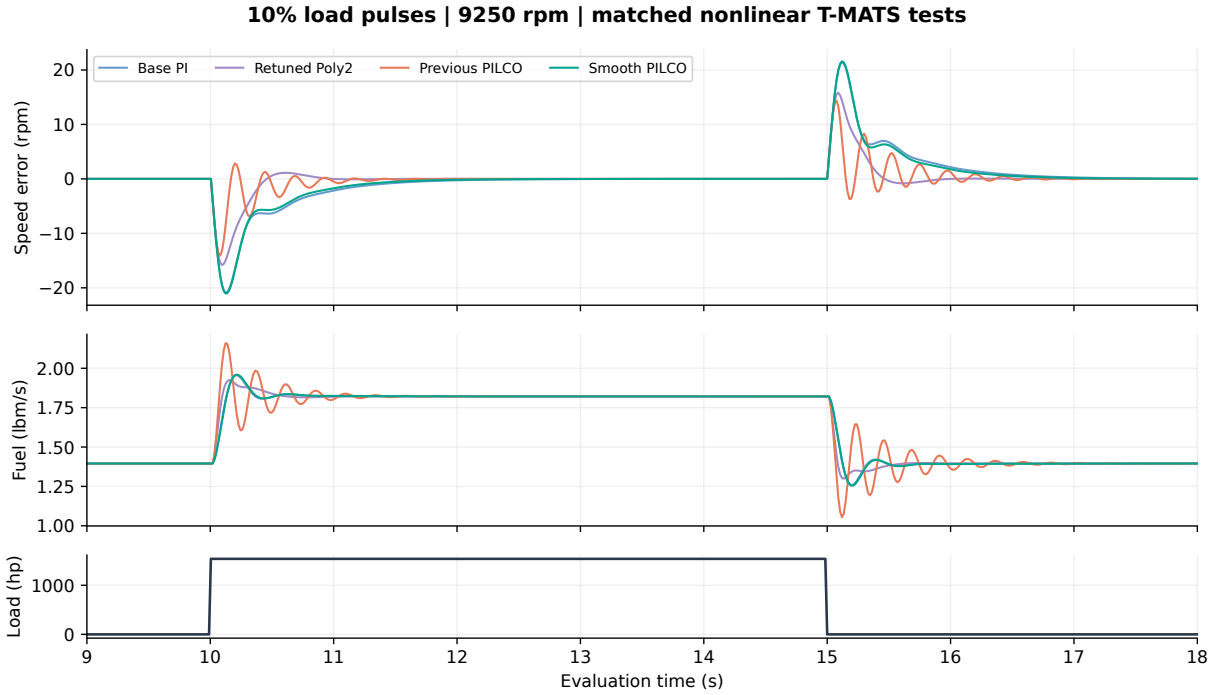


Figure 11.7: First load pulse, showing the effect of the ringing penalty on physical speed and fuel command.

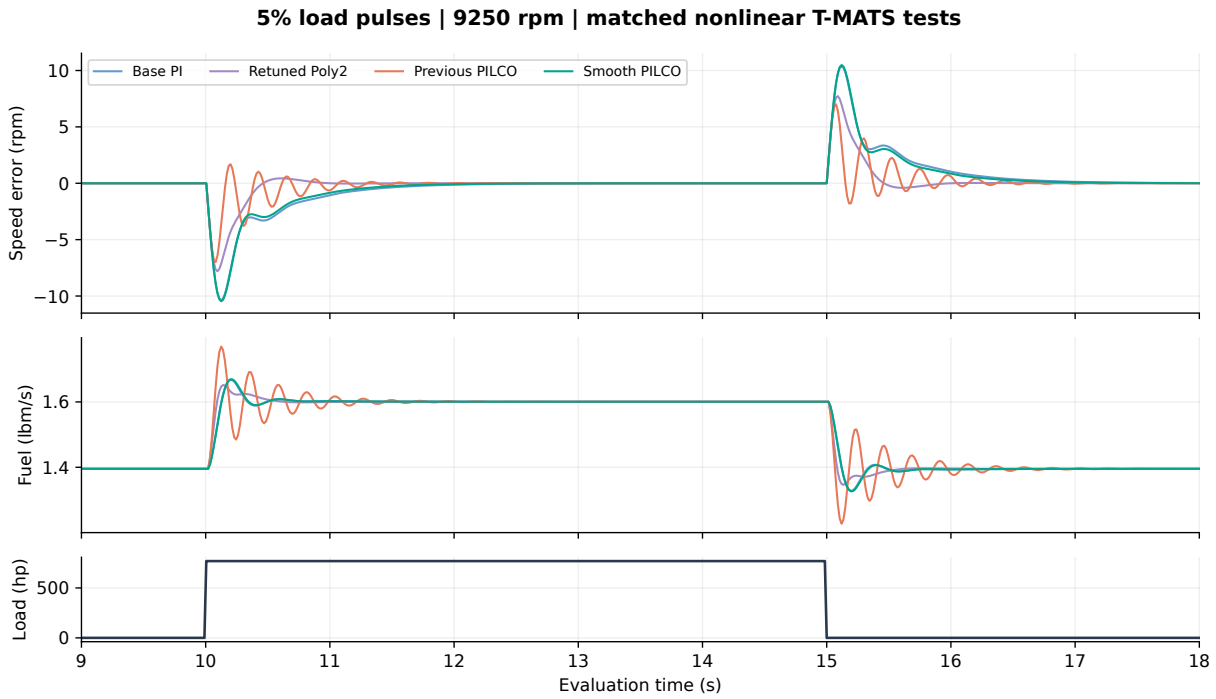


Figure 11.8: Frozen-policy 5% amplitude validation with the same timing and plant.

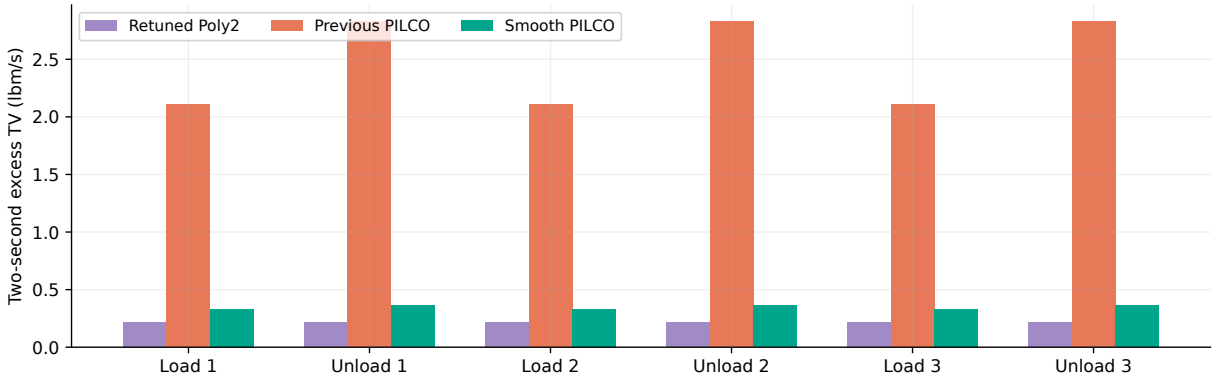


Figure 11.9: Ringing decomposed into six individual 10% load edges. Each bar is total fuel movement minus net movement in the same two-second window.

11.2.10 Step 8: numerical checks and metric definitions

All selected traces pass the shared plant acceptance checks. Maximum independent command replay discrepancy is 1.46e-13 lbm/s; maximum integral update discrepancy is 5.38e-15 lbm/s. The folded-normal formula passes 250,000-sample scalar Monte Carlo checks, including zero-variance and nonzero-variance cases. For the selected policy, a 5000-particle GP rollout predicts normalized excess TV 2.317963, compared with 2.317930 from Gaussian moment closure; their relative difference is 0.0014%. The particle standard error is 0.000037. This checks prediction consistency on the same surrogate; it does not independently validate the physical GP uncertainty.

The plant acceptance test requires finite signals, positive fuel, positive surge margin, flow residual at most 10^{-9}, fewer than 200 solver iterations, full 150 s duration and preparation speed within 0.1 rpm. Speed RMSE and peak error use all evaluation samples. IAE uses the previous study's rectangular

sum. Ringing sums TV minus endpoint motion in six two-second load-edge windows. Fine settling requires staying inside ±0.1 rpm until the next event; it is not a broad percentage-of-speed settling criterion. Reported peak slew is the maximum sampled fuel change divided by 0.015 s in those windows.

Independent replay reconstructs $u^* = u_{\text{ff}} + K_p(r - N_s) + I$ and every eligible integral update, then checks the saved command. Metrics are recomputed from CSV and compared against MATLAB's shared implementation. Identical time, reference and load vectors are checked across compared runs. The extra previous-command and anchor states exist only in offline prediction: the runtime controller still requires just its integral state.

11.2.11 Limits and how to use the result

The result is smoother than the earlier tracking-only PILCO, but the complete comparison must also account for tracking, settling and surge margin. It does not dominate the retuned Poly2 controller on both speed RMSE and ringing. That comparator has derivative/acceleration feedback, whereas the new policy is restricted to PI. The chosen weight is the best under the declared score among the tested candidates, not a universal tuning value.

This is an offline PILCO-style method, not a reproduction of the original full collect–refit–improve loop. The original PILCO work motivates GP dynamics and uncertainty-aware policy evaluation [11]. This implementation uses a fitted linear mean, a subset GP with shared covariance hyperparameters, a restricted PI policy, numerical gradients, and a custom expected excess-TV cost. It assumes an unsaturated affine policy during planning; plant runs apply the actual saturation and anti-windup. Five-percent testing checks amplitude transfer at one operating point and is not proof of robustness to noise, speed changes, ambient changes or hardware dynamics.

To prioritize still less actuator activity, select a larger penalty only after examining its tracking cost. To require a hard slew bound, add and validate an explicit actuator-rate constraint or a different policy architecture; an excess-TV penalty alone does not provide such a bound. To claim advantages for the quadratic feedforward map itself, perform separate multi-speed and ambient validation.

11.2.12 Reproduction

Run `pilco_smooth.py` to load the frozen GP and optimize the coarse weights. Run `run_tmats_pilco_smooth('tune')` in MATLAB. Run `refine_pilco_smooth.py`, then repeat the MATLAB tuning call, which resumes saved candidates and evaluates the additions. Run `select_pilco_smooth.py` to write `selected_policy.json`, then `run_tmats_pilco_smooth('validate')` only after selection. The ready-to-open model is `GasTurbine_PILCO_Smooth_PI.mdl`; its preload callback loads the frozen selected policy. Run `check_pilco_smooth_mc.py` for the particle cross-check and the report/video builders for final artifacts. Outputs are in `results/pilco_smooth_20261010`. This is an overlay for the existing T-MATS workspace and adjacent library; earlier studies remain unchanged.

Chapter 12

Frozen-controller acceleration and the combined cycle

These tests change the operating task after the earlier local controller development. The no-load ramp and common-sensor audit distinguish actual mechanical speed from the feedback measurement. The first combined cycle is a transfer test for the earlier frozen controllers; the same cycle later becomes a selection case for DOB98. Its evidential role therefore depends on the controller being discussed.

12.1 Five-second no-load acceleration and the common-sensor audit

Archived October study. All simulations, searches, and checks described below predate this report.

Chapter overview

Four existing controllers are tested on the same nonlinear plant with zero external shaft extraction. The target rises from 9000 to 10000 rpm in five seconds. Base PI has the lowest ramp tracking error in this experiment. Ringing-aware PILCO improves ramp tracking relative to the previous PILCO, but does not outperform base PI. This is a transfer test of frozen policies, not a new training result.

12.1.1 Identical-sensor audit and fresh rerun

All four models explicitly use the same T-MATS first-order speed-sensor library block: time constant 0.05 s, unity gain, limits 0–20000 rpm, initial value from the same plant initial speed, noise disabled, unity fault gain and zero drift. The sensor sample-time setting is the same plant solve interval. Model wiring assertions verify that both the base PI and each active advanced controller receive this sensor output directly. The sensor input comes from the same plant-output block in every model.

The earlier models already had these settings. The new simulations explicitly enforce them in separate model copies and reproduce the earlier speed, sensed-speed and fuel traces. Thus the prior ranking was not caused by unequal sensor dynamics. Controllers respond differently to the same lag. During a ramp, actual-speed error and sensed-speed error differ; the primary tables continue to score actual shaft speed. Individual audit JSON files and a sensor-comparison CSV accompany the results.

12.1.2 Reproducible test protocol

Rated speed is defined as $N_{100} = 10000$ rpm. With $\tau = t - 60$ s, the reference is

$$r(\tau) = 9000 + 200 \min\{5, \max\{0, \tau\}\} \quad \text{rpm}. \tag{12.1}$$

Thus the reference acceleration is 200 rpm/s, or two percentage points of rated speed per second. External PTO load is identically zero. The compressor still consumes shaft power: no external load does not mean zero internal torque.

The plant starts at 10000 rpm, ramps down during absolute time 5–10 s, and settles at 9000 rpm before the evaluation begins at 60 s. Controller handover is at 45 s. All four traces are within 0.1 rpm of 9000 rpm throughout the final five seconds before the test. The requested acceleration occurs at absolute time 60–65 s; simulation continues to 80.01 s. Sea-level, ISA-zero, Mach-zero settings are shared. The reference governor permits 200 rpm/s.

Logged samples are spaced by 0.015 s. The continuous target ramp lasts exactly 5 s; its first post-end sample is at $\tau = 5.01$ s. Maximum logged-reference discrepancy from the equation above is below 2×10^{-12} rpm. Every saved sample passes the existing model-validity flag, and every external-load sample is zero.

12.1.3 Controllers held fixed

Controller	K_p	K_i	K_d
Base PI	.025	.050	0
Retuned Poly2 + PID	.025	.140	.00125
Previous PILCO	.06460568	.17102610	0
Smooth PILCO	.02479762	.05385920	0

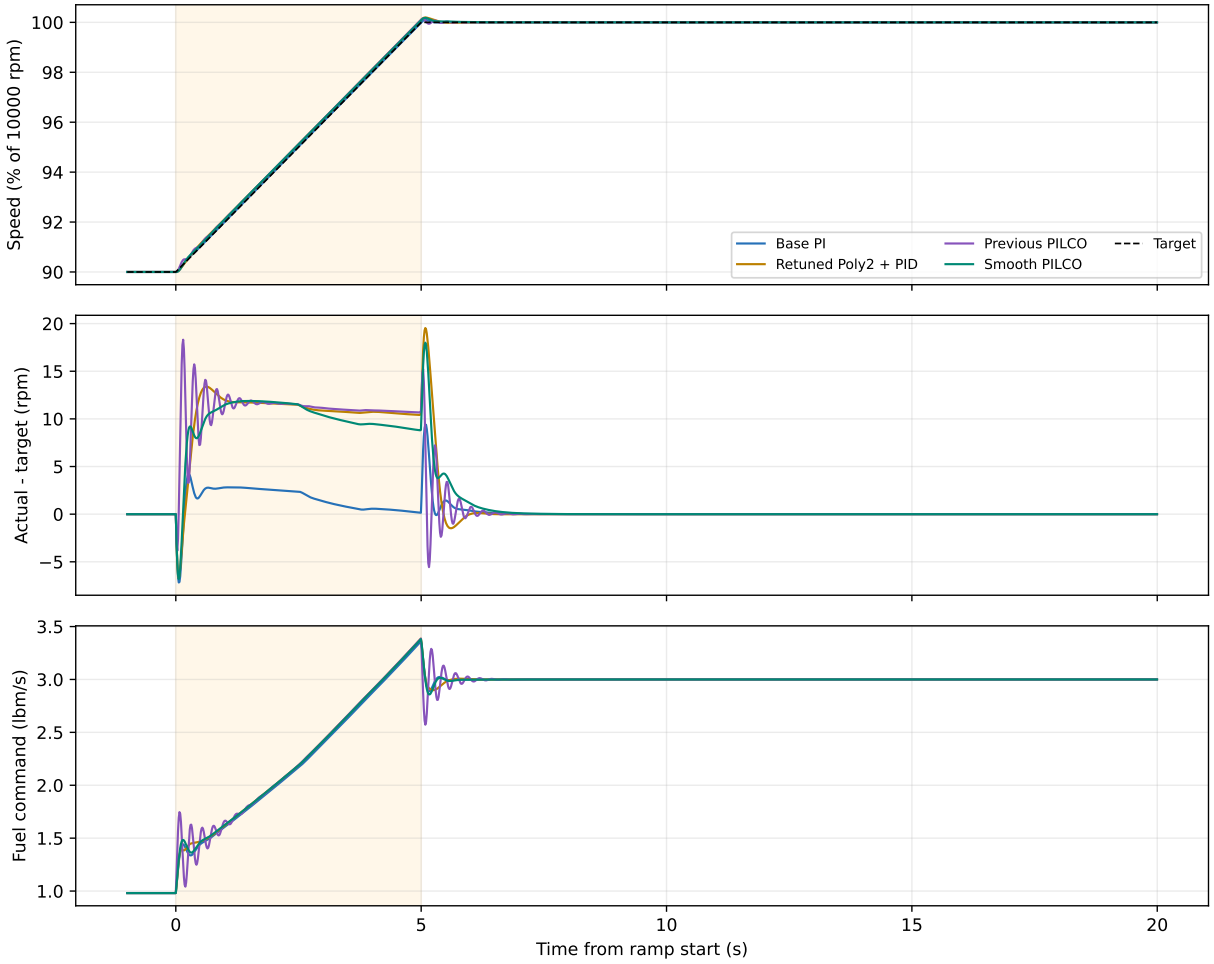
Both PILCO-style policies retain the prior quadratic feedforward, PI feedback, saturation and anti-windup implementation. Their nominal command structure is

$$u_k^* = p_2(r_k, 0, T_k, P_k) + K_p(r_k - N_{s,k}) + I_k, \tag{12.2}$$

$$u_k = \mathrm{clip}(u_k^*, 0.2, 4.0), \qquad I_{k+1} = I_k + hK_i(r_k - N_{s,k}), \tag{12.3}$$

with conditional integration at saturation. The zero second feedforward argument is retained from the prior controller, including during acceleration. No acceleration feedforward term is newly fitted for this test. The Poly2 PID comparator retains its previous inverse-model/derivative implementation; it is not replaced by the PILCO PI law.

12.1.4 Measured response



The pale band marks the five-second target ramp. Curves are raw recorded simulation traces; no display smoothing is used. Speed curves nearly overlap at the full-speed scale, so the signed error panel is needed to distinguish tracking performance.

12.1.5 Quantitative comparison

For each sample, $e_k = N_k - r_k$. Ramp RMSE uses $0 \le \tau < 5$ s; full RMSE uses $0 \le \tau \le 20.01$ s. Settling is measured after ramp completion: the first recorded time after which every remaining sample stays within ± 1 rpm of 10000 rpm, minus 5 s. It is a finite-record measurement.

Controller	Ramp RMSE rpm	Full RMSE rpm	Peak $\lvert e \rvert$ rpm	Settle s after ramp
Base PI	2.161	1.309	9.42	0.610
Retuned Poly2 + PID	10.949	5.799	19.54	0.760
Previous PILCO	11.224	5.710	18.31	0.820
Smooth PILCO	10.252	5.392	18.00	1.045

Fuel excess total variation is

$$R_u = \sum_{k=a+1}^{b} |u_k - u_{k-1}| - |u_b - u_a|. \tag{12.4}$$

It measures movement beyond net endpoint change, in lbm/s, not fuel consumption. For acceleration, some excess variation is physically useful: fuel may rise to accelerate and then drop to its steady value. Therefore the whole-test measure is not a pure measure of unwanted oscillation. A separate post-ramp value is included. These intervals differ from the earlier load-pulse study; their totals must not be compared directly.

Controller	Whole R_u lbm/s	Post-ramp R_u lbm/s	Overshoot rpm	Min. surge %
Base PI	1.2188	0.2895	9.42	13.98
Retuned Poly2 + PID	1.0004	0.2189	19.54	13.74
Previous PILCO	6.2413	2.6423	15.23	13.74
Smooth PILCO	1.3489	0.3454	18.00	13.78

12.1.6 Interpretation and limits

Base PI gives the smallest ramp and full-evaluation RMSE. The smooth PILCO ramp RMSE is 10.252 rpm versus 11.224 rpm for the previous PILCO, an 8.7% reduction. This does not establish a universal ranking: the earlier load-disturbance objective and the present acceleration objective are different.

PILCO gains were learned around 9250 rpm under load pulses. This ramp moves beyond that local operating condition; portions also lie beyond the approximately 9264–9750 rpm polynomial fitting range. The present result is evidence for this single deterministic nominal simulation, not a robustness guarantee. It does not validate the Gaussian-process model across the entire ramp or prove that a ringing penalty alone explains all differences between the gain sets.

The previous derivation remains applicable to how the gains were obtained: expected tracking and fuel-magnitude costs were augmented by expected excess fuel variation, using previous-command and segment-anchor memory. This run does not retrain that objective. A dedicated acceleration-training study would be needed to claim optimized ramp performance.

12.1.7 Reproduction

Run `ramp90_equal_sensor_batch.m` in MATLAB with the existing T-MATS dependencies. The four `GasTurbine_EqualSensor_*.mdl` copies preserve previous studies. Raw CSV/MAT traces and summary metrics are in `results/pilco_ramp90_equal_sensor_20261010`. Run `build_ramp90_equal_sensor.py` for this analysis and `build_ramp90_equal_sensor.py` for the synchronized video.

12.2 Combined acceleration and a five-second load pulse

Archived October study. All simulations, searches, and checks described below predate this report.

12.2.1 Requested cycle

Define relative time $\tau = t - 60$ s. The shaft-speed target and external load are

$$r(\tau) = 9000 + 200 \min(5, \max(0, \tau)) \quad \text{rpm}, \tag{12.5}$$

$$P_L(\tau) = \begin{cases} 0.1 P_{\text{ref}}, & 10 \le \tau < 15, \\ 0, & \text{otherwise.} \end{cases} \tag{12.6}$$

The speed target ramps from 90% to 100% in five seconds, then remains at 10000 rpm. A common load is applied at second 10 and removed at second 15. Recovery is recorded through second 25.005. The five-second pulse is programmed as a zero-order-held workspace signal, with interpolation disabled.

The reference is the mean turbine shaft power at settled 10000 rpm from seconds 8–10 of the preceding no-load base-PI run: $P_{\text{ref}} = 27896.501$ hp. Therefore every controller receives exactly the same pulse amplitude, 2789.650 hp. This is 10% of that common turbine-power reference, not 10% of each controller's instantaneous power. Internal compressor power is still present.

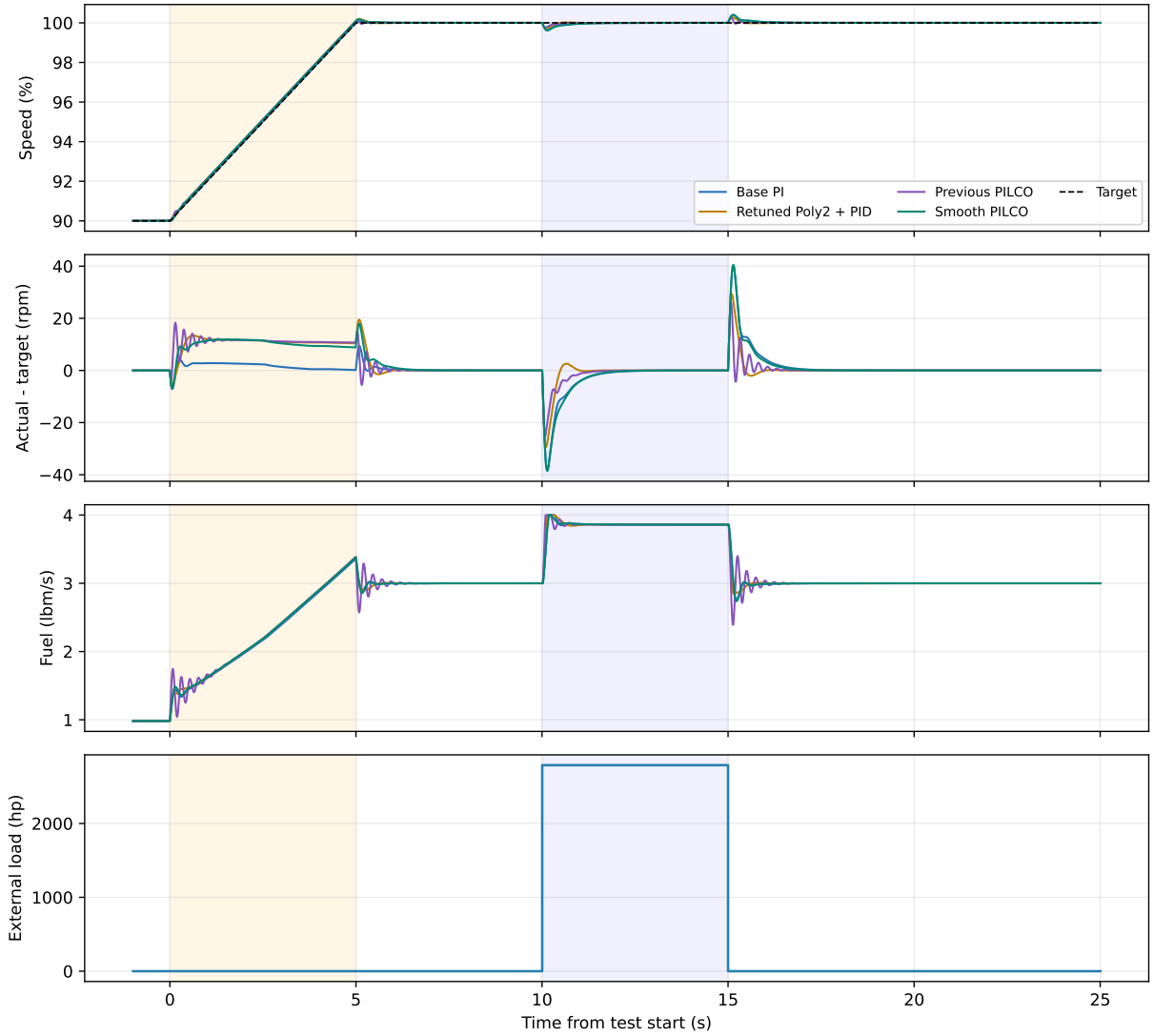
The preparation stage begins at 10000 rpm, moves to 9000 rpm during absolute seconds 5–10, and settles before the cycle starts at absolute second 60. All cases use sea-level ISA-zero, Mach-zero conditions. Controller gains remain fixed. All four sensors have a 0.05 s time constant, unity gain, identical limits and initial conditions, and noise disabled. Wiring assertions confirm that each active controller receives the sensor output.

The model sample interval is 0.015 s. The first logged loaded sample is at relative 10.005 s, and the first unloaded sample is at 15.000 s. The programmed event interval is exactly 10–15 s; observed event timing is resolved on the sample grid. All simulations pass the existing model-validity checks.

12.2.2 Comparison definitions

Error means actual shaft speed minus target, $e_k = N_k - r_k$. Ramp RMSE uses $0 \le \tau < 5$ s; load/recovery RMSE uses $10 \le \tau \le 25.005$ s; full-cycle RMSE uses $0 \le \tau \le 25.005$ s. These are sample-weighted metrics. Scoring the two phases separately prevents the long recovery interval from concealing ramp differences.

12.2.3 Recorded responses



Orange shading marks the target ramp; blue shading marks the external-load pulse. These are raw nonlinear simulation traces, with no visual smoothing.

12.2.4 Numerical results

Controller	Ramp RMSE rpm	Load/recovery RMSE rpm	Full RMSE rpm
Base PI	2.161	6.815	5.408
Retuned Poly2 + PID	10.949	4.416	6.214
Previous PILCO	11.224	3.393	5.745
Smooth PILCO	10.252	6.913	7.207

Controller	Load peak $\lvert e\rvert$ rpm	Fuel excess TV lbm/s	Min. surge %	Saturation % samples
Base PI	40.13	0.945	7.64	0.7
Retuned Poly2 + PID	29.39	0.696	7.75	0.1
Previous PILCO	26.03	4.489	7.77	0.7
Smooth PILCO	40.47	0.919	7.63	0.2

Fuel excess total variation is $\sum |\Delta u| - |u_{\text{end}} - u_{\text{start}}|$, computed separately over 10–15 s and 15–25.005 s and then summed. It describes excess command movement, not fuel consumption. Minimum surge margin and fuel saturation are evaluated over the full cycle.

The previous and smooth PILCO policies use quadratic setpoint feedforward plus PI feedback. Retuned Poly2 retains its inverse-model/PID implementation; base PI remains the original baseline. No gains, feedforward coefficients or ringing weights were fitted to this combined cycle. These are nominal transfer-test results, not proof of robustness or global superiority. In particular, all controllers share the same sensor dynamics; differences in actual-speed tracking must not be attributed to unequal sensor time constants.

12.2.5 Reproduction and saved outputs

Run `combined_cycle_batch.m` in the existing MATLAB/T-MATS workspace. New models are named `GasTurbine_Combined_*.mdl`; earlier studies are preserved. Raw traces, model audit JSON, numerical metrics and the exact protocol are saved under `results/pilco_combined_cycle_20261010`. Analysis and video builders are `build_combined_cycle.py` and `build_combined_video.py`.

Chapter 13

Local DOB98 design, PI retuning, and attribution

The observer is identified around a physical speed of 9800 rpm; the number 98 does not describe model-fit accuracy. The independent multisine assesses the local predictor. The combined ramp/load cycle selects PI gains with the observer enabled and must be reported as selection evidence. The original-gain observer result is retained as a partial attribution check.

13.1 A new inverse model at 9800 rpm and PI tuning after DOB engagement

Archived October study. All simulations, searches, and checks described below predate this report.

13.1.1 Scope and identification experiment

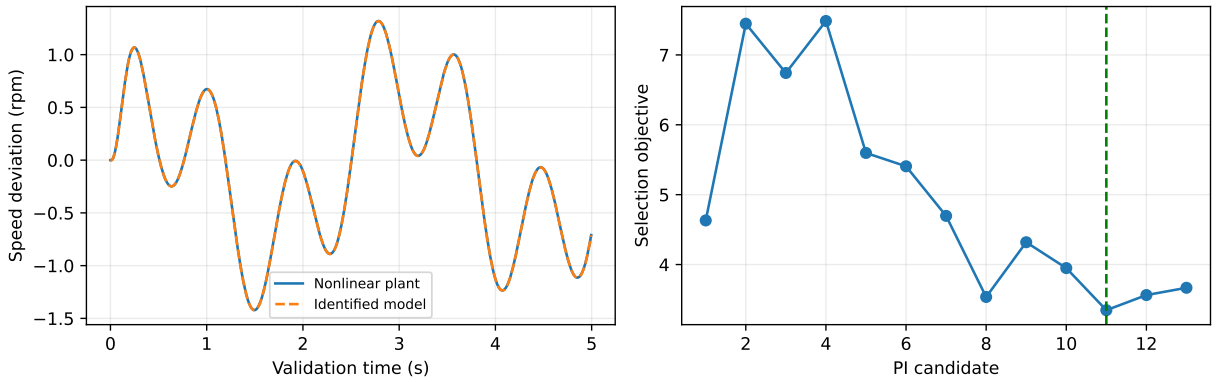
Previous PILCO is replaced in this comparison by DOB98 + tuned PI. Base PI, retuned Poly2 + PID and smooth PILCO retain their earlier gains and traces. The new controller uses no quadratic feedforward. Its structure is PI minus a filtered input-disturbance estimate, followed by the common fuel limits.

A new nominal model is identified specifically at 9800 rpm (98% rated speed), with zero external shaft load. After preparation, training applies a deterministic binary fuel perturbation of amplitude 0.035 lbm/s, switched approximately every 0.18 s for 30 s. The base PI stabilizes the experiment; identification uses actual applied fuel, including the excitation, and actual shaft speed. Independent validation uses the sum of 0.37 and 1.17 Hz sinusoids, with amplitudes 0.025 and 0.015 lbm/s. No validation samples enter the least-squares fit.

Let $y_k = N_k - 9800$ and $v_k = u_k - u_0$, with $u_0 = 2.530351198$ lbm/s and $T_s = 0.015$ s. A parsimonious local model is

$$y_{k+1} = a y_k + b v_k, \qquad a = 0.9789097769, \quad b = 9.3039586704. \tag{13.1}$$

The parameters minimize the squared one-step residual over the excited training interval. The identified pole is stable, and the numerator has no finite nonminimum-phase zero. The model is local: the training speed varies only a few rpm around 9800. Independent free-run fit is 99.7829%, with 0.001406 rpm RMSE. Fit means $100(1 - \|y - \hat{y}\|/\|y - \bar{y}\|)$.



13.1.2 Causal filtered inverse, step by step

1. Nominal speed dynamics. With $q = z^{-1}$ denoting a one-sample delay,

$$P_n(q) = \frac{bq}{1 - aq}. \tag{13.2}$$

2. Include the sensor. Every controller retains the same physical sensor: time constant 0.05 s, unity gain, common initial conditions and noise disabled. The DOB uses its discrete approximation

$$H_n(q) = \frac{(1-p)q}{1-pq}, \qquad p = e^{-T_s/0.05}. \tag{13.3}$$

Thus the inverse is formed for sensed-speed dynamics P_nH_n. This approximation does not change the physical sensor blocks.

3. Make the inverse causal and filtered. Choose a unity-DC, two-pole low-pass filter with two delays:

$$Q(q) = \frac{(1-r)^2 q^2}{(1-rq)^2}, \qquad r = e^{-2\pi f_c T_s}, \quad f_c = 1 \text{ Hz}. \tag{13.4}$$

Then the two nominal delays cancel analytically:

$$\frac{Q}{P_nH_n} = \frac{(1-r)^2(1-aq)(1-pq)}{b(1-p)(1-rq)^2}. \tag{13.5}$$

This is proper, causal and stable. A numerical frequency-response check verifies the identity, and an independent Python filter replay is compared with the recorded Simulink observer output.

4. Estimate input-equivalent disturbance. Use the known saturated command, without access to the external load signal:

$$\hat{d}_{\text{raw}} = \frac{Q}{P_nH_n}(N_s - N_0) - Q(u_{\text{cmd}} - u_0). \tag{13.6}$$

The last estimate before absolute time 45 s is latched as a trim bias d_0. Compensation engages over the following two seconds, before the evaluation starts at 60 s:

$$c = \text{clip}\{0.7(\hat{d}_{\text{raw}} - d_0), -1, 1\}, \qquad u_{\text{cmd}} = \text{clip}\{u_{\text{PI}} - c, 0.2, 4\}. \tag{13.7}$$

The compensation and fuel-command bounds are in lbm/s. The observer gain 0.7, cutoff 1 Hz and compensation limit 1 lbm/s are fixed engineering choices during the PI search; they are not claimed optimal.

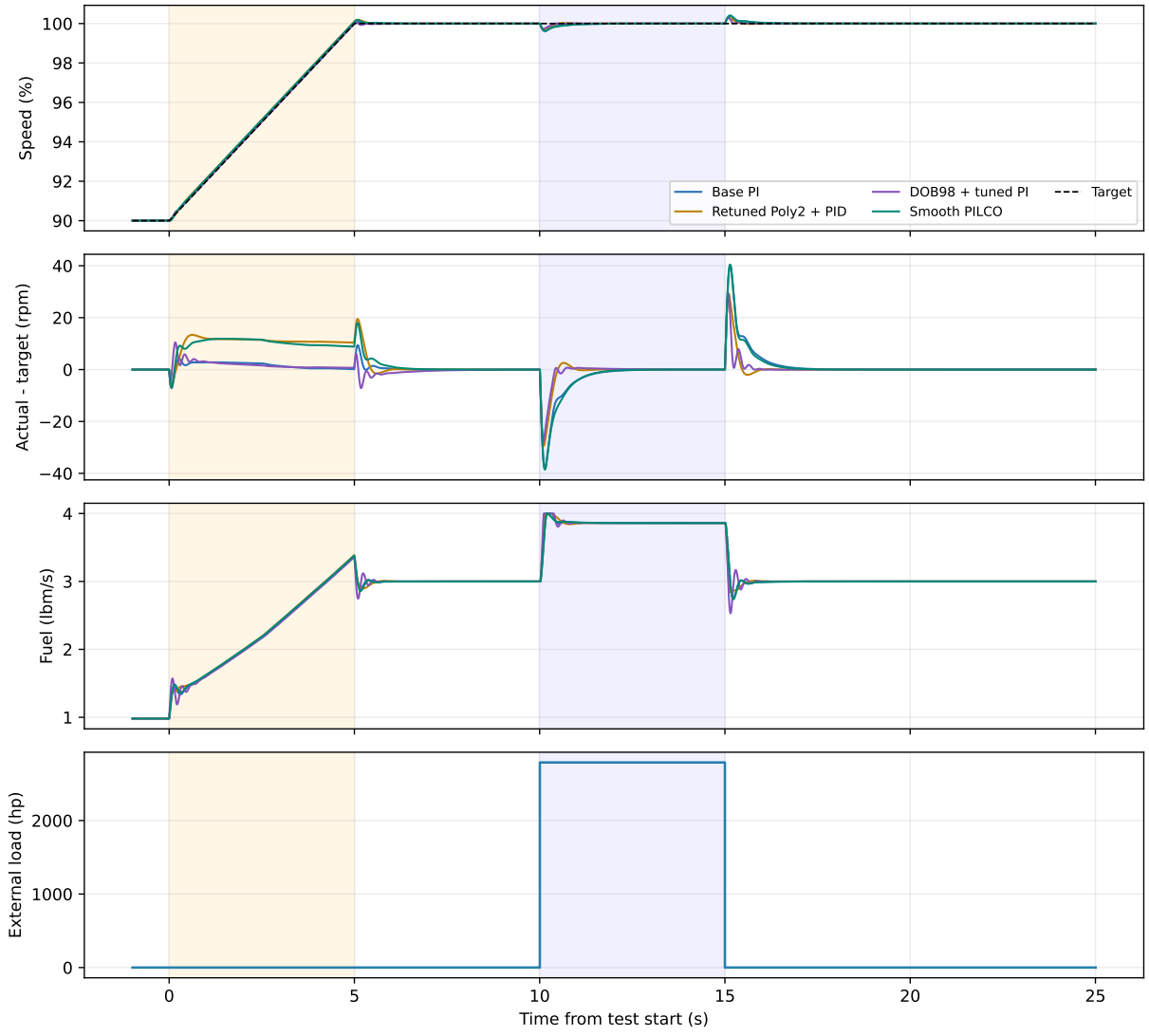
5. Tune PI after enabling the DOB. The existing PI block uses $K_p(r - N_s) + K_i \int (r - N_s)dt$. Nine gain pairs, including the original (.025,.05), are followed by four local refinements. All candidates are evaluated with the DOB enabled on the requested combined cycle. Selection minimizes

$$J = \mathrm{RMSE}_{0:25.005} + 0.05 R_{\mathrm{seg}}, \tag{13.8}$$

where R_{seg} sums excess fuel total variation over 0–5, 5–10, 10–15 and 15–25.005 s. The weight is a numerical tuning weight with units rpm/(lbm/s). Invalid trajectories are rejected. This finite search selects $K_p = 0.048$ and $K_i = 0.05$; it does not establish a global optimum.

13.1.3 Same combined cycle, revised four-controller comparison

The target ramps from 9000 to 10000 rpm during relative seconds 0–5. A common 2789.650 hp external load is applied at second 10 and removed at second 15, followed by recovery through 25.005 s. The load is 10% of the previously measured settled turbine power at 10000 rpm. Physical plant, sensor settings, reference, external load and fuel limits are matched. Saved comparator traces are reused because their implementations are unchanged.



13.1.4 Results and limits

Controller	Ramp RMSE rpm	Load/recovery RMSE rpm	Full RMSE rpm
Base PI	2.161	6.815	5.408
Retuned Poly2 + PID	10.949	4.416	6.214
DOB98 + tuned PI	2.459	3.710	3.142
Smooth PILCO	10.252	6.913	7.207

Controller	Load peak $\|e\|$ rpm	Load excess TV lbm/s	Min. surge %
Base PI	40.13	0.945	7.64
Retuned Poly2 + PID	29.39	0.696	7.75
DOB98 + tuned PI	29.10	2.184	7.74
Smooth PILCO	40.47	0.919	7.63

Errors use actual shaft speed, not sensed speed. Ramp RMSE covers 0–5 s; load/recovery RMSE covers 10–25.005 s; full RMSE covers the entire cycle. Load excess TV sums the 10–15 and 15–25.005 s segments. Sample interval is 0.015 s; first logged load-on is 10.005 s, and load-off is 15.000 s.

Contribution of each change. Base PI full-cycle RMSE is 5.408 rpm. Adding the DOB while retaining the original PI gains gives 4.546 rpm. Retuning PI with the DOB enabled gives 3.142 rpm. This ablation separates observer addition from gain tuning.

Independent scenario check. The selected gains are frozen and tested on a four-second speed ramp with a 5% load pulse at seconds 10–15. Full-cycle RMSE is 2.024 rpm; minimum surge margin is 11.14%. This check does not enter selection and is not a comparison against retuned baselines on that second scenario.

The requested combined cycle is a tuning case for DOB+PI, while the other gains were frozen from earlier studies. Consequently this is a comparison of the resulting controllers, not an equal-budget optimization contest. A local nominal model plus a DOB can tolerate some mismatch, but the small-excitation fit alone does not prove accuracy over 9000–10000 rpm. These nominal simulations are not a robust-stability proof. Fuel limits and compensation limits remain active parts of the implementation; the CSV records saturation and observer traces.

13.1.5 Reproduction

Run `dob98_identify_batch.m`, then `fit_dob98_model.py`, then `dob98_tune_batch.m`. The saved controller is `GasTurbine_DOB98_PI.mdl`, initialized by `tmats_dob98_setup.m`. Run `validate_dob98.py` for independent replay checks and `build_dob98_deliverables.py` for the revised figures and video. Data and selected parameters are under `results/dob98_pi_20261010`. Existing T-MATS workspace dependencies are required.

13.2 Detailed interpretation of the final sensor and DOB results

13.2.1 Actual and sensed tracking answer different questions

The sensor audit gives a particularly clear matched comparison because every maximum difference between the old and explicitly equalized trace is zero for actual speed, sensed speed, and fuel. The audit therefore rules out different sensor settings as the reason for the observed ordering. It does not imply that actual and sensed tracking errors are interchangeable. The original PILCO policy is

Controller	Actual-speed ramp RMSE rpm	Sensed-speed ramp RMSE rpm
Base PI	2.160858	10.051133
Retuned Poly2 + PID	10.948833	3.198864
Original PILCO-style PI	11.223721	1.547154
Smooth PILCO	10.251905	3.081877

Table 13.1: The same ramp scored using different signals. Values are from the archived sensor-comparison CSV; no new simulation is involved.

closest to the target in sensed-speed RMSE and farthest in actual-speed RMSE among these four. A reader examining only the feedback error could therefore reach a different conclusion from the primary actual-speed comparison. For physical shaft regulation, the actual quantity is the relevant evaluation signal in this ideal simulation. In hardware it would need an independently justified measurement or state estimate.

13.2.2 Deriving the local 9800-rpm identification explicitly

Define $y_k = N_k - N_0$ and $v_k = u_k - u_0$. For each accepted identification transition, the equation $y_{k+1} = ay_k + bv_k + \epsilon_k$ supplies one row of the regression matrix:

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix} y_0 & v_0 \\ y_1 & v_1 \\ \vdots & \vdots \\ y_{M-1} & v_{M-1} \end{bmatrix}}_{X} \underbrace{\begin{bmatrix} a \\ b \end{bmatrix}}_{\theta} + \epsilon. \tag{13.9}$$

Least squares chooses $\hat{\theta}$ to minimize $\|Y - X\theta\|^2$. The formula $(X^\top X)^{-1}X^\top Y$ explains the solution when full rank holds; numerically, a linear least-squares solver avoids unnecessary explicit inversion. The input must be the applied fuel including the excitation, because identifying against the injected perturbation alone would omit the stabilizing PI action.

One-step prediction uses the true y_k in each row. Free-run prediction starts from an initial value and recursively uses $\hat{y}_k$. The archived model reports training one-step RMSE 1.7911×10^{-4} rpm and free-run RMSE 0.003258 rpm. On the separate multisine record these are 9.8925×10^{-5} and 0.001406 rpm. The small errors describe a speed range of only 9796.735–9803.056 rpm. A tiny local prediction error is entirely compatible with substantial mismatch at 9000 rpm or under a large external load.

13.2.3 A directly implementable filtered inverse

Write $r_f = e^{-2\pi f_c T_s}$ for the filter pole, to avoid confusing it with the speed reference, and define

$$c_f = \frac{(1-r_f)^2}{b(1-p)}, \quad F(q) = \frac{Q}{P_n H} = c_f \frac{1-(a+p)q+apq^2}{1-2r_f q+r_f^2 q^2}. \tag{13.10}$$

If $w_k = N_{s,k} - N_0$ and $z = Fw$, then

$$z_k = 2r_f z_{k-1} - r_f^2 z_{k-2} + c_f[w_k - (a+p)w_{k-1} + apw_{k-2}]. \tag{13.11}$$

The parallel filtered-command branch $h = Qv$ follows

$$h_k = 2r_f h_{k-1} - r_f^2 h_{k-2} + (1-r_f)^2 v_{k-2}. \tag{13.12}$$

Thus the raw estimate is $z_k - h_k$. These recurrences explain why the two nominal delays are necessary in Q: the speed-inverse branch can be calculated from current and previous measurements, while the command branch explicitly retains the two delays. The construction uses the known saturated command; it does not read the external load schedule. Its poles are the filter poles, inside the unit circle, after the analytical cancellation specified by the model.

The trim latch subtracts the settled bias at handover. It removes a constant initial mismatch; it does not adapt the nominal model as speed changes. The compensation gain 0.7 and bound of 1 lbm/s are fixed during the PI search. The recorded maximum compensation in the selected main run is about 0.428413 lbm/s, below this compensation bound. The fuel command can nevertheless saturate because its total includes PI action as well as observer correction.

13.2.4 What PI retuning bought and what it cost

Adding the observer with the original gains reduces full-cycle RMSE from 5.407708 to 4.546279 rpm. Retuning to $(K_p, K_i) = (0.048, 0.05)$ reduces it further to 3.142353 rpm. However, the original-gain DOB has ramp RMSE 1.441616 rpm, compared with 2.458988 rpm after retuning. The selected policy therefore trades some ramp performance for improved load/recovery behavior under the declared full-cycle objective.

The objective adds $0.05R_{\text{seg}}$ to full RMSE. Its excess-variation windows include both the ramp and load phases. The load-only excess TV in the comparison table is a different subtotal. For the selected candidate these values are 4.085550 and 2.184455 lbm/s respectively. Confusing them would make the objective appear inconsistent with the displayed comparison. The selected controller also spends 1.139% of full-cycle samples at a fuel limit, compared with 0.659% for base PI. Better tracking is accompanied by additional actuator activity.

13.2.5 The complete saved PI search

Every candidate in the saved table is numerically accepted. This table describes a finite nine-pair search plus four refinements. It is not a proof that no untested pair could improve the objective, nor an equal tuning-budget comparison with the frozen Poly2 and PILCO methods. Candidate 1 is particularly useful because it isolates adding the DOB while retaining the original PI gains.

Case	K_p	K_i	Full RMSE	Ramp RMSE	R_{seg}	Objective
1	0.025	0.050	4.5463	1.4416	1.6970	4.6311
2	0.015	0.025	7.3993	7.9607	0.9555	7.4470
3	0.015	0.050	6.6614	2.0759	1.6012	6.7414
4	0.015	0.100	7.2924	5.7493	3.8650	7.4857
5	0.025	0.025	5.5285	6.6983	1.3702	5.5970
6	0.025	0.100	5.2704	5.4682	2.7144	5.4061
7	0.040	0.025	4.5578	5.3052	2.7683	4.6962
8	0.040	0.050	3.3854	2.0343	3.0062	3.5357
9	0.040	0.100	4.1326	5.5323	3.7298	4.3191
10	0.032	0.050	3.8400	1.6350	2.1835	3.9492
11	0.048	0.050	3.1424	2.4590	4.0855	3.3466
12	0.040	0.040	3.4183	1.7692	2.8927	3.5630
13	0.040	0.060	3.5107	2.9731	3.1287	3.6672

Table 13.2: All archived DOB98 PI candidates. RMSE is rpm; R_{seg} is lbm/s. Candidate 11 is selected by the stated objective.

13.2.6 The independent scenario and replay checks

The frozen selected controller is then evaluated with a four-second 9000–10000 rpm ramp, corresponding to 250 rpm/s, and a 5% load pulse. Its full RMSE is 2.024232 rpm, ramp RMSE 3.289699 rpm, and load/recovery RMSE 1.702970 rpm; minimum recorded surge margin is 11.135%. The ramp metric uses the four-second interval, not the five-second interval of the tuning case. Other controller families were not retuned and compared on this independent scenario, so this result supports transfer of the selected controller but not a new ranking.

Independent filter replay matches the recorded observer branch within approximately 2.4×10^{-13} lbm/s. The nominal filtered-inverse identity has maximum numerical error around 6.5×10^{-14}. These checks support correct coefficient ordering, delay handling, and filter realization. They do not establish uncertainty calibration, noise robustness, or a global stability margin for the nonlinear plant. Their proper role is to remove implementation ambiguity before interpreting the accepted plant trajectories.

Chapter 14

Cross-study answers, attribution, and validity

14.1 RQ1: prediction quality is necessary for some uses, but not sufficient

The identification studies distinguish three information patterns. An online one-step predictor is corrected by each newly observed output. A frozen free-running model must propagate its own past predictions. An inverse controller additionally needs suitable zeros, local sensitivity and signal timing. The same numerical fit cannot establish all three properties.

The ten-term RLS has a reported pre-update one-step fit of 99.8729 percent, but frozen held-out ramp fit is 62.6191 percent. The sparse model improves the held-out result to 65.2666 percent and lowers the reported regression conditioning problem, but it does not remove the gap between online prediction and free-running transfer. The raw acceleration model's near-unit nonminimum-phase zero also makes inversion a separate problem from prediction. The physical equilibrium correction and causal filter are therefore substantive design steps, not details justified by a high fit score.

For nominal GP regression, accurate output prediction again fails to guarantee good feedback. An inverse's derivative controls how a small speed or acceleration deviation changes command. Two models can give close predictions at sampled points while having different local slopes between them. In this archive, weak or negative slopes motivate normalization, guards and fallback. This establishes a control-design concern in the tested inverse, not a theorem that GPs have poor derivatives.

The DOB98 local model supplies an especially clear scope example. Its independent free-run error is about 0.001406 rpm, but training speed covers only 9796.735–9803.056 rpm. A 9000–10000 rpm ramp spans a much larger region. The combined-cycle success is an observed closed-loop result; the local fit cannot be converted into a global model-error bound.

14.2 RQ2: what the existing ablations identify

14.2.1 Constant feedforward and integral state

At fixed reference and ambient conditions, let c be a constant feedforward value. Compare $u_A = K_p e + I_A$ with $u_B = c + K_p e + I_B$. If $I_B(0) = I_A(0) - c$ and both integrators follow the same update, then $u_A = u_B$ for every sample for which the same conditional-integration decisions apply. Equal commands imply equal plant responses, preserving the equality inductively under matched implementations.

This is a coordinate shift in controller memory, not extra disturbance information. It explains the same-gain FF ablations. When feedforward becomes time varying, the shifted integral would require an additional compensating update, and the equivalence no longer follows automatically.

The covered-cycle FF result must therefore be considered separately from fixed-reference load rejection.

14.2.2 Inverse coordinates and equivalent PID action

Locally, an inverse difference behaves approximately as $g_N e - g_a \hat{a}$ at a constant reference. PI action on that difference creates proportional, integral and acceleration-sensitive terms. Under the stated linearization, the effective coefficients include

$$K_{p,\text{eff}} = K_p g_N + K_i g_a, \quad K_{i,\text{eff}} = K_i g_N, \quad K_{d,\text{eff}} = K_p g_a. \tag{14.1}$$

The exact deployed realization also contains filtering, delay, saturation and possibly derivative variation, so these coefficients are a local explanation. They show why changing inverse slopes changes feedback behavior even if nominal fuel prediction remains accurate.

At the R3 target, RMSE is 1.75198 rpm and ringing is 1.45899; the matched filtered PID gives 1.74912 rpm and 1.42916. The correct conclusion is that no additional GP-specific advantage is demonstrated at that target. The environmental transfer record remains useful, but it does not erase this attribution result. A GP controller can perform well without its nonlinear representation being the cause of the local improvement.

14.2.3 Changing representation before retuning

On the covered cycle, the polynomial replacement gives full RMSE of about 0.676972 rpm compared with 0.702711 rpm for the GP version under the retained architecture. Base PI remains lower at about 0.591513 rpm. The polynomial's poorer regression score therefore does not prevent a competitive controller configuration. This is evidence against using nominal prediction error alone as a controller-selection criterion. It is not proof of a universal polynomial advantage or a statistical population difference.

The later 9250-rpm retune is a different comparison because gains change and the target is slightly below the old nominal training-speed range. Its lower local RMSE and ringing must be read alongside its larger peak error relative to original Poly2 and higher peak fuel slew than base PI. The fixed-gain substitution and the later tuning result should not be merged into one isolated "model simplification benefit."

14.2.4 Observer addition and PI retuning

The final combined-cycle results permit the following sequential comparison:

Configuration	Full RMSE (rpm)	Ramp RMSE (rpm)
Inherited PI	5.407708	2.160858
DOB98 with original PI	4.546279	1.441616
DOB98 with selected PI	3.142353	2.458988

The first change is a reduction of about 15.9 percent in full RMSE; the next is about 30.9 percent relative to the original-gain DOB result. The final reduction relative to inherited PI is about 41.9 percent. These percentages have different denominators and must not be added. They describe the

recorded path through configurations, not independent causal shares that would remain unchanged under another gain choice.

An equally tuned PI without DOB is missing. Consequently, the evidence does not answer whether a comparable conventional gain search could achieve the same full-cycle result. The observer improves the inherited-gain branch, and tuning improves the observer branch, but the interaction between architecture and tuning is not fully identified.

14.3 RQ3: objective choice and sensing change the result

14.3.1 Repairing a policy is not baseline dominance

On the 10 percent pulse case, original PILCO-style PI has RMSE about 1.19728 rpm and ringing 14.82523. The smooth version has RMSE 2.38773 rpm and ringing 2.08375. This is a large reduction in reversal, about 85.9 percent, purchased with worse tracking than the aggressive policy. Base PI ringing is approximately 2.07023. The smooth version therefore repairs the original policy's excessive reversal without establishing dominance over the conventional baseline.

The selected smooth gains, approximately 0.02480 and 0.05386, are close to the inherited PI values 0.025 and 0.05. This is consistent with the regularizer favoring a less aggressive controller. It does not prove that the probabilistic optimizer was unnecessary, because an equally budgeted direct search was not compared. It does limit a claim that the final behavior demonstrates an intrinsically superior learned control law.

The expectation calculation is exact for an absolute value of a Gaussian scalar with the supplied moments. The propagated trajectory distribution is an approximation, and the deployed controller includes saturation absent from the affine planning policy. "Analytic expected penalty" must therefore not be interpreted as exact expected fuel behavior of the nonlinear plant.

14.3.2 Actual speed versus feedback measurement

The no-load ramp is 200 rpm/s. A first-order sensor with time constant 0.05 s has a continuous steady-ramp lag of approximately 10 rpm; discrete timing contributes to the observed signal difference. A controller that keeps the lagged measurement close to the moving target may place actual speed ahead of it. Thus low sensed-error RMSE does not automatically mean low actual-error RMSE.

The archived actual/sensed ramp RMSE pairs are 2.160858/10.051133 rpm for base PI and 11.223721/1.547154 rpm for original PILCO-style PI. They order those controllers in opposite ways. The explicit equal-sensor rerun produces zero maximum trace differences for actual speed, sensed speed and fuel, so unequal sensor settings are ruled out as the explanation for the previous ordering.

The hypothesis that the base controller's response partly offsets sensor lag is consistent with this evidence. A zero-lag or compensated-sensor ablation was not performed. The causal role of that cancellation therefore remains an interpretation, not a completed isolated experiment. Version 2 makes this distinction explicit.

14.3.3 No single actuator metric tells the whole story

The final DOB's load-phase excess variation is 2.184455 compared with base PI's 0.944543, approximately a 131 percent increase. Its full segmented value is 4.085550, a different quantity used in the selection objective. Fuel-limit occupancy also increases from 0.659 to 1.139 percent. Lower full-cycle speed RMSE is therefore not evidence of uniformly smoother or less demanding actuation.

These numbers do not establish actuator damage. No actuator model or endurance experiment connects this metric directly to wear. They establish increased command activity under the stated definition and motivate reporting multiple constraints before declaring a practical winner.

14.4 RQ4: transfer is limited and multidimensional

The environmental grid, reduced-amplitude pulses and later ramps are useful precisely because they differ from local development cases. However, the broad cycle exhibits extensive out-of-domain and fallback exposure. The subsequently covered cycle is an informative revised scenario, not a test selected before the broad-cycle outcome was known. Its better coverage does not imply that the final 9250-rpm gains or the later DOB98 inherit validation over the entire grid.

The final DOB frozen-gain four-second ramp/5 percent pulse check supplies a second scenario. Because the complete comparator lineup was not evaluated there, it supports successful transfer of that controller configuration rather than superior transfer relative to all alternatives. The local predictor's independent multisine is also a different validation target: it tests local model prediction, not the controller's wide-range ranking.

No result in this archive tests sensor bias, noisy differentiation, uncertain component maps or a new engine family in a controlled campaign. The correct answer to RQ4 is therefore a map of demonstrated local transfers and known limits, not a single yes/no claim of robustness.

14.5 Threats to validity and the remaining effect on claims

Threat	Existing mitigation	Remaining consequence
Unequal tuning opportunity	Original-gain DOB, unchanged-gain model substitution and matched PID retained	No general architectural superiority claim.
Selection on reported scenario	Explicit evidence-role ledger; held-out amplitude records distinguished	Main selection scores remain optimistic for generalization.
Narrow model support	Speed ranges, ambient coverage and fallback recorded	No global error or stability guarantee.
Measurement/metric ambiguity	Actual/sensed audit; common time step and local windows	Sensor-cancellation mechanism not isolated by a new ablation.
Numerical invalidity	Residual/iteration checks; rejected cases preserved	Validity conditional on solver and model conventions.
Statistical dependence	No pseudo-replicate confidence intervals or significance claims	Small numerical differences are descriptive.

Threat	Existing mitigation	Remaining consequence
Missing physical uncertainty	Explicit exclusion of noise, actuator and deterioration robustness claims	Nominal simulation cannot qualify hardware performance.
AI-assisted authorship and review	Disclosure, source preservation, defence questions	Independent mastery and acceptance require human examination.

Chapter 15

Conclusions, practical interpretation, and future work

15.1 Answer to the primary question

Identified and learned models improve some tested configurations, but their benefit is conditional on how the model is used, which gains are selected, which scenario is applied and which signal is scored. The archive does not support one controller ranking across all tasks. It also does not support attributing every improvement over inherited PI to a learned representation.

The most discriminating existing evidence points to feedback architecture and tuning as major explanations. Constant-reference feedforward can be absorbed into an integral offset. A matched filtered PID reproduces the R3 target behavior. A polynomial inverse remains competitive despite worse nominal prediction. The final DOB improves the inherited-gain configuration, while its subsequent PI search further improves the full score and worsens some other metrics. These observations support a comparative engineering conclusion, not a claim that learning or observer methods are universally ineffective.

15.2 Conclusions organized by contribution

C1: Evidence discipline. The revised ledger makes the data's role explicit for each model and controller. Numerically accepted trajectories, model coverage, parameter selection, frozen transfer and algebraic checks are treated separately. Failed runs remain visible. This enables a reader to determine what a result can support before interpreting its size.

C2: Attribution. Existing substitutions and ablations support several local mechanisms and expose missing comparisons. The thesis can explain why some apparent feedforward or GP benefits are not independently attributable to those elements. It cannot quantify an architecture-only improvement over an equally tuned PI for the final DOB case.

C3: Performance tradeoffs. Lower tracking error can accompany greater reversal, larger peak error, higher slew or more saturation. The no-load ramp also demonstrates that actual and sensed tracking are different objectives. A controller must therefore be assessed against a stated application requirement, not a single favorable metric selected after seeing the result.

C4: Technical adaptation. The Gaussian expected excess-variation calculation provides a reproducible way to include a particular reversal objective in the restricted offline policy search. It substantially reduces reversal relative to the aggressive original policy. Its novelty and superiority over simpler smoothness objectives are not established, and zero excess variation is not a bound on slew.

15.3 Implications for choosing among the existing configurations

For the recorded five-second no-load ramp scored on actual speed, inherited PI has the lowest error among the tested frozen policies. For the 9250-rpm load experiments, retuned Poly2 offers a useful local tracking/reversal tradeoff. Original PILCO-style PI emphasizes tracking at a large reversal cost; the smooth version moves toward the baseline tradeoff. For the final combined cycle used in its tuning, DOB98 plus selected PI has the lowest full-cycle RMSE of the revised lineup, with higher command activity than base PI.

These statements identify suitable starting points for further engineering evaluation. They are not deployment recommendations for a real engine. If an application imposes a hard slew, peak-speed, actuator or protection constraint, the preferred configuration may change. The archive lacks a quantified real-world requirement against which to make that decision.

15.4 A future protocol for the unresolved architecture question

The first new study should answer the missing comparison, rather than introduce another controller name. Choose a common plant/sensor/actuator setup and a declared training set of ramp and load scenarios. Compare PI, filtered PID, DOB plus PI and the selected inverse architecture with comparable gain-search budgets. Define the objective, bounds, numerical acceptance and operating-domain criteria before examining test results. Tune each architecture on the training scenarios only.

Next, freeze every controller and evaluate a separate set of speeds, ramp rates, pulse amplitudes and ambient conditions. Include the same reference and applied load for every comparator and report both full and phase-specific errors. Save the true and sensed outputs, requested and applied fuel, controller states, clipping and convergence flags. Retain failed cases in the reported acceptance rate.

To isolate components, include observer off/on at matched gains and PI-only/DOB-plus-PI after separate equal-budget tuning. This creates the missing comparison between the effect of adding an observer and the best result achievable by tuning within each family. It does not require claiming that one gain pair is universally optimal.

These are proposed experiments. They were not performed for this version, and the present conclusions do not depend on pretending that they were.

15.5 A future protocol for the penalty and robustness questions

For the ringing objective, compare excess variation, ordinary total variation and squared command increments using the same policy class, dynamics data, optimizer budget and constraints. Preserve peak slew and saturation as separate measurements. Test whether an event-window penalty merely moves aggression outside the scored window. If uncertainty propagation is claimed to improve policy selection, compare it with a matched mean-dynamics planning procedure.

For robustness, introduce measurement noise, bias, actuator lag and plant perturbations separately before combining them. The disturbance observer's filter must be evaluated against the resulting noise-versus-rejection tradeoff. A local-model schedule or richer nominal model can then be compared with the single 9800-rpm model. Any stability analysis should state its model, uncertainty class, switching assumptions and saturation treatment explicitly.

Hardware or hardware-in-the-loop work would address additional implementation and physical questions. It is not automatically required to make the present descriptive findings valid; it is required before treating those findings as qualification of a physical installation. None of these future tasks is counted as a completed contribution here.

15.6 Master's revision status

Version 2 addresses the jury's documentary concerns by specifying one primary question, distinguishing established methods from contributions, classifying evidence roles, foregrounding ablations, and limiting claims. The detailed technical account is preserved in a thematic structure rather than presented as an automatic progression toward a superior controller.

The missing equal-budget experiments and physical robustness evidence remain limitations. Because universal superiority and robustness are no longer claimed, their absence does not contradict the revised scope. An actual master's examination must still evaluate the candidate's independent understanding, ownership of the work and compliance with institutional requirements. This document is a revised submission candidate, not a self-issued acceptance certificate.

Appendix A

Response to the jury-style examination

A.1 Purpose and status conventions

The response addresses the written assessment dated 11 October 2026 [19]. "Addressed in the manuscript" means the framing, explanation or reporting has been changed. "Claim narrowed" means missing evidence is explicitly acknowledged and the stronger claim is not made. "Requires examination" means the issue cannot be resolved by writing a report. No status implies an institutional degree decision.

Jury issue	Revision and evidence	Status	Location
Unfocused research question	One primary question with four answerable subsidiary questions; contributions C1–C4 bounded to the archive	Addressed in manuscript	Chs. 1, 14
Literature gap and novelty unclear	Targeted primary-source comparison; established methods separated from application findings and unestablished penalty originality	Addressed; no first-in-literature claim	Ch. 2
Unequal baseline tuning	Missing equally retuned PI explicitly stated; final result described as configured-system improvement	Claim narrowed; comparison not performed	Chs. 3, 14
Selection and validation mixed	Study-by-study ledger distinguishes predictor fit, controller selection, frozen transfer and replay	Addressed in manuscript	Ch. 3
Strong local fit overinterpreted	DOB98 range 9796.735–9803.056 rpm stated alongside wider ramp; no global error bound	Claim narrowed	Chs. 13, 14
Verification confused with robustness	Algebra, GP moment checks, nominal trajectories and physical uncertainty assigned separate roles	Addressed; robustness untested	Chs. 3, 14
PILCO terminology too broad	Consistent description as offline two-gain PILCO-style adaptation; planning/deployment differences identified	Addressed in manuscript	Chs. 2, 11
Ringing repair portrayed as superiority	Original/smooth/base comparison and monotonic-command limitation made explicit	Claim narrowed	Chs. 11, 14
Matched PID and FF ablations underemphasized	Added central attribution argument; fixed-reference equivalence and local PID interpretation explained	Addressed in manuscript	Ch. 14
Sensor explanation overstated	Equal-sensor replay separated from untested causal cancellation hypothesis; actual/sensed scoring shown	Claim narrowed	Chs. 12, 14
Archive breadth obscures argument	Technical material consolidated into thematic chapters; explicit cross-study answers precede final conclusions	Addressed in manuscript	Chs. 1–15

Jury issue	Revision and evidence	Status	Location
Candidate's independent mastery	Authorship disclosure and defence questions preserved; no claim that text proves independence	Requires human examination	Front matter; this appendix

A.2 Response to the six master's requirements

A.2.1 Focused question and contribution statement

The revised question concerns attribution within the archived campaign. C1–C4 define the contributions without claiming that any standard component is newly invented. The abstract reports both favorable and limiting evidence. The conclusion answers the same question rather than presenting the last controller as the inevitable winner.

A.2.2 Literature comparison

The new literature chapter explains the relevance of orthogonal identification, established DOB methods, prior GP engine feedforward, original PILCO and smooth-action regularization. It identifies the expected excess-variation construction as an adaptation whose originality has not been established. External papers provide methodological context, not substitute data for this turbine. The review's targeted scope is disclosed.

A.2.3 Training, selection and held-out evaluation

The ledger records dataset roles for the complete experimental progression. In particular, the 10 percent smooth-policy case is a selection case even though part of the original transition record was withheld from GP fitting. The final combined cycle is selection evidence for DOB98 gains and transfer evidence for frozen comparators. The four-second/5 percent check is not called comparative superiority because the full comparator lineup is absent.

A.2.4 Conclusions limited to the tested conditions

Architectural dominance, global robustness, GP uncertainty as a physical error bound, and fuel-economy benefits are not claimed. The local model range and the actual/sensed speed distinction appear in the main argument. Rounding in synthesis tables is explicitly descriptive; full-precision archived results remain available.

A.2.5 Central discussion of discriminating evidence

The synthesis uses the matched-PID result, constant-FF equivalence, GP-to-polynomial substitution, sensor audit and original-gain DOB branch as its principal attribution evidence. It explains the missing comparisons as carefully as the successful ones. This changes the thesis from a catalogue of algorithms into an argument about what the experiments can identify.

A.2.6 Independent defence

The report can provide equations, records and questions, but cannot conduct an oral examination or certify the candidate's independent mastery. This requirement is left open. The candidate should be able to reconstruct the reasoning below and trace numerical claims to the source rather than merely repeat the revised conclusion.

A.3 Defence questions and the reasoning an answer should contain

Why can matched PID reproduce R3 locally? An adequate answer should linearize the inverse map, identify how its derivatives scale static and acceleration terms, and explain the effect of normalization. It should also state why local equivalence does not imply global identity under changing derivatives, fallback or saturation.

What uncertainty does the GP represent? The answer should distinguish posterior uncertainty conditional on nominal data and a kernel from unknown shaft-load mismatch, sensor bias and physical parameter variation. It should explain why low posterior variance cannot alone choose a safe gain direction.

What does the DOB ablation identify? The answer should state the original-gain improvement and the subsequent tuning improvement, then identify the missing tuned-PI-only branch. It should not add percentage reductions with different denominators or interpret sequential differences as architecture-independent causal shares.

Can zero ringing conceal an aggressive command? The answer should give a monotone step as a counterexample, derive zero excess variation from equality between total variation and net change, and distinguish that result from peak slew or fuel mass.

Why is the inverse causal? The answer should account for the delays in the nominal plant and sensor, the two delays included in the Q filter, and the resulting implementable recurrence. It should distinguish stability of that filtered nominal inverse from robust stability of the complete nonlinear loop.

Why do actual and sensed rankings differ? The answer should derive the approximate ramp lag, identify which signal enters feedback and which is scored, and state that identical sensor settings do not imply identical responses to the lag.

What contribution remains after prior art is acknowledged? The answer should identify the comparative findings and their application-specific limits, rather than asserting invention of GP, PI, DOB or smooth-action costs. It should state the limited novelty status of the event-window adaptation.

A.4 Preservation and reproducibility of this revision

Version 1 remains unchanged. The V2 build assembles the archived derivations and figures with new framing, a literature chapter, an evidence ledger, a synthesis and this response. The source bundle retains the original evidence manifest and adds a revision manifest. Compilation and PDF checks do not invoke MATLAB, fit a model or tune a controller. Rebuilding the document is therefore distinct from rerunning the experiments.

The directory register describes the full workspace history, not the contents of this portable document package. The repository study map and companion archive identify the raw experiments and simulation dependencies.

Appendix B

Complete early and fixed-controller result tables

The following tables preserve rejected full trajectories as dashes. They are not imputed as zero, assigned a finite failure penalty, or silently omitted from acceptance counts. R and S denote ramped and stepped *shaft load*; the reference is constant during evaluation. These tables should not be pooled with the later moving-speed-cycle reversal metric.

rpm/%/type	Initial	FF retune	SD down	SD up
9000/5/R	0.3706	0.2512	0.3378	0.2501
9000/10/R	0.7441	0.5045	0.6797	0.5021
9000/20/R	1.581	1.072	1.452	1.066
9000/30/R	2.467	1.671	2.284	1.659
9500/5/R	0.5222	0.3547	0.4601	0.3546
9500/10/R	1.091	0.7414	0.9616	0.7413
9500/20/R	2.284	1.552	2.012	1.552
9500/30/R	3.681	–	–	–
9500/20/S	3.095	–	3.758	–
9500/30/S	–	–	–	–

Table B.1: All early scenarios: speed RMSE (rpm). R: ramped load; S: stepped load. A dash is a rejected full trajectory, not zero (D02).

rpm/%/type	Initial	FF retune	SD down	SD up
9000/5/R	0.369	0.6762	0.2579	0.6901
9000/10/R	0.7169	1.328	0.5075	1.356
9000/20/R	1.09	2.601	0.9559	2.666
9000/30/R	1.345	3.329	1.243	3.419
9500/5/R	0.5508	0.8865	0.442	0.8868
9500/10/R	0.9796	1.961	0.892	1.962
9500/20/R	1.629	4.128	1.74	4.13
9500/30/R	1.958	–	–	–
9500/20/S	61.84	–	13.34	–
9500/30/S	–	–	–	–

Table B.2: All early scenarios: load-edge fuel ringing (lbm/s). R: ramped load; S: stepped load. A dash is a rejected full trajectory, not zero (D02).

B.1 Four fixed controllers

The inverse-controller gains are 2/30 in all four-way comparisons. Their near-equal no-direct-FF and direct-FF scores provide the repeated constant-reference equivalence check. Base PI and base PI + FF retain the inherited 0.025/0.05 gains.

rpm/%/type	P	PF	V	VF
9000/5/R	0.5942	0.6014	0.499	0.499
9000/10/R	1.193	1.208	1.002	1.002
9000/20/R	2.54	2.571	2.131	2.131
9000/30/R	3.973	4.022	3.327	3.327
9500/5/R	1.017	1.049	0.7035	0.7035
9500/10/R	2.125	2.192	1.47	1.47
9500/20/R	4.453	4.593	3.078	3.078
9500/30/R	–	–	–	–
9500/20/S	6.43	6.712	4.054	4.054
9500/30/S	–	–	–	–
9500/5/S	1.484	1.548	0.9304	0.9304

Table B.3: Four fixed controllers: speed RMSE (rpm). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04).

rpm/%/type	P	PF	V	VF
9000/5/R	0.138	0.1397	0.1007	0.1007
9000/10/R	0.2756	0.2791	0.2003	0.2003
9000/20/R	0.5898	0.5987	0.3934	0.3934
9000/30/R	0.9456	0.9571	0.5517	0.5517
9500/5/R	0.1977	0.2056	0.1575	0.1575
9500/10/R	0.417	0.4339	0.3186	0.3186
9500/20/R	0.8916	0.9262	0.629	0.629
9500/30/R	–	–	–	–
9500/20/S	4.705	4.315	9.169	9.169
9500/30/S	–	–	–	–
9500/5/S	1.195	1.089	2.488	2.488

Table B.4: Four fixed controllers: fuel ringing (lbm/s). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04).

rpm/%/type	P	PF	V	VF
9000/5/R	4.095	4.08	5.055	5.055
9000/10/R	4.44	4.395	5.67	5.67
9000/20/R	4.77	4.755	6.33	6.33
9000/30/R	4.965	4.935	6.705	6.705
9500/5/R	4.38	4.26	5.1	5.1
9500/10/R	4.725	4.56	5.7	5.7
9500/20/R	5.055	4.875	6.285	6.285
9500/30/R	–	–	–	–
9500/20/S	2.985	2.745	4.335	4.335
9500/30/S	–	–	–	–
9500/5/S	2.265	2.1	3.165	3.165

Table B.5: Four fixed controllers: persistent recovery to 0.1 rpm (s). P: base PI; PF: base PI + FF; V: inverse PI; VF: inverse PI + FF (D04).

Appendix C

Environmental transfer and inverse-slope audit

Table C.1 gives the complete environmental grid. The reference speed is 9500 rpm, load is 10% of the condition-specific unloaded reference, and R3 includes static FF. Ratios below one favor the challenger for that metric. Fallback percentages must be interpreted with the separate map-range discussion in the main text. A 35/35 paired improvement in this table is not a robustness theorem outside the simulated plant family.

Table C.1: Complete 35-condition transfer audit, 9500 rpm / 10% load steps. The four middle ratios are speed RMSE relative to base PI. V: previous inverse PI; R3 includes static FF. A dash means rejected. FB: R3 fallback percentage (D07).

h	ISA	P RMSE	P ring	V/P	R1/P	R2/P	R3/P	R3 ring/P	FB%
0	+0	3.08	2.371	0.6304	0.8388	0.6832	0.6931	0.7633	0
2500	+0	2.883	2.246	0.6772	0.9051	0.8088	0.692	0.724	0
5000	+0	2.485	2.189	1.035	1.402	2.014	0.6888	0.6968	0
7500	+0	1.874	1.705	1.128	1.528	2.483	0.6878	0.6884	0
10000	+0	1.247	1.281	5.018	6.966	–	0.6825	0.6972	100
0	-10	3.529	2.554	0.5924	0.7858	0.6118	0.6942	0.7504	0
2500	-10	3.223	2.461	0.6668	0.8921	0.7919	0.6967	0.7384	0
5000	-10	2.576	2.283	1.018	1.377	1.92	0.688	0.6997	0
7500	-10	1.86	1.869	16.72	22.1	73.57	0.6837	0.6788	100
10000	-10	1.173	1.26	6.057	8.275	–	0.6801	0.7319	100
0	+10	2.72	2.17	0.6583	0.8777	0.7518	0.6911	0.7578	0
2500	+10	2.517	1.998	0.6862	0.917	0.8203	0.6913	0.7324	0
5000	+10	2.266	1.854	0.7719	1.037	1.069	0.6907	0.7109	0
7500	+10	1.807	1.635	1.17	1.586	2.73	0.6873	0.686	0
10000	+10	1.32	1.301	16.43	19.01	–	0.6847	0.6706	100
0	+20	2.354	1.934	0.6712	0.895	0.7815	0.6897	0.7757	0
2500	+20	2.21	1.854	0.7472	1.002	0.9716	0.6898	0.7374	0
5000	+20	2.008	1.656	0.7728	1.038	1.062	0.6905	0.711	0
7500	+20	1.742	1.557	1.065	1.442	2.261	0.6898	0.6924	0
10000	+20	1.289	1.192	1.318	1.789	3.861	0.6871	0.6815	0
0	-20	4.007	2.866	0.5945	0.7893	0.6185	0.6946	0.7547	0
2500	-20	3.457	2.956	0.8852	1.197	1.393	0.6913	0.7163	0
5000	-20	2.705	2.553	1.461	2.011	–	0.6224	0.9895	98.65
7500	-20	1.756	1.84	8.034	10.94	–	0.6815	0.7089	100
10000	-20	1.111	1.251	6.17	8.438	–	0.678	0.7532	100
0	+30	2.003	1.73	0.6872	0.9162	0.7986	0.688	0.7981	0
2500	+30	1.935	1.65	0.7522	1.008	0.9929	0.689	0.7424	0
5000	+30	1.747	1.488	0.7927	1.065	1.105	0.6899	0.7227	0
7500	+30	1.55	1.334	0.9076	1.225	1.514	0.6892	0.6924	0
10000	+30	1.243	1.141	1.319	1.79	3.932	0.6872	0.6751	0
0	-30	4.532	3.703	0.7908	1.061	1.064	0.6779	0.7823	0
2500	-30	3.583	3.075	0.8843	1.194	1.383	0.6889	0.7107	0
5000	-30	2.577	2.637	15.87	22.36	83.79	0.6828	0.6873	100

h	ISA	P RMSE	P ring	V/P	R1/P	R2/P	R3/P	R3 ring/P	FB%
7500	-30	1.649	1.809	13.46	18.31	–	0.6788	0.7429	100
10000	-30	1.052	1.251	4.714	6.386	–	0.676	0.7828	100

The raw D08 derivative file contains all 36 target/grid operating-point diagnostics, including the nine locations with negative nominal static speed slope. It records both analytic-versus-numerical derivative checks and the corresponding ideal equivalent gains. Negative slopes are a feedback-coordinate concern even where the output fit is accurate; they are not automatically evidence of a wrong numerical derivative.

Appendix D

Polynomial coefficients and full local gain search

The coefficient ordering is intercept, four linear terms, four squares, then six cross terms. Use the input means and scales stated in the full-quadratic-inverse chapter before evaluating the expression. Coefficients have fuel-flow units and are not gains. Full precision is retained in D14 and in the original JSON model; the printed table is rounded for readability.

Term	Coefficient (lbm/s)
1	1.60113254059
z_1	0.119416371262
z_2	0.0540225588644
z_3	-0.342237138763
z_4	0.711670738338
z_1^2	-0.00361356608798
z_2^2	0.000977366025356
z_3^2	-0.0800375317654
z_4^2	0.0232444919616
$z_1 z_2$	0.00100931606766
$z_1 z_3$	0.0366434140839
$z_1 z_4$	0.0629030714395
$z_2 z_3$	0.00136341822536
$z_2 z_4$	-0.000707278500721
$z_3 z_4$	-0.20459203567

Table D.1: Full quadratic coefficients in standardized coordinates. All 15 terms are retained (D14).

D.1 All 9250 rpm tuning candidates

The original gains form candidate 1. Candidates 2–28 are the coarse Cartesian grid; candidates 29–34 are the single coordinate-refinement round. All candidates use the same frozen model and pulse profile. Base PI is excluded from the objective and therefore omitted from this candidate table.

Table D.2: All 34 tested Poly2 gain settings at 9250 rpm. All were numerically accepted; candidate 34 minimizes the stated joint score (D15).

ID	K_p	K_i	K_d	RMSE	Ringing	J
1	0.025	0.1	0.002	1.6658	1.6968	1
2	0.015	0.08	0.001	2.47654	1.05239	1.48669
3	0.025	0.08	0.001	1.9035	0.940295	1.14269
4	0.035	0.08	0.001	1.61308	1.96343	1.15714
5	0.015	0.14	0.001	2.13254	2.30021	1.35562
6	0.025	0.14	0.001	1.58529	1.45065	0.951666
7	0.035	0.14	0.001	1.33132	2.19236	1.29206
8	0.015	0.2	0.001	2.07613	4.27141	2.51734
9	0.025	0.2	0.001	1.46863	2.19493	1.29357
10	0.035	0.2	0.001	1.21753	2.48137	1.46239
11	0.015	0.08	0.002	2.33302	1.26257	1.40054
12	0.025	0.08	0.002	1.82436	1.64788	1.09519
13	0.035	0.08	0.002	1.55644	2.72951	1.60862
14	0.015	0.14	0.002	1.90057	1.72301	1.14094
15	0.025	0.14	0.002	1.46746	1.84422	1.08688
16	0.035	0.14	0.002	1.24958	2.81321	1.65795
17	0.015	0.2	0.002	1.72149	2.34619	1.38272
18	0.025	0.2	0.002	1.30742	2.05752	1.21259
19	0.035	0.2	0.002	1.10962	2.90497	1.71203
20	0.015	0.08	0.003	2.25979	2.44295	1.43974
21	0.025	0.08	0.003	1.78329	3.22949	1.90329
22	0.035	0.08	0.003	1.5279	4.6707	2.75266
23	0.015	0.14	0.003	1.79248	2.71685	1.60116
24	0.025	0.14	0.003	1.40908	3.3398	1.9683
25	0.035	0.14	0.003	1.2097	4.72939	2.78725
26	0.015	0.2	0.003	1.5755	3.01185	1.77502
27	0.025	0.2	0.003	1.23212	3.44905	2.03268
28	0.035	0.2	0.003	1.05902	4.77544	2.81438
29	0.01875	0.14	0.001	1.86401	1.59225	1.11898
30	0.03125	0.14	0.001	1.40929	1.77466	1.04589
31	0.025	0.105	0.001	1.72922	1.09514	1.03807
32	0.025	0.175	0.001	1.50354	1.86286	1.09787
33	0.025	0.14	0.00075	1.63519	1.7329	1.02128
34	0.025	0.14	0.00125	1.54614	1.29904	0.928166

Appendix E

Algorithm registry and reproducibility

E.1 Controller identities

Name in this paper	Defining implementation distinction
Base PI	Direct sensed-speed error; fixed gains 0.025/0.05.
Base PI + FF	Early state-scheduled, equilibrium-subtracted, limited GP correction.
Initial inverse PI	Two inverse queries; dynamic fuel error in P and I; gains 4/40.
FF-retuned inverse	Direct setpoint inverse FF; residual integral; gains 3/60.
SD down / SD up	Same 3/60 base gains multiplied by the stated SD schedule; gain-change integral transfer.
Low-ringing inverse	Fixed gains 2/30 selected at 9500 rpm / 5% steps.
Environmental inverse	Four-input exact GP; same 2/30 gains and direct static FF.
R1	Dynamic virtual error retained; gains 1.5/22.
R2	Static error integrated; attenuated acceleration fuel in proportional path; 16/40, $\beta = 0.05$.
Historical R3	Derivative normalization and guarded fallback, 0.025/0.1/0.002, including static FF.
R3-FB	Same R3 feedback but no direct static FF in command sum.
R3-GP	R3 plus $g(r, 0, T, P)$; equals historical R3 on the covered cycle.
R3-P2	Same R3 architecture and initial gains, replacing inverse values/derivatives by full quadratic regression.
Retuned Poly2	R3-P2 with 0.025/0.14/0.00125 selected at 9250 rpm.

E.2 Primary evidence register

The source archive includes the following numerical tables under `data/`, together with a JSON manifest containing the original relative path and SHA-256 hash. The local archive paths refer to the shared T-MATS workspace. Native MAT/CSV trajectories remain in the original study directories; the paper source archive contains the compact evidence tables and figures required to reproduce the manuscript, not the complete engine installation or every simulation trace.

ID	Archived evidence file
D01	`results/inverse_gpr_20260913_150824_698/metrics.csv`
D02	`results/virtual_fuel_uqup_20260913_231743/comparison_all_methods.csv`
D03	`results/ff_lowgain_20260913_234549/tuning.csv`
D04	`results/four_fixed_20260913_235253/summary.csv`
D05	`results/environment_gpr_20260914_201325/gpr_metrics.csv`
D06	`results/environment_gpr_20260914_201325/comparison_summary.csv`
D07	`results/gpr_remedies_20260914_220633/remedy_comparison.csv`
D08	`results/gpr_remedies_20260914_220633/inverse_slope_diagnosis.csv`
D09	`results/gpr_remedies_20260914_220633/ablation_worker_01.csv`
D10	`results/r3_ambient_cycle_20260915/summary.csv`
D11	`results/r3_covered_cycle_20260915_final/coverage_summary.csv`

ID	Archived evidence file
D12	results/r3_poly2_challenger_20260915/summary.csv
D13	results/poly2_inverse_20260915/fit_metrics.csv
D14	results/poly2_inverse_20260915/coefficients.csv
D15	results/poly2_tune9250_20260915/tuning_scores.csv
D16	results/poly2_tune9250_20260915/validation_summary.csv
D17	results/poly2_tune9250_20260915/selected.csv
D18	results/environment_gpr_20260914_201325/test_200_points.csv
D19	results/r3_poly2_challenger_20260915/coverage_summary.csv
D20	results/poly2_tune9250_20260915/verification.csv
D21	results/virtual_fuel_ff_20260913_194955/same_gains_ablation.csv
D22	results/poly2_inverse_20260915/poly2_model.json

E.3 September-paper V2 review evidence

The companion response records three separate AI role reviews and the author dispositions. The review files are preserved under `reviews/` in the source package. The supplementary `metric_sensitivity.csv` and its trace-hash manifest document the calculations in Section 10.2.4. The six supporting archived traces are included as `review_traces/`; they are unchanged simulator outputs, not additional experiments. The revision changes exposition and derived reporting only. The original paper and original numerical source tables remain preserved.

E.4 Reproducing the historical September paper

This subsection describes the September manuscript, not the present master's revision. To typeset this report, compile `TMATS_Masters_Thesis_V2.tex` three times from the supplied V2 source directory. The older filenames below are preserved for historical reproducibility.

The assembled LaTeX source is `control_evolution_paper.tex`. The original tables are generated from archived CSV files by `build_control_evolution_paper.py`; V2 assembly is performed by `revise_control_evolution_paper.py`. The supplementary sensitivity tables are generated by `review_control_evolution_metrics.py`. New early scheduling figures are generated by `plot_control_evolution.m`; other figures are copied from the finalized stage-specific reports and hashed. Compile the assembled source at least twice with `pdflatex` to resolve chapter, table, figure, and bibliography references. No shell escape or network access is required for LaTeX compilation.

To reproduce simulations rather than typesetting, use the corresponding runners in the existing MATLAB/T-MATS workspace: `run_tmats_virtual_fuel_study.m`, `run_tmats_virtual_fuel_ff_study.m`, the UQ/UQUp runners, `run_tmats_four_fixed_comparison.m`, `run_tmats_environment_comparison.m`, `run_tmats_gpr_remedies.m`, `run_tmats_r3_cycle.m`, and `run_tmats_poly9250.m`. The study-specific setup files determine frozen model paths, load references, startup timing, and gains. Some runners resume existing results; use a new directory when a fresh numerical rerun is intended.

E.5 Evidence exclusions and reader guidance

The preliminary broad-cycle governor diagnostic, the 9400–9600 rpm hull-excursion attempt, and the superseded dynamic-acceleration FF attempt are historical diagnostics, not members of the

final comparisons. Their exclusion is stated because they answer different implementation questions. Rejected plant trajectories in the final scenario banks remain represented in validity counts and as missing full-run scores.

The literature references support method definitions and intellectual context. They do not supply the numerical turbine performance claims, which come from D01–D22. The original doctoral work is cited for lineage only; its engine-airpath results are not transferred to this turbine. The paper makes no claim of external peer review, hardware validation, global optimization, or closed-loop robustness certification.

Appendix F

Complete study-directory register

The register covers every direct directory under the project results folder at assembly time. Counts are file counts, not independent simulations. Aggregate reports and repeated attempts are retained for provenance. A timestamp alone does not establish acceptance. The detailed reports and validity columns control interpretation.

baseline_20260907_202021_071

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

baseline_20260907_202114_161

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

baseline_20260907_202213_678

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

chirp_9000_10000_01_1Hz_60s_20260907_203056_686

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

chirp_request_20260907_203021_394

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 10. Entry files: `chirp_data.csv`, `diagnostic_signals.csv`, `README.md`, `requested_command.csv`, `summary.json`.

Comprehensive_Observer_Tutorial

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 4. Entry files: `parameter_derivation.json`, `PARAMETER_DERIVATION.md`, `source_sha256.json`, `TUTORIAL_TEXT.md`.

dob98_pi_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 53. Entry files: `base.csv`, `case_01.csv`, `case_02.csv`, `case_03.csv`, `case_04.csv`.

dob_final_20260912_175308_755

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 20. Entry files: `case_01.csv`, `case_02.csv`, `case_03.csv`, `case_04.csv`, `case_05.csv`.

dob_pilot_20260912_174947_383

Empty pilot directory; no empirical result. Archived file count: 0.

dob_pilot_20260912_175035_035

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: `case_01.csv`, `case_02.csv`, `case_03.csv`, `case_04.csv`, `metrics.csv`.

dob_pto_20260912_182940_001

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 14. Entry files: `case_01.csv`, `case_02.csv`, `events.csv`, `metrics.csv`, `metrics.json`.

dob_pto_20pct_20260912_183919_021

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: `case_01.csv`, `case_02.csv`, `events.csv`, `metrics.csv`, `metrics.json`.

dob_pto_30pct_20260912_184142_273

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: `case_01.csv`, `case_02.csv`, `events.csv`, `metrics.csv`, `metrics.json`.

dob_pto_9000rpm_05pct_20260912_201537_806

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: `case_01.csv`, `case_02.csv`, `events.csv`, `metrics.csv`, `metrics.json`.

dob_pto_9000rpm_10pct_20260912_201600_313

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: case_01.csv, case_02.csv, events.csv, metrics.csv, metrics.json.

dob_pto_9000rpm_20pct_20260912_201611_189

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: case_01.csv, case_02.csv, events.csv, metrics.csv, metrics.json.

dob_pto_9000rpm_30pct_20260912_201622_343

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: case_01.csv, case_02.csv, events.csv, metrics.csv, metrics.json.

dob_pto_9500rpm_20pct_20260912_195059_980

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: case_01.csv, case_02.csv, events.csv, metrics.csv, metrics.json.

dob_pto_9500rpm_30pct_20260912_195120_747

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: case_01.csv, case_02.csv, events.csv, metrics.csv, metrics.json.

dob_refined_20260912_175532_449

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 29. Entry files: case_01.csv, case_02.csv, case_03.csv, case_04.csv, case_05.csv.

environment_gpr_20260914_201325

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 305. Entry files: comparison_01_BasePI.csv, comparison_01_InvGPR_PI_GPRFF.csv, comparison_02_BasePI.csv, comparison_02_InvGPR_PI_GPRFF.csv, comparison_03_BasePI.csv.

exact_gpr_20260913_145742_458

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: metrics.csv, REPORT.md, test_100_points.csv, TMATS_EXACT_GPR_USAGE.md, training_points.csv.

ff_lowgain_20260913_234549

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 33. Entry files: candidate_01.csv, candidate_02.csv, candidate_03.csv, candidate_04.csv, candidate_05.csv.

four_fixed_20260913_235253

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 127. Entry files: 9000rpm_05pct_ramps_BasePI.csv, 9000rpm_05pct_ramps_BasePI_GPRFF.csv, 9000rpm_05pct_ramps_InvGPR_PI.csv, 9000rpm_05pct_ramps_InvGPR_PI_GPRFF.csv, 9000rpm_10pct_ramps_BasePI.csv.

gpr_ff_benchmarks_20260913_173407_036

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 136. Entry files: summary.csv, summary.json.

gpr_remedies_20260914_220633

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 403. Entry files: ablation_e36_c78.csv, ablation_implementation_checks.csv, ablation_worker_01.csv, inverse_slope_diagnosis.csv, remedy_comparison.csv.

inverse_gpr_20260913_150824_698

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: metrics.csv, REPORT.md, test_100_points.csv, TMATS_INVERSE_GPR_USAGE.md, training_points.csv.

literature_observers_9000rpm_10pct_20260912_211427_977

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: configuration.json, DOB.csv, GPIO.csv, LESO.csv, metrics.csv.

literature_observers_9000rpm_10pct_20260912_211600_355

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 22. Entry files: `configuration.json`, `DOB.csv`, `events.csv`, `events.json`, `GPIO.csv`.

literature_observers_9500rpm_10pct_20260912_213607_364

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 13. Entry files: `configuration.json`, `DOB.csv`, `GPIO.csv`, `LESO.csv`, `metrics.csv`.

Observer_9000_9500_Comparison

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 22. Entry files: `events.csv`, `events.json`, `events_before_GPR_FF.csv`, `events_before_GPR_FF.json`, `parameter_derivation.json`.

Observer_9000_9500_GPR_FF_Comparison

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 21. Entry files: `all_scenarios.csv`, `all_scenarios.json`, `events.csv`, `events.json`, `failures.csv`.

pilco_combined_cycle_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 24. Entry files: `base.csv`, `base_sensor_audit.json`, `comparison.csv`, `poly.csv`, `poly_sensor_audit.json`.

pilco_quadratic_pi_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 45. Entry files: `comparison.csv`, `gp_holdout.csv`, `moment_checks.json`, `pilco_10pct.csv`, `pilco_5pct.csv`.

pilco_ramp90_100_5s_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 19. Entry files: `base.csv`, `comparison.csv`, `poly.csv`, `previous.csv`, `simulation_summary.csv`.

pilco_ramp90_equal_sensor_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 22. Entry files: `base.csv`, `base_sensor_audit.json`, `comparison.csv`, `poly.csv`, `poly_sensor_audit.json`.

pilco_smooth_20261010

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 70. Entry files: `candidates.csv`, `candidate_00.csv`, `candidate_01.csv`, `candidate_02.csv`, `candidate_03.csv`.

poly2_inverse_20260915

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 7. Entry files: `coefficients.csv`, `fit_metrics.csv`, `implementation_checks.csv`, `normalization.csv`, `poly2_model.json`.

poly2_tune9250_20260915

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 90. Entry files: `candidate_00.csv`, `candidate_01.csv`, `candidate_02.csv`, `candidate_03.csv`, `candidate_04.csv`.

prbs_9000_10000_60s_20260907_234637_780

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

prbs_9000_10000_60s_20260907_234849_576

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

prbs_validation_20260907_234554_997

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 3. Entry files: `prbs_command.csv`.

prbs_validation_20260907_234805_753

Excitation request or diagnostic record. Full-amplitude runs fail; only explicitly accepted prefixes are usable. Archived file count: 13. Entry files: `all_signals.csv`, `diagnostic_signals.csv`, `prbs_command.csv`, `README.md`, `summary.json`.

PTO_20_30_comparison

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 5. Entry files: `REPORT.md`, `source_sha256.json`, `summary.csv`, `summary.json`.

PTO_9000rpm_ramp_comparison

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 11. Entry files: `events.csv`, `events.json`, `REPORT.md`, `source_sha256.json`, `summary.csv`.

PTO_9500rpm_comparison

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 5. Entry files: `REPORT.md`, `source_sha256.json`, `summary.csv`, `summary.json`.

r3_ambient_cycle_20260915

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 19. Entry files: `BasePI.csv`, `BasePI_request_diagnostic.csv`, `BasePI_segments.csv`, `R3.csv`, `R3_segments.csv`.

r3_covered_cycle_20260915

Intermediate or superseded interpretation/scenario; use the final covered-cycle and static-feedforward studies for conclusions. Archived file count: 20. Entry files: `BasePI.csv`, `BasePI_coverage_queries.csv`, `BasePI_segments.csv`, `coverage_summary.csv`, `R3.csv`.

r3_covered_cycle_20260915_final

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 20. Entry files: `BasePI.csv`, `BasePI_coverage_queries.csv`, `BasePI_segments.csv`, `coverage_summary.csv`, `R3.csv`.

r3_dynamic_ff_20260915

Intermediate or superseded interpretation/scenario; use the final covered-cycle and static-feedforward studies for conclusions. Archived file count: 7. Entry files: `BasePI.csv`, `BasePI_segments.csv`, `R3.csv`, `R3_segments.csv`, `summary.csv`.

r3_poly2_challenger_20260915

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 30. Entry files: `BasePI.csv`, `BasePI_coverage_queries.csv`, `BasePI_segments.csv`, `coverage_summary.csv`, `R3.csv`.

r3_static_ff_20260915

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 26. Entry files: `BasePI.csv`, `BasePI_coverage_queries.csv`, `BasePI_segments.csv`, `coverage_summary.csv`, `R3.csv`.

ramp_selection_stretch1_20260908_214642_454

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

ramp_validation_stretch1_20260908_214653_124

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

rls_acceleration_20260908_225810_094

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 24. Entry files: `final_arx_coefficients.json`, `forgetting_factor_sensitivity.csv`, `frozen_ramp_validation.csv`, `full_chirp_gate_diagnostics.csv`, `metrics.json`.

rls_blocks_acceleration_20260909_063720_926

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 1.

rls_blocks_acceleration_20260909_063846_871

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 33. Entry files: `basic_blocks_audit.json`, `final_arx_coefficients.json`, `forgetting_factor_sensitivity.csv`, `frozen_ramp_validation.csv`, `full_chirp_gate_diagnostics.csv`.

step_10rpm_20260907_202219_989

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

step_smoke_20260907_202121_527

Plant preparation, numerical acceptance, or identification input record. See the early experiment narrative for accepted and rejected variants. Archived file count: 4. Entry files: `data.csv`, `metadata.json`.

Turkce_Gozlemci_Makale

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 10. Entry files: `parameter_derivation.json`, `RAPOR_TR.md`, `source_sha256.json`.

Turkce_Gozlemci_Ogretici_Rapor

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 7. Entry files: `parameter_derivation.json`, `RAPOR_TR.md`, `source_sha256.json`.

Turkce_Gozlemci_V2

Derived comparison, tutorial, or presentation collection; not an independent set of plant trials. Archived file count: 10. Entry files: `audit.json`, `findings.json`, `HAKEM_YANITLARI.md`, `parameter_derivation.json`, `RAPOR_TR.md`.

two_stage_rls_20260912_124851_753

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 10. Entry files: `candidate_path.csv`, `coefficients.json`, `comparison.csv`, `metrics.json`, `REPORT.md`.

two_stage_rls_20260912_125027_844

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 14. Entry files: `candidate_path.csv`, `coefficients.json`, `comparison.csv`, `metrics.json`, `REPORT.md`.

virtual_fuel_20260913_182229_028

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 121. Entry files: `9000rpm_05pct_ramps_PI.`

csv, 9000rpm_05pct_ramps_VirtualFuel.csv, 9000rpm_10pct_ramps_PI.csv, 9000rpm_10pct_ramps_VirtualFuel.csv, 9000rpm_20pct_ramps_PI.csv.

virtual_fuel_ff_20260913_194955

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 89. Entry files: 9000rpm_05pct_ramps_PI.csv, 9000rpm_05pct_ramps_VirtualFuelFF.csv, 9000rpm_10pct_ramps_PI.csv, 9000rpm_10pct_ramps_VirtualFuelFF.csv, 9000rpm_20pct_ramps_PI.csv.

virtual_fuel_uq_20260913_201212

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 46. Entry files: 9000rpm_05pct_ramps_VirtualFuelUQ.csv, 9000rpm_10pct_ramps_VirtualFuelUQ.csv, 9000rpm_20pct_ramps_VirtualFuelUQ.csv, 9000rpm_30pct_ramps_VirtualFuelUQ.csv, 9500rpm_05pct_ramps_VirtualFuelUQ.csv.

virtual_fuel_uqup_20260913_231743

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 42. Entry files: 9000rpm_05pct_ramps_VirtualFuelUQUp.csv, 9000rpm_10pct_ramps_VirtualFuelUQUp.csv, 9000rpm_20pct_ramps_VirtualFuelUQUp.csv, 9000rpm_30pct_ramps_VirtualFuelUQUp.csv, 9500rpm_05pct_ramps_VirtualFuelUQUp.csv.

wiener_ramps_20260908_214558_458

Study evidence; acceptance and tuning/validation roles are defined in the corresponding narrative and saved metrics. Archived file count: 26. Entry files: candidate_scores.csv, fit_metrics.json, independent_checks.json, model_coefficients.json, REPORT.md.

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